
UNIT 1 FUNDAMENTALS OF DATA WAREHOUSE

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1.0 INTRODUCTION

A database often contains information or data collection that is generally stored electronically in a computer system. It is easy to access, manage, modify, update, monitor, and organize the data. Data is stored in the tables of the database.

The process of consolidating data and analyzing it to obtain some insights has been around for centuries, but we just recently began referring to this as data warehousing. Any operational or transactional system is only designed with its own functionality and hence, it could handle limited amounts of data for a limited amount of time. The operational systems are not designed or architected for long term data retention as the historical data is little to no importance to them. However, to gain a point-in-time visibility and understand the high-level operational aspects of any business, the historical data plays a vital role. With the emergence of matured Relational Database Management Systems (RDBMS) in 1960s, engineers across various enterprises started

architecting ways to copy the data from the transactional systems over to different databases via manual or automated mechanism and use it for reporting and analysis. As the data in the transactional systems would get purged periodically, it would not be the case in these analytical repositories as their purpose was to store as much data as possible; hence the word “data warehouse” came into existence because these repositories would become a warehouse for the data.

Data Warehousing (DW) as a practice became very prominent during late 80s when the enterprises started building decision support systems that were mainly responsible to support reporting. As there was a rapid advancement in the performance of these relational database during late 1990s and early 2000s, Data Warehousing became a core part of the Information Technology group across large enterprises. In fact, some of the vendors like *Netezza*, *Teradata* started offering customized hardware to manage data warehouse architectures within state-of-the-art machines. Data Warehousing had evolved to be on top of the list of priorities since mid 2000s. Data supply chain ecosystem has grown exponentially in the current world and so is the way enterprises architect their data warehouses.

A well architected data warehouse serves as an extended vision for the enterprise where multiple departments can gain actionable insights to manage key business decisions that could drive operational excellence or revenue generating opportunities for the enterprise.

This unit covers the basic features of data warehousing, its evolution, characteristics, online transaction processing (OLTP), online analytical processing, popular platforms and applications of data warehouses.

1.1 OBJECTIVES

After going through this unit, you shall be able to:

- understand the evolution of data warehouse;
- describe various characteristics of data warehouse;
- describe the benefits and applications of a data warehouse;
- discuss the significance of metadata in data warehouse;
- list and discuss the types of data warehouses, and
- identify the popular data warehouse platforms;

1.2 EVOLUTION OF DATA WAREHOUSE

The relational database revolution in the early 1980s ushered in an era of improved access to the valuable information contained deep within data. It was soon discovered that databases modeled to be efficient at transactional processing were not always optimized for complex reporting or analytical needs.

In fact, the need for systems offering decision support functionality predates the first relational model and SQL. But the practice known today as Data Warehousing really saw its genesis in the late 1980s. An IBM Systems Journal article published in 1988, *An architecture for a business information system* coined the term “business data warehouse,” although a future progenitor of the practice, Bill Inmon, used a similar term in the 1970s. Considered by many to be the Father of Data Warehousing, Bill Inmon, an American Computer Scientist is first began to discuss the principles around the Data Warehouse and even coined the term. Throughout the latter 1970s into the 1980s, Inmon worked extensively as a data professional, honing his expertise in all manners of relational Data Modeling. Inmon’s work as a Data Warehousing pioneer took off in the early 1990s when he ventured out on his own, forming his first company, Prism Solutions. One of Prism’s main products was the Prism Warehouse Manager, one of the first industry tools for creating and managing a Data Warehouse.

In 1992, Inmon published *Building the Data Warehouse*, one of the seminal volumes of the industry. Later in the 1990s, Inmon developed the concept of the Corporate Information Factory, an enterprise level view of an organization’s data of which Data Warehousing plays one part. Inmon’s approach to Data Warehouse design focuses on a centralized data repository modeled to the third normal form. Inmon’s approach is often characterized as a top-down approach. Inmon feels using strong relational modeling leads to enterprise-wide consistency facilitating easier development of individual data marts to better serve the needs of the departments using the actual data. This approach differs in some respects to the “other” father of Data Warehousing, Ralph Kimball.

While Inmon’s *Building the Data Warehouse* provided a robust theoretical background for the concepts surrounding Data Warehousing, it was Ralph Kimball’s *The Data Warehouse Toolkit*, first published in 1996, that included a host of industry-honed, practical examples for OLAP-style modeling. Kimball, on the other hand, favors the development of individual data marts at the departmental level that get integrated together using the Information Bus architecture. This bottom up approach fits-in nicely with Kimball’s preference for star-schema modeling. Both approaches remain core to Data Warehousing architecture as it stands today. Smaller firms might find Kimball’s data mart approach to be easier to implement with a constrained budget. Dimensional modeling in many cases is easier for the end user to understand.

According to Bill Inmon, “A warehouse is a subject-oriented, integrated, time-variant and non-volatile collection of data in support of management’s decision making process”.

According to Ralph Kimball, “Data warehouse is the conglomerate of all data marts within the enterprise. Information is always stored in the dimensional model”.

1.3 DATA WAREHOUSING AND ITS NEED

Data Warehouse is used to collect and manage data from various sources, in order to provide meaningful business insights. A data warehouse is usually used for linking and analyzing heterogeneous sources of business data. The data warehouse is the center of the data collection and reporting framework developed for the BI system. Data warehouse systems are real-time repositories of information, which are likely to be tied to specific applications. Data warehouses gather data from multiple sources (including databases), with an emphasis on storing, filtering, retrieving and in particular, analyzing huge quantities of organized data. The data warehouse operates in information-rich environment that provides an overview of the company, makes the current and historical data of the company available for decisions, enables decision support transactions without obstructing operating systems, makes information consistent for the organization, and presents a flexible and interactive information source.

1.3.1 Need for Data Warehouse

Data warehouses are used extensively in the largest and most complex businesses around the world. In demanding situations, good decision making becomes critical. Significant and relevant data is required to make decisions. This is possible only with the help of a well-designed data warehouse. Following are some of the reasons for the need of Data Warehouses:

Enhancing the turnaround time for analysis and reporting: Data warehouse allows business users to access critical data from a single source enabling them to take quick decisions. They need not waste time retrieving data from multiple sources. The business executives can query the data themselves with minimal or no support from IT which in turn saves money and time.

Improved Business Intelligence: Data warehouse helps in achieving the vision for the managers and business executives. Outcomes that affect the strategy and procedures of an organization will be based on reliable facts and supported with evidence and organizational data.

Benefit of historical data: Transactional data stores data on a day to day basis or for a very short period of duration without the inclusion of historical data. In comparison, a data warehouse stores large amounts of historical data which enables the business to include time-period analysis, trend analysis, and trend forecasts.

Standardization of data: The data from heterogeneous sources are available in a single format in a data warehouse. This simplifies the readability and accessibility of data. For example, gender is denoted as Male/ Female in Source 1 and m/f in Source 2 but in a data warehouse the gender is stored in a format which is common across all the businesses i.e. M/F.

Immense ROI (Return On Investment): Return On Investment refers to the additional revenues or reduces expenses a business will be able to realize from any project.

Now, let us study the benefits.

1.3.2 Benefits of Data Warehouse

Several enterprises adopt data warehousing as it offers many benefits, such as streamlining the business and increasing profits. Following are some of the benefits of having a data warehouse:

Scalability - Businesses today cannot survive for long if they cannot easily expand and scale to match the increase in the volume of daily transactions. DW is easy to scale, making it easier for the business to stride ahead with minimum hassle.

Access to Historical Insights - Though real-time data is important, historical insights cannot be ignored when tracing patterns. Data warehousing allows businesses to access past data with just a few clicks. Data that are months and years old can be stored in the warehouse.

Works On-Premises and on Cloud - Data warehouses can be built on-premises or on cloud platforms. Enterprises can choose either option, depending on their existing business system and the long-term plan. Some businesses rely on both.

Better Efficiency - Data warehousing increases the efficiency of the business by collecting data from multiple sources and processing it to provide reliable and actionable insights. The top management uses these insights to make better and faster decisions, resulting in more productivity and improved performance.

Improved Data Security - Data security is crucial in every enterprise. By collecting data in a centralized warehouse, it becomes easier to set up a multi-level security system to prevent the data from being misused. Provide restricted access to data based on the roles and responsibilities of the employees.

Increase Revenue and Returns - When the management and employees have access to valuable data analytics, their decisions and actions will strengthen the business. This increases the revenue in the long run.

Faster and Accurate Data Analytics - When data is available in the central data warehouse, it takes less time to perform data analysis and generate reports. Since the data is already cleaned and formatted, the results will be more accurate.

Let us study the various approaches in detail in the following section.

1.4 DATA WAREHOUSE DESIGN APPROACHES

Data Warehouse design approaches are very important aspect of building data warehouse. Selection of right data warehouse design could save lot of time and project cost.

There are two different Data Warehouse Design Approaches normally followed when designing a Data Warehouse solution and based on the requirements of your project you can choose which one suits your particular scenario. These methodologies are a result of research from Bill Inmon (Top-Down Approach) and Ralph Kimball(Bottom up Approach).

1.4.1 Top-down Approach

Bill Inmon's design methodology is based on a top-down approach which is illustrated in the Figure 1. In the top-down approach, the data warehouse is designed first and then data mart is built on top of data warehouse.

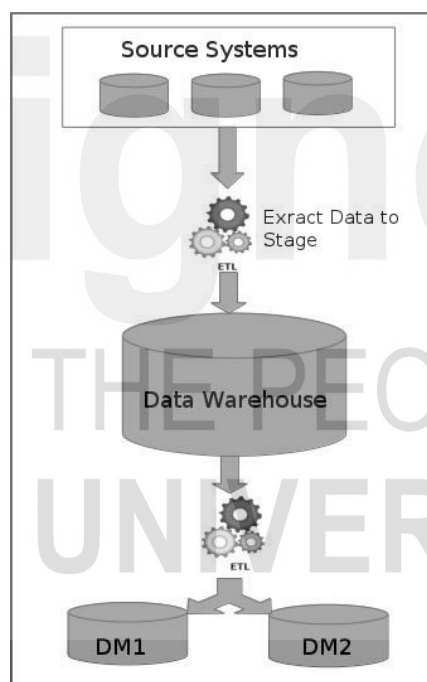


Figure 1: Top-Down DW Design Approach

Below are the steps that are involved in top-down approach:

- Data is extracted from the various source systems. The extracts are loaded and validated in the stage area. Validation is required to make sure the extracted data is accurate and correct. You can use the ETL tools or approach to extract and push to the data warehouse.
- Data is extracted from the data warehouse in regular basis in stage area. At this step, you will apply various aggregation, summarization techniques on extracted data and loaded back to the data warehouse.
- Once the aggregation and summarization is completed, various data marts extract that data and apply the some more transformation to make the data structure as defined by the data marts.

1.4.2 Bottom-up Approach

Ralph Kimball's data warehouse design approach is called dimensional modelling or the Kimball methodology which is illustrated in Figure 2. This methodology follows the bottom-up approach.

As per this method, data marts are first created to provide the reporting and analytics capability for specific business process, later with these data marts enterprise data warehouse is created.

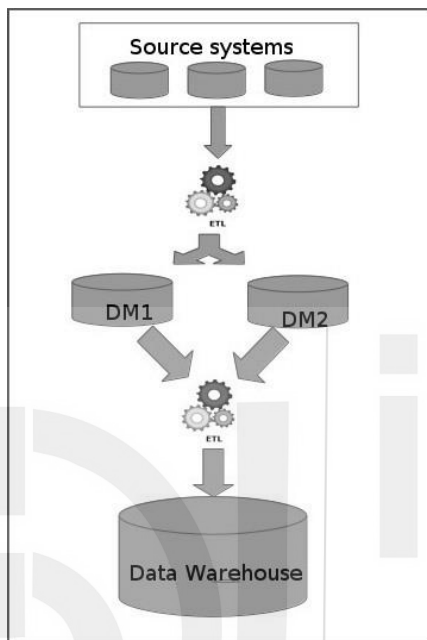


Figure 2: Bottom-Up DW Design Approach

Basically, Kimball model reverses the Inmon model i.e. Data marts are directly loaded with the data from the source systems and then ETL process is used to load in to Data Warehouse. The above image depicts how the top-down approach works.

Below are the steps that are involved in bottom-up approach:

- The data flow in the bottom up approach starts from extraction of data from various source systems into the stage area where it is processed and loaded into the data marts that are handling specific business process.
- After data marts are refreshed the current data is once again extracted in stage area and transformations are applied to create data into the data mart structure. The data is the extracted from Data Mart to the staging area is aggregated, summarized and so on loaded into EDW and then made available for the end user for analysis and enables critical business decisions.

Having discussed the data warehouse design strategies, let us study the characteristics of the DW in the next section.

1.5 CHARACTERISTICS OF A DATA WAREHOUSE

Data warehouses are systems that are concerned with studying, analyzing and presenting enterprise data in a way that enables senior management to make decisions. The data warehouses have four essential characteristics that distinguish them from any other data and these characteristics are as follows:

- **Subject-oriented**

A DW is always a subject-oriented one, as it provides information about a specific theme instead of current organizational operations. On specific themes, it can be done. That means that it is proposed to handle the data warehousing process with a specific theme (subject) that is more defined. Figure 3 shows Sales, Products, Customers and Account are the different themes.

A data warehouse never emphasizes only existing activities. Instead, it focuses on data demonstration and analysis to make different decisions. It also provides an easy and accurate demonstration of specific themes by eliminating information that is not needed to make decisions.

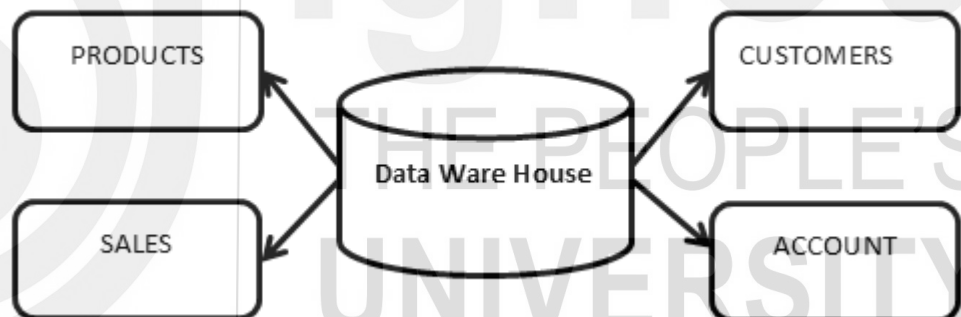


Figure 3: Subject-oriented Characteristic Feature of a DW

- **Integrated**

Integration involves setting up a common system to measure all similar data from multiple systems. Data was to be shared within several database repositories and must be stored in a secured manner to access by the data warehouse. A data warehouse integrates data from various sources and combines it in a relational database. It must be consistent, readable, and coded.. The data warehouse integrates several subject areas as shown in the figure 4.

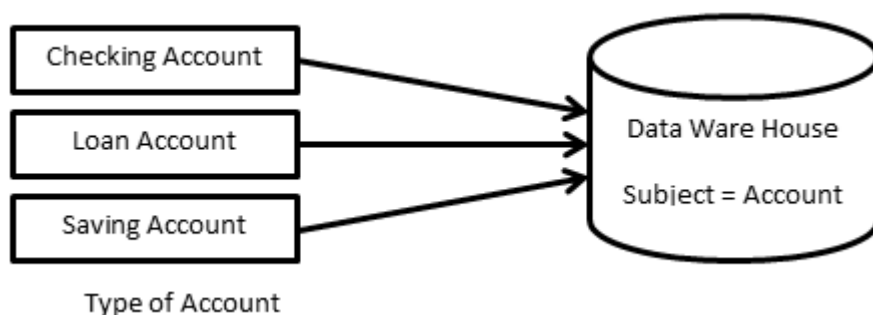


Figure 4: Integrated Characteristic Feature of a DW

- **Time-Variant**

Information may be held in various intervals such as weekly, monthly, and yearly as shown in Figure 5. It provides a series of limited-time, variable rate, online transactions. The data warehouse covers a broader range of data than the operational systems. When the data stored in the data store has a certain amount of time, it can be predictable and provide history. It has aspects of time embedded within it. One other facet of the data warehouse is that the data cannot be changed, modified or updated once it is stored.

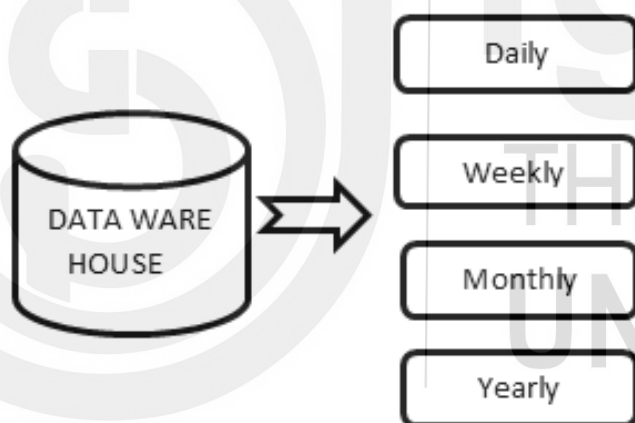


Figure 5: Time- Variant Characteristic Feature of a DW

- **Non-Volatile**

The data residing in the data warehouse is permanent, as the name non - volatile suggests. It also ensures that when new data is added, data is not erased or removed. It requires the mammoth amount of data and analyses the data within the technologies of warehouse. Figure 6 shows the non-volatile data warehouse vs operational database. A data warehouse is kept separate from the operational database and thus the data warehouse does not represent regular changes in the operational database. Data warehouse integration manages different warehouses relevant to the topic.

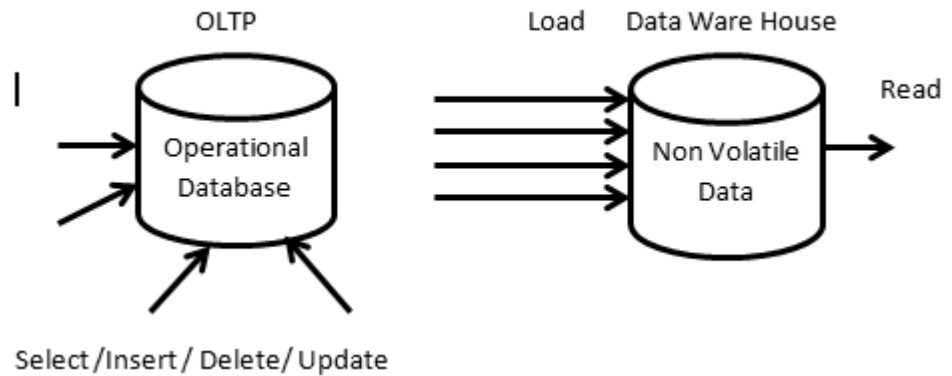


Figure 6: Non –Volatile Characteristic Feature of DW

1.5.1 How Data Warehouse Works?

A data warehouse is a central repository in which one or more sources of information are collected. Data in the data warehouse may be Structured, Semi-structured, or Unstructured. Data are processed, transformed, and accessed by end users for use in business intelligence reporting and decision-making. A data warehouse integrates disparate primary sources into a comprehensive source. Through the integration of all this information, an organization can maintain a more holistic level of customer service. This ensures that all available data is properly considered. Data warehouse enables data mining to find patterns of information that increases profits.

The figure 7 shows the important components of the data warehouse.

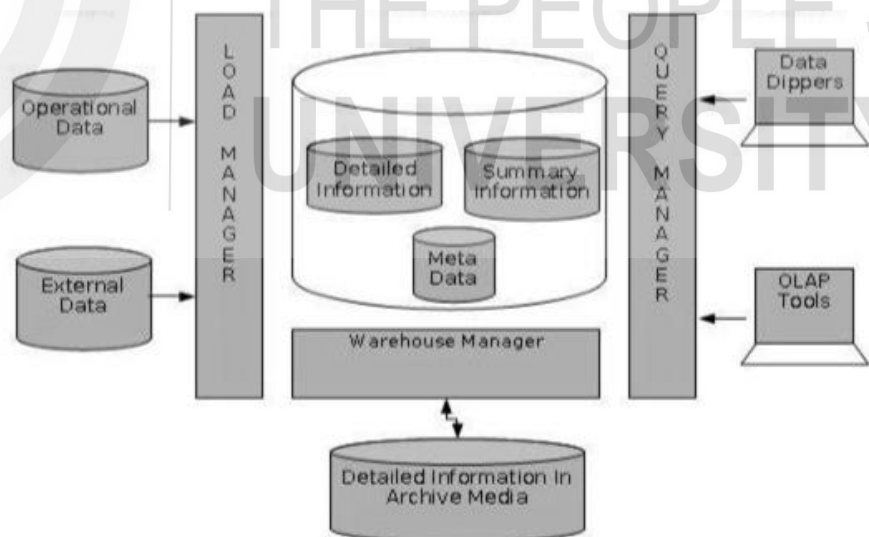


Figure 7: Components of a Data Warehouse

- **Load Manager**

Load Manager Component of data warehouse is responsible for collection of data from operational system and converts them into usable form for the users. This component is responsible for importing and exporting data from operational systems. This component includes all of the programs and

applications interfaces that are responsible for pooling the data out of the operational system, preparing it, loading it into warehouse itself it performs the following tasks such as identification of data, validation of data about the accuracy, extraction of data from original source, cleansing of data by eliminating meaningless values and making it usable, data formatting, data standardization by getting them into a consistent form, data merging by taking data from different sources and consolidating into one place and establishing referential integrity.

- **Warehouse Manager**

The warehouse manager is the center of data-warehousing system and is the data warehouse itself. It is a large, physical database that holds a vast amount of information from a wide variety of sources. The data within the data warehouse is organized such that it becomes easy to find, use and update frequently from its sources.

- **Query Manager**

Query Manager Component provides the end-users with access to the stored warehouse information through the use of specialized end-user tools. Data mining access tools have various categories such as query and reporting, on-line analytical processing (OLAP), statistics, data discovery and graphical and geographical information systems.

- **End-user access tools**

This is divided into the following categories, such as:

- Reporting Data
- Query Tools
- Data Dippers
- Tools for EIS
- Tools for OLAP and tools for data mining.

☞ Check Your Progress 1

1) What is a Data Warehouse and why is it important?

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2. Mention the characteristics of a Data Warehouse.

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1.6 OLTP AND OLAP

Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP) are the two terms which look similar but refer to different kinds of systems. Online transaction processing (OLTP) captures, stores, and processes data from transactions in real time. Online analytical processing (OLAP) uses complex queries to analyze aggregated historical data from OLTP systems.

1.6.1 Online Transaction Processing (OLTP)

An OLTP system captures and maintains transaction data in a database. Each transaction involves individual database records made up of multiple fields or columns. Examples include banking and credit card activity or retail checkout scanning.

In OLTP, the emphasis is on fast processing, because OLTP databases are read, written, and updated frequently. If a transaction fails, built-in system logic ensures data integrity.

1.6.2 Online Analytical Processing (OLAP)

Online Analytical Processing (OLAP) applies complex queries to large amounts of historical data, aggregated from OLTP databases and other sources, for data mining, analytics, and business intelligence projects. In OLAP, the emphasis is on response time to these complex queries. Each query involves one or more columns of data aggregated from many rows.

Examples include year-over-year financial performance or marketing lead generation trends. OLAP databases and data warehouses give analysts and decision-makers the ability to use custom reporting tools to turn data into information. Query failure in OLAP does not interrupt or delay transaction processing for customers, but it can delay or impact the accuracy of business intelligence insights.

OLTP is operational, while OLAP is informational. A glance at the key features of both kinds of processing illustrates their fundamental differences, and how they work together. The table (Table 1) below summarizes differences between OLTP and OLAP.

Table 1: OLTP Vs OLAP

	OLTP	OLAP
Characteristics	Handles a large number of small transactions	Handles large volumes of data with complex queries
Query types	Simple standardized queries	Complex queries
Operations	Based on INSERT, UPDATE, DELETE commands	Based on SELECT commands to aggregate data for reporting

Response time	Milliseconds	Seconds, minutes, or hours depending on the amount of data to process
Design	Industry-specific, such as retail, manufacturing, or banking	Subject-specific, such as sales, inventory, or marketing
Source	Transactions	Aggregated data from transactions
Purpose	Control and run essential business operations in real time	Plan, solve problems, support decisions, discover hidden insights
Data updates	Short, fast updates initiated by user	Data periodically refreshed with scheduled, long-running batch jobs
Space requirements	Generally small if historical data is archived	Generally large due to aggregating large datasets
Backup and recovery	Regular backups required to ensure business continuity and meet legal and governance requirements	Lost data can be reloaded from OLTP database as needed in lieu of regular backups
Productivity	Increases productivity of end users	Increases productivity of business managers, data analysts, and executives
Data view	Lists day-to-day business transactions	Multi-dimensional view of enterprise data
User examples	Customer-facing personnel, clerks, online shoppers	Knowledge workers such as data analysts, business analysts, and executives
Database design	Normalized databases for efficiency	Denormalized databases for analysis

OLTP provides an immediate record of current business activity, while OLAP generates and validates insights from that data as it's compiled over time. That historical perspective empowers accurate forecasting, but as with all business intelligence, the insights generated with OLAP are only as good as the data pipeline from which they emanate.

☞ Check Your Progress 2

1) Why a data warehouse is separated from Operational Databases?

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2) Mention the key differences between a database and a data warehouse.

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1.7 DATA GRANULARITY

Granularity is one of the main elements in the modeling of DW data. Granularity of data refers to detail levels. Multiple levels of detail may be available depending on the requirements. At least two granular levels exist for many data warehouses. The relation between detailing and granularity is important to understand. It means greater detail of the data (less summary) when you speak of less granularity or fine granularity. Greater granularity means fewer details or gross granularity (greater summarization). The operational data is stored at the lowest level of information. The sale units will be stored and stored in the outlet system at the unit level of product per transaction. The amount ordered is collected and stored by the customer at unit level per order in the order entry system. You can add up the individual transactions whenever you need the summary data. When you search for items in a product ordered this month and add them together, you have the total of all product orders entered in that month. Referral data is generally not tracked in an operating system. When a user requests an analysis in the data warehouse, the user has to view summary data first. A user can successfully promote the entire product unit throughout a given area. The user may then wish to examine the region's breakdowns. The next step might be to look at the following levels of the sales units in each store. The analysis often starts at a high level, and is reduced in detail.

Therefore, in a data store, the summary of data at different levels can be maintained effectively. You can provide answers at the simplest level or the most detailed level. The detail level of data is the level of granularity in an existing data warehouse. The more detail in the data, the finer the size of the data. You need to save a lot of permanent data in the data warehouse in order to safely store data. The type of granularity depends on how data will be processed, and the performance expectations. For example, each year details of each month, day, hour, minute, second, and so forth are available.

1.8 METADATA AND DATA WAREHOUSING

In a data warehouse, data is stored using a common schema controlled by a common dictionary. Within the data dictionary, data is kept about the logical data structures, file and address data, index information and others. The metadata should contain the following data warehouse information:

- The data structure based on the programmer's view
- Data structure based on DSS analysts' view
- The DW's data sources
- The data transformation at the moment of its migration to DW
- Model of data
- The connection between the data model and the DW
- Data extraction history

In the DW environment, metadata is a major component. Metadata helps to control reporting accuracy, validates the transformation of data and ensures calculation accuracy. Metadata also complies with the company end-users' definition of business terms. More details on metadata are provided in the next Unit.

1.9 DATA WAREHOUSING APPLICATIONS

In different sectors, there are numerous applications such as e-commerce, telecommunication, transport, marketing, distribution and retail. Given below are some of the applications of data warehouses:

Investment and Insurance: In this sector, data warehousing is used to analyze the customer, market trends and other patterns of data. The two sub-sectors where data warehousing plays an important role are Forex and stock markets.

Healthcare: A data warehousing system is used to forecast outcomes of a treatment generate its reports and share the data with different units. These units can be the research labs, medical units, and insurance providers. Enterprise data warehouses serve as the backbone of healthcare systems as they are updated with recent information which is crucial for saving lives.

Retail: Be it distribution, marketing, examining pricing policies, keeping a track of promotional deals, and finding the pattern in the customer buying trends: data warehousing solves it all. Many retail chains incorporate enterprise data warehousing for business intelligence and forecasting.

Social Media Websites: Social networking sites such as *Facebook*, *Twitter*, *LinkedIn* etc. are based on large data sets analyses. These sites collect data on members, groups; locations etc. and store this information in a single central repository. Data warehouse is necessary to implement the same data, because of its high volume of data.

Banking: Most banks are now using warehouses to see account/cardholder spending patterns. They use this to make special offers, deals, etc. available.

Government: In addition to store and analyze taxes used to detect tax theft, government uses the data warehouse.

Airlines: It is used in the airline system for operational purposes such as crew assignments, road profitability analyses, flight frequency programs promotions, etc.

Public sector: Information is collected in the public sector's data warehouse. It helps government agencies and departments manage their data and records.

1.10 TYPES OF DATA WAREHOUSES

There are three different types of traditional Data Warehouse models as listed below:

- i. Enterprise
- ii. Operational
- iii. Data Mart

(i) Enterprise Data Warehouse

An enterprise provides a central repository database for decision support throughout the enterprise. It is a central place where all business information from different sources and applications are made available. Once it is stored, it can be used for analysis and used by all the people across the organization. The enterprise goal is to provide a complete overview of any particular object in the data model.

(ii) Operational Data Warehouse

These features have a sizable enterprise-wide scope, but unlike the substantial enterprise warehouse, data is refreshed in near real-time and used for routine commercial activity. It assists in obtaining data straight from the database, which also helps data transaction processing. The data present in the Operational Data Store can be scrubbed, and the duplication which is present can be reviewed and fixed by examining the corresponding market rules.

(iii) Data Mart

Data Mart may be a subset of knowledge warehouse, and it supports a specific region, business unit, or business function. Data Mart focuses on storing data for a particular functional area, and it contains a subset of data saved in a memory. Data Marts help in enhancing user responses and also reduces the volume of data for data analysis. It makes it more comfortable to go forward with the report. More on Data Marts can be studied in the next Unit.

1.11 POPULAR DATA WAREHOUSE PLATFORMS

A data warehouse is a critical database for supporting data analysis and acts as a conduit between analytical tools and operational data stores. The most popular data warehousing solutions include a range of useful features for data management and consolidation.

You can use them to extract/curate data from a range of environments, transform data and remove duplicates, and ensure consistency in your analytics.

Google BigQuery

BigQuery is a cost-effective data warehousing tool with built-in machine learning capabilities. You can integrate it with *Cloud ML* and *TensorFlow* to create powerful AI models. It can also execute queries on petabytes of data for real-time analytics. This scalable and serverless cloud data warehouse is ideal for companies that want to keep costs low. If you need a quick way to make informed decisions through data analysis, *BigQuery* is one of the solutions.

AWS Redshift

Redshift is a cloud-based data warehousing tool for enterprises. The platform can process petabytes of data quite fast. That's why it's suitable for high-speed data analytics. It also supports automatic concurrency scaling. The automation increases or decreases query processing resources to match workload demand.

Although tooling provided by Amazon reduces the need to have a database administrator full time, it does not eliminate the need for one. Amazon *Redshift* is known to have issues with handling storage efficiently in an environment prone to frequent deletes.

Snowflake

Snowflake is a data warehousing solution that offers a variety of options for public cloud technology. With *Snowflake*, you can make your business more data-driven. You may use *Snowflake* to set up an enterprise-grade cloud data warehouse. With *Snowflake*, you can analyze data from various unstructured and structured sources. However, *Snowflake* is dependent on *Azure*, *Amazon Web Services (AWS)*, *Google Cloud Services (GCS)*. The support can be a problem whenever one of those cloud servers has an independent outage.

Microsoft Azure Synapse

Microsoft Azure is a robust platform for data management, analytics, integration, and more, with solutions spanning AI, blockchain, and more than a dozen unique databases for varying use cases. Among them is *Azure Synapse*, formerly known as *Azure SQL Data Warehouse*, a platform built for analytics, providing you the ability to query data using either serverless or provisioned resources at scale.

Azure Synapse brings together the two worlds of data warehousing and analytics with a unified experience to ingest, prepare, manage, and serve data for immediate BI and machine learning. The broader *Azure* platform includes thousands of tools, including others that interface with the various *Azure* databases.

1.12 SUMMARY

In this unit you have studied about the evolution, characteristics, benefits and applications of data warehouse.

Operational database system provides day-to-day information, although strategic decision-making cannot be used easily. Data Warehouse is a concept designed to aid strategic information. Data Warehouse allows people to make decisions and provides flexible, convenient and interactive sources of strategic intelligence. A data warehouse combines several technologies because it collects data from various operational database systems and external sources such as magazines, newspapers and reports from the same industry, removes contradictions, transforms the data and then stores them in formats suited to easy access for decision-making purposes. The defining characteristics of the data warehouse are: Subject oriented, integrated, time-variant, and non-volatile.

Data warehouses are meant to be used by executives, managers, and other people at higher managerial levels who may not have much technical expertise in handling the databases.

Advantages of data warehouses include better decisions, increased productivity, lower operational costs, enhanced asset and liability management, and better CRM.

1.13 SOLUTIONS/ANSWERS

Check Your Progress 1

- 1) Data Warehousing (DW) is a process for collecting and managing data from diverse sources to provide meaningful insights into the business. A Data Warehouse is typically used to connect and analyze heterogeneous sources of business data. The data warehouse is the centerpiece of the BI system built for data analysis and reporting.

It is amalgam of technologies and components which helps to use data strategically. Instead of transaction processing, it is the automated collection of a vast amount of information by a company that is configured for demand and review. It's a process of transforming data into information and making it available for users to make a difference in a timely way.

The archive of decision support (Data Warehouse) is managed independently from the operating infrastructure of the organization. The data warehouse, however, is not a product but rather an environment. It is an organizational framework of an information

system that provides consumers with knowledge regarding current and historical decision help that is difficult to access or present in the conventional operating data store.

Data storage platforms also sort data on a variety of subjects like customers, products or business.

- Data storage is a tool that companies can use increasingly important for corporate intelligence:
- Make uniformity possible. All research data gathered and shared to decision makers worldwide should be used in a uniform format. Standardization of data from various sources reduces the risk of misinterpretation as well as overall accuracy of interpretation.
- Take better business decisions. Successful entrepreneurs have a thorough understanding of data, and are good at predicting future trends. The data storage system helps users access various data sets at speed and efficiency.
- Data storage platforms allow companies to access their business' past history and evaluate ideas and projects. This gives managers an idea of how they can improve their sales and management practices.

2). Following are the four main characteristics of a data warehouse:

i) Subject oriented

A data warehouse is subject-oriented, as it provides information on a topic rather than the ongoing operations of organizations. Such issues may be inventory, promotion, storage, etc. Never does a data warehouse concentrate on the current processes. Instead, it emphasized modeling and analyzing decision-making data. It also provides a simple and succinct description of the particular subject by excluding details that would not be useful in helping the decision process.

(ii) Integrated

Integration in Data Warehouse means establishing a standard unit of measurement from the different databases for all the similar data. The data must also get stored in a simple and universally acceptable manner within the Data Warehouse. Through combining data from various sources such as a mainframe, relational databases, flat files, etc., a data warehouse is created. It must also keep the naming conventions, format, and coding consistent. Such an application assists in robust data analysis. Consistency must be maintained in naming conventions, measurements of characteristics, specification of encoding, etc.

(iii) Time-variant

Compared to operating systems, the time horizon for the data warehouse is given period and provides historical information. It contains a temporal element, either explicitly or implicitly. One such

location in the record key system where Data Warehouse data shows time variation is. Each primary key contained with the DW should have an element of time either implicitly or explicitly. Just like the day, the month of the week, etc.

(iv) Non-volatile

Also, the data warehouse is non-volatile, meaning that prior data will not be erased when new data are entered into it. Data is read-only, only updated regularly. It also assists in analyzing historical data and in understanding what and when it happened. The transaction process, recovery, and competitiveness control mechanisms are not required. In the Data Warehouse environment, activities such as deleting, updating, and inserting that are performed in an operational application environment are omitted.

Check Your Progress 2

- 1) Data Warehouse systems are segregated from production databases so that they aren't intermingled and cause conflicts.
 - There is a database available for tasks such as searching records, indexing, and digital archiving. Data warehouse queries are often complex due to their varied and complex nature.
 - It is possible to manage multiple transactions simultaneously through business databases. Concurrency control and recovery mechanisms are needed to ensure that the database in operational databases is robust and consistent.
 - The operational database query allows for reading and modification of operations, whilst the read access to stored information is required for OLAP queries only.
 - A database of operations maintains current information. In contrast, historical data is kept in a warehouse.
- 2) A database stores the current data required to power an application. A data warehouse stores current and historical data from one or more systems in a predefined and fixed schema, which allows business analysts and data scientists to easily analyze the data. The table below summarizes differences between databases, data warehouses:

Table 2: Database Vs Data Warehouse

Characteristic Feature	Database	Data Warehouse
Workloads	Operational and transactional	Analytical
Data Type	Structured or semi-structured	Structured and/or semi-structured

Schema Flexibility	Rigid or flexible schema depending on database type	Pre-defined and fixed schema definition for ingest (schema on write and read)
Data Freshness	Real time	May not be up-to-date based on frequency of ETL processes
Users	Application developers	Business analysts and data scientists
Pros	Fast queries for storing and updating data	The fixed schema makes working with the data easy for business analysts
Cons	May have limited analytics capabilities	Difficult to design and evolve schema Scaling compute may require unnecessary scaling of storage, because they are tightly coupled

1.14 FURTHER READINGS

1. William H. Inmon, Building the Data Warehouse, Wiley, 4th Edition, 2005.
2. Data Warehousing Fundamentals, Paulraj Ponnaiah, Wiley Student Edition
3. Data Warehousing, Reema Thareja, Oxford University Press

UNIT 2 DATA WAREHOUSE ARCHITECTURE

Structure

- 2.0 Introduction
- 2.1 Objectives
- 2.2 Data Warehouse Architecture and its Types
 - 2.2.1 Types of Data Warehouse Architectures
- 2.3 Components of Data Warehouse Architecture
- 2.4 Layers of Data Warehouse Architecture
 - 2.4.1 Best Practices for Data Warehouse Architecture
- 2.5 Data Marts
 - 2.5.1 Data Mart Vs Data Warehouse
- 2.6 Benefits of Data Marts
- 2.7 Types of Data Marts
- 2.8 Structure of a Data Mart
- 2.9 Designing the Data Marts
- 2.10 Limitations with Data Marts
- 2.11 Summary
- 2.12 Solutions / Answers
- 2.13 Further Readings

2.0 INTRODUCTION

In the previous unit we had studied about the data warehousing and related topics. Despite numerous advancements over the last five years in the arena of Big Data, cloud computing, predictive analysis, and information technologies, data warehouses have only gained more significance. For the success of any data warehouse, its architecture plays an important role. Since three decades, the data warehouse architecture has been the pillar of the corporate data ecosystems.

This unit presents various topics including the basic concept of data warehouse architecture, its types, significant components and layers of data warehouse architecture, data marts and their designing.

2.1 OBJECTIVES

After going through this unit, you shall be able to:

- Understand the purpose of data warehouse architecture;
- Describe the process of storing the data in a data warehouse;
- List and discuss the various types of data warehouse architectures;
- Discuss various components and layers of data warehouse architecture;
- To summarize the functionality of data marts, their benefits and various types, and
- To know the ways of structuring and designing the data marts.

2.2 DATA WAREHOUSE ARCHITECTURE AND ITS TYPES

Data warehouse architecture is a data storage framework's design of an organization. It takes information from raw data sets and stores it in a structured and easily digestible format.

A data warehouse architecture plays a vital role in the data enterprise. As databases assist in storing and processing data, and data warehouses help in analyzing that data.

Data warehousing is a process of storing a large amount of data by a business or organization. The data warehouse is designed to perform large complex analytical queries on large multi-dimensional datasets in a straightforward manner. Data warehouses extract data from different resources, which are in different fonts, convert it into a unique form, and place data in Data Warehouse.

2.2.1 Types of Data Warehouse Architectures

Data warehouse architecture defines the arrangement of the data in different databases. As the data must be organized and cleansed to be valuable, a modern data warehouse structure identifies the most effective technique of extracting information from raw data.

Using a dimensional model, the raw data in the staging area is extracted and converted into a simple consumable warehousing structure to deliver valuable business intelligence. When designing a data warehouse, there are three different types of models to consider, based on the approach of number of *tiers* the architecture has.

- (i) Single-tier data warehouse architecture
- (ii) Two-tier data warehouse architecture
- (iii) Three-tier data warehouse architecture

The details of each of the architecture are given below:

(i) Single-tier data warehouse architecture

The single-tier architecture (Figure 1) is not a frequently practiced approach. The main goal of having such architecture is to remove redundancy by minimizing the amount of data stored. Its primary disadvantage is that it doesn't have a component that separates analytical and transactional processing.

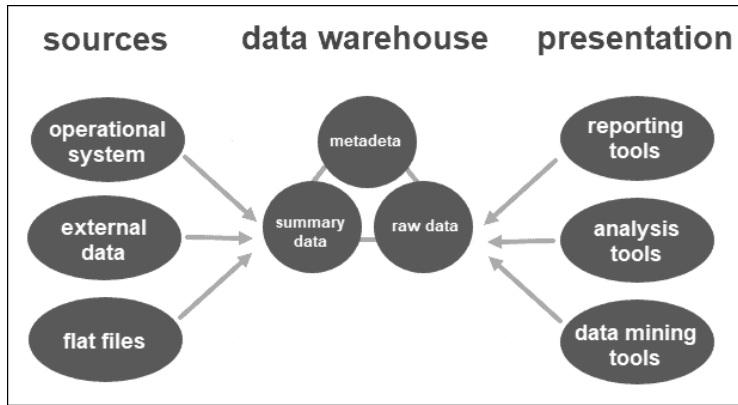


Figure 1: Single Tier Data Warehouse Architecture

(ii) Two-tier data warehouse architecture

The two-tier architecture (Figure 2) includes a staging area for all data sources, before the data warehouse layer. By adding a staging area between the sources and the storage repository, you ensure all data loaded into the warehouse is cleansed and in the appropriate format.

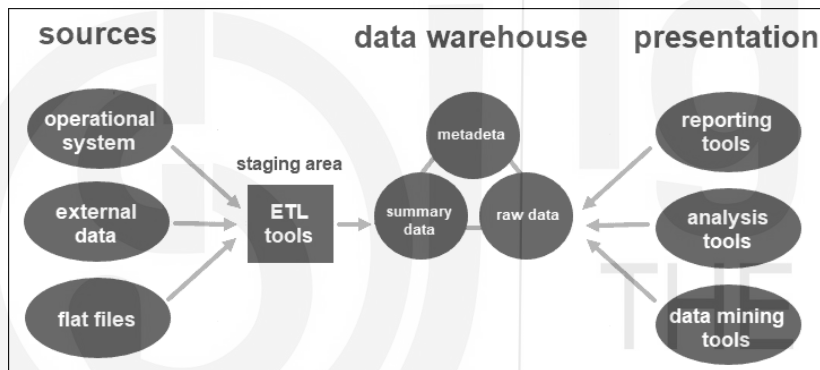


Figure 2: Two-Tier Data Warehouse Architecture

(iii) Three-tier data warehouse architecture

The three-tier approach (Figure 3) is the most widely used architecture for data warehouse systems.

Essentially, it consists of three tiers:

1. **The bottom tier** is the database of the warehouse, where the cleansed and transformed data is loaded.
2. **The middle tier** is the application layer giving an abstracted view of the database. It arranges the data to make it more suitable for analysis. This is done with an OLAP server, implemented using the ROLAP or MOLAP model.
3. **The top-tier** is where the user accesses and interacts with the data. It represents the front-end client layer. You can use reporting tools, query, analysis or data mining tools.

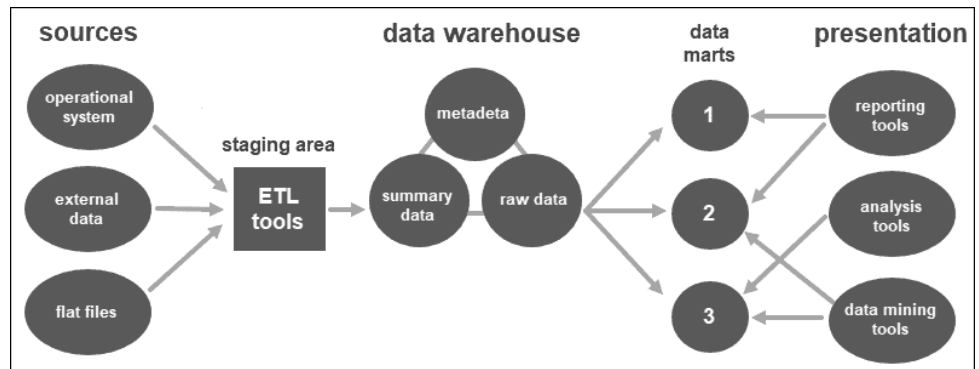


Figure 3: Three- Tier Data Warehouse Architecture

Figure 4 illustrates the complete data warehouse architecture with the three tiers:

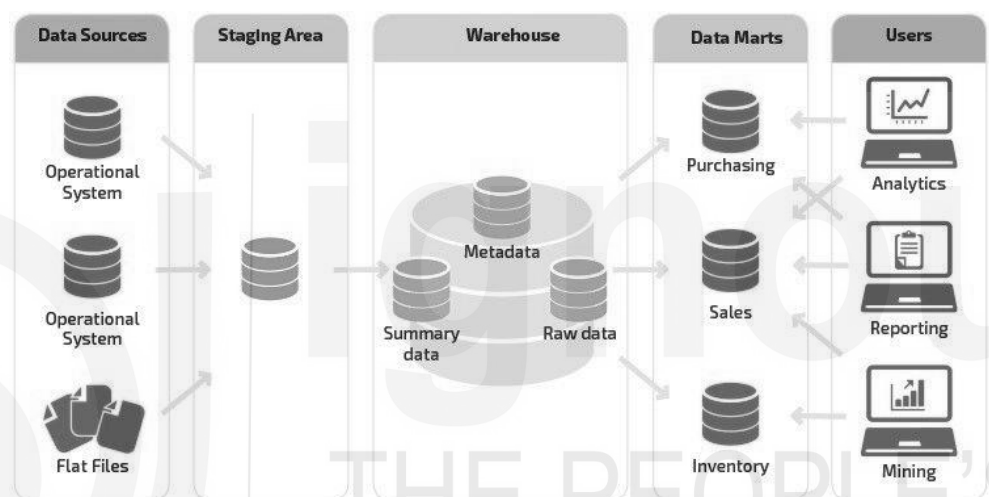


Figure 4: 3-Tiers of Data Warehouse

2.2.2 Cloud-based Data Warehouse Architecture

Cloud-based data warehouse architecture is relatively new when compared to legacy options. This data warehouse architecture means that the actual data warehouses are accessed through the cloud. There are several cloud based data warehouses options, each of which has different architectures for the same benefits of integrating, analyzing, and acting on data from different sources. The difference between a cloud-based data warehouse approach compared to that of a traditional approach include:

- **Up-front costs:** The different components required for traditional, on-premises data warehouses mandate pricey up-front expenses. Since the components of cloud architecture are accessed through the cloud, these expenses don't apply.
- **Ongoing costs:** While businesses with on-prem data warehouses must deal with upgrade and maintenance costs, the cloud offers a low, pay-as-you-go model.

- **Speed:** Cloud-based data warehouse architecture is substantially speedier than on-premises options, partly due to the use of ELT — which is an uncommon process for on-premises counterparts.
- **Flexibility:** Cloud data warehouses are designed to account for the variety of formats and structures found in big data. Traditional relational options are designed simply to integrate similarly structured data.
- **Scale:** The elastic resources of the cloud make it ideal for the scale required of big datasets. Additionally, cloud-based data warehousing options can also scale down as needed, which is difficult to do with other approaches.

Cloud-based platforms make it possible to create, share, and store massive data sets with ease, paving the way for more efficient and effective data access and analysis. Cloud systems are built for sustainable business growth, with many modern Software-as-a Service (SaaS) providers separating data storage from computing to improve scalability when querying data.

Some of the more notable cloud data warehouses in the market include Amazon Redshift, Google BigQuery, Snowflake, and Microsoft Azure SQL Data Warehouse.

Now, let's learn about the major components of a data warehouse and how they help build and scale a data warehouse in the next section.

2.3 COMPONENTS OF DATA WAREHOUSE ARCHITECTURE

A data warehouse design consists of six main components:

- Data Warehouse Database
- ETL
- Metadata
- Data Warehouse Access Tools
- Data Warehouse Bus
- Data Warehouse Reporting Layer

The details of all the components are given below.

2.3.1 Data Warehouse Database

The central component of a DW architecture is a data warehouse database that stocks all enterprise data and makes it manageable for reporting. Obviously, this means you need to choose which kind of database you'll use to store data in your warehouse.

The following are the four database types that you can use:

- Typical relational databases are the row-centered databases you perhaps use on an everyday basis—for example, Microsoft SQL Server, SAP, Oracle, and IBM DB2.
- Analytics databases are precisely developed for data storage to sustain and manage analytics, such as Teradata and Greenplum.
- Data warehouse applications aren't exactly a kind of storage database, but several dealers now offer applications that offer software for data management as well as hardware for storing data. For example, SAP Hana, Oracle Exadata, and IBM Netezza.
- Cloud-based databases can be hosted and retrieved on the cloud so that you don't have to procure any hardware to set up your data warehouse—for example, Amazon Redshift, Google BigQuery, and Microsoft Azure SQL.

2.3.2 Extraction, Transformation, and Loading Tools (ETL)

ETL tools are central components of enterprise data warehouse architecture. These tools help extract data from different sources, transform it into a suitable arrangement, and load it into a data warehouse.

The ETL tool you choose will determine:

- The time expended in data extraction
- Approaches to extracting data
- Kind of transformations applied and the simplicity to do so
- Business rule definition for data validation and cleansing to improve end-product analytics
- Filling mislaid data
- Outlining information distribution from the fundamental depository to your BI applications

2.3.3 Metadata

Before we delve into the different types of metadata in data mining, we first need to understand what metadata is. In the data warehouse architecture, metadata describes the data warehouse database and offers a framework for data. It helps in constructing, preserving, handling, and making use of the data warehouse.

There are two types of metadata in data mining:

- **Technical Metadata** comprises information that can be used by developers and managers when executing warehouse development and administration tasks.
- **Business Metadata** comprises information that offers an easily understandable standpoint of the data stored in the warehouse.

Metadata plays an important role for businesses and the technical teams to understand the data present in the warehouse and convert it into information.

2.3.4 Data Warehouse Access Tools

A data warehouse uses a database or group of databases as a foundation. Data warehouse corporations generally cannot work with databases without the use of tools unless they have database administrators available. However, that is not the case with all business units. This is why they use the assistance of several no-code data warehousing tools, such as:

- **Query and reporting tools** help users produce corporate reports for analysis that can be in the form of spreadsheets, calculations, or interactive visuals.
- **Application development tools** help create tailored reports and present them in interpretations intended for reporting purposes.
- **Data mining tools for data warehousing** systematize the procedure of identifying arrays and links in huge quantities of data using cutting-edge statistical modeling methods.
- **OLAP tools** help construct a multi-dimensional data warehouse and allow the analysis of enterprise data from numerous viewpoints.

2.3.5 Data Warehouse Bus

It defines the data flow within a data warehousing bus architecture and includes a data mart. A data mart is an access level that allows users to transfer data. It is also used for partitioning data that is produced for a particular user group.

2.3.6 Data Warehouse Reporting Layer

The reporting layer in the data warehouse allows the end-users to access the BI interface or BI database architecture. The purpose of the reporting layer in the data warehouse is to act as a dashboard for data visualization, create reports, and take out any required information.

Constructing a data warehouse is primarily dependent on a particular business. And so, every data warehouse architecture has four layers. Let's study them in following section.

2.4 LAYERS OF DATA WAREHOUSE ARCHITECTURE

In general, the data warehouse architecture can be divided into four layers. They are:

- i. Data Source Layer
- ii. Data Staging Layer
- iii. Data Storage Layer
- iv. Data Presentation Layer

Let us study the various layers and their functionality.

(i) Data source layer

The data source layer is the place where unique information, gathered from an assortment of inner and outside sources, resides in the social database.

Following are the examples of the data source layer:

- **Operational Data** — Product information, stock information, marketing information, or HR information
- **Social Media Data** — Website hits, content fame, contact page completion
- **Outsider Data** — Demographic information, study information, statistics information

While most data warehouses manage organized data, thought ought to be given to the future utilization of unstructured data sources, for example, voice accounts, scanned pictures, and unstructured text. These floods of data are significant storehouses of information and ought to be viewed when building up your warehouse.

(ii) Data Staging Layer

This layer dwells between information sources and the data warehouse. In this layer, information is separated from various inside and outer data sources. Since source data comes in various organizations, the data extraction layer will use numerous technologies and devices to extricate the necessary information. Once the extracted data has been stacked, it will be exposed to high-level quality checks. The conclusive outcome will be perfect and organized data that you will stack into your data warehouse. The staging layer contains the given parts:

- **Landing Database and Staging Area**

The landing database stores the information recovered from the data source. Before the data goes to the warehouse, the staging process does stringent quality checks on it. Arranging is a basic step in architecture. Poor information will add up to inadequate data, and the result is poor business dynamic. The arranging layer is where you need to make changes in accordance with the business process to deal with unstructured information sources.

- **Data Integration Tool**

Extract, Transform and Load tools (ETL) are the data tools used to extricate information from source frameworks, change, and prepare information and load it into the warehouse.

(iii) Data Storage Layer

This layer is the place where the data that was washed down in the arranging zone is put away as a solitary central archive. Contingent upon your business and your warehouse architecture necessities, your data storage might be a data

warehouse center, data mart (data warehouse somewhat recreated for particular departments), or an Operational Data Store (ODS).

(iv) **Data Presentation Layer**

This is where the users communicate with the scrubbed and sorted out data. This layer of the data architecture gives users the capacity to query the data for item or service insights, break down the data to conduct theoretical business situations, and create computerized or specially appointed reports.

You may utilize an OLAP or reporting instrument with an easy to understand Graphical User Interface (GUI) to assist users with building their queries, perform analysis, or plan their reports.

2.4.1 Best Practices for Data Warehouse Architecture

Designing the data warehouse with the designated architecture is an art. Some of the best practices are shown below:

- Create data warehouse models that are optimized for information retrieval in both dimensional, de-normalized, or hybrid approaches.
- Select a single approach for data warehouse designs such as the top-down or the bottom-up approach and stick with it.
- Always cleanse and transform data using an ETL tool before loading the data to the data warehouse.
- Create an automated data cleansing process where all data is uniformly cleaned before loading.
- Allow sharing of metadata between different components of the data warehouse for a smooth retrieval process.
- Always make sure that data is properly integrated and not just consolidated when moving it from the data stores to the data warehouse. This would require the 3NF normalization of data models.
- Monitor the performance and security. The information in the data warehouse is valuable, though it must be readily accessible to provide value to the organization. Monitor system usage carefully to ensure that performance levels are high.
- Maintain the data quality standards, metadata, structure, and governance. New sources of valuable data are becoming available routinely, but they require consistent management as part of a data warehouse. Follow procedures for data cleaning, defining metadata, and meeting governance standards.
- Provide an agile architecture. As the corporate and business unit usage increases, they will discover a wide range of data mart and warehouse needs. A flexible platform will support them far better than a limited, restrictive product.
- Automate the processes such as maintenance. In addition to adding value to business intelligence, machine learning can automate data warehouse technical management functions to maintain speed and reduce operating costs.
- Use the cloud strategically. Business units and departments have different deployment needs. Use on-premise systems when required,

and capitalize on cloud data warehouses for scalability, reduced cost, and phone and tablet access.

2.5 DATA MARTS

A data mart is a subset of a data warehouse focused on a particular line of business, department, or subject area. Data marts make specific data available to a defined group of users, which allows those users to quickly access critical insights without wasting time searching through an entire data warehouse. For example, many companies may have a data mart that aligns with a specific department in the business, such as finance, sales, or marketing.

2.5.1 Data Mart Vs Data Warehouse

Data marts and data warehouses are both highly structured repositories where data is stored and managed until it is needed. However, they differ in the scope of data stored: data warehouses are built to serve as the central store of data for the entire business, whereas a data mart fulfills the request of a specific division or business function. Because a data warehouse contains data for the entire company, it is best practice to have strictly control who can access it. Additionally, querying the data you need in a data warehouse is an incredibly difficult task for the business. Thus, the primary purpose of a data mart is to isolate—or partition—a smaller set of data from a whole to provide easier data access for the end consumers.

A data mart can be created from an existing data warehouse—the top-down approach—or from other sources, such as internal operational systems or external data. Similar to a data warehouse, it is a relational database that stores transactional data (time value, numerical order, reference to one or more object) in columns and rows making it easy to organize and access.

On the other hand, separate business units may create their own data marts based on their own data requirements. If business needs dictate, multiple data marts can be merged together to create a single, data warehouse. This is the bottom-up development approach.

In a nut-shell, following are the differences:

- Data mart is for a specific company department and normally a subset of an enterprise-wide data warehouse.
- Data marts improve query speed with a smaller, more specialized set of data.
- Data warehouses help make enterprise-wide strategic decisions, data marts are for department level, tactical decisions.
- Data warehouse includes many data sets and takes time to update, data marts handle smaller, faster-changing data sets.

- Data warehouse implementation can take many years, data marts are much smaller in scope and can be implemented in months.

2.6 BENEFITS OF DATA MARTS

Data marts are designed to meet the needs of specific groups by having a comparatively narrow subject of data. And while a data mart can still contain millions of records, its objective is to provide business users with the most relevant data in the shortest amount of time.

With its smaller, focused design, a data mart has several benefits to the end user, including the following:

- **Cost-efficiency:** There are many factors to consider when setting up a data mart, such as the scope, integrations, and the process to extract, transform, and load (ETL). However, a data mart typically only incurs a fraction of the cost of a data warehouse.
- **Simplified data access:** Data marts only hold a small subset of data, so users can quickly retrieve the data they need with less work than they could when working with a broader data set from a data warehouse.
- **Quicker access to insights:** Intuition gained from a data warehouse supports strategic decision-making at the enterprise level, which impacts the entire business. A data mart fuels business intelligence and analytics that guide decisions at the department level. Teams can leverage focused data insights with their specific goals in mind. As teams identify and extract valuable data in a shorter space of time, the enterprise benefits from accelerated business processes and higher productivity.
- **Simpler data maintenance:** A data warehouse holds a wealth of business information, with scope for multiple lines of business. Data marts focus on a single line, housing under 100GB, which leads to less clutter and easier maintenance.
- **Easier and faster implementation:** A data warehouse involves significant implementation time, especially in a large enterprise, as it collects data from a host of internal and external sources. On the other hand, you only need a small subset of data when setting up a data mart, so implementation tends to be more efficient and include less set-up time.

2.7 TYPES OF DATA MARTS

There are three types of data marts that differ based on their relationship to the data warehouse and the respective data sources of each system.

- **Dependent data marts** are partitioned segments within an enterprise data warehouse. This top-down approach begins with the storage of all business data in one central location. The newly created data marts extract a defined subset of the primary data whenever required for analysis.
- **Independent data marts** act as a standalone system that doesn't rely on a data warehouse. Analysts can extract data on a particular subject or business process from internal or external data sources, process it, and then store it in a data mart repository until the team needs it.
- **Hybrid data marts** combine data from existing data warehouses and other operational sources. This unified approach leverages the speed and user-friendly interface of a top-down approach and also offers the enterprise-level integration of the independent method.

2.8 STRUCTURE OF A DATA MART

A data mart is a subject-oriented relational database that stores transactional data in rows and columns, which makes it easy to access, organize, and understand. As it contains historical data, this structure makes it easier for an analyst to determine data trends. Typical data fields include numerical order, time value, and references to one or more objects.

Companies organize data marts in a multidimensional schema as a blueprint to address the needs of the people using the databases for analytical tasks. The three main types of schema:

Star

Star schema is a logical formation of tables in a multidimensional database that resembles a star shape. In this blueprint, one fact table—a metric set that relates to a specific business event or process—resides at the center of the star, surrounded by several associated dimension tables.

There is no dependency between dimension tables, so a star schema requires fewer joins when writing queries. This structure makes querying easier, so star schemas are highly efficient for analysts who want to access and navigate large data sets.

Snowflake

A snowflake schema is a logical extension of a star schema, building out the blueprint with additional dimension tables. The dimension tables are normalized to protect data integrity and minimize data redundancy.

While this method requires less space to store dimension tables, it is a complex structure that can be difficult to maintain. The main benefit of using snowflake schema is the low demand for disk space, but the caveat is a negative impact on performance due to the additional tables.

Vault

Data vault is a modern database modeling technique that enables IT professionals to design agile enterprise data warehouses. This approach enforces a layered structure and has been developed specifically to combat issues with agility, flexibility, and scalability that arise when using the other schema models.

Data vault eliminates star schema's need for cleansing and streamlines the addition of new data sources without any disruption to existing schema.

2.9 DESIGNING THE DATA MARTS

Data marts guide important business decisions at a departmental level. For example, a marketing team may use data marts to analyze consumer behaviors, while sales staff could use data marts to compile quarterly sales reports. As these tasks happen within their respective departments, the teams don't need access to all enterprise data.

Typically, a data mart is created and managed by the specific business department that intends to use it. The process for designing a data mart usually comprises the following steps:

(i) Essential Requirements Gathering

The first step is to create a robust design. Some critical processes involved in this phase include collecting the corporate and technical requirements, identifying data sources, choosing a suitable data subset, and designing the logical layout (database schema) and physical structure.

(ii) Build/Construct

The next step is to construct it. This includes creating the physical database and the logical structures. In this phase, you'll build the tables, fields, indexes, and access controls.

(iii) Populate/Data Transfer

The next step is to populate the mart, which means transferring data into it. In this phase, you can also set the frequency of data transfer, such as daily or weekly. This usually involves extracting source information, cleaning and transforming the data, and loading it into the departmental repository.

(iv) Data Access

In this step, the data loaded into the data mart is used in querying, generating reports, graphs, and publishing. The main task involved in this phase is setting up a meta-layer and translating database structures and item names into corporate expressions so that non-technical operators can easily use the data mart. If necessary, you can also set up API and interfaces to simplify data access.

(v) Manage

The last step involves management and observation, which includes:

- Controlling ongoing user access.
- Optimization and refinement of the target system for improved performance.
- Addition and management of new data into the repository.
- Configuring recovery settings and ensuring system availability in the event of failure.

2.10 LIMITATIONS WITH DATA MARTS

Prospective builders of data warehouses are frequently advised to “start small” with a data mart and use that kernel to expand gradually into a full blown data warehouse. This approach to warehousing generally leads to failed projects for several reasons.

Sometimes the new data mart is so successful that the configuration is overrun by user demands. The databases grow too large too fast, response times become unacceptably long, and user frustration leads to searching for other ways to get the answers.

The more common reason for failure is that the data mart is immediately unsuccessful because it is designed in such a way that users are unable to retrieve the sort of information they want and need to extract from the data. Databases are highly denormalized to respond to a small set of canned queries; summaries, rather than detail data, comprise the database so that fine-grained exploratory data analysis is not possible; and support for ad hoc queries is either absent or so poor as to discourage users from bothering with them.

The very factors that frequently defeat data mart projects are also the most commonly recommended approaches to designing data marts and data warehouses in the popular data warehousing literature:

- Denormalization (dimensional modeling)
- Storing aggregates at the expense of detail data
- Skewing performance toward a small, preselected set of queries at the expense of all other exploratory analyses

Check your Progress 1

1. Define data warehouse architecture.

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2. What is the correct flow of the data warehouse architecture?

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3. Mention some Data Mart Use Cases.

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2.11 SUMMARY

Data warehouse architecture is the design and building blocks of the modern data warehouse. In this unit we have studied the basic building blocks of the data warehouse, data warehouse architecture, its types, architecture models, data marts, designing of data marts and limitations.

In this next unit we will study about Dimensional Modeling.

2.12 SOLUTIONS / ANSWERS**Check Your Progress 1:**

1. The method for defining the entire architecture of data communication processing as well as the presentation that exists for end-clients is the data warehouse architecture. Every data warehouse is different, and each of them

is characterized based on the standard vital components.

In simple words, a data warehouse is an information system that consists of commutative and historical data from single or multiple sources. The process of reporting and analysis of data in the organizations is simplified with the help of different data warehousing concepts. There are different approaches to constructing a data warehouse architecture. Any approach is used based on the requirements of the organizations.

2. On every operational database, there are a certain fixed number of operations that have to be applied. There are different well-defined techniques for delivering suitable solutions. Data warehousing is found to be more effective when the correct flow of the data warehouse architecture is completely followed.

The four different processes that contribute to a data warehouse are extracting and loading the data, cleaning and transforming the data, backing up and archiving the data, and carrying out the query management process by directing them to the appropriate data sources.

3. Data marts are used to solve specific organizational problems, especially those that are unique to one department. Typical use cases for a data mart include:

Focused Analytics

Analytics is perhaps the most common application of data marts. The data in these repositories is entirely relevant to the requirements of the business department, with no extraneous information, resulting in faster and more accurate analysis. For example, financial analysts will find it easier to work with a financial data mart, rather than working with an entire data warehouse.

Fast Turnaround

Data marts are generally faster to develop than a data warehouse, as the developers are working with fewer sources and a limited schema. Data marts are ideal for data projects operating under challenging time constraints.

Permission Management

Data marts can be a risk-free way to grant limited data access without exposing the entire data warehouse. For example, dependent data mart contains a segment of warehouse data, and users are only able to view the contents of the mart. This prevents unauthorized access and accidental writes.

Better Resource Management

Data marts are sometimes used where there is a disparity in resource usage between different departments. For example, the logistics department might perform a high volume of daily database actions, which causes the marketing team's analytics tools to run slow. By

providing each department with its own data mart, it's easier to allocate resources according to their needs.

2.13 FURTHER READINGS

1. William H. Inmon, Building the Data Warehouse, Wiley, 4th Edition, 2005.
2. Data Warehousing Fundamentals, Paulraj Ponnaiah, Wiley Student Edition, 2001.
3. Data Warehousing, Reema Thareja, Oxford University Press, 2009.



UNIT 3 DIMENSIONAL MODELING

Structure

- 3.0 Introduction
- 3.1 Objectives
- 3.2 Dimensional Modeling
 - 3.2.1 Key Features of Dimensional Modeling
 - 3.2.2 Steps Involved in Dimensional Modeling
- 3.3 Identifying Facts and Dimensions
- 3.4 Star Schema
 - 3.4.1 Features of Star Schema
- 3.5 Advantages and Limitations of Star Schema
 - 3.5.1 Advantages of Star Schema
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- 3.8 Aggregate Tables
 - 3.8.1 Need for Building Aggregate Fact Tables
 - 3.8.2 Aggregate Fact Tables and Derived Dimension Tables
- 3.9 Fact Constellation Schema
 - 3.9.1 Advantages of Fact Constellation Schema
 - 3.9.2 Limitations of Fact Constellation Schema
- 3.10 Summary
- 3.11 Solutions/Answers
- 3.12 Further Readings

3.0 INTRODUCTION

In the earlier unit on Data Warehouse Architecture we have studied that many companies transform the data through an ETL (Extract, Transform and Load) process and store this data in a Data Warehouse for further analysis. In order to access the data from this Data Warehouse, companies use a process called Dimensional Modeling.

Dimensional Data Modeling is a data structure that helps optimize a Data Warehouse to retrieve data quickly. Ralph Kimball developed this technique that could read, analyze and summarize data in a Data Warehouse for further analysis.

Dimensional Modeling is a technique that uses Dimensions and Facts to store the data in a Data Warehouse efficiently. It optimizes the database for faster retrieval of the data. Dimensional Models have a specific structure and organize the data to generate reports that improve performance.

It stores the data in the most optimized way to ensure there is no redundancy of the data and to improve performance.

3.1 OBJECTIVES

After going through this unit, you shall be able to:

- understand the purpose of dimension modeling;
- identifying the measures, facts, and dimensions;
- discuss the fact and dimension tables and their pros and cons;
- discuss the Star and Snowflake schema;
- explore comparative analysis of star and snowflake schema;
- describe Aggregate facts, fact constellation, and
- discuss various examples of star and snowflake schemas.

3.2 DIMENSIONAL MODELING

Dimensional modeling is a data model design adopted when building a data warehouse. Simply, it can be understood that dimension modeling reduces the response time of query fired unlike relational systems. The concept behind dimensional modeling is all about the conceptual design. Firstly let's see the introduction to dimensional modeling and how it is different from a traditional data model design. A data model is a representation of how data is stored in a database and it is usually a diagram of the few tables and the relationships that exist between them. This modeling is designed to read, summarize and compute some numeric data from a data warehouse.

A data warehouse is an example of a system that requires small number of large tables. This is due to many users using the application to read lot of data a characteristic of a data warehouse is to write the data once and read it many times over so it is the read operation that is dominant in a data warehouse. Now let's look at the data warehouse containing customer related information in a single table this makes it a lot easier for analytics just to count the number of customers by country but this time the use of tables in the data warehouse simplify the query processing.

The main objective of dimension modeling is to provide an easy architecture for the end user to write queries and also, to reduce the number of relationships between the tables and dimensions hence providing efficient query handling.

Dimensional modeling populates data in a cube as a logical representation with OLAP data management. The concept was developed by Ralph Kimball. It has *fact* and *dimension* as its two important measures. The transaction record is divided into either *facts*, which consists of business numerical transaction data, or *dimensions*, which are the reference information that gives context to the facts. The main objective of dimension modeling is to provide an easy architecture for the end user to write queries. Also it will reduce the number of relationships between the tables and dimensions, hence providing efficient query handling.

3.2.1 Key Features of Dimensional Modeling

Data Dimensional Modeling has gained popularity because of its unique way of analyzing data present in different Data Warehouses. The main features of dimensional modeling are as follows:

- **Easy to Understand:** It helps developers create and design databases and Schemas easily interpreted by business users. The relationship between Dimensions and Facts are pretty simple to read and understand.
- **Promote Data Quality:** Data modeling schemas enforce data quality before loading into Facts and Dimensions. Dimension and Fact are tied up by foreign keys that act as a constraint for referential integrity check to prevent fraudulent data from being loaded onto Schemas.
- **Optimise Performance:** Data modeling breaks the data into Dimensions and Facts and links them with foreign keys, thereby reducing the data redundancy. The data is stored in the optimized form and hence occupies less storage and can be retrieved faster.
- It reduces the number of relationships between different data elements.
- The aggregate functions used in the schemas optimize the query performance posted by the customers. Since data warehouse size keeps on increasing and with this increased size, the optimization becomes the concern which dimension modeling makes it easy.

3.2.2 Steps Involved in Dimensional Modeling

The following are the steps involved in Dimension modeling as shown in figure1.

- Identify Business objectives
- Identify Granularity
- Identify dimensions and attributes
- Build the Schema

The model should describe the Why, How much, When/Where/Who and What of your business process.

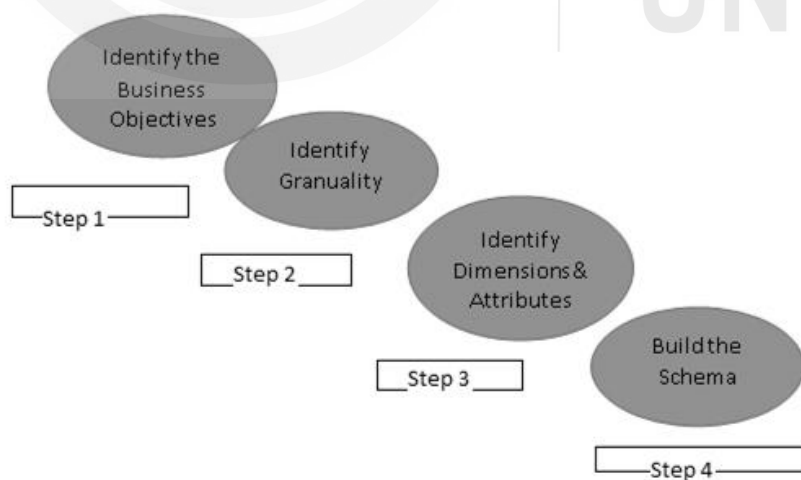


Figure 1: Steps involved in Dimension Modeling

Step 1: Identify the Business Objectives

Selection of the right business process to build a data warehouse and identifying the business objectives is the first step in dimension modeling. This

is very important step otherwise this can lead to repeated process and software defects.

Step 2: Identifying Granularity

The grain literally means each minute detail of the business problem. This is decomposing of the large and complex problem into the lowest level information. For example, if there is some data month-wise. So, the table would contain details of all the months in a year. It depends on the report to be submitted to the management. This affects the size of the data warehouse.

Step 3: Identifying Dimensions and attributes

The dimensions of the data warehouse can be understood by the entities of the database. like, items, products, date, stocks, time etc. The identification of the primary keys and the foreign keys specifications all are described here.

Step 4: Build the Schema

The database structure or arrangement of columns in a database table, decides the schema. There are various popular schemas like, star, snowflake, fact constellation schemas - summarizing, from the selection of business process to identifying each and every finest level of detail of the business transactions. Identifying the significant dimensions and attributes would help to build the schema.

3.3 IDENTIFYING FACTS AND DIMENSIONS

We have just studied the steps involved in dimension modeling in the previous section. The last step is to build the schema which has greater significance in the overall modeling. So, let's see the elementary measures to build a schema.

Facts: A fact is an event. It is a measure which represents business items or transactions of items having association and context data.

Fact table: The Fact table contains the description of all the primary keys of all the tables used in the business processes which acts as a foreign key in the fact table. It also has an aggregate function to compute the business process on some entity. It is a numeric attribute of a fact, representing the performance or behavior of the business relative to the dimensions. The number of columns in the fact table is less than the dimension table. It is more normalized form.

Dimensions: It is a collection of data which describe one business dimension. Dimensions decide the contextual background for the facts, and they are the framework over which OLAP is performed.

Dimension table: Dimension tables establish the context of the facts. The table stores fields that describe the facts. The data in the table are in de normalized form. So, it contains large number of columns as compared to fact table. The attributes in a dimension table are used as row and column headings in a document or query results display.

Example: In the example of student registration case study to any particular course can have attributes like student_id, course_id, program_id, date_of_registration, fee_id in fact table. Course summary can have course name, duration of the course etc. Student information can contain the personal details about the student like name, address, contact details etc.

Case Study or Scenario: Student Registration

Fact Table (student_id, course_id, program_id, date_of_registration, fee_id,
Measure: Sum (Fee_amount))

Dimension Tables (Student_details,
Course_details
Program_details,
Fee_details,
Date)

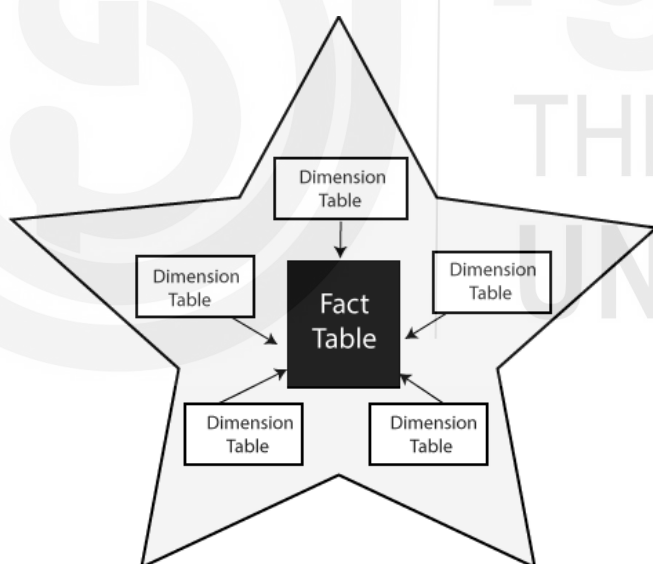
In the context of Relational context, there are two basic models which are used in dimensional modeling:

- Star Model
- Snowflake Model

In the next section, let us study the details of these modeling techniques.

3.4 STAR SCHEMA

It represents the multidimensional model. In this model the data is organized into facts and dimensions. The star model is the underlying structure for a dimensional model. It has one broad central table (fact table) and a set of smaller tables (dimensions) arranged in a star design around the primary table. This design is logically shown in the below figure 2.



Star Schema

Figure 2: Star Schema

3.4.1 Features of Star Schema

- The data is in denormalized database.
- It provides quick query response
- Star schema is flexible can be changed or added easily.
- It reduces the complexity of metadata for developers and end users.

3.5 ADVANTAGES AND LIMITATIONS OF STAR SCHEMA

Star schemas are easy for end users and applications to understand and navigate. With a well-designed schema, users can quickly analyze large, multidimensional data sets.

3.5.1 Advantages of Star Schema

The main advantages of star schemas in a decision-support environment are:

- **Query performance**

Because a star schema database has a small number of tables and clear join paths, queries run faster than they do against an OLTP system. Small single-table queries, usually of dimension tables, are almost instantaneous. Large join queries that involve multiple tables take only seconds or minutes to run.

In a star schema database design, the dimensions are linked only through the central fact table. When two dimension tables are used in a query, only one join path, intersecting the fact table, exists between those two tables. This design feature enforces accurate and consistent query results.

- **Load performance and administration**

Structural simplicity also reduces the time required to load large batches of data into a star schema database. By defining facts and dimensions and separating them into different tables, the impact of a load operation is reduced. Dimension tables can be populated once and occasionally refreshed. You can add new facts regularly and selectively by appending records to a fact table.

- **Built-in referential integrity**

A star schema has referential integrity built in when data is loaded. Referential integrity is enforced because each record in a dimension table has a unique primary key, and all keys in the fact tables are legitimate foreign keys drawn from the dimension tables. A record in the fact table that is not related correctly to a dimension cannot be given the correct key value to be retrieved.

- **Easily understood**

A star schema is easy to understand and navigate, with dimensions joined only through the fact table. These joins are more significant to the end user, because they represent the fundamental relationship between parts of the underlying business. Users can also browse dimension table attributes before constructing a query.

3.5.2 Limitations of Star Schema

As mentioned before, improving read queries and analysis in a star schema could involve certain challenges:

- **Decreased data integrity:** Because of the denormalized data structure, star schemas do not enforce data integrity very well. Although star schemas use countermeasures to prevent anomalies from developing, a simple insert or update command can still cause data incongruities.

- Less capable of handling diverse and complex queries: Databases designers build and optimize star schemas for specific analytical needs. As denormalized data sets, they work best with a relatively narrow set of simple queries. Comparatively, a normalized schema permits a far wider variety of more complex analytical queries.
- No Many-to-Many Relationships: Because they offer a simple dimension schema, star schemas don't work well for "many-to-many data relationships"

Example: Suppose a star schema is composed of a fact table as shown in Figure 3, SALES, and several dimension tables connected to it for time, branch, item, and geographic locations.

The TIME table has a column for each day, month, quarter, and year. The ITEM table has columns for each item_Key, item_name, brand, type, supplier_type. The BRANCH table has columns for each branch_key, branch_name, branch_type. The LOCATION table has columns of geographic data, including street, city, state, and country.

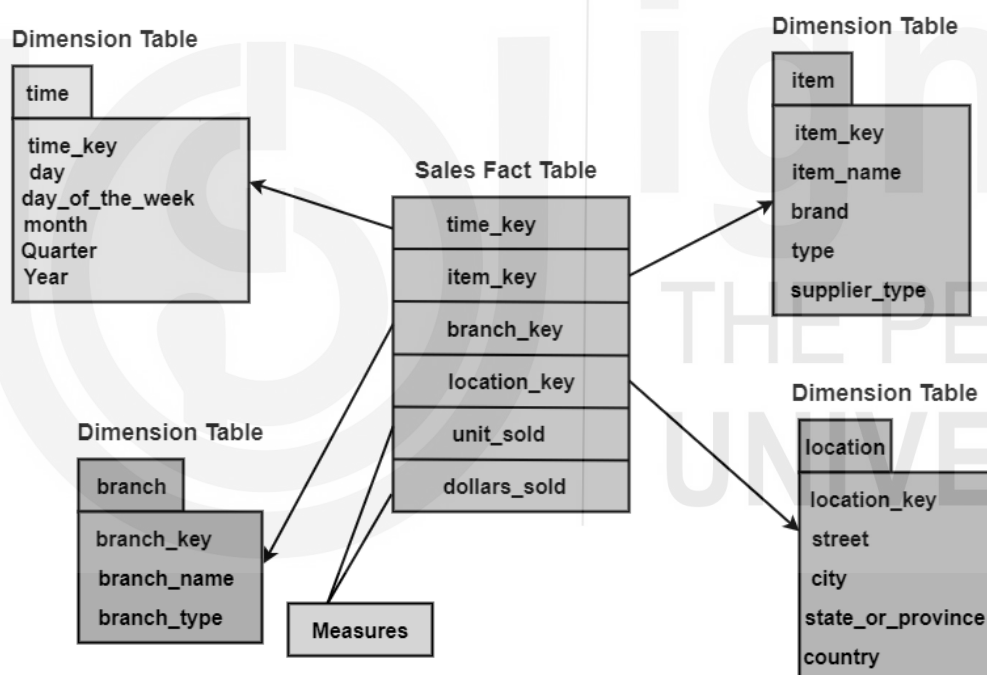


Figure 3: Example of Star Schema

☞ Check Your Progress 1:

- 1) Discuss the characteristics of star schema?

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2) Mention the benefits and limitations of Dimensional Modeling.

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3.6 SNOWFLAKE SCHEMA

The other modeling technique is Snowflake Schema. You can understand the term flakes as chocolate flakes on the pastry and ice-creams. These flakes add additional tastes to the chocolate. Similarly, snowflake schema is the extension of star schema which adds more dimensions to give more meaning to the logical view of the database. These additional tables are more normalized than star schema. The arrangement of data is like that the centralized fact table relates to multiple related dimensional tables. This can become more complex if the dimensions are more detailed and at multiple levels. In the conceptual hierarchy child table has multiple parent tables. You must keep in mind that we are just extending or flaking the dimension tables not the fact tables.

Snowflake Model: The snowflake model is the conclusion of decomposing one or more of the dimensions. Snowflake Schema in data warehouse is a logical arrangement of tables in a multidimensional database such that the ER diagram resembles a snowflake shape. A Snowflake Schema is an extension of a Star Schema, and it adds additional dimensions. The dimension tables are normalized which splits data into additional tables.

In the following Snowflake Schema example, Country is further normalized into an individual table.

3.6.1 Features of Snowflake Schema

- It has normalized tables
- Occupy less disk space.
- It requires more lookup time as many tables are interconnected and extending dimensions.

Example: In the below Figure 4 , the snowflake schema is shown of a case study of customers, sales, products, location wise quantity sold, and number of items sold are calculated. The customers, products, date, store are saved in the fact table with their respective primary keys acting in fact table as a foreign key. You will observe that the two aggregate functions can be applied to calculate quantity sold and amount sold. Further, the some dimensions are extended to the type of customer and also store information territory wise too. Note, date has been expanded into date, month, year. This schema will give you more opportunity to perform query handling in detail.

Snowflake Schema

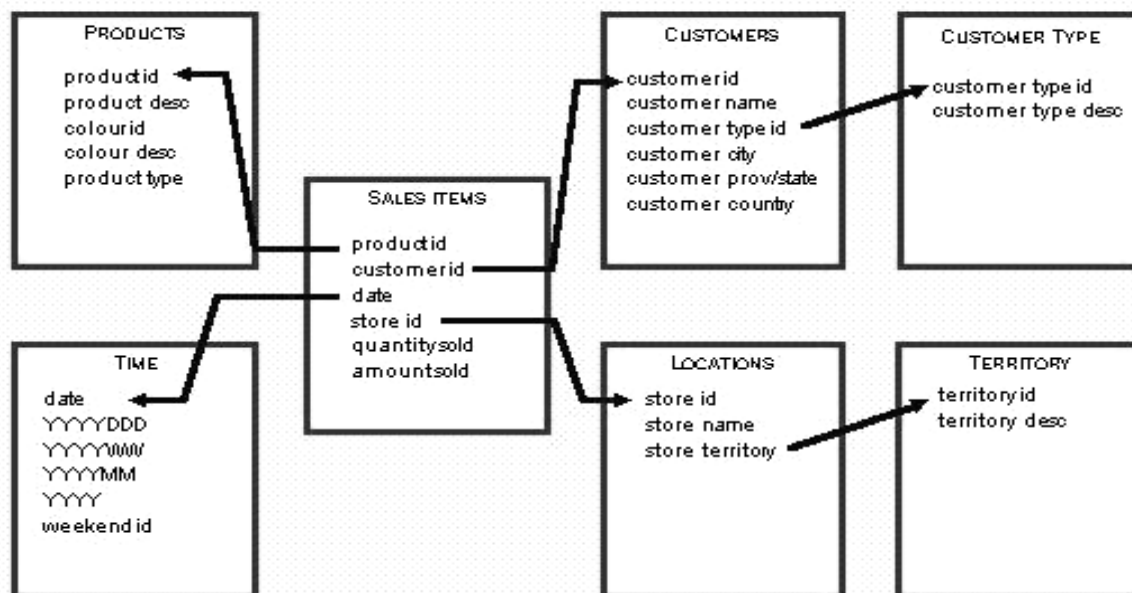


Figure 4: Snowflake Schema

3.7 ADVANTAGES AND LIMITATIONS OF SNOWFLAKE SCHEMA

Following are the advantages and limitations of Snowflake Schema:

3.7.1 Advantages of Snowflake Schema

- A Snowflake schema occupies a much smaller amount of disk space compared to the Star schema. Lesser disk space means more convenience and less hassle.
- Snowflake schema of small protection from various Data integrity issues. Most people tend to prefer the Snowflake schema because of how safe it is.
- Data is easy to maintain and more structured.
- Data quality is better than star schema.

3.7.2 Limitations of Snowflake Schema

- Complex data schemas: As you might imagine, snowflake schemas create many levels of complexity while normalizing the attributes of a star schema. This complexity results in more complicated source query joins. In offering a more efficient way to store data, snowflake can result in performance declines while browsing these complex joins. Still, processing technology advancements have resulted in improved snowflake schema query performance in recent years, which is one of the reasons why snowflake schemas are rising in popularity.

- Slower at processing cube data: In a snowflake schema, the complex joins result in slower cube data processing. The star schema is generally better for cube data processing.
- Lower data integrity levels: While snowflake schemas offer greater normalization and fewer risks of data corruption after performing UPDATE and INSERT commands, they do not provide the level of transnational assurance that comes with a traditional, highly-normalized database structure. Therefore, when loading data into a snowflake schema, it's vital to be careful and double-check the quality of information post-loading.

3.7.3 Star Schema Vs Snowflake Schema

Let us study the differences between the Star and Snowflake schemas. The major differences are summarized in the following Table 1:

Table 1: Star Schema Vs Snowflake Schema

	Star Schema	Snowflake Schema
Dimension Table	The dimension tables in star schema are not normalized so they may contain redundancies	This schema has normalized dimension tables
Queries	The execution of queries is relatively faster as there are less joins needed in forming a query.	The execution of snowflake schema complex queries is slower than star schema as many joins and foreign key relations are needed to form a query. Thus performance is affected.
Performance	Star schema model has faster execution and response time	It has slow performance as compared to star schema
Storage Space	This type of schema requires more storage space as compared to snowflake due to unnormalised tables.	Snowflake schema tables are easy to maintain and save storage space due to normalized tables.
Usage	Star schema is preferred when the dimension tables have lesser rows	If the dimension table contains large number of rows, snowflake schema is preferred
Type of DW	This schema is suitable for 1:1 or 1: many relationships such as data marts.	It is used for complex relationships such as many: many in enterprise Data warehouses.
Dimension Tables	Star schema has a single table for each dimension	Snowflake schema may have more than one dimension table for each dimension.

3.8 AGGREGATE TABLES

Since, in the data warehouse the data is stored in multidimensional cube. In the information technology industry, there are various tools available to process the queries posted on the data warehouse engine. These tools are called business intelligence (BI) tools. These tools help to answer the complex queries and to take decisions. Aggregate word is very similar to the aggregation of the database schemas of relational tables that you must be familiar with. Aggregate fact tables roll up the basic fact tables of the schema to improve the query processing. The business tools smoothly select the level of aggregation to improve the query performance. Aggregate fact tables contain foreign keys referring to dimension tables.

Points to note about Aggregate tables:

- They are also called summary tables.
- It contains pre-computed queries of the data warehouse schema.
- It reduces the dimensionality of the base fact tables.
- It can be used to respond to the queries of the dimensions that are saved.

3.8.1 Need for Building Aggregate Fact Tables

Let us understand the need of building aggregate table. Aggregate tables also referred to pre-computed tables having partially summarized data. Simply putting in one word, it's about speed or quick response to queries. This you can understand as an intermediate table which stores the results of the queries on I/O disk space. It uses aggregates functionality.

For example, there is a company ABC corporation limited which takes orders online and it there are millions of customer transactions placing orders. So, the dimension tables for the company could be Customer, Product and Order_date. In the fact table it maintains all the orders placed say, Fact_Orders. To generate a report of monthly orders by product type and by particular region, it needs aggregate (which are summary tables can be obtained) by Group by SQL query.

- It occupies less space than atomic fact tables. It nearly takes the half time of a general query processing.
- One of the more popular uses of aggregates is to adjust the granularity of a dimension. When the granularity of a dimension is changed, the fact table must be partially summarized to match the current grain of the new dimension, resulting in the creation of new dimensional and fact tables that fit this new grain standard.
- The Roll-up OLAP operation of the base fact tables generates aggregate tables. Hence the query performance increases as it reduces the number of rows to be accessed for the retrieval of data of a query.

3.8.2 Aggregate Fact Tables and Derived Dimension Tables

Aggregate facts are produced by calculating measures from more atomic fact tables. These tables contain computational SQL aggregate functions like AVERAGE, MIN, MAX, COUNT etc. It also contains function that helps to find output using group by. The aggregate fact tables produce summary statistics. Whenever, the speedy query handling is required the aggregate fact tables is the best option.

- Basically, aggregates allow you to store the intermediate results or pre-calculate the subqueries or queries fired on a data warehouse by summing data up to higher levels and storing them in a separate star.
- You can understand aggregate fact tables as the conformed copy of the fact table as it should provide you the same result of the query as the detailed fact table.
- This aggregate fact tables can be used in the case of large datasets or when there are large number of queries. It reduces the response time of the queries fired by users or customers. It is very useful in business intelligence application tools.

When you have complicated questions of multiple facts in multiple tables that are stored at different levels from one another, and when a reporting request includes yet another level, the levels at which facts are stored become even more relevant. You must be able to meet users' need for fact reporting at the business level. There's nothing wrong with improving the overall intelligence. The levels at which facts are stored become especially important when you begin to have complex queries with multiple facts in multiple tables that are stored at levels different from one another, and when a reporting request involves still a different level.

You must be able to support fact reporting at the business levels which users require. There is nothing wrong with enhancing an aggregate with new facts or deriving new dimension. For measures, the only issue is if the new measures are atomic in the context of the aggregate fact. If, however, the new measures are received at a lower grain, you would be better off creating a new atomic fact for those measures prior to incorporating summarized measures into the aggregate. This would allow the new measures to be used for other purposes without having to go back to the source.

Let's say we have a fact table, **FactBillReceipt** has monthly transactions. There can be different types of transaction receipts during a month for each supplier. This huge data would result in lot of calculations. So, we would build another aggregate table which is derived of base table.

FactBillMonthReceipt: It contains aggregated receipts per month, per supplier. But the problem is it has additional foreign keys like supplier_status for the month. To solve this, we have the concept of derived tables which contains additional measures and foreign keys that are not present in the base fact table.

Conformed Dimension

A conformed dimension is the dimension that is shared across multiple data mart or subject area. An organization may use the same dimension table across different projects without making any changes to the dimension tables.

Derived Tables

It is the significant addition to the Datawarehouse. Derived tables are used to create a second-level data marts for cross functional analysis.

Consolidated Fact tables: It is the fact table which has data from different fact tables used to form a schema with a common grain.

For example, to design a Sales department Datawarehouse schema assuming there are following entities and respective grains in them.

Sales: Employee, date, and product.

Budget: Department, Financial Year, Quarter-wise

Product can have various attributes like, product size, product _category and so on.

One thing to notice here is that the product attributes keep on changing as per the requirements, but product dimension remains the same. So, it is better to keep Product as a separate dimension. Let's design the tables and its grains.

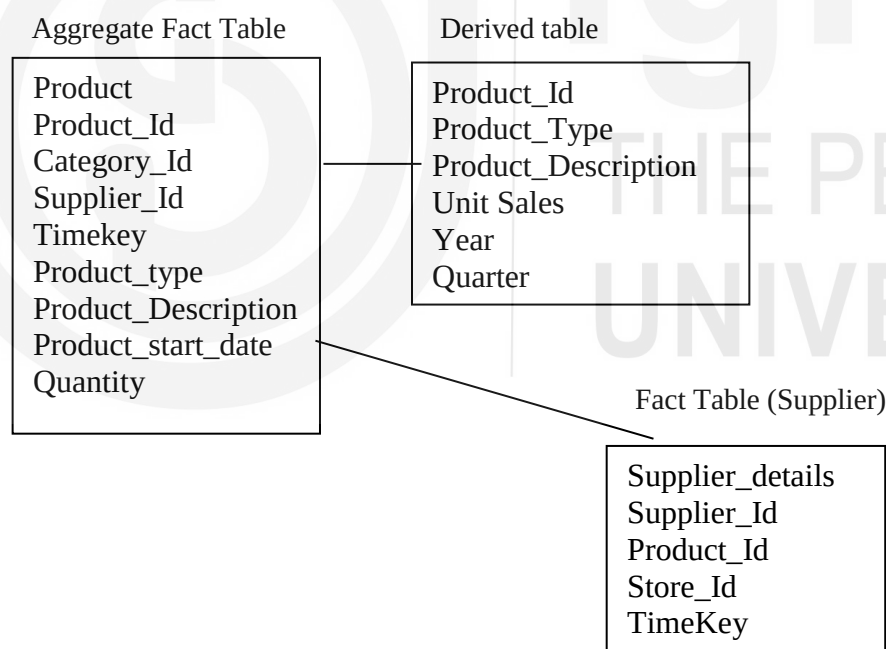


Figure 6: Aggregate Tables and Derived tables

The derived tables are very useful in terms of putting fewer loads on the Data warehouse engine for calculation.

☞ Check Your Progress 2

- 1) Write down an example of aggregate SQL query of a store Sales fact table(Date, Product, Store, Promotion, Transaction ID) Date aggregates(Month, Year), Product aggregates(Brand, Manufacturer),

Promotion aggregates(Ad, Discount, Coupon, In-Store Display) Store aggregates (District, State and Measurements Total_amt, Quantity?

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2) What are fact and fact less tables in data warehouse?

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3) Discuss the limitations of Aggregate Fact tables.

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3.9 FACT CONSTELLATION SCHEMA

There is another schema for representing a multidimensional model. This term fact constellation is like the galaxy of universe containing several stars. It is a collection of fact schemas having one or more-dimension tables in common as shown in the figure below. This logical representation is mainly used in designing complex database systems.

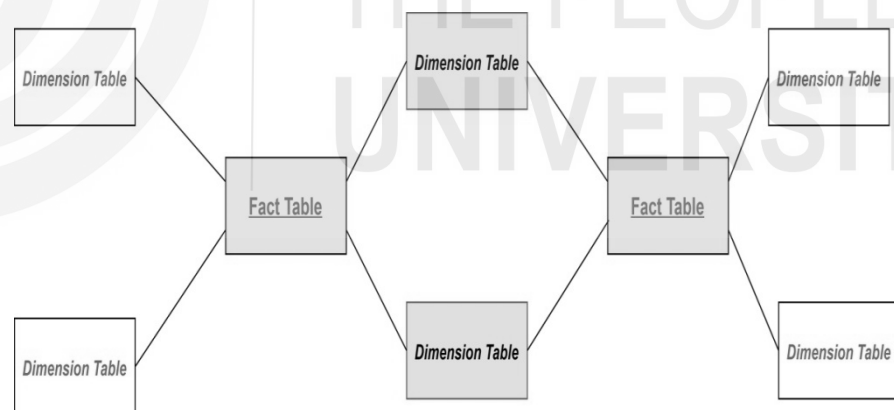


Figure 7: Fact Constellation Schema

In the above figure, it can be observed that there are two fact tables and two-dimension tables in the pink boxes are the common dimension tables connecting both the star schemas.

For example, if we are designing a fact constellation schema for University students. In the problem it is given that their fact table as

Fact tables

Placement (Stud_roll, Company_id, TPO_id) , need to calculate the number of students eligible and number of students placed.

Workshop (Stud_roll, Institute_id, TPO_id) need to find out the facts about number of students selected, number of students attended the workshop)

So, there are two fact tables namely, Placement and Workshop which are part of two different star schemas having dimension tables – Company, Student and TPO in Star schema with fact table Placement and dimension tables – Training Institute, Student and TPO in Star schema with fact table Workshop.

Both the star schema has two-dimension tables common and hence, forming a fact constellation or galaxy schema.

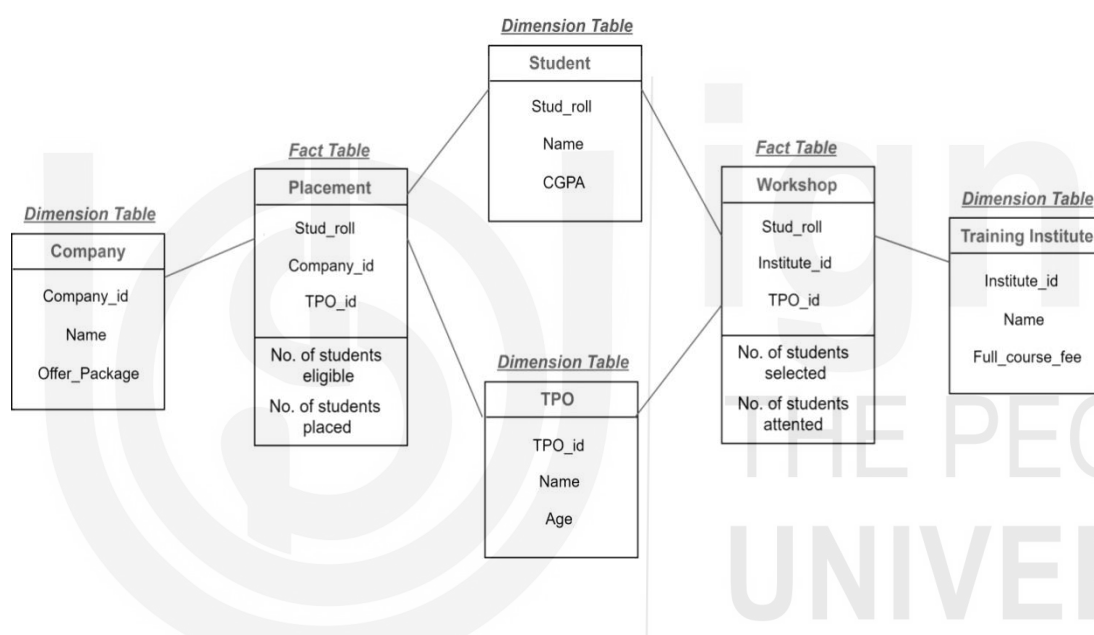


Figure 7: Fact Constellation of student and placement records

3.9.1 Advantages of Fact Constellation Schema

- Different fact tables are explicitly assigned to the dimensions.
- Provides a flexible schema for implementation

3.9.2 Limitations of Fact Constellation Schema

- Complexity of the schema involved because of several aggregations
- Fact constellation solution is hard to maintain and support

☞ Check Your Progress 3:

1. Suppose that a data warehouse consists of dimensions time, doctor, ward and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit. Draw (a) Star and (b) Snowflake schemas:

3.10 SUMMARY

This unit presented the basic designing of data warehouse. These topics are more focused on the various kind of modeling and schemas. It explored the grains, facts, and dimensions of the schemas. It is important to know about the dimensional modeling .as the appropriate modeling technique would yield the correct respond the queries.

A dimensional modeling is a kind of data structure used to optimize design of Data warehouse for the query retrieval operations. There are various schema designs. Here, it discussed star, snowflake, and fact constellations. From denormalized to normalized schemas uses dimension, fact, derived and aggregate fact table. Every table has some purpose and used for efficient designing in terms of space and query handling. This unit discusses the pros and cons of every tables. The number of examples used to explain the designing in different scenarios.

In the next unit, we will focus on Data Extraction, Transforming and Loading (ETL) process in Data Warehousing.

3.11 SOLUTIONS / ANSWERS

Check Your Progress 1:

- 1) Characteristics of Star Schema:
 - Every dimension in a star schema is represented with only one-dimension table.
 - The dimension table should contain the set of attributes.
 - The dimension table is joined to the fact table using a foreign key
 - The dimension table are not joined to each other
 - Fact table would contain key and measure
 - The Star schema is easy to understand and provides optimal disk usage.
 - The dimension tables are not normalized. For instance, in the above figure, Country ID does not have Country lookup table as an OLTP design would have.
 - The schema is widely supported by BI Tools
- 2) Following are some of the benefits of dimensional modeling:
 - The Dimension Table stores the history information and a standard Dimension Table holds good quality data and allows easy access across the business.
 - You can introduce new Dimensions without affecting other Dimensions and Facts in the Schema.
 - Dimension and Fact Tables are easier to read and understand as compared to a normal table.
 - Dimensional Models are built based on business terms, and hence it is quite understandable by the business.

- Dimensional Data Modeling in a Data Warehouse creates a Schema which is optimized for high performance. It means fewer joins between tables and it also helps with minimized data redundancy.
- The Dimensional Data Model also helps to boost query performance. It is more denormalized; therefore, it is optimized for querying.
- Dimensional Data Models can comfortably accommodate the change. Dimension Tables can have more columns added to them without affecting existing Business Intelligence applications using these tables.

Limitations of Dimensional Modeling

Although Dimensional Data Modeling is very crucial to any organization, it has a few limitations that companies need to take care of when incorporating the concept into their applications. Some of those limitations are given below:

- Designing and creating Schemas require domain knowledge about the data.
- To maintain the integrity of Facts and Dimensions, loading the Data Warehouses with a record from various operational systems is complicated.
- It is severe to modify the Data Warehouse operations if the organization adopts the Dimensional technique and changes the method in which they do business.
- Despite these limitations, the DDM technique has proved to be one of the simplest and efficient techniques to handle data in Data Warehouses till date

Check Your Progress 2:

- 1) Sales Fact Table
 - Dimensions(Date, Product, Store, Promotion, TransactionID)
 - Measurements Total_amt, Quantity
 - Create aggregate table with (Date, Store)
 - `SELECT Datakey, StoreKey, SUM(Total_amt), SUM(Qunatity)`
`FROM Sales`
`GROUP BY DateKey, StoreKey`
 - Store the result in Sales2 table
 - Queries that only reference Date and Store attributes can use the aggregate table instead.
 - `SELECT Store.District, SUM(Qunatity)`
`FROM Sales, Store, Date`
`WHERE Sales.DateKey = Date.Date_key`
`AND Sales.Store_key = Store.Store_key`
`AND Date.Month='April2021'`
`GROUP BY Store.District`
 - Replaces “Sales” by “Sales2” → Same query result !
- 2) Fact less tables don't contain any facts. No measure or aggregate functions. It only contains foreign keys. It has data which has textual

information. Fact less tables does not refer to the data which requires calculations.

For example: Fact Table (Day_Id, Employee_ID,leave_Id, Time_Id
Dimension Table (Employee, leave, time, Day)

- 3) **Limitations of Aggregate fact tables:** Aggregate tables take lot of time to scan the rows of the base fact table. So, there will be more tables to manage. The size of aggregates in computing can be costly. Based on the greedy approach the size of aggregates is decided using hashing technique. If there are n dimensions in the table, then there can be 2^n possible aggregates. The load on the data warehouse becomes more complex.

Check Your Progress 3:

a) Star Schema

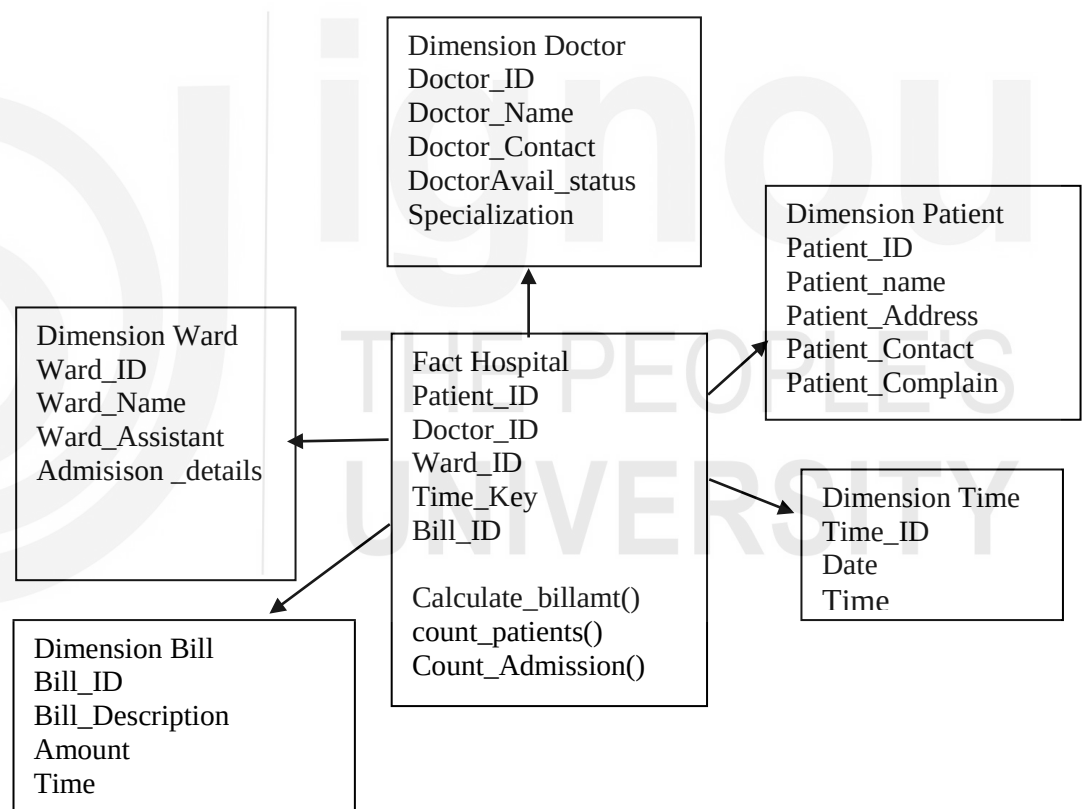


Figure 9 : Star Schema of Hospital Management System

a) Snowflake Schema

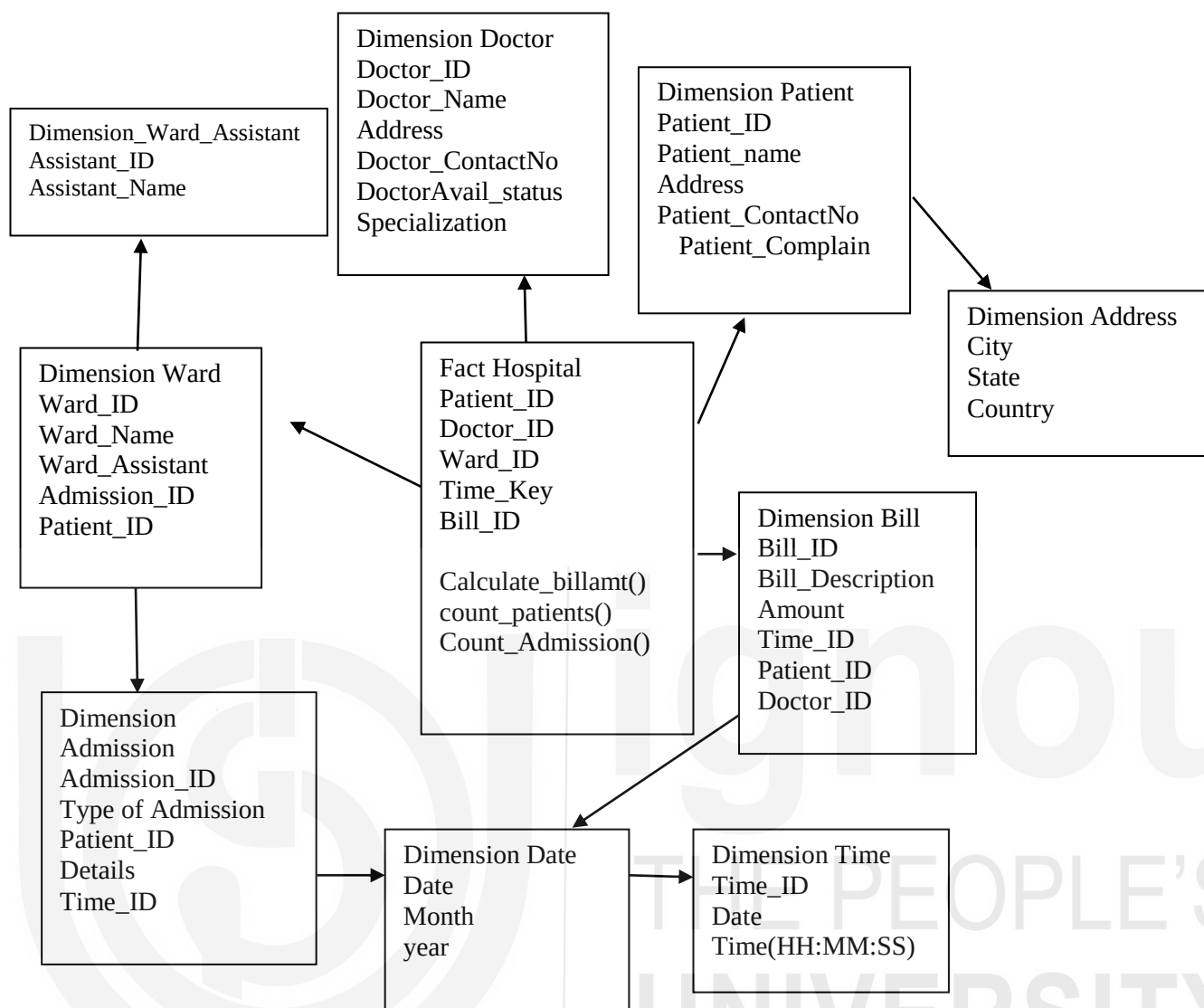


Figure 10: Snowflake Schema of Hospital Management System

3.12 FURTHER READINGS

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UNIT 4 EXTRACT, TRANSFORM AND LOADING

- 4.0 Introduction
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- 4.2 ETL and its Need
 - 4.2.1 Why do You Need ETL?
- 4.3 ETL Process
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4.0 INTRODUCTION

A data warehouse is a digital storage system that connects and harmonizes large amounts of data from many different sources. Data warehouses store current and historical data in one place and act as the single source for an organization. A typical data warehouse has four main components namely:

Central database: A database serves as the foundation of your data warehouse. Traditionally, these have been standard relational databases running on premise or in the cloud. But because of Big Data, the need for true, real-time performance, and a drastic reduction in the cost of RAM, in-memory databases are rapidly gaining in popularity.

Data integration: Data is pulled from source systems and modified to align the information for rapid analytical consumption using a variety of data integration approaches such as ETL (extract, transform, load) and ELT as well as real-time data replication, bulk-load processing, data transformation, and data quality and enrichment services.

Metadata: Metadata is data about your data. It specifies the source, usage, values, and other features of the data sets in your data warehouse. There is business metadata, which adds context to your data, and technical metadata,

which describes how to access data – including where it resides and how it is structured.

Data warehouse access tools: Access tools allow users to interact with the data in your data warehouse. Examples of access tools include: query and reporting tools, application development tools, data mining tools, and OLAP tools. All these components are engineered for speed so that you can get results quickly and analyze the data within no time.

In this unit, we will study about Data Integration component approach such as Extract, Transform and Load (ETL) in detail.

4.1 OBJECTIVES

After going through this unit, you shall be able to:

- Understand the purpose ETL;
- Describe the ETL process, benefits and ETL tools;
- Know the complete working of the ETL
- Discuss various layers involved in the ETL implementation;
- To summarize the functionality of ELT, its need and benefits, and
- To compare and contrast the ETL with ELT.

4.2 ETL AND ITS NEED

Extract, Transform, Load (ETL) as shown in Figure 1, is a process of data integration that encompasses three steps - **extraction**, **transformation**, and **loading**. In a nutshell, ETL systems take large volumes of raw data from multiple sources, convert it for analysis, and load that data into your warehouse.

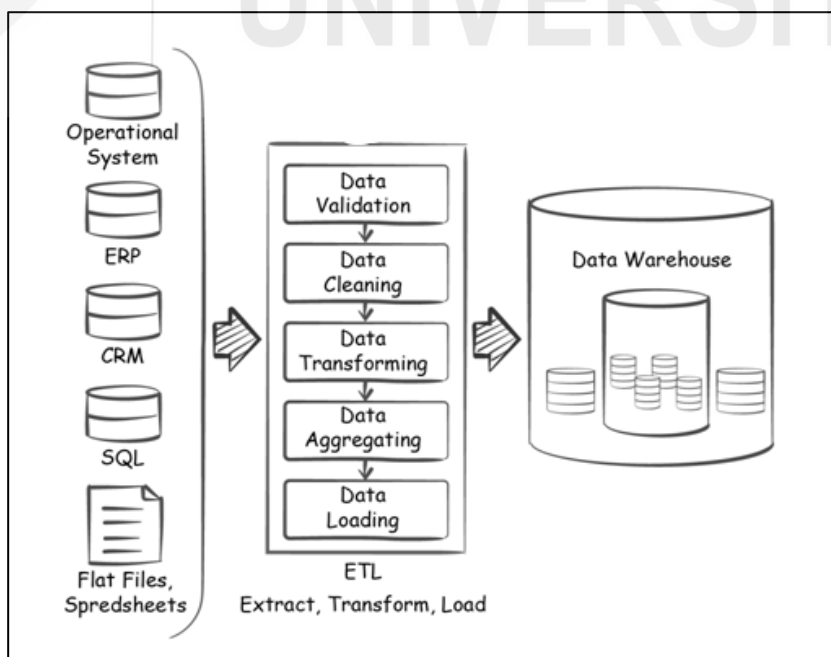


Figure 1: ETL in a Data Warehouse

4.2.1 Why Do You Need ETL?

ETL saves you significant time on data extraction and preparation - time that you can better spend on evaluating your business. Practicing ETL is also part of a healthy data management workflow, ensuring high data quality, availability, and reliability. Each of the three major components in the ETL saves time and development effort by running just once in a dedicated data flow:

Extract: In ETL, the first link determines the strength of the chain. The extract stage determines which data sources to use, the refresh rate (velocity) of each source, and the priorities (extract order) between them — all of which heavily impact your time to insight.

Transform: After extraction, the transformation process brings clarity and order to the initial data swamp. Dates and times combine into a single format and strings parse down into their true underlying meanings. Location data convert to coordinates, zip codes, or cities/countries. The transform step also sums up, rounds, and averages measures, and it deletes useless data and errors or discards them for later inspection. It can also mask personally identifiable information (PII) to comply with GDPR, CCPA, and other privacy requirements.

Load: In the last phase, much as in the first, ETL determines targets and refresh rates. The load phase also determines whether loading will happen incrementally, or if it will require “upsert” (updating existing data and inserting new data) for the new batches of data.

4.3 ETL PROCESS

ETL collects and processes data from various sources into a single data store (a data warehouse or data lake), making it much easier to analyze. The three steps in ETL process are mentioned below:

4.3.1 Data Extraction

Data extraction involves the following four steps:

Identify the data to extract: The first step of data extraction is to identify the data sources you want to incorporate into your data warehouse. These sources might be from relational SQL databases like MySQL or non-relational NoSQL databases like MongoDB or Cassandra. The information could also be from a SaaS platform like Salesforce or other applications. After identifying the data sources, you need to determine the specific data fields you want to extract.

Estimate how large the data extraction is: The size of the data extraction matters. Are you extracting 50 megabytes, 50 gigabytes, or 50 petabytes of data? A larger quantity of data will require a different ETL

strategy. For example, you can make a larger dataset more manageable by aggregating it to month-level rather than day-level, which reduces the size of the extraction. Alternatively, you can upgrade your hardware to handle the larger dataset.

Choose the extraction method: Since data warehouses need to update continually for the most accurate reports, data extraction is an ongoing process that may need to happen on a minute-by-minute basis. There are three principal methods for extracting information:

- (a) *Update notifications:* The preferred method of extraction involves update notifications. The source system will send a notification when one of its records has changed, and then the data warehouse updates with only the new information.
- (b) *Incremental extraction:* The second method, which you can use when update notifications aren't possible, is incremental extraction. This involves identifying which records have changed and performing extraction of only those records. A potential setback is that incremental extraction cannot always identify deleted records.
- (c) *Full extraction:* When the first two methods won't work, a complete update of all the data through full extraction is necessary. Keep in mind that this method is likely only feasible for smaller data sets.

Assess your SaaS platforms: Businesses formerly relied on in-house applications for accounting and other record-keeping. These applications used OLTP transactional databases that they maintained on an on-site server. Today, more businesses use SaaS (software as a service) platforms like Google Analytics, HubSpot, and Salesforce. To pull data from one of these, you'll need a solution that integrates with the unique API of the platform. Xplenty is one such solution.

4.3.2 Data Transformation

In traditional ETL strategies, data transformation that occurs in a staging area (after extraction) is "multistage data transformation". In ELT, data transformation that happens after loading data into the data warehouse is "in-warehouse data transformation". You may need to perform some of the following data transformations:

Deduplication (normalizing): Identifies and removes duplicate information.

Key restructuring: Draws key connections from one table to another.

Cleansing: Involves deleting old, incomplete, and duplicate data to maximize data accuracy - perhaps through parsing to remove syntax errors, typos, and fragments of records.

Format revision: Converts formats in different datasets - like date/time, male/female, and units of measurement - into one consistent format.

Derivation: Creates transformation rules that apply to the data. For example, maybe you need to subtract certain costs or tax liabilities from business revenue figures before analyzing them.

Aggregation: Gathers and searches data so you can present it in a summarized report format.

Integration: Reconciles diverse names/values that apply to the same data elements across the data warehouse so that each element has a standard name and definition.

Filtering: Selects specific columns, rows, and fields within a dataset.

Splitting: Splits one column into more than one column.

Joining: Links data from two or more sources, such as adding spend information across multiple SaaS platforms.

Summarization: Creates different business metrics by calculating value totals. For example, you might add up all the sales made by a specific salesperson to create total sales metrics for specific periods.

Validation: Sets up automated rules to follow in different circumstances. For instance, if the first five fields in a row are NULL, then you can flag the row for investigation or prevent it from being processed with the rest of the information.

4.3.3 Data Loading

Data loading is the process of loading the extracted information into your target data repository. Loading is an ongoing process that could happen through “full loading” (the first time you load data into the warehouse) or “incremental loading” (as you update the data warehouse with new information). Because incremental loads are the most complex, we'll focus on them in this section.

4.3.3.1 Types of Incremental Loads

Incremental loads extract and load information that has appeared since the last incremental load. This can happen in two ways: (a) Batch incremental loads and (b) Streaming incremental loads.

(a) Batch incremental loads: The data warehouse ingests information in packets or batches. If it's a large batch, it's best to carry out a batch load during off-peak hours - on a daily, weekly, or monthly basis - to prevent system slowdowns. However, modern data warehouses can also ingest small batches of information on a minute-by-minute basis with an ETL platform like Xplenty. This allows them to achieve an approximation of real-time updates for the end-user.

(b) Streaming incremental loads: The data warehouse ingests new data as it appears in real-time. This method is particularly valuable when the end-user requires real-time updates (for example: for up-to-the-minute decision-making). Further, streaming incremental loads are only possible when the updates involve a very small amount of data. In most cases, minute-by-minute batch updates offer a more robust solution than real-time streaming.

4.3.3.2 Challenges in Incremental Loading

Incremental loads can disrupt system performance and cause a host of problems, including:

Data structure changes: Data formats in your data sources or data warehouse may need to evolve according to the needs of your information system. However, changing one part of the system could lead to incompatibilities that interfere with the loading process. To prevent problems relating to inconsistent, corrupt, or incongruent data, it's important to zoom out and review how slight changes affect the total ecosystem before making the appropriate adjustments.

Processing data in the wrong order: Data pipelines can follow complex trajectories that result in your data warehouse processing, updating, or deleting information in the wrong order. That can lead to corrupt or inaccurate information. For this reason, it's vital to monitor and audit the ordering of data processing.

Failure to detect problems: Quick detection of any problems with your ETL workflow is crucial: e.g. when an API goes down, when your API access credentials are out-of-date, when system slowdowns interrupt dataflow from an API or when the target data warehouse is down. The sooner you detect the problem, the faster you can fix it, and the easier it is to correct the inaccurate/corrupt data that results from it.

4.4 WORKING OF ETL

In this section, we'll dive a little deeper, taking an in-depth look at each of the three steps in the ETL process.

You can use scripts to implement ETL (i.e. custom *do it yourself* code) or you can use a dedicated ETL tool. An ETL system performs a number of important functions, including:

(a) Parsing/Cleansing: Data generated by applications may be in various formats like JSON, XML, or CSV. The parsing stage maps data into a table format with headers, columns, and rows, and then extracts specified fields.

- (b) **Data enrichment:** Preparing data for analytics usually requires certain data enrichment steps, including injecting expert knowledge, resolving discrepancies, and correcting bugs.
- (c) **Setting velocity:** “Velocity” refers to the frequency of data loading, i.e. inserting new data and updating existing data.
- (d) **Data validation:** In some cases, data is empty, corrupted, or missing crucial elements. During data validation, ETL finds these occurrences and determines whether to stop the entire process, skip the data or set the data aside for human inspection.

4.4.1 Layered Implementation of ETL in a Data Warehouse

When an ETL process is used to move data into a data warehouse, a separate layer represents each phase:

- (a) **Mirror/Raw layer:** This layer is a copy of the source files or tables, with no logic or enrichment. The process copies and adds source data to the target mirror tables, which then hold historical raw data that is ready to be transformed.
- (b) **Staging layer:** Once the raw data from the mirror tables transform, all transformations wind up in staging tables. These tables hold the final form of the data for the incremental part of the ETL cycle in progress.
- (c) **Schema layer:** These are the destination tables, which contain all the data in its final form after cleansing, enrichment, and transformation.
- (d) **Aggregating layer:** In some cases, it's beneficial to aggregate data to a daily or store level from the full dataset. This can improve report performance, enable the addition of business logic to calculate measures, and make it easier for report developers to understand the data.

4.5 ETL AND OLAP DATA WAREHOUSES

Data engineers have been using ETL for over two decades to integrate diverse types of data into online analytical processing (OLAP) data warehouses. The reason for doing this is simple: to make data analysis easier. Normally, business applications use online transactional processing (OLTP) database systems. These are optimized for writing, updating, and editing the information inside them. They are not good at reading and analysis. However, online analytical processing database systems are excellent at high-speed reading and analysis. That's why ETL is necessary to transform OLTP information, so it can work with an OLAP data warehouse.

During the ETL process, information is:

- i. Extracted from various relational database systems (OLTP or RDBMS) and other sources.

- ii. Transformed within a staging area, into a compatible relational format, and integrated with other data sources.
- iii. Loaded into the online analytical processing (OLAP) data warehouse server.

In the past, data engineers hand-coded ETL pipelines in R, Python, and SQL - a laborious process that could take months to complete. Today, hand-coded ETL continues to be necessary in many cases. However, modern ETL solutions like Xplenty allow data teams to skip hand-coding and automatically integrate the most popular data sources into their data warehouses. This has dramatically increased the speed of setting up an ETL pipeline, while eliminating the risk of human error.

In the next section let us focus on the benefits of the ETL tools.

4.6 ETL TOOLS AND THEIR BENEFITS

ETL tools come in a wide variety both in open source or proprietary categories. There are ETL frameworks and libraries that you can use to build ETL pipelines in Python. There are tools and frameworks you can leverage for GO and Hadoop. Really, there is an open-source ETL tool out there for almost any unique ETL need. The downside, of course, is that you'll need lots of custom coding, setup, and man-hours getting the ETL operational. Even then, you may find that you need to tweak your ETL stack whenever you introduce additional tasks. Following are some of the benefits of the ETL tools:

- **Scalability:** Trying to scale-out hand-coded ETL solutions is difficult. As schema complexity rises and your tasks grow more complex and resource-hungry, establishing solid pipelines and deploying the necessary ETL resources can become impossible. With cloud-based ETL tools like Xplenty, you have unlimited scalability at the click of a button.
- **Simplicity:** Going from a hand-coded ETL solution using SQLAlchemy and pandas with rpy2 and parse to something as simple as a cloud-based ETL can be lifechanging. The benefits of having all of your needs layered into one tool saves you time, resources, and lots of headaches.
- **Out-of-the-box:** While open source ETL tools like Apache Airflow require some customization, cloud-based ETL tools like Xplenty work out-of-the-box.
- **Compliance:** The overwhelming nature of modern data compliance can be frightening. Between GDPR, CCPA, HIPAA, and all of the other compliance and privacy nets, using an ETL tool that bakes compliance into its framework is an easy way to skip difficult and risky compliance setups.

- **Long-term costs:** Hand-coded solutions may be cheaper up-front, but they will cost you in the long run. The same thing could be said about open source ETL tools. Since you have to spend time and energy on modification, you're forced to onboard early or risk delaying project launches. Cloud-based ETL tools handle maintenance and back-end caretaking for you.

4.7 IMPROVING THE PERFORMANCE OF ETL

Ultimately tuning is very much required for the ETL to perform better. Following are some of the factors to be considered to improve the ETL performance:

(i) *Tackle the Bottlenecks*

Before anything else, make sure you log metrics such as time, the number of records processed, and hardware usage. Check how many resources each part of the process takes and address the heaviest one. Usually, it will be the second part, building facts, and dimensions in the staging environment.

(ii) *Load Data Incrementally*

Loading only the changes between the previous and the new data saves a lot of time as compared to a full load. It's more difficult to implement and maintain, but difficult doesn't mean impossible, so do consider it. Loading incrementally can definitely improve the ETL performance.

(iii) *Partition Large Tables*

If you use relational databases and you want to improve the data processing window, you can partition large tables. That is, cut big tables down to physically smaller ones, probably by date. Each partition has its own indices and the indices tree is shallower thus allowing for quicker access to the data. It also allows switching data in and out of a table in a quick metadata operation instead of actual insertion or deletion of data records.

(iv) *Cut Out Extraneous Data*

It's important to collect as much data as possible, but not all of it is worthy enough to enter the data warehouse. To improve the ETL performance, define exactly which data should be processed and leave irrelevant rows/columns out. Better to start small and grow as you go as opposed to creating a giant data that takes much time to process.

(v) *Cache the Data*

Caching data can greatly speed things up since memory access performs faster than do hard drives. Note that caching is limited by the maximum amount of memory your hardware supports.

(vi) *Process in Parallel using Hadoop*

Instead of processing serially, optimize resources by processing in parallel. Sort and aggregate functions (count, sum, etc.) block processing because they must end before the next task can begin. Even if you can process in parallel, it won't help if the machine is running on 100% CPU the entire time. You could scale up by upgrading the CPU, but it would scale only to a limit. Hadoop is a much better solution.

Apache Hadoop is designed for the distributed processing of large data over a cluster of machines. It uses HDFS, a dedicated file system that cuts data into small chunks and optimally spreads them over the cluster. Duplicate copies are kept and the system maintains integrity automatically.

MapReduce is used to process tasks (Hadoop 2 or YARN allows more applications). Each MapReduce job works in 2 stages:

- (a) Map - filtering and sorting data - tasks are divided into sub-tasks and processed in parallel by the cluster machines.
- (b) Reduce - summary operations - data from the previous stage is combined.

Hadoop is optimized for distributed processing analytics. Sort and aggregate functions execute in parallel on an entire cluster.

4.8 ELT AND ITS NEED

Extract/load/transform (ELT) is the process of extracting data from one or multiple sources and loading it into a target data warehouse. Instead of transforming the data before it's written, ELT takes advantage of the target system to do the data transformation. This approach requires fewer remote sources than other techniques because it needs only raw and unprepared data. The process is illustrated in Figure 2.

ELT is an alternative to the traditional extract/transform/load (ETL) process. It pushes the transformation component of the process to the target database for better performance. This capability is very useful for processing the massive data sets needed for business intelligence (BI) and big data analytics.

Because it takes advantage of the processing capability already built into a data storage infrastructure, ELT reduces the time data spends in transit and boosts efficiency.

It is becoming increasingly common for data to be extracted from its source locations, then loaded into a target data warehouse to be transformed into actionable business intelligence. ELT process consists of three steps:

- a. Extract - This step works similarly in both ETL and ELT data management approaches. Raw streams of data from virtual infrastructure, software, and applications are ingested either in their entirety or according to predefined rules.
- b. Load – The ELT differs here with the ETL. Rather than deliver this mass of raw data and load it to an interim processing server for transformation, ELT delivers it directly to the target storage location. This shortens the cycle between extraction and delivery.
- c. Transform - The database or data warehouse sorts and normalizes the data, keeping part or all of it on hand and accessible for customized reporting. The overhead for storing this much data is higher, but it offers more opportunities to mine it for relevant business intelligence in near real-time.

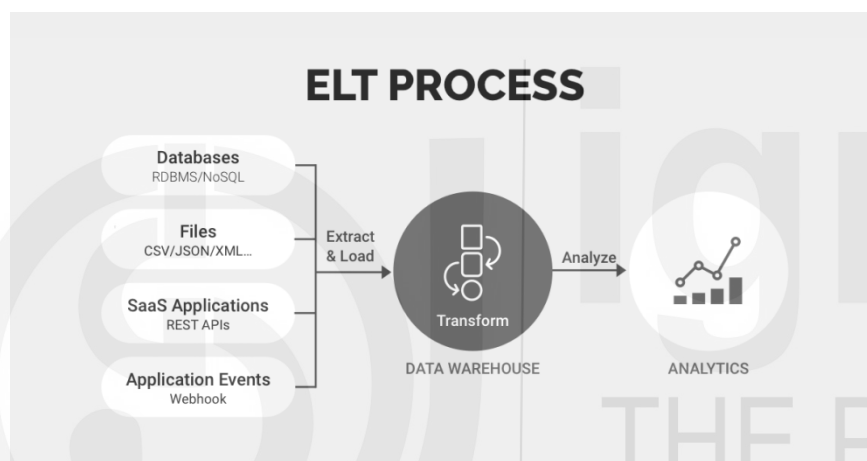


Figure 2: ELT Process

4.8.1 Why Do You Need ELT?

Transforming data after uploading it to modern cloud ecosystems is most effective for:

- Large enterprises with vast data volumes
- Businesses that collect data from multiple source systems or in dissimilar formats
- Companies that require quick or frequent access to integrated data
- Data scientists who rely on business intelligence
- IT departments and data stewards interested in a low-maintenance solution

The ELT process improves data conversion and manipulation capabilities due to parallel load and data transformation functionality. This schema allows data to be accessed and queried in near real time.

However, you might want to stick with ETL if you have dirty data (e.g., duplicate records, incomplete or inaccurate data) that will require data engineers to clean and format it prior to data loading.

4.8.2 Benefits of ELT

With traditional ETL, relevant data is transformed before it is uploaded to a data warehouse, and then it must be pushed out of the warehouse for analysis or processing. This data pipeline works, but it can take more time to migrate data from the source to the target system.

The ELT process saves you steps and time. Data is first loaded into the target ecosystem, such as a data warehouse, and then transformed. Authorized users can securely access the data without returning it to source systems. No downloading is necessary for it. There are reasons to continue using ETL tools. For example, some companies want to keep all their data on-premises. If there is a small amount of data, and it is relational and structured, traditional ETL is effective for businesses that favor hands-on data integration. However, the ELT approach has several benefits for most industries which are listed below:

a) Get better results with more efficient effort

ELT allows you to integrate and process large amounts of data, both structured and unstructured from multiple servers. And, both raw and cleansed data can be accessed with artificial intelligence (AI) and machine learning (ML) tools in addition to SQL and NoSQL processing.

b) Transform your data faster

ELT doesn't have to wait for the data to be transformed and then loaded. The transformation process happens where the data resides, so you can access your data in a few seconds, a huge benefit when processing time-sensitive data.

c) Combine data from different sources and formats

Larger enterprises typically have multiple, disparate data sources like onsite servers, cloud warehouses and log files. Using ELT means you can combine data from various data sets regardless of the source or whether it is structured or unstructured, related or unrelated.

d) Manage data at scale

Technological advances allow organizations to collect petabytes (a million gigabytes!) of data. ELT streamlines the management of massive amounts of data by allowing raw and cleansed data to be stored and accessed. If you're planning to use cloud-based data warehousing or high-end data processing engines like Hadoop, ELT can take advantage of the native processing power for greater scalability.

e) Save time and money

ELT reduces the time data spends in transit and doesn't require an interim data system or additional remote resources to transform the data outside the cloud. Plus, there's no need to move data in and out of cloud ecosystems for analysis. The more your data moves around, the more the costs add up. The scalability of ELT makes it cost-effective for businesses of any size.

4.8.3 ETL Vs ELT

The primary differences between ETL and ELT are how much data is retained in data warehouses and where data is transformed. With ETL, the transformation of data is done before it is loaded into a data warehouse. This enables analysts and business users to get the data they need faster, without building complex transformations or persistent tables in their business intelligence tools. Using the ELT approach, data is loaded into the warehouse as is, with no transformation before loading. This makes jobs easier to configure because it only requires an origin and a destination.

The ETL and ELT approaches to data integration differ in several key ways as listed below:

- **Load time**

It takes significantly longer to get data from source systems to the target system with ETL.

- **Transformation time**

ELT performs data transformation on-demand, using the target system's computing power, reducing wait times for transformation.

- **Complexity**

ETL tools typically have an easy-to-use GUI that simplifies the process. ELT requires in-depth knowledge of BI tools, masses of raw data, and a database that can transform it effectively.

- **Data warehouse support**

ETL is a better fit for legacy on-premise data warehouses and structured data. ELT is designed for the scalability of the cloud.

- **Maintenance**

ETL requires significant maintenance for updating data in the data warehouse. With ELT, data is always available in near real-time.

Both ETL and ELT processes have their importance in the data warehouse architecture as per the understanding of business unique needs and strategies which is key to determining which process will deliver the best results.

Check your Progress 1

1. Define Data Extraction process of ETL along the variety of sources of data accounted for this process. Also, mention the challenge(s) for an ETL tool during the extracting process.

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2. Describe the Data Transformation process involved in the ETL.

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3. Discuss briefly the Data Loading process of ETL.

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4.9 SUMMARY

The process of extracting data from source systems and bringing it into the data warehouse is commonly called ETL process, which stands for extraction, transformation, and loading. In this unit, we had studied that ETL refers to a broad process. The methodology and tasks of ETL have also been studied.

Apart from the ETL, we had studied a new approach known as ELT and its benefits too, in this unit.

In the next unit, we will be study about Online Analytical Processing (OLAP).

4.10 SOLUTIONS / ANSWERS

Check your Progress 1:

1. Extracting data is the act of pulling data from one or more data sources. During the extraction phase of ETL, you may handle a variety of sources with data, such as:
 - Relational and non-relational databases
 - Flat files (e.g. XML, JSON, CSV, Microsoft Excel spreadsheets, etc.)

- SaaS applications, such as CRM (customer relationship management) and ERP (enterprise resource planning) systems
- APIs (application programming interfaces)
- Websites
- Analytics and monitoring tools
- System logs and metadata

In the first step, extracted data sets come from a source (*e.g., Salesforce, Google AdWords, etc.*) into a staging area. The staging area acts as a buffer between the data warehouse and the source data. Since data may be coming from multiple different sources, it's likely in various formats, and directly transferring the data to the warehouse may result in corrupted data. The staging area is used for data cleansing and organization.

A big challenge during the data extraction process is how your ETL tool handles structured and unstructured data. All of those unstructured items (*e.g., emails, web pages, etc.*) can be difficult to extract without the right tool, and you may have to create a custom solution to assist you in transferring unstructured data if you chose a tool with poor unstructured data capabilities.

2. It's rarely the case that your extracted data is already in the exact format that you need it to be. For example, it may require the following:

- Rearrange unstructured data into a structured format.
- Limit the data you've extracted to just a few fields.
- Sort the data so that all the columns are in a certain order.
- Join multiple tables together.
- Clean the data to eliminate duplicate and out-of-date records.

The data cleaning and organization stage is the transformation stage. All of that data from multiple source systems will be normalized and converted to a single system format — improving data quality and compliance. ETL yields transformed data through these methods:

- Cleaning
- Filtering
- Joining
- Sorting
- Splitting
- Deduplication
- Summarization

3. Once the process has transformed, sorted, cleaned, validated, and prepared the data, you need to load it into data storage somewhere and the most common target database is a data warehouse. Depending upon your business needs, data can be loaded in batches or all at once. The exact nature of the loading will depend upon the data source, ETL tools, and various other factors.

ETL architecture often consists of a diagram that outlines the flow of information from start to finish in your ETL pipeline. Data originates from sources files and databases, where it's taken up and entered into the ETL transformation engine. From here, the data may be loaded into any or all of the following locations:

- **Landing area:** The landing area is where data first arrives after being extracted from a source location. Few, if any, transformations are applied to the data in the landing area. If you are performing ETL batch processing, the landing area may store multiple batches of data before moving it through the ETL pipeline.
- **Staging area:** The staging area is a temporary, intermediate location for performing ETL transformations. This area may take the form of a relational database, or binary or text files. The transformations performed in this area include joining and consolidating multiple data sources, cleansing and validating source data, and standardizing and aligning information.
- **Data warehouse area:** The data warehouse area is the final destination for data in an ETL pipeline. From here, your data can easily be queried and analyzed in order to obtain valuable insights and make better business decisions. This area may consist of an enterprise data warehouse spanning the entire organization, or a data mart that has been set up to serve the needs of a single team or department.

2.11 FURTHER READINGS

1. William H. Inmon, Building the Data Warehouse, Wiley, 4th Edition, 2005.
2. Data Warehousing Fundamentals, Paulraj Ponnaiah, Wiley Student Edition, 2001.
3. Data Warehousing, Reema Thareja, Oxford University Press, 2009.

UNIT 5 INTRODUCTION TO ONLINE ANALYTICAL PROCESSING

Structure

- 5.0 Introduction
- 5.1 Objectives
- 5.2 Online Analytical Processing and its Need
- 5.3 Characteristics of OLAP
- 5.4 OLAP and Multidimensional Analysis
 - 5.4.1 Multidimensional Logical Data Modeling and its Users
 - 5.4.2 Multidimensional Structure
 - 5.4.3 Multidimensional Operations
- 5.5 OLAP Functions
- 5.6 Data Warehouse and OLAP: Hypercube and Multicubes
- 5.7 Applications of OLAP
- 5.8 Steps involved in OLAP Creation
- 5.9 Advantages of OLAP
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- 5.11 Online Transaction Processing
- 5.12 Summary
- 5.13 Solutions/Answers
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5.0 INTRODUCTION

In the earlier unit, we had studied on Extract, Transform and Loading (ETL) component of data warehousing. Online Analytical Processing is the core component of the data warehouse implementation, providing fast and flexible multi-dimensional data analysis for business intelligence (BI) and decision support applications.

In this unit we will study the need of Online Analytical Processing, its functions, applications and Online Transaction Processing Systems.

5.1 OBJECTIVES

After going through this unit, you should be able to:

- understand the purpose of a OLAP;
- describe the motivation and benefits of OLAP;
- discuss Multidimensional Modeling Structure;
- describe various OLAP operations;
- list the applications of Multi cube and steps to create OLAP server;
- differentiate between various types of OLAP architectures, and
- understand the online transaction processing, its need, applications and characteristics

5.2 ONLINE ANALYTICAL PROCESSING AND ITS NEED

Online Analytical Processing (OLAP) is the technology to analyze and process data from multiple sources at the same time. It accesses the multiple databases at the same time. It helps the data analysts to collect data from different perspective for developing effective business strategies. The query operations like grouping, join or aggregation can be easily done with OLAP using pre-calculated or pre-aggregated data hence making it much faster than simple relational databases. You can understand OLAP as a multi cubic structure, which has many cubes - each cube is pertaining to some database. The cubes are designed in such a way that generates reports effectively and efficiently.

OLAP is a software used to perform high-speed, multivariate analysis of large amounts of data in data warehouses, data markets, or other unified and centralized data warehouses. The data is broken down for display, monitoring or analysis. For example, sales figures can be related to location (region, country, state/province, company), time (year, month, week, day), product (clothing, male/female/child, brand, type), etc., but In a data warehouse, records are stored in tables, and each table can only sort data on two of the dimensions at a time. Recording and reorganizing them into a multi-dimensional format allows very fast processing and very in-depth analysis.

The primary objective of OLAP is not just data processing. For instance, if a company might compare their sales in the month of January with the month of February then compare those results with another location which may be stored in a separate database. In this case, it needs a multi-view of database design storing all the data categories. Another example is in Amazon it analyzes purchases made by its customers to recommend the customers with a personalized home page of products which are likely to be interested by them. It creates a single platform for all types of business analytical means which includes planning budgeting, forecasting and analysis.

The main benefit of OLAP is the consistency of information and calculations using OLAP systems and we can easily apply security restrictions on users and objects to comply with regulations and protect sensitive data. OLAP assists managers in making decisions by giving multidimensional record views that are efficient to provide, hence enhancing their productivity. Due to the inherent flexibility support provided by organized databases, OLAP functions are self-contained. Through extensive control of analysis-capabilities, it permits simulation of business models and challenges.

Let's see the need to use OLAP to have better understanding of OLAP over relational databases or online transaction processing (OLTP):

- **Efficient and Effective methods to improve the sales of an Organization:** In a retail, having multiple products with different number of channels for selling the product across the globe. OLAP makes it effective and efficient to search for a product in a different region within a specified time period (like, excluding weekdays sales or just weekend sales or festival duration sales very specific from a very large data distributed).
- **It improves the sales of a business:** The data analysis power of OLAP brings effective results in sales. It helps in identifying expenditures which produces a high return of investments (ROI).

Usually, data operations and analysis are performed using the simple spreadsheets, where data values are arranged in row and column format. This is ideal for two-

dimensional data. However, OLAP contains multidimensional data, with data usually obtained from a different and unrelated source. Using a spreadsheet is not an optimal option. The cube can store and analyze multidimensional data in a logical and orderly manner.

5.3 CHARACTERISTICS OF OLAP

The main characteristics of OLAP are as follows:

- **Fast:** OLAP act as bridge between data warehouse and front-end. Hence helps in the better accessibility of data yielding faster results.
- **Analysis:** OLAP data analysis and computational measure and their results are stored in separate data files. OLAP distinguishes better zero and missing values. It should ignore missing values and performs the correct aggregate values. OLAP facilitates interactive query handling and complex analysis for the users.
- **Shared:** OLAP operations drill-down or roll-up, it navigates between various dimensions in multidimensional cube making it effective and efficient reporting system.
- **Multidimensional:** OLAP has Multidimensional conceptual view and access of data to different users at different levels. The increasing number of dimensions and report generation performance of the OLAP system does not significantly degrade.
- **Data and Information:** OLAP has calculation power for complex queries and data. It does data visualization using graphs and charts.

5.4 OLAP AND MULTIDIMENSIONAL ANALYSIS

The multi-dimensional data model stores data in the form of data cube in a data warehouse. Generally, it supports two or three-dimension cubes. It gives the data different views and perspectives. Practically in a retail store the data is maintained month-wise, item-wise, region-wise thus involving many different dimensions.

5.4.1 Multidimensional Logical Data Modeling and its Users

The multidimensional data modeling provides:

- **Different views and perspectives to the data from different angles:** The business users have a dimensional and logical view of the data in the data warehouse.
- **Multidimensional conceptual view:** It allows users to have a dimensional and logical view of the data.
- **Multidimensional Modeling creates environment for multiuser:** Since the OLAP techniques are shared, the OLAP and database operations, containing retrieval, update, adequacy control, integrity, and security can be easily performed.

For example, in the figure 1, it is shown that the dimensions Time, Regions and Products of a company can be logically saved in a cube. In figure 2, the cross tabular form in every quarter products quantity is shown. In figure 1, Products, Time and

regions these dimensions can be combined into cubes you can imagine what two dimensions would look like by using a spreadsheet metaphor with the time dimension as the columns and the products dimension as the rows if we add data to this view such as units sold that would be a measure. **Measures** can be any quantity such as revenues expenses unit's statistics or any text or numerical value if we consider adding the third dimension regions then you can imagine each region being represented as an additional spreadsheet this is how it works when you're limited to a two-dimensional spreadsheet. however, an OLAP cube can represent all three dimensions as a single data set which allows users to fluidly explore all the data from any perspective and despite its name, a cube can hold many more than three dimensions so what's the value of using all that to illustrate this:

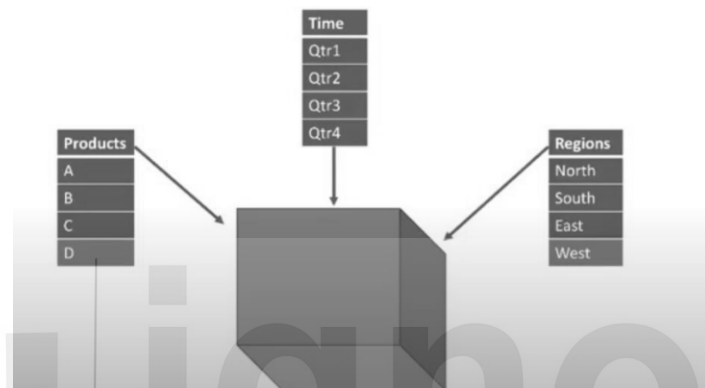


Figure 1: Cube Representation

Figure 2 shows a 2D spreadsheet view of the OLAP cube data. The columns are Time (Qtr1, Qtr2, Qtr3, Qtr4) and the rows are Products (A, B, C, E). The data values are:

	Qtr1	Qtr2	Qtr3	Qtr4
A	300	200	100	600
B	500	200	300	100
C	100	400	100	300
E	300	400	700	200

Figure 2: Measurable data shown

Let's say that a manager is tracking sales units with three different spreadsheets with three different dimensions products, quarters and regions from looking at these spreadsheets, it appears that everything is equal as the manager of these stores would probably stock them with the same number of items for each product quarter and region. The manager of a store house makes very different decisions to generate a report with just one or two dimensions or by adding more dimensions and reveal more detail which would allow to make better decisions on managing the inventory of the stores. Hence, you can view OLAP facilitates business oriented multidimensional data having lot of calculations. The data saved in multidimensional structure is very significant in speed thought analysis to companies to take better decisions. OLAP provides the flexibility of data retrieval to generate reports.

5.4.2 Multidimensional Structure

In a multidimensional cube there are dimensions. In each of the dimension of the cube usually we have something called hierarchy which means some of the attributes in a single dimension may form a hierarchy.

For instance let's just use the time dimension we can have years and two quarters into month so kind of we can drill down and look at the more detailed aspects of the dimension. If we use the time dimension as an example, we can have multiple levels of attributes which can help us to form the hierarchy break it down to quarter and break it down to month. Let's use on our grocery database as an example the cells fact table on which has multiple dimensions one of the dimensions is the product dimension and in the product dimension, we can find naturally the following hierarchy. The data has been organized into multiple dimensions and at each level of dimension, contains multiple levels of abstraction defining the Concept Hierarchy. It provides flexibility to view data from different angles.

Concept Hierarchy of a Product

Department → Category → Subcategory → Brand → Product

It is important to identify the hierarchy from multi-dimensional cube in terms of query. Then we must look at the performance measure on which attribute or dimension the query is focused on. For instance, department we have grocery and house, so we have two possible values under grocery we have food and then drink obviously based on this relationship.

We know that Department can be broken down into category, so in the grocery department we sell food and drinks and for fresh food if we look at the value, food category can break down into frozen food and fresh food and frozen food can be broken down into two brand codes. So that's kind of the hierarchy you shouldn't count on the sequence of the column within the table to determine the hierarchy you need to really understand your data and sometime study the value stored in the dimensional table that will help you determining hierarchy as well so how this multi-dimensional cube can be used for different queries. Also, during getting the answers, performance measure should be considered by looking at the performance driver for different dimensions.

5.4.3 Multidimensional Operations

Typically, there are four types of OLAP operations are:

- i. Roll-up
- ii. Drill-down
- iii. Slice and Dice
- iv. Pivot (rotate)

In practical daily life we come across operations where the manager is interested in knowing the aggregate of data from the concept hierarchy. It can use the concept hierarchy to roll the data up so for instance instead of a daily aggregated data we have monthly aggregate data and quarterly and then annual year.

Concept hierarchy of Time dimension

Year → Quarter → Month → Week → Daily

So, to perform this operation, we can roll-up and store the result of roll-up and subtotal those aggregated data. So, if the manager is interested in going down the concept hierarchy or interested in the minute details to find out the driving attribute responsible for the increase or decrease of sales. For this OLAP operation drill down can be performed.

(i) **Roll-up:**

- It is an OLAP operation to move from low level of concept hierarchy to higher level known as aggregation.
- It can be performed by reducing the dimensions as well.

In the following example, it is shown a multidimensional cube containing the products of a Home appliances like laptop, furniture, mobile and kitchen appliances, if the manager wants to view the sales of all the products quarterly. The Roll-up operation can be performed on these categories. In this aggregation process, hierarchy moves up and becomes Home Appliance Electronics, as shown in figure 3 below.

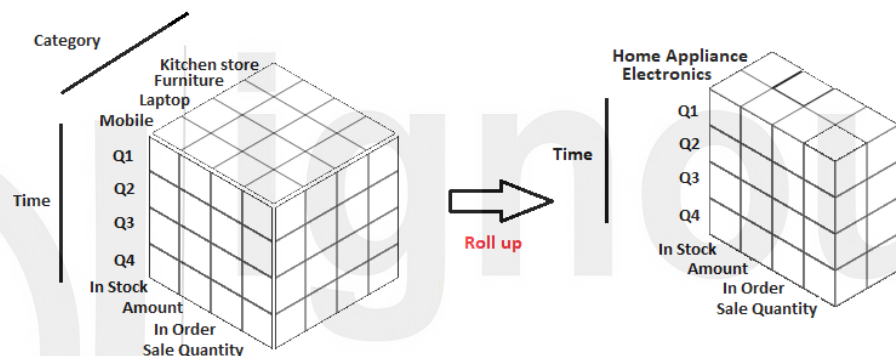


Figure 3: Roll-up on (Category from Home Appliances and Electronics)

(ii) **Drill-down:**

In this process of scaling down from higher level to the lower level the data is broken into smaller parts.

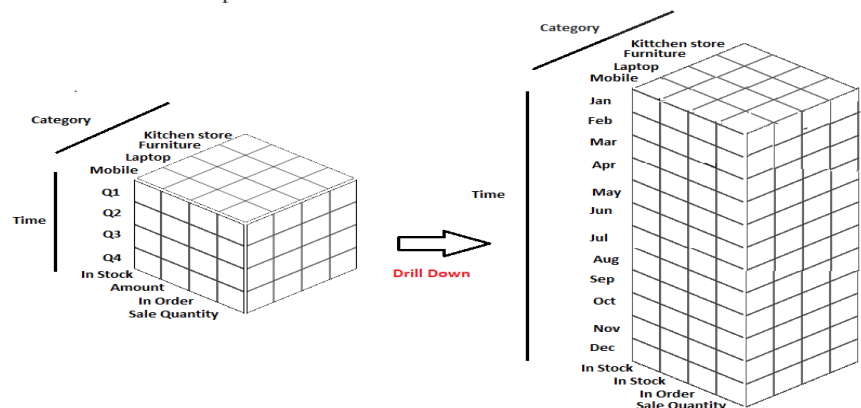


Figure 4: Drill down from Time to Months

You will observe in the above example as shown in figure 4, a multidimensional cube containing products and time. The Time dimension has been drilled-down from

Quarter → Months to observe the sales month-wise.

(iii) Slice:

It is another OLAP operation to fetch the data. In this the query on one dimension is triggered in the database and a new sub cube is created.

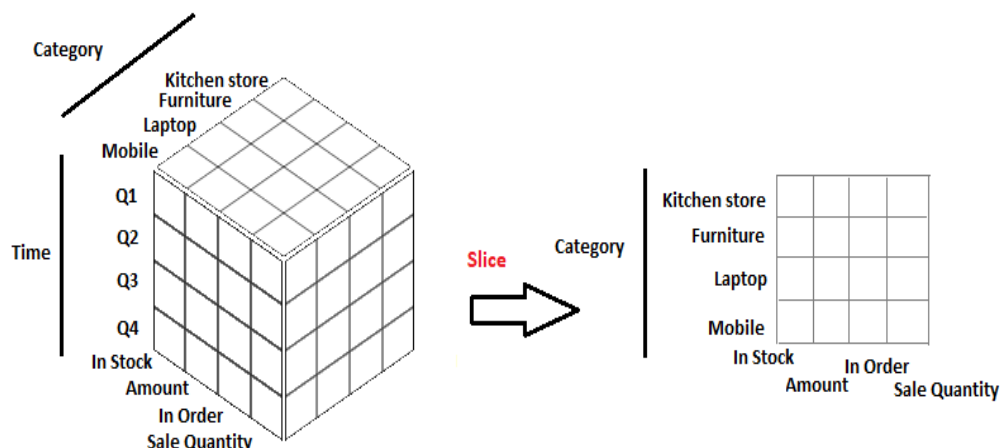


Figure 5: Slice OLAP Operation

In the above figure 5, it can be observed that slice operation is performed on “Time” dimension and a new sub cube is created to retrieve the results.

Slice for Time = “Q1”

(iv) Dice:

This OLAP Dice operation as shown in figure 6 is just like the Projection relational query you have read in RDBMS. In this technique, you select two or more dimensions that results in the creation of a sub cube as shown in the figure.

Dice for (Category= “Laptop” or “Mobile”) and (Time = “Q1” or “Q2”) and (Stock = “Amount” or “Sale Quantity”)

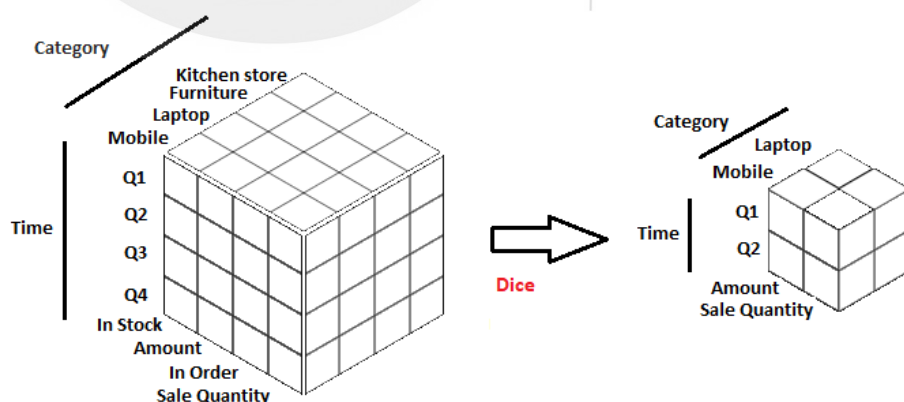


Figure 6: Dice OLAP Operation

(v) Pivot:

This OLAP operation fixes one attribute as a Pivot and rotate the cube to fetch the results, just like an inverted spreadsheet, giving it a different perspective. You can observe in the figure 7, the presentation of the dimensions has been changed to impart a different perspective of the data cube for data analysis.

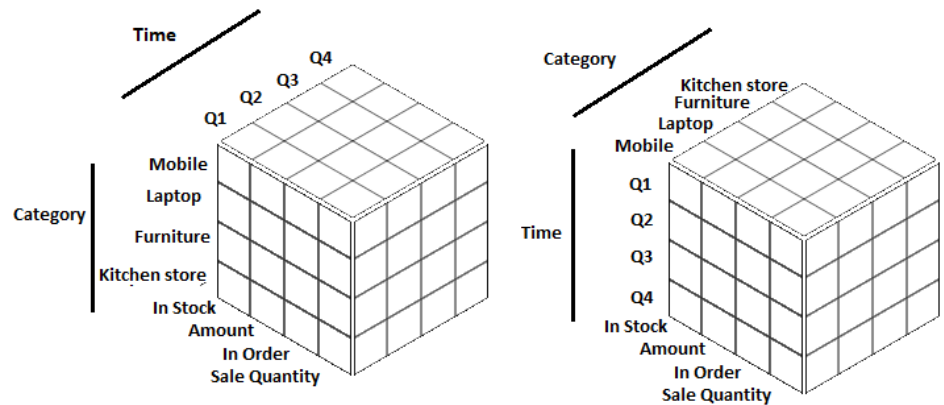


Figure 7: Pivot OLAP operation

☞ Check Your Progress 1:

- 1) Who are the users of the Multidimensional data modeling?

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- 2) What are the five categories of decision support tool?

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5.5 OLAP FUNCTIONS

Online Analytical Processing functions can return the ranking, row numbering. It is very similar to the SQL aggregate functions, but aggregate functions return atomic value.

- The OLAP function returns a scalar value of a query. OLAP functions can be performed at the individual row levels too.
- An OLAP function provides data mining functionalities and data analysis. The detailed data analysis and values are supported with OLAP functions.
- The exhaustive and comprehensive data analysis can be achieved row wise unlike simple SQL functions which produces results in the form of reports like WITH. OLAP runs on rows of the data warehouse.
- OLAP functions uses SQL commands like INSERT/SELECT/ POPULATE on tables or Views.

5.6 DATA WAREHOUSE AND OLAP: HYPERCUBE AND MULTICUBES

The OLAP cube is a data structure optimized for very quick data analysis. The OLAP Cube consists of numeric Facts called Measures which are categorized by dimensions. OLAP Cube is also called the hypercube. So, we can say that multidimensional databases can be seen as hypercube and multi cube. Multidimensional cubes have

smaller multiple cubes and in hypercube it seems there is one cube as logically all the data seems to be as one unit of cube.

Table 1: Differences between Multi cube and Hyper cube

	Multi Cube	Hyper Cube
Metadata	Each dimension can belong to many cubes	Each dimension belongs to one cube only
Dimension	Not necessary all the dimensions should belong to some cube	Every dimension owned by a hypercube
Measure Computation	Complex, data can be retrieved from the all the cubes	Simple, as all the numerical facts are available at one place
Multiple	Multicube system, if there are two rows in the DIMENSIONS rowset for which the DIMENSION_NAME value is the same (and the CUBE_NAME value is different), these two rows represent the same dimension. As, sub cubes are built from the same pool of available dimensions.	in a multiple hypercube scenario, it is possible for two hypercubes to have a dimension of the same name, each of which has different characteristics. In this case, the DIMENSION_UNIQUE_NAME value is guaranteed to be different.

5.7 APPLICATIONS OF OLAP

OLAP reporting system is widely used in business applications like:

- Sales and Marketing
- Retail Industry
- Financial Organizations – Budgeting
- Agriculture
- People Management
- Process Management

Examples are Essbase from Hyperion Solution and Express Server from Oracle.

Check Your Progress 2:

- 1) Explain the OLAP application reporting system in Marketing?

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- 2) What is the purpose of hyper cube. Show slice and dice operation on the sub-cube/hypercube?

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3) List the features of an OLAP

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5.8 STEPS INVOLVED IN OLAP CREATION

The basic unit of OLAP is an OLAP cube. It is a data structure designed for better and faster retrieval of results from the data analysis. It has dimensions with numeric facts. The data arrangement in rows and columns in multidimensional is the logical view not the physical view. The building of an OLAP cube uses multidimensional array so that the data can be viewed in all the directions, analysis of data and respond to queries can become efficient. For example, the dimensions of cube are customer, time and products and measure count and total sales. The steps involved in the creation of OLAP are as follows:

Step 1: <i>Extract data from variety of sources like text, excel sheets, multimedia files, Online Transaction Processing data in flat files.</i>
Step 2: <i>Transformation and Standardization of data:</i> Since, the data is distributed and incompatible to each other. It involves the data preprocessing or cleaning part where the semantics of databases are changed into a standard form.
Step 3: <i>Loading of data:</i> After all the database nomenclature has been followed then the data is loaded onto the OLAP server or OLAP multidimensional cube.
Step 4: <i>Building of a Cube for data analysis</i> • Select the dimensions means set of subsets of significant attributes. • Select the concept hierarchies. • Populate the cube with the relevant data • Select the numeric attribute to apply aggregate function.
Step 5: <i>Report Generation</i>

The steps to create OLAP shown in the below figure 8:

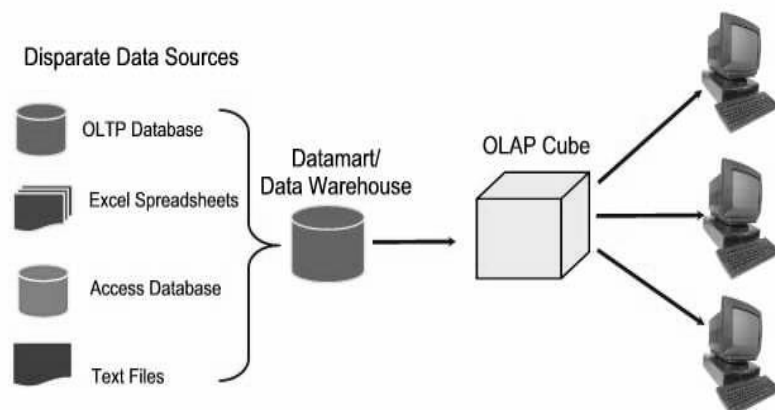


Figure 8 : Steps to create OLAP Cube

5.9 ADVANTAGES OF OLAP

The SQL functions like Group By, Aggregating functions are quite complex to operate in relational databases as compared to multidimensional databases. OLAP can pre-compute the queries can save in sub cubes. The hypercubes also make the computation task faster and saves time. OLAP has proved to an extremely scalable and user – friendly method which is able to perfectly cater to its entire customer needs ranging from small to large companies.

Some listed benefits of using OLAP are as follows:

- ***Data Processing at a faster speed***

The speed of query execution has been tremendous since the use of OLAP technology and is now counted as one of the primary benefits for it. This prevents the customers from spending a lot of time and money on heavy calculations and creating complex reports.

- ***Accessibility***

The cube enables the various kinds of data like – transactional data from various resources, information about every supplier and consumer, etc. all is saved in a concise one location which is easy to operate.

- ***Concise and Fine Data***

OLAP works on the principle of combining multiple and similar records together, which are saved in multiple tables forming a schema between them as a source of connection. These tables combine to form the cube to make the massive information concise and yet finely available to the user. Records can be elemental right down to a single element by “drill down” and back to the cube by “drill up” operations.

- ***Data Representation in Multi-Dimension***

OLAP cube is the center of all the data. Each element of the cube contains various attributes and the number of processes performed on it. The cube axes are outlined by the measure and dimension of the cube which is mostly three - dimensional system. This allows the user to take the information from various slices of the cube. A cube slice is a two – dimensional in nature which gives a clear image of the knowledge trying to be represented.

- ***Business Expressions commonly used***

The size of an OLAP cube consisting of data portrays the company’s economic and financial conditions. The end user does not manipulate the database files; they deal with end processes like products, salesmen, employees, customers, etc. This gives a reason to even user with less to zero technical background to use OLAP technology.

- ***Situational Scenarios***

The way the cube can cover almost all parts of a data item is through creating various what – if situations; these what – if situations help in extraction of cube information without tampering the original information on the cube. This feature of OLAP technology is responsible for providing the customers the ability to update the values to look at the consequences brought in the cube’s situation. Through this feature business intelligence can deeply examine the possible factors of driving a situation in a company and prevent them if necessary.

- ***Easily Understood Technology***

Most of the users or customers working on OLAP technology come from a background of less to minimum technology skills. They mostly do not need any unique training to use this technology, which in return helps the company save some

money. Moreover, OLAP technology providers provide their end users with enough tutorial, documents and some start off technical assistance particularly in case of web based OLAP operations. The end customers are given sessions to continuously work with a group of technical experts so that they do not have to solve all the OLAP issues by themselves.

5.10 OLAP ARCHITECTURE: MOLAP, ROLAP, HOLAP AND DOLAP

There are various types of OLAP architecture: ROLAP, MOLAP, HOLAP and others as shown in the below figure 9.

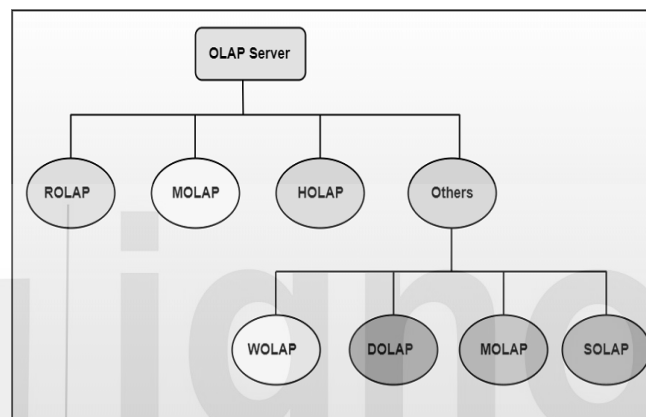


Figure 9: Types of OLAP Architecture

5.10.1 ROLAP Architecture

ROLAP implies Relational OLAP, an application based on relational DBMSs. It performs dynamic multidimensional analysis of data stored in a relational database. The architecture is like three-tier. It has three components viz. front end (User Interface), ROLAP server (Metadata request processing engine) and the back end (Database Server) as shown in the figure 10.

- Database server
- ROLAP server
- Front-End tool

In this three-tier architecture the user submits the request and ROLAP engine converts the request into SQL and submits to the backend database. After the processing of request by the engine, it presents the resulting data into multidimensional format to make the task easier for the client to view it.

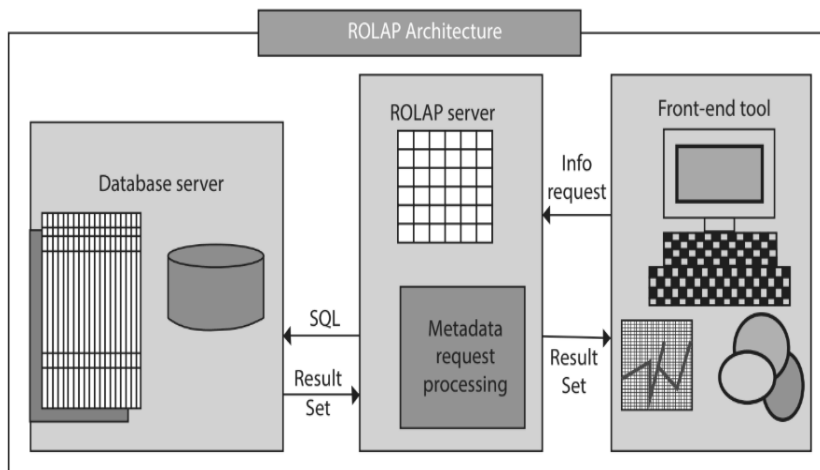


Figure 10 : ROLAP Architecture

The characteristics of ROLAP are:

- ROLAP utilizes the more processing time and disk space.
- ROLAP enables and supports larger user group in the distributed environment.
- ROLAP processes complex queries utilizing the greater amounts of data.

Popular ROLAP products include Metacube by Stanford Technology Group, Red Brick Warehouse by Red Brick Systems.

5.10.2 MOLAP Architecture

MOLAP stands for Multidimensional Online Analytical Processing. It processes the data using the multidimensional cube using various combinations. Since, the data is stored in multidimensional structure the MOLAP engine uses the pre-computed or pre-stored information. The architecture has three components:

- Database server
- MOLAP server
- Front-End tool

As shown in Figure 11, MOLAP engine processes pre-compiled information. It has dynamic abilities to perform aggregation of concept hierarchy. MOLAP is very useful in time-series data analysis and economic evaluation.

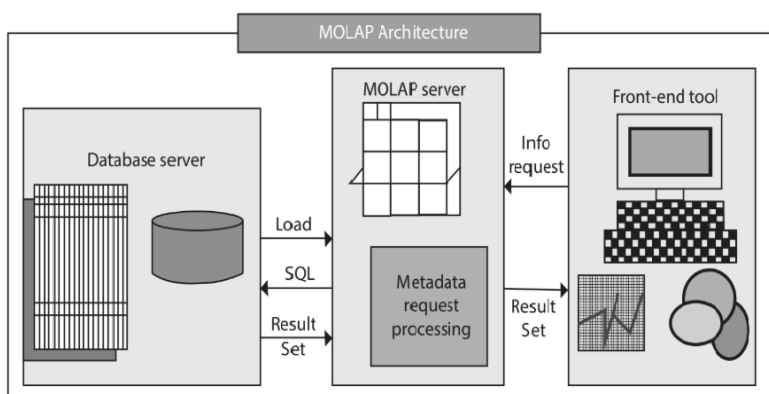


Figure 11 : MOLAP Architecture

The characteristics of MOLAP

- It is a user-friendly architecture, easy to use.
- The OLAP operations slice and dice speeds up the data retrieval.
- It has small pre-computed hypercubes.

Tools that incorporate MOLAP include Oracle Essbase, IBM Cognos, and Apache Kylin.

5.10.3 HOLAP Architecture

It defines Hybrid Online Analytical Processing. It is the hybrid of ROLAP and MOLAP technologies. It connects both the dimensions together in one architecture. It stores the intermediate or part of the data in ROLAP and MOLAP. Depending on the query request it accesses the databases. It stores the relational tables in ROLAP structure, and the data requires multidimensional views, stored and processed using MOLAP architecture as shown in figure 12.

It has the following components:

- Database server.
- ROLAP and MOLAP server.
- Front-end tool

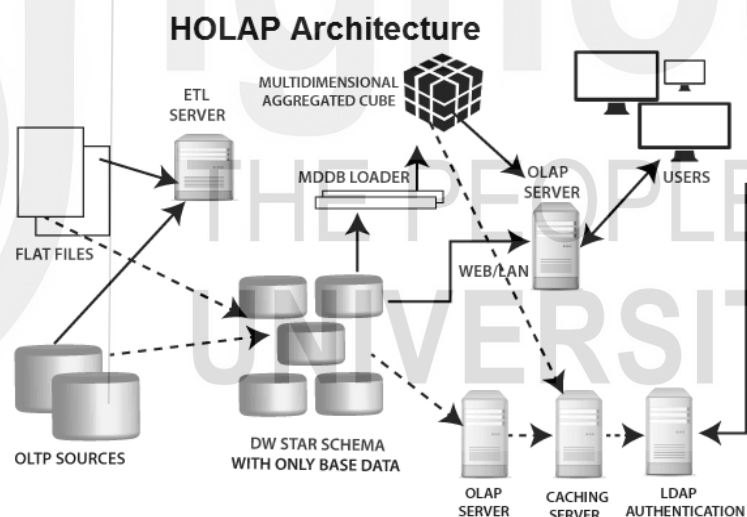


Figure 12 : HOLAP Architecture

The characteristics of HOLAP are:

- Flexible handling of data.
- Faster aggregation of data.
- HOLAP can drill down the hierarchy of data and can access to relational database for any relevant and stored information in it.

Popular HOLAP products are Microsoft SQL Server 2000 presents a hybrid OLAP server.

5.10.4 DOLAP Architecture

Desktop Online Analytical Processing (DOLAP) architecture is most suitable for local multidimensional analysis. It is like a miniature of multidimensional database or it's like a sub cube or any business data cube. The components are:

- Database Server
- DOLAP server
- Front End

The characteristics of DOLAP are:

- The three-tier architecture is designed for low-end, standalone user like a small shop owner in the locality.
- The data cube is locally stored in the system so, retrieval of results is faster.
- No load on the backend or at the server end.
- DOLAP is relatively cheaper to deploy.

5.11 ONLINE TRANSACTION PROCESSING

Online Transactional Processing (OLTP) enables the real-time execution of large numbers of database transactions by large numbers of people, typically over the internet.

A database transaction is a change, insertion, deletion, or query of data in a database. OLTP systems (and the database transactions they enable) drive many of the financial transactions we make every day, including online banking and ATM transactions, eCommerce and in-store purchases, and hotel and airline bookings, to name a very few. In each of these cases, the database transaction also remains as a record of the corresponding financial transaction. OLTP can also drive non-financial database exchanges, including password changes and text messages.

In OLTP, the common, defining characteristic of any database transaction is its *atomicity* (or indivisibility) – a transaction either succeeds as a whole or fails (or is canceled). It cannot remain in a pending or intermediate state.

Examples:

Since the inception of the internet and the e-commerce era, OLTP systems have grown ubiquitous. They're found in nearly every industry or vertical market and in many consumer-facing systems. Everyday examples of OLTP systems include the following:

- ATM machines (this is the classic, most often-cited example) and online banking applications
- Credit card payment processing (both online and in-store)
- Order entry (retail and back-office)
- Online bookings (ticketing, reservation systems, etc.)
- Record keeping (including health records, inventory control, production scheduling, claims processing, customer service ticketing, and many other applications)

5.11.1 Characteristics of OLTP Systems

The most common architecture of an OLTP system that uses transactional data is a three-tier architecture that typically consists of a presentation tier, a business logic tier, and a data store tier. The presentation tier is the front end, where the transaction originates via a human interaction or is system-generated. The logic tier consists of rules that verify the transaction and ensure all the data required to complete the transaction is available. The data store tier stores the transaction and all the data related to it.

The main characteristics of an online transaction processing system are the following:

- **ACID compliance:** OLTP systems must ensure that the entire transaction is recorded correctly. A transaction is usually an execution of a program that may require the execution of multiple steps or operations. It may be complete when all parties involved acknowledge the transaction, or when the product/service is delivered, or when a certain number of updates are made to the specific tables in the database. A transaction is recorded correctly only if all the steps involved are executed and recorded. If there is any error in any one of the steps, the entire transaction must be aborted and all the steps must be deleted from the system. Thus OLTP systems must comply with atomic, consistent, isolated, and durable (ACID) properties to ensure the accuracy of the data in the system.
 - **Atomic:** Atomicity controls guarantee that all the steps in a transaction are completed successfully as a group. That is, if any steps between the transactions fail; all other steps must also fail or be reverted. The successful completion of a transaction is called commit. The failure of a transaction is called abort.
 - **Consistent:** The transaction preserves the internal consistency of the database. If you execute the transaction all by itself on a database that's initially consistent, then when the transaction finishes executing the database is again consistent.
 - **Isolated:** The transaction executes as if it were running alone, with no other transactions. That is, the effect of running a set of transactions is the same as running them one at a time. This behavior is called serializability and is usually implemented by locking the specific rows in the table.
 - **Durable:** The transaction's results will not be lost in a failure.
- **Concurrency:** OLTP systems can have enormously large user populations, with many users trying to access the same data at the same time. The system must ensure that all these users trying to read or write into the system can do so concurrently. Concurrency controls guarantee that two users accessing the same data in the database system at the same time will not be able to change that data, or that one user has to wait until the other user has finished processing before changing that piece of data.
- **Scale:** OLTP systems must be able to scale up and down instantly to manage the transaction volume in real time and execute transactions concurrently, irrespective of the number of users trying to access the system.
- **Availability:** An OLTP system must be always available and always ready to accept transactions. Loss of a transaction can lead to loss of revenue or may have legal implications. Because transactions can be executed from anywhere in the world and at any time, the system must be available 24/7.
- **High throughput and short response time:** OLTP systems require nanosecond or even shorter response times to keep enterprise users productive and to meet the growing expectations of customers.
- **Reliability:** OLTP systems typically read and manipulate highly selective, small amounts of data. It is paramount that at any given point of time the data

in the database is reliable and trustworthy for the users and applications accessing that data.

- **Security:** Because these systems store highly sensitive customer transaction data, data security is critical. Any breach can be very costly for the company.
- **Recoverability:** OLTP systems must have the ability to recover in case of any hardware or software failure.

☞ Check Your Progress 3:

- 1) Compare ROLAP, MOLAP and HOLAP.

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.....

- 2) Write limitations of OLAP cube.

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5.12 SUMMARY

OLAP has proven to be an asset in the field of Business Intelligence as it helps in relieving the large amount of data handling along adding the cost benefits of working with this very technique. Furthermore, OLAP providers normally offer their clients with significant documentation, tutorials, and spark off technical assistance in terms of web-primarily based totally OLAP clients. The customers are continuously loose to deal with the group of tech experts while not having to control all the troubles tied to the software program themselves. The concept hierarchies help to organize the dimensions into logical levels. The various OLAP operations help to extract information across sub cubes. The creation of cube and types of OLAPs helps to understand the architecture and usage of various applications of OLAP.

5.13 SOLUTIONS/ANSWERS

Check Your Progress 1

- 1) Knowledge workers such as data analysts, business analysts, and Executives are the users of OLAP.
- 2) Decision making Tool features are:
- Report Generation
 - Query Handling
 - EIS (Executive Information System)
 - OLAP (Online Analytical Processing)
 - Data Mining

Check Your Progress 2

- 1) In Marketing, OLAP can be used for various purposes as it helps like planning, budgeting, financial marketing, sales data analysis and forecasting. The customer experience is very important to all the companies. So, OLAP works very efficiently in analyzing the data of customers, market research analysis, cost-benefit analysis of any project considering all the dimensions.

There are various OLAP tools available. The OLAP tool should have the ability to analyze large amounts of data, data analysis, fast response to the queries and data visualization. For example, IBM Cognos is a very powerful OLAP marketing tool.

- 2) Purpose of Hypercube in OLAP: The cube is basically used to represent data with some meaningful measure to compute. Hypercube logically has all the data at one place as a single unit or spreadsheet which makes the computation of queries faster. Each dimension logically belongs to one cube.

For example, a multidimensional cube contains data of the cities of India, Product, Sales and Time with conceptual hierarchy (Delhi→2018→Sales). As, shown in below figures.

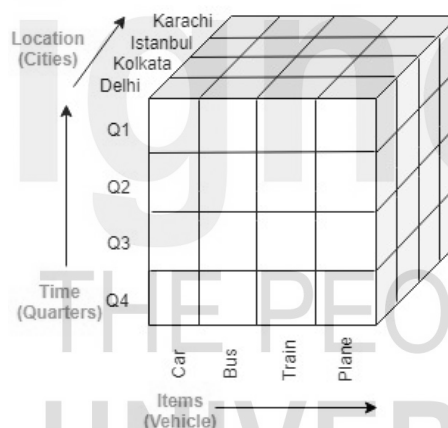


Figure 13: Multidimensional Cube

In the cube given in the overview section, a sub-cube (hypercube) is selected with the following conditions:

Location = “Delhi” or “Kolkata” Time = “Q1” or “Q2”, Item = “Car” or “Bus”

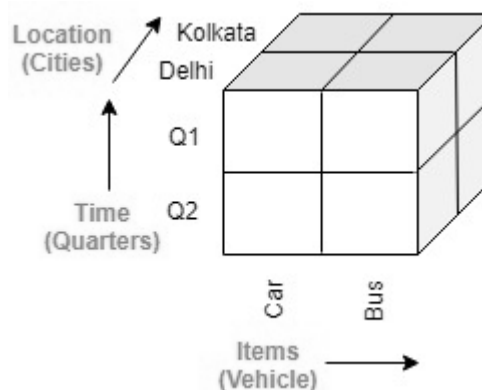


Figure 14: Hypercube or sub-cube

Slice is performed on the dimension Time = “Q1”.

Karachi				
Istanbul				
Kolkata				
Delhi				
	Car	Bus	Train	Plane

Figure 15: Slice

On the sub-cube, pivot operation is performed.

Car				
Bus				
Train				
Plane				
	Delhi	Kolkata	Istanbul	Karachi

Figure 16: Pivot operation

3) The features of OLAP are:

- Conceptual multidimensional view
- Accessibility of data
- Efficient and flexible Reporting system
- Client/Server architecture
- Supports unrestricted dimensions and aggregation levels
- Uses dynamic sparse matrix handling for faster query results
- Multiuser support

Check Your Progress 3

- 1) Limitations of OLAP cube are:
 - OLAP requires a star/snowflake schema
 - There is a limited number of dimensions (fields) in a single OLAP cube
 - It is nearly impossible to access transactional data in the OLAP cube
 - Changes to an OLAP cube requires a full update of the cube – a lengthy process

2) Following table 2 summarizes the comparative analysis of ROLAP, MOLAP and HOLAP:

Table 2: Differences between ROLAP, MOLAP and HOLAP

FEATURES	ROLAP	MOLAP	HOLAP
Accessibility of data and Processing time	Very slow because of join operation between tables. The data is fetched from data warehouse.	Fast because of multidimensional storage. The data is fetched from multidimensional data cube.	Fast.
Storage space requirement	Data is stored in relational tables. Comparatively Large storage space requirement	Data is stored in multidimensional tables. Medium storage space requirements	It uses both ROLAP, MOLAP. Small storage space requirements. No duplicate data.
Latency	Low latency	High latency	Medium latency
Query response time	Slow query response time	Fast query response time.	Medium query response time
Volume of data	Used for large volumes of data	Limited volume of data	Can be used in both scenarios
Retrieval of data	Complex SQL queries are used	Sparse Matrix is used	Both
Data View	Static view of data	Dynamic view of data	Both static and dynamic view of data

5.14 FURTHER READINGS

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UNIT 6 TRENDS IN DATA WAREHOUSE

Structure

- 6.0 Introduction
- 6.1 Objectives
- 6.2 Data Warehouse – Key Challenges
- 6.3 Data Lakes
 - 6.3.1 Data Lake Maturity
 - 6.3.2 Successful Creation of Data Lake
 - 6.3.3 Data Swamp
 - 6.3.4 Data Lake Architectures
 - 6.3.5 Data Lake Organization
 - 6.3.6 Data Governance
 - 6.3.7 Self Service Enablement
- 6.4 Complex Data
 - 6.4.1 Introduction
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6.0 INTRODUCTION

In the earlier units, we have learned the conceptual aspects of a data warehouse including its types and their underlying architectures. You have also seen how new forms of databases evolved viz. Relational, Object-oriented, etc. which, in turn influenced the architectures of data warehouses to encompass the need for accommodating various forms of data. The earlier challenges of data warehousing were successfully addressed through research like innovative architectures, algorithms, modeling techniques, methodologies, etc..

Researchers have been trying to find better solutions for data warehousing and analytics for past few decades. The focus has been limited to relational technologies. Data management solutions began with databases (initially relational), then as the magnitude of data increased the world saw the advent of very-large-database systems (VLDBs). As data originated from various sources and consecutively possessed the directional attribute (via data-sharing and distributed storage, etc.) the research and technological focus evolved towards non-relational systems. Presently, we are in the phase of data-explosion called Big-Data and consequently there have been new innovations in this direction towards managing extremely huge magnitude of data without losing the essence of performance, scalability, reliability, security, etc..

In this unit we will discuss the trends in data warehousing namely data-lakes, complex data-marts, migration of data warehouses to cloud environment, real-time data warehouses, Hadoop software and how it supports data-warehouse and automated data warehouses.

6.1 OBJECTIVES

After going through this unit, you shall be able to:

- understand changing technological landscape related to data warehouse;
- describe data lakes, complex data marts and real time data warehousing;
- understand the concepts of big-data and Cloud-based data warehouse;
- describe how Hadoop supports the data warehouse design pattern;
- understand Data Warehouse automation and its necessity, and
- list examples that support the above data warehousing trends.

6.2 DATA WAREHOUSE – KEY CHALLENGES

When we attempt to understand a data warehouse from a technical perspective we comprehend that it is a database of considerable size ranging from several hundreds of gigabytes to several terabytes and in some cases, few petabytes too. The sheer enormity of database size, the complexity of analytical queries, data mining algorithms and the heterogeneous aspect of data loaded into a data warehouse makes the entire scenario incomprehensible and complicated. These results in significantly acute technological challenges for researchers while raising critical questions for which, solutions appear to be equally intricate or pragmatically irresolute.

It is evident that exhaustive research in data warehousing is going on in the areas namely, conceptual modeling of DWs and logical data models, data warehouse loading (data-refreshing), execution efficacy of OLAP queries and data mining algorithms, materialized views, data analysis techniques, metadata management, evolution management of DWs, stream-based, real-time and active data warehouses, and complex data warehousing (for example spatial, XML, object, multimedia).

Some of the research and technological challenges of utmost significance are dealt through research on topics like amalgamation of heterogeneous data sources, i.e., ETL, cleaning, source characterization, and data integration. Data source discovery, metadata management and standardization, handling data source evolution at an integration layer as well as data flow (ETL) performance optimization are topics which still lack sufficient attention in research. Inadequate research has been done on topics like: data management architectures for standard (table-like) data with unfinished solutions extent. Big data management has exposed many new challenges due to data-explosion, caused mainly by the diversity and velocity of data-generation. Solution development for the assimilation and storage of big data is imperative for

design and implementation of a fully functional big data architecture that can be used by analytics, recommendation systems, visualization systems, etc.

We have discussed the challenges faced by researchers pertinent to data warehousing issues like complications in utilizing semi-structured and unstructured data from various sources, etc. It is evident that the exponential growth in data, including the numerous attributes inherited had a profound impact on the architectural elements of data warehouse. There are yet many capabilities and extensions of functionalities that necessitate their integration into a state-of-the-art system. The additional functionalities if developed severally, are required to function seamlessly as an integral part of the primary system. Architectural impact of issues requires critical thought for implementing solutions that offer optimizing total cost of ownership (TCO) with substantial return on investment (ROI). Since researchers have already embarked upon the journey to build newer architectures to address the proliferation of data and its capture to produce meaningful results, we are likely to see new trends emerging in the form of solutions and challenges for the future.

Let us look at the key challenges faced by researchers which have been categorized based on their technical significance.

Computational techniques: A primary candidate for this research challenge is the computational scalability required with drastic increase in data volumes. Emerging technological trends like Cloud data warehouses (CDWs) may address this issue. Parallel and In-memory computational techniques are being developed and perfected.

Design: Efficacy of design has been of particular significance for researchers. Due to performance implications involved design techniques need to consider the following questions carefully:

- a. How can design dependent latencies for cubes be minimized? For example, the computational costs of aggregated data in case of huge datasets may prove unreasonable.
- b. How can the update and refresh times for data warehouses (specifically cubes) be improved upon? For example, there exist various techniques and strategies for accomplishing these tasks which tend to fail for unstructured/heterogeneous data.

Quality: The word has different connotations in different contexts and is understood accordingly. For example if the query is complicated the underlying cubes would be overwhelmed with number of *dimensions* and *measures* required to build the cubes, especially if the data is unstructured/heterogeneous in nature. This can be attributed to the skill level of the data warehouse user. On the other hand, if even a simple query cannot fetch data in a fair amount of time then the modeling techniques may be questioned. If the modeling is impeded by lack of functionality then the architecture or design of data warehouse software itself is questionable. Therefore, quality plays a significant role at every stage of the data warehousing process. Good governance and best practices may partially resolve the quality related issues but a holistic scrutiny would reveal any gaps if extant.

Size: The essence of practicality is eliminated when a data warehouse's computational capabilities are throttled. This is caused by the cumulative size of fact tables which tend to expand exponentially over huge datasets.

Operability: A data warehouse needs to provide seamless integration with external data sources for data capture and collection. Interoperability is also another challenge when communicating with various devices and software components. Data warehouses usually make use of Representational State Transfer Application Programming Interfaces (REST APIs) to overcome these issues but security risks increase with the use of such components.

In-memory representation: Distinct from the in-memory computation techniques mentioned above this issue deals with the efficacy of memory-based representation of the data warehouse and its components like OLAP cubes, datasets and result sets. For example, the increasing number of dimensions causes explosive increase in cardinal values for cells. Consequently, secondary and tertiary memory based solutions require critical analysis. SAP HANA which is a High-performance ANalytic Appliance multi-model database that stores data in its memory instead of keeping it on a disk. This results in data processing that is magnitudes faster than that of disk-based data systems, allowing for advanced, real-time analytics). SAP HANA is exemplar of such in-memory database implementation utilizing a cloud platform.

User Centric Interface: Like other tools mentioned above that are required for specific functions it is imperative that the interface allows the user to make optimal use of the data warehousing software. In order to extract maximum benefit (ROI) from the data warehouse implementation the user should not be exposed to the complexities of the system. This challenge has led to a trend of developing autonomous databases equipped with self-monitoring mechanisms and self-healing techniques. There is also a trend towards making as many data warehouse processes automated as possible with a goal of achieving zero maintenance. This is to reduce the maintenance burden on individuals and organizations. It also has other advantages like reduction in human error rates, saving time on maintenance tasks, etc. It also allows organizations to cut down on resources and maintain a lean environment. The user would only intervene if a business decision is involved. A user centric interface also reduces training expenditure while putting the controls at user's fingertips.

Pioneering Infrastructure foundation: Converged, Hyper-converged and Dynamic Infrastructures are deployed in data-centers. These are deployed either on-premises / onsite (private cloud) or on the Cloud (public cloud, e.g. Amazon Web Services (AWS), Google Cloud Platform GCP), Microsoft Azure (MSA), etc.) Or a combination of both Private and Public cloud also called as Hybrid cloud platforms. These platforms can provide an innovative foundation for medium to exceptionally large data warehouse implementations. Also availability of specialized computing infrastructure components like graphic processing units, Artificial Intelligence (AI) or Machine Learning (ML) based Processor chips are available (example NVIDIA's Tesla or Drive PX series processors, AMD's Fire Stream or Radeon Instinct processors). The data warehouse software also needs to improve its architecture and design constructs to derive the maximum benefit from such advanced infrastructures.

Complexity: For example building OLAP cubes over exceptionally large datasets attracts the penalty of rising complexity concerns that are usually unheard of in relational data warehouses. For example, there is an exponential increase in the number of *dimensions* in case of unstructured data in addition to several *measures* due to unstructured data.

Optimization and Innovations in Query Language: Traditional query languages like Multidimensional Expressions or MDX developed for OLAP databases was later wrapped in XML to be called mdXML to include XML data retrieval. With data undergoing more diverse forms and originating from equally varied sources the query languages require serious thought to improve their performance and capabilities to handle sufficiently large data from disparate sources while being able to build and populate cubes through query plan optimizations.

Data Visualization: Many organizations and institutions capture data from varied sources to assist in strategizing, business decisions, weather forecasts, predicting climate changes, knowledge visualization, cognitive maps, etc. In all these instances the sheer amount of data collected needs to be managed efficiently on various fronts to produce viable reports in order to visualize data. In healthcare systems that are dynamic in nature a patient's health parameters captured via various sensor devices need to be analyzed and evaluated in real-time. Other dynamic models like Climate

change prediction systems need to monitor data captured via Internet of Things (IoT) devices are used to predict weather/climate anomalies or impending catastrophes. These scenarios require state-of-the-art software and infrastructure components. The algorithms implemented via software and computed using the infrastructure has to make sense of the data and at times discover hidden meanings and therefore new knowledge. This requires specialized query tools for example GaphQL, etc. while at the same time the semi / unstructured aspect of data requires appropriate management software that is architecturally robust. There are other critical issues with AI / ML based data visualization like necessity for innovative algorithms, multi-parameter based optimizations, accuracy of results, etc.

In the next section, we will learn about the Data lakes.

6.3 DATA LAKES

A data lake is a central location that holds a large amount of data in its native, raw format. Compared to a hierarchical data warehouse, which stores data in files or folders, a data lake uses a flat architecture and object storage to store the data. Object storage stores data with metadata tags and a unique identifier, which makes it easier to locate and retrieve data across regions, and improves performance. By leveraging inexpensive object storage and open formats, data lakes enable many applications to take advantage of the data.

Data lakes were developed in response to the limitations of data warehouses. While data warehouses provide businesses with highly performant and scalable analytics, they are expensive, proprietary and can't handle the modern use cases most companies are looking to address. Data lakes are often used to consolidate all of an organization's data in a single, central location, where it can be saved "as is," without the need to impose a schema (i.e. a formal structure for how the data is organized) up front like a data warehouse does. Data in all stages of the refinement process can be stored in a data lake: raw data can be ingested and stored right alongside an organization's structured, tabular data sources (like database tables), as well as intermediate data tables generated in the process of refining raw data. Unlike most databases and data warehouses, data lakes can process all data types — including unstructured and semi-structured data like images, video, audio and documents — which are critical for today's machine learning and advanced analytics use cases.

6.3.1 Need for a Data Lake

First and foremost, data lakes are open format, so users avoid lock-in to a proprietary system like a data warehouse, which has become increasingly important in modern data architectures. Data lakes are also highly durable and low cost, because of their ability to scale and leverage object storage. Additionally, advanced analytics and machine learning on unstructured data are some of the most strategic priorities for enterprises today. The unique ability to ingest raw data in a variety of formats (structured, unstructured, semi-structured) along with the other benefits mentioned, make a data lake the clear choice for data storage.

6.3.2 Data Warehouse Vs Data Lake

A data warehouse stores structured business data in its processed form. This approach requires fairly rigid schemas for well-understood types of data. While data warehouses are an important tool for enterprises to manage their important business data as a source for business intelligence, they don't work well with unstructured data.

Data lakes allow the storage of raw data, both relational, as well as non-relational that is intended to be used by data scientists and developers along with the business analysts. They take the data out of the silos and make it accessible to all business users promoting centralization of data.

The key differences are summarized in the Table1 given below:

Table 1: Data Warehouse Vs Data Lake

Attribute	Data Warehouse	Data Lake
Type of Data	Structured data from sources like transactional systems and operational databases.	Raw Data from varied sources like websites, mobile apps, IoT devices, social media channels etc.
Schema	Schema-on-write	Schema-on-read
Intended users	Primarily business analysts	Data scientists, developers and business analysts
Price/Performance	Fastest query results using higher cost storage	Query results getting faster using low-cost storage
Data Quality	Highly curated data that serves as the central version of the truth	Any data that may or may not be curated (ie. raw data)
Type of analytics	Business intelligence, visualization and batch reporting	Machine learning, predictive analytics, profiling and data discovery.
Agility	Fixed configuration, less agile	Highly agile, can be configured and reconfigured as per requirements.
Security	More secure storage	Higher accessibility makes ensuring security a challenge

6.3.2 Data Lake Maturity

A data lake is said to go through various stages of maturity during its life-cycle, considering that it is a fairly new design pattern. Bill (William) Inmon (the father of data warehouse) proposes data lake maturity process (or stages) as below.

According to **Bill Inmon**, native or raw data is classified into three discreet categories viz. i) Analog data, ii) Application data and iii) Textual data. He states as below:

“In order to organize the different types of data into a structure that can be analyzed, it is necessary to create a high-level structure of data within the data lake. As data enters the lake it first enters the raw data pond. The purpose of the raw data pond is to serve as a holding cell. There is little or no analysis or other organized activity of the data while in the raw data pond.

Once it is time for analysis, the information in the raw data pond is sent to one of three different ponds based on the kind of data entailed. For example, analog, application and textual data all require a unique data pond.” – **Bill Inmon**

When the data has passed its useful life in the data pond it is moved from its respective data pond to an archival data pond as illustrated in Figure 1 below.

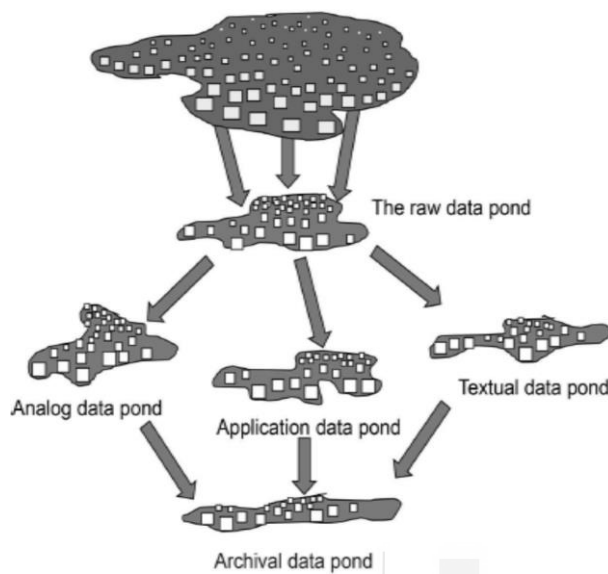


Figure 1: Data Lake Architecture by Bill Inmon

Alternatively **Alex Gorelik** introduces additional data lake stages, as part of data lake's maturity aspect, as below:

- a. **Data Puddle:** A data mart built using big data technology with a sole purpose or as a single-project with data being loaded into it for the consumption of a single project or team. The data mart concept is well known and use of big data technology is mandated to reduce cost and improve performance.
- b. **Data Pond:** Similar to that proposed by Bill Inmon, except in this case it is a collection of data puddles in the form of collocated data marts or similar to a poorly designed data warehouse with minimal transformation of source data. As such these data ponds restrict data usage solely to the projects that necessitate it.
- c. **Data Lake:** It caters to business users in two significant aspects. Firstly as a self-support service through which business users are able to utilize the data and secondly, provisioning data to business even when not required by the project.
- d. **Data Ocean:** Facilitate enterprise-wide availability of self-service data and data driven decision making, irrespective of the data location or its existence within the data lake.

The above components are shown in Figure 2 depicting data lake maturity process proposed by Alex Gorelik.

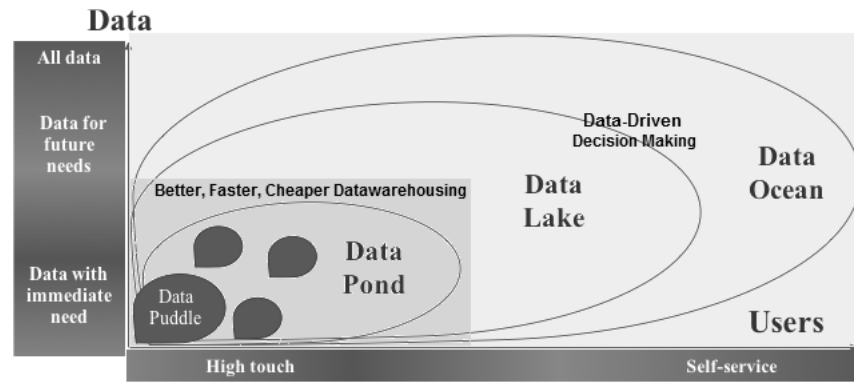


Figure 2: The Enterprise Big Data Lake by Alex Gorelik

6.3.3 Data Lake Architecture

The below given Figure 3 shows a standard proposal for an architecture of a data lake system. The system consists of four layers which are i) Ingestion Layer, ii) Storage Layer, iii) Transformation Layer, and iv) Interaction Layer.

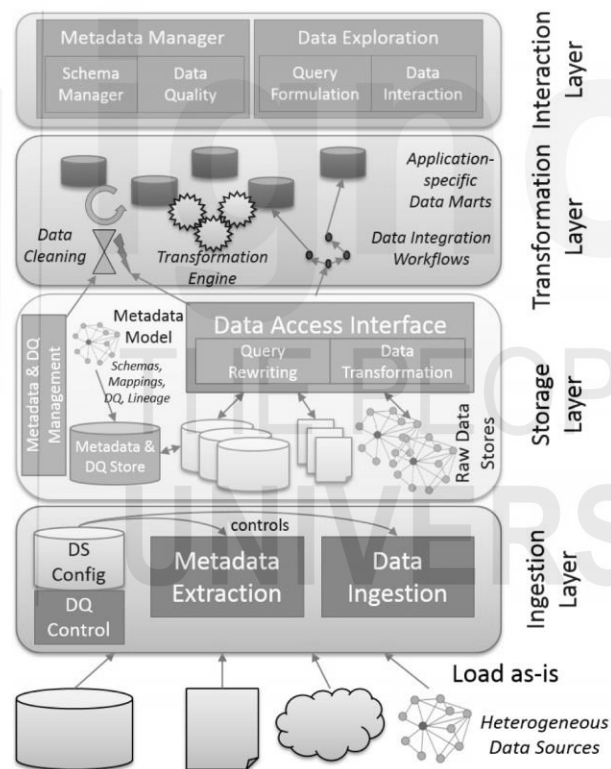


Figure 3: Layered Architecture of a Data Lake

i) Ingestion Layer

The Ingestion Layer is in charge of bringing data into the data lake system from various sources. One of the main features of the data lake concept is the ease with which any type of data can be ingested and loaded. Data lakes, on the other hand, have been repeatedly reported as requiring governance in order to avoid becoming data swamps. Administrators of data lakes are in charge of this critical topic.

The metadata extractor is the most key component of the ingestion layer. It should make it easier for data lake administrators to configure new data sources and make them accessible in the data lake. To accomplish this, the metadata extractor should extract as much metadata as possible from the data source (for example, schemas from

relational or XML sources) and store it in the data lake's metadata storage. The raw data, in addition to the meta- data, must be absorbed into the data lake. Since the raw data are kept in their original format, this is more like a “copy” operation, which is certainly less complex than an ETL (Extract-Transform-Load) process in data warehouses.

ii) Storage Layer

The *metadata repository* and the *raw data repositories* are the two key components of the storage layer. Since raw data repositories must be stored in their native format, data lake environments must support a variety of storage systems for relational, graph, XML, and JSON data. Hadoop appears to be a strong candidate for the storage layer's basic platform. To support data fidelity, however, additional components such as Tez or Falcon are needed.

iii) Transformation Layer

Data can be transformed from storage to user experiences using the transformation layer. It requires operations such as cleansing, format transformations, and so on.

iv) Interaction Layer

All of these functionalities that are needed to work with the data should be covered by the interaction layer. These functionalities should include visualization, annotation, selection and filtering of data, as well as basic analytical methods. It's also worth noting that more advanced analytics like machine learning and data mining should not be regarded as part of a data lake system's capabilities.

6.3.4 Data Swamp

A data swamp is a data pond that has expanded to the size of a data lake in the absence of self-service and governance facilities. At best, the data swamp is used like a data pond and at worst it is left unused. While various teams may frequently use areas of the lake for their projects the bulk of the data is dark/obfuscated, undocumented and therefore, unusable.

Data swamps can be accessed by those who are technologically proficient. This is accomplished by incising or chiseling small puddles for themselves and their teams. At the advent of data lakes many enterprises hastily bought Hadoop implementations and loaded them with petabytes of raw data, from various sources, without clarity on the manner of its utilization. In the absence of any systematic approaches to organize the data while loading it turned into an incomprehensible mess. Additionally, enterprises were prohibited by governance regulations against permitting data swamps' access to a wider audience without obfuscating sensitive data. Since the precise location of sensitive data was dubious, access was interdicted. Consequently, the data essentially remained unusable and unutilized.



Check Your Progress 1

- 1) What are present challenges in data-management?

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- 2) Describe in your own words the logical data lake concept?

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- 3) What is your interpretation of a successful data lake and a data swamp?

6.4 COMPLEX DATA

6.4.1 Introduction

We have already seen how big data has become pervasive across enterprises. We have also seen different ways in which data is generated through various sources like emails, files, IoT, Logs, etc. Enterprises. With the scale and speed of data adding to the type of data being generated, the term Complex data was coined to define all data that does not conform to standard data types like dates, currency, alphabets, numbers, etc. Complex data is still essential, as we have seen when dealing with raw data, in building data-insights and big data analytics. Complex data is processed as complex datasets and is used by pattern matching algorithms in Machine Learning models. Here we take a quick look of how the complex data trend has influenced the data-driven technologies and enterprises.

6.4.2 Complex data modeling

With data streams traversing many hubs and myriad technologies complex data modeling have become a standard practice across the industry. Complex data travels through data processing tools, ETLs, ERP software, data lakes and other paths. Complex data has its intrinsic traits like its own syntax, schema, technology, terminology and type. This diversity of traits complicates the work of data modelers.

Various models like statistical, polyglot, no payload, multi-level, etc. exist to store, process and analyze complex data. Some standard models are also available like Anchor models and Data Vault Models. Both of these models provide advantages like Scalability, Temporal data handling and are resilient to structural and content-based changes.

6.4.3 Complex data models

6.4.3.1 Anchor Model:

An example of the Anchor Model is shown in Figure 10a.

Anchor modeling has four elements to it:

Anchors

The anchors are used to model entities and events

Attributes

Attributes are used to model properties of anchors.

Ties

Ties are used to model the relationships between anchors.

Knots

Knots are used in the modeling of shared properties, such as states.

Attributes and ties can be versioned (termed as historization), when changes in the information the model need to be retained.

The different graphical symbols, representing the elements of the model, are similar to those used in entity-relationship modeling with a few extensions. An outline on a tie or attribute depicts versioning of changes.

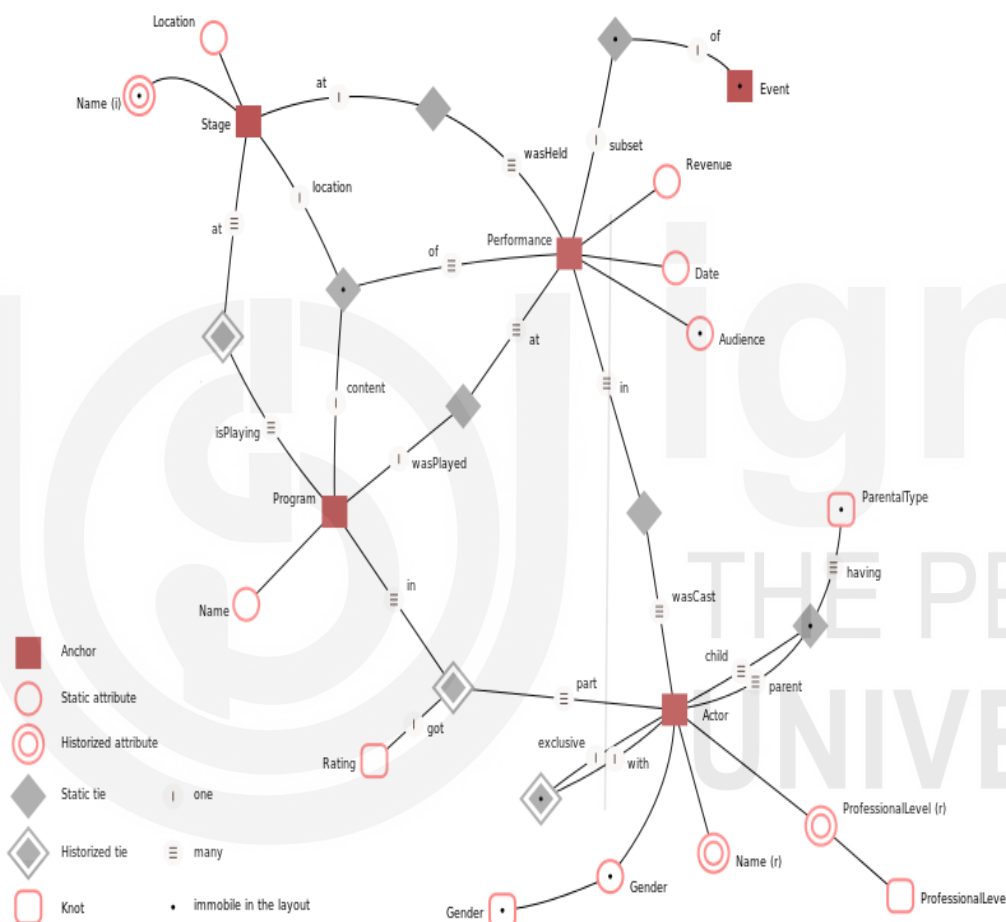


Figure-10a. An example of the Anchor Model

6.4.3.2 Data Vault Model

Data Vault is a database modeling technique offering long-term storage of historical data arriving from multiple operational systems. The model also allows coping with issues such as auditing, data traceability, data loading performance and resilience to change. Data traceability implies that every row of data in a data vault must be associated with a record source and date-of-load attributes. This information allows an auditor to trace values to their respective sources. Daniel (Dan) Linstedt developed the Data Vault model in 2000.

Figure 10b illustrates the Data vault model. The components of this model are briefly discussed below:

Hubs:

Hubs are comprised of a list of distinct business keys with low affinity to change and contain a surrogate key per Hub item. The Hub also contains metadata designating the source of the business key. Satellite tables; discussed below, store descriptive attributes for the data on the Hub (for example, a multiple languages description for the key). A Hub should have at least one satellite and at the least, should comprise of the following attributes:

Surrogate key: Connects the other structures to the table comprising the surrogate key
Business key: The driver attributes of the Hub structure may contain multiple fields.
Record source, Allows determining which system was the first to load a business key
Metadata (optional) Contains information on manual updates (user/time) and data extraction date.

Existence of multiple business keys is precluded in a Hub, except when two systems provide the same business key but having different meanings to resolve conflict.

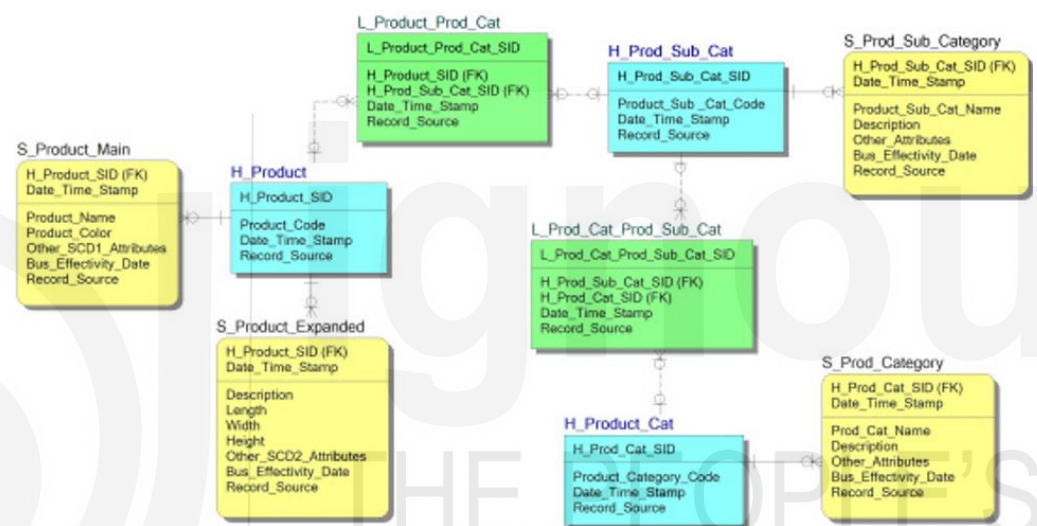


Figure-10b. An example of a simple Data Vault model with two hubs (blue), four satellites and a link (green)

Links

Link tables are models providing relations between business keys and are fundamentally many-to-many join tables, containing some metadata.

Links can connect to other links, to handle granularity variations. Using link references within another link is bad modeling practice, as it creates dependencies between links making parallel loading of links more challenging.

Sometimes links connect hubs to data that is not adequate for hub creation. This scenario happens when a non-real business key is associated with the link. This happens only when using the business key for another link or as key for attributes in a satellite is not possible. This technique has been called a 'peg-legged link' by Dan Linstedt on his (now defunct) forum.

Links also comprise of surrogate keys for linked hubs. The surrogate keys of the hubs, for the link and metadata describe the relationship source. As in hubs, the descriptive attributes for the relationship information gets stored in the satellite table.

Satellites

The hubs and link's temporal attributes and descriptive attributes stored in separate constructs called satellite tables or simply satellites. The satellites also comprise metadata connecting them to their parent link or hub, metadata designating the relationship source and attributes, also including a time-series with the attribute's start and end dates. While the links and hubs deliver the skeletal structure of the model, the satellites deliver the "meat or substance" of the model in the form of business processes contexts that are stored in links and hubs. Storage of these attributes is done relative to the timeline and content details and span from relatively simple (a linked satellite with only a timeline and a valid indicator) to relatively complex (all fields designating a client's entire profile).

Typically, the source system groups attributes in satellites. Rate of change for descriptive attributes like amount, size, color, speed or cost may vary. These attributes can be separated into various satellites based on the rate at which they change.

Metadata is contained within all the tables, designating the validation-date of the entry and system from which it originated, providing a holistic historical perception of the data as it is loaded into a data warehouse.



Check Your Progress 2

- 1) Explain your understanding of complex data and the need for complex data modeling

- 2) Write a note on the Data Vault model.

6.5 CLOUD DATA WAREHOUSING

6.5.1 Introduction

The transmogrification of raw data into insights is a convoluted challenge and the solution is not simple. That challenge is made worse by the accelerated pace of environmental dynamics affecting the enterprises. The pace at which business is done is in fact accelerating. Things that we expected to be done in a week or overnight or an hour need to be done in real time. Innovations in various industries are taking place simultaneously or at an overwhelming pace.

The challenge for most enterprises is to discover opportunities to exploit this change rather than being overpowered by it. The answer, though superficially simple, lies in leveraging data and transform into a data-driven enterprise. This transformation should not be limited to a technological perspective, rather it needs to be assimilated and trusted in by people across the enterprise as the crux of organizational health and wellbeing.

Enterprises have to confront demands of space and location (capacity to store and utilize voluminous data. Further, new demands imposed by regulatory and compliance mandates, data velocity, new data sources, performance, scalability, QoS, etc. have compelled the enterprises to rethink their data-leveraging strategies.

Enterprises have realized that a data warehouse implementation is no more a silver bullet for their problems considering the environmental dynamics and other issues discussed above. Therefore, we have recently witnessed the trend of extant data warehouses migrating to cloud while new ones are adopting cloud as their data warehouse implementation platform. This forethought is a consequence of realizing issues of capacity planning, infrastructure costs, usage policies, governance, flexibility of utilization based expenditure and many more.

6.5.2 Reasons for Migrating to the cloud

On-premises data warehouse implementations are reducing and consequently its business is diminishing too. Vendors like Google BigQuery, AWS Redshift, Snowflake, Azure Synapse Analytics, etc. are popular ones for cloud hosted data warehouses. These vendors offer near-instantaneous installations with storage, computing and network resources with increased performance and scalability benefits. The supporting software and platform are always maintained up to date. Most enterprises or a new data lake tend to choose a cloud-hosted data repository.

Enterprises are migrating to cloud data warehouses because they get instant access to all the infrastructure resources necessary to scale that solution. They also derive a pay for use benefit as they pay only for additional memory, compute, storage, networking or other resources they might need to scale. Best-of-breed applications can also be availed through SaaS to their cloud-implementations, whereby their entire business intelligence and analytics stack can be provided as a service. They can scale up or scale down as required with minimal installation or maintenance charges. The trend towards cloud-based data warehousing is clearly evident and that is where this market is progressing.

To sum up, enterprises are adopting the cloud migration path due to the following advantages offered by a cloud platform:

- i. Improved performance
- ii. Flexibility of cost through flexibility of cloud resources utilization
- iii. Migration of existing products to the cloud environment
- iv. Derive benefits that are inherent to a cloud platform

6.5.3 Challenges of cloud data warehouses

Although cloud-based data warehouses are an essential component of the future of enterprise-data, there are additional tasks to be performed and therefore, challenges involved. The significant challenges of hosting data warehouses in a cloud environment are as below:

- i. Data extraction, transformation and subsequent loading
- ii. Context and User based Data access
- iii. Management of heterogeneous data velocity
- iv. Ensuring instant availability of new data sources
- v. Ensuring data quality
- vi. Management of sensitive data and compliance to regulatory mandates
- vii. Interoperability with tools and infrastructure external to cloud environment
- viii. Communicating with legacy systems that could not be migrated to cloud for technical and organizational reasons

- ix. Data governance and obfuscation of sensitive data
- x. Automation of data offload for sophisticated analytics and ML

6.5.4 Building a successful cloud data warehouse

Building a successful CDW entails overcoming the challenges mentioned in the previous section. Since cloud data warehouses deal with enterprise data at scale, eliminating or reducing the impediments in its path to derive maximum business value forms a crucial element in building a successful CDW. To ensuring a successful CDW implementation the CDW has to go through the following stages.

Stage 1 - Formulating an approach for data curation at scale and data integration

Firstly, enterprise resource scalability is imperative with increasing volumes of data that results from business growth. A precise and pertinent data curation and integration approach is a deciding factor in an enterprise's ability to succeed as a data-driven enterprise. This success in turn is based on an enterprise's ability to leverage data analytics based insights ahead of its competition.

In order to carefully plan the approach to data integration the following considerations are essential:

- i. Preparedness for enterprise changes, besides technological changes, or the ability to prepare for such scenarios.
- ii. Enterprise readiness to adopt new technologies including unknown ones developed in the future
- iii. Future objectives of your enterprise that would have a critical impact on your data integration architecture already in place
- iv. Enterprise expectations from this approach in terms of adding measurable and tangible value to its business growth

The above considerations entail due diligence in solution implementations as below: Transformation into a data-driven enterprise involves changes on the technological front but also at various other levels. These changes affect both the technical and business stakeholders alike. To list a significant few, the changes involve enterprise risks, costs, dependencies, ROI, benefit-amendments, assumptions, and cultural issues. For large enterprises, this necessitates the presence of change management professionals, who educate, train and influence stakeholders mindsets to the imminent changes. Enterprises planning to undergo a data-driven transformation may hire change-management agencies or professionals based on the scope and dynamics of their businesses growth.

Architecture of systems employing data integration technologies should be flexible enough to accommodate new tools in the future with minimal impact on the architecture and consequently the system services. Conceptualization of future-proof architectures is essential to make a system capable to adapt to yet unknown technologies that might be developed in the future.

Enterprises may have plans to expand their data-leveraging capabilities, for example moving their Enterprise Resource Planning (ERP) data i.e. data from ERP modules like Customer Relationship Management (CRM) / Supply Chain Management (SCM) / Human Resource Management (HRM), etc. to the cloud data warehouse. However, the actual objectives of the enterprise might involve efficient master data or workflow management to emerge as a leader in customer services landscape. Such objectives need to be stated upfront in order to formulate the best approach that caters effectively to present and future business requirements.

Wisdom dictates that expenditure on the latest and greatest technological tools might not be adequate for business growth if they cannot cater to the specific functions of an enterprise. Also, availability of great feature-sets in a tool does not necessarily guarantee business growth if only a few of them can be truly exploited to an enterprise's advantage. Tools that offer customization or flexibility in terms of specific functionality needs, choice of features, etc., in a data integration platform are available. A clear understanding of the customizations involved allows technology decision makers to understand the implementation components required to collate data and ensure its availability across the enterprise. It also helps the enterprise understand TCO of the data integration platform, which helps to optimize and improvise the approach to extract maximum ROI. Further, the approach also enables a precise understanding of the business benefits expected from a data integration platform and subsequently, their measurability and tangibility.

Stage 2 – Leveraging Data Integration platform for on-demand data provisioning

It is imperative for an enterprise to develop faster decision making capabilities. However, the vast volumes and heterogeneous characteristics of data make it extremely difficult to do so if it is not able to scale its infrastructure and tool-sets to match the rate of data explosion (i.e. rate of increase of data velocity, diversity and volume). Data-explosion also triggers the need to assimilate and utilize data at equivalently faster rates i.e. before its business worth diminishes. The fresher the data is, the most up-to-date the data visualizations, which give an almost instantaneous view of the most recent business landscape. These representations allow an enterprise to make informed decisions to strategize, improvise, adapt and steer their business on the path to success.

To extract maximum value from data, the right data should be available to right people, and at the right time. After the enterprise has sourced huge volumes of data it may develop the capability to store. This data needs to be transformed based on business requirements so that it is available to various people with specific expertise related to business areas. For example, data engineers should have access to raw data, analysts should have access to curated and organized data (e.g. in relational format), and other business functions like human resources, finance, manufacturing etc. should have access to their respective data sets. Sensitive data would also require access controls and obfuscation for cross functional requirements. All this advanced work will necessitate time and resources, creating latencies at different levels. The latencies at different levels like data extraction, data storage, organization and management, access control, access to desensitized data, transformation, loading, etc. Also, if the data cannot be visualized at scale through the use of appropriate tools the business decisions get delayed. It is easy for us to understand now; that in order to reduce the machine level latencies a meticulously planned data integration approach is imperative.

An approach to a fully optimized data integration platform to supplement business data needs of business will allow data-sharing across the enterprise, irrespective of functional domains and thereby, enhancing data visualizations through combined insights. Data provisioning puts data at the fingertips of stakeholders with very little dependence on technical personnel thus allowing for self-service. As a result of self-service capabilities operational latencies are drastically reduced while at the same time increasing innovative data insights. As ideas and representations are shared across functional domains, they can be improved upon through collective inputs and critical thinking. A thoughtfully planned approach should consider the advantages within the boundaries of data governance.

Stage 3 – Ensuring high quality of heterogeneous data across CDWs

As already mentioned, in Stage 1 above, utilizing a vast collection of the best tools does not guarantee best results if the tools cannot be customized to the business requirements of an enterprise. The technical dependencies cannot be resolved in case of some tools that offer deep customizations and a host of features (of which, all might not be applicable). Conversely, tools that are easy to implement may have limited capabilities and therefore cannot be classified as enterprise applications and their implementation, despite, is likely to cause more operational latencies than if not implemented at all. Therefore, it is necessary to scrupulously evaluate and scrutinize various tools, for your data platform so that they can meet the current and future needs of the business. Ensuring high quality of data availability, ubiquitously across your CDWs, is now possible through cloud-aware ETL tools which people with basic technical knowledge can easily use. The cloud based ETL tools are compliant to security and regulatory requirements and employ state-of-the art AI/ML techniques making the data reliable and accessible. As discussed in Stage 2, data sharing is necessary for timely decision making through mitigation of operational latencies. Implementing the right set of tools across different functional domains can expedite analytics and decision support capabilities.

Check Your Progress 3

- 1) What are the reasons for migrating to a Cloud data warehouse and what are the challenges involved for a CDW?

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6.6 REAL TIME DATA WAREHOUSING (RTDWg)

6.6.1 Introduction

What follows on from the discussion on various stages of implementing a successful cloud data warehouse, in the above section, is the most essential and significant attribute of a data-driven enterprise, which is timely availability of data across the enterprise. Timely or instantaneous data availability combined with appropriate access has many connotations, based on performance, context, relevance, perspective, etc. The rationale behind this expectation is that in a data-driven enterprise, data undergoes various stages before it can be available to the management in the form of meaningful representations. The representations or data visualization reports need to be based on trustworthy and most current data to be of any worth to the business. To make this possible the underlying processes – of sourcing the data, transforming it and making it available to various functional domains across the enterprise – need to happen as optimally as possible within the shortest possible time.

A birds-eye view or a look at the bigger picture makes it evident that the data platform that drives the enterprise should essentially work on-time across every stage of the journey of data. If we look at the detailed level, we realize that every element expects data to move immediately or in real-time. The laws of science do not make the real-time concept possible; therefore, real-time terminology is sometimes altered to near-real-time. Enterprises set performance thresholds for their real-time interpretation, and if data warehousing capability falls within these thresholds, it is termed a real-time data warehouse (RTDW). The proliferation of new data or updates to actual data across a data warehouse immediately has become synonymous with real-time data warehousing and is the favored definition in the industry.

CDWs are an excellent candidate for enterprises wanting to implement real-time data warehousing because of their intrinsic feature to scale resources e.g. computational, memory-related, storage, networking, etc. Despite the best resources available, processes like data transformation, and data loading into a dimensional model are distinctively complex. Also, taking into account the massive amounts of data involved, the complexity gets further compounded. Therefore, the processes for updating new data coming in from different sources require specialized handling to make the latest data available in real-time.

We already know that the primary component of data warehousing i.e. the ETL process takes care of data movement from its source to the data warehouse storage and is extremely resource intensive. The mandates of real-time data warehousing require drastic alterations to the way ETL process takes place in addition to other components of the data warehouse. Researchers have come up with some unconventional approaches e.g. novel ETL architectures, update methods – cached, partial, separate synchronized DWHs, etc., to ensure data availability in real-time. Figure 11 is a conceptual representation of real-time big data analytics which gives us a glimpse of what it means to implement a real-time data warehouse.

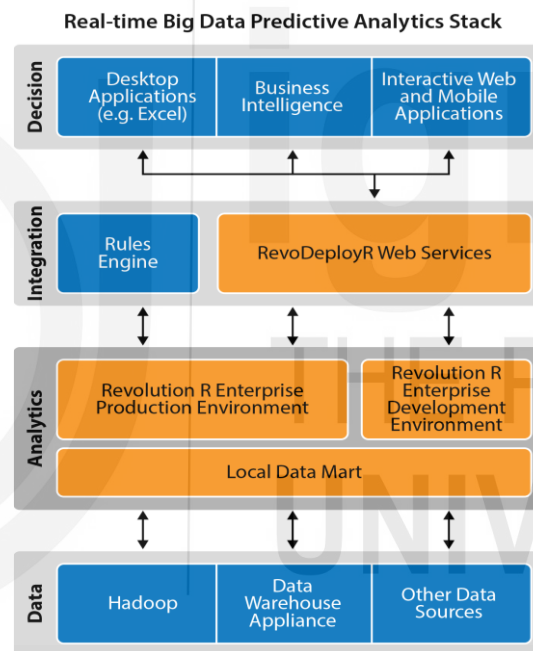


Figure-11. From David Smith's presentation "Real-Time Big Data Analytics: From Deployment to Production"

Hadoop is a popular choice at the data-layer as it offers many tools for ingesting and transforming data in real-time. It also allows for integration of other tools like Spark for replacing the built-in MapReduce tool. For example, the Spark engine that powers the Shark (a RTDW) data warehouse gives a 10-100 times faster performance than MapReduce. The Analytics and Decision layers are bound together by the Integration Layer and this composite collection of layers is fed through various components of the data layer.

6.6.2 Real-Time Data Warehouse architecture

The Integration Layer binds together the Analytics and Decision layers and this composite collection of layers is fed through various components of the data layer. We can see that these changes expose atypical constructs that are essential for the way a) data sources are added/updated, b) data is propagated across the warehouse and c) data is loaded into various repositories for heterogeneous data. We discuss a few

architectures that were proposed over the past decade and have greatly influenced some of the popular data warehousing components today.

The proposed construct comprises of following component blocks:

- i) Data production systems hosted on *data sources* and responsible for populating the data warehouse. This block comprises of a source flow regulator (SFlowR) that periodically (based on preset or customized policy) transmits data to the data warehouse after the data has been identified for pertinent changes.
- ii) The *data processing area* (DPA) responsible for data-quality and transformation. This block contains the data flow regulator (DFlowR) used for detecting the transmission-ready source.
- iii) After cleaning, an ETL workflow collects the records from the transmitting data source and transforms them to data warehouse format. The DPA also has other myriad responsibilities like ensuring QoS, checkpoint generation, caching data into reservoirs if the data warehouse pipelines are full, etc. Post-processing, the data transmission to the data warehouse, is orchestrated by a workflow regulator (WFlowR) component.

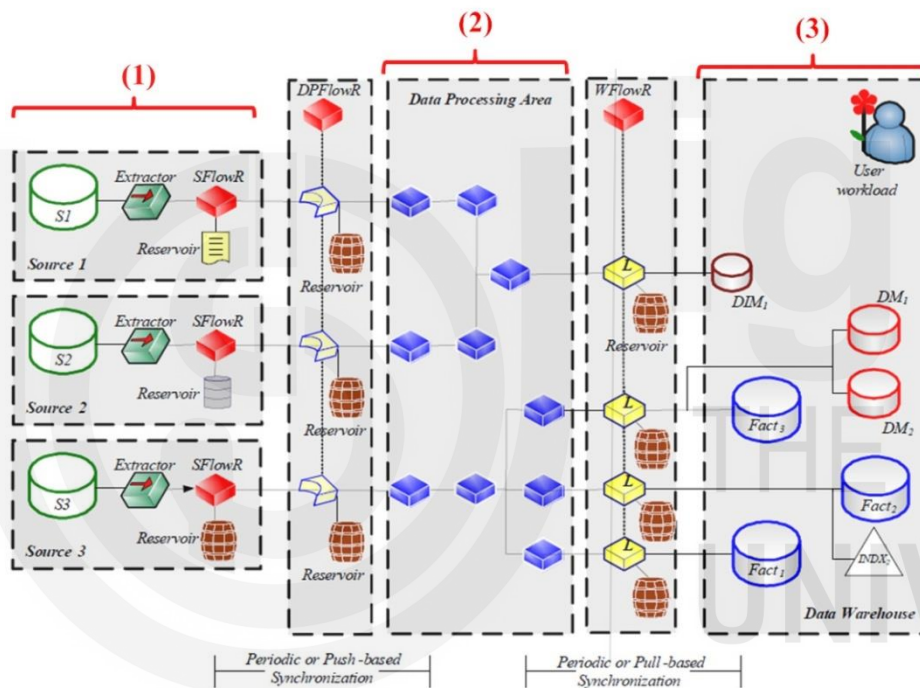


Figure-12. Near-Real-Time data warehouse architecture.
(courtesy - Vassiliadis, P., Simitsis, A.: Near Real Time ETL)

- iv) The *data warehouse* itself. The data warehouse is comprised of dimensions, fact tables, materialized views and indexes (primarily bitmaps and B+ trees) which makes it a complex repository. Data propagation through this block depends on the flow regulation components and the real-time aspect is ensured through the successful performance of these components.

An alternative architectural construct is proposed by Obali and his team is shown in Figure-13.

The architectural components are: a) Metadata b) Web Service (WS) client, c) Web Service provider, d) ETL e) Real-time Partition, f) Real-time data integration and g) Data warehouse.

Change Data Capture (CDC) designating the modified data is initiated by the WS client and sent to the WS provider through a related web service. CDC involves use of various techniques for identifying changes to data like: direct extraction of source data if it is new, timestamp-based, application aided, file comparison based and trigger

criteria based data capture. Data is populated in the real-time partition through a WS provider web service based on the data transmitted by WS client. The received data is separated into data and metadata. A structure query language (SQL) is generated, by a web service through SQL generator, using metadata and inserting data into data warehouse log tables. The generated SQL is executed to populate the data warehouse data repository (database).

The data orchestration between the data warehouse and real-time partition is handled by the real-time data integration component which acts as glue between the two. When there is a user-request for historical data the real-time data integration component generates a query that is sent to the data warehouse. In case of a hybrid data request (both historical and recent) a query is generated which integrates the data from the real-time partition and the data warehouse.

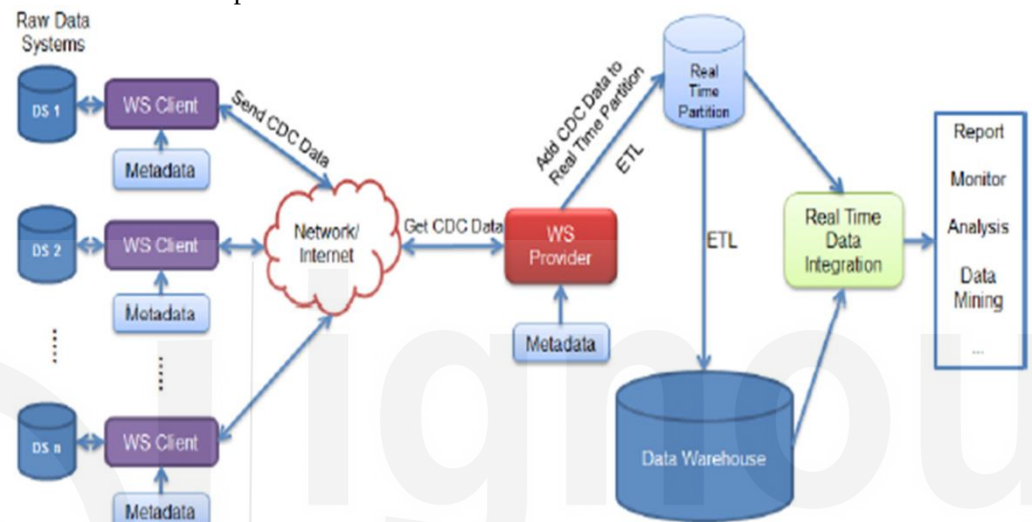


Figure-13. Real-time data warehouse architecture based on web services
(courtesy: Obali, M., Dursun, B., Erdem, Z., Grr, A.K.: A real time data warehouse approach for data processing.)

Yet another architecture by Ferreira and his team can be seen in Figure-14.

The architectural components involve: Data brought into an Input Stream Buffer component in a packaged format from the data sources component. The packaging is meant to alleviate the burden of identifying the data for changes. A transform ETL component brings cleaned data to a Dynamic DW (DDW) component from where the historical data is moved to the Static DW component. A Merger component, used for querying data from Static and Dynamic data warehouse (M-SDDW) handles the query generation directed towards the appropriate data repository for relevant user-request.

Based on the integration policies the data from either of the independent repositories (D-DW or S-DW) can be integrated just in time. Also data completeness can be achieved, for example through foreign-key resolutions or existence of related transactional information.

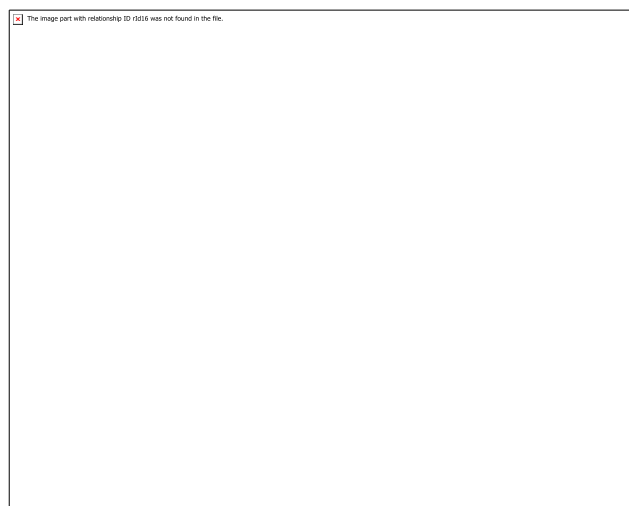


Figure-14. Real-time data warehouse architecture model proposed by Ferreira et al
(courtesy: Ferreira, N., Martins, P., Furtado, P.: Near real-time with traditional data ware-house architectures)

6.6.3 Real-time Data Warehouse Architecture Tradeoffs

It is understood from the above architectures, few changes required across the entire data warehouse ecosystem to implement a real time data warehouse. Consequently, there are a few architectural tradeoffs that users of RTDWs need to understand.

The principles of ETL system are presumed to remain unchanged on the journey to a RTDW. Also responsibilities towards data integration, backup, recovery, archival, regulatory compliance, security, quality, governance, etc. are to be maintained. The trade-offs that ensure on the RTDW journey is briefly explained as below:

6.6.3.1 Batch file replacement

Use of message queues and transaction log files are recommended by Ralph Kimball, replacing batch files. From a batch file's perspective, the data is complete with all keys resolved. Data from message queues and transaction logs is raw data that does not fall within the purview of business rules or corrective processes and is instantaneous.

6.6.3.2 Restrict scrutiny of data quality

Here the scrutiny is applied to columnar data. As data requests start to become more demanding, some data quality will need to be compromised by decreasing mandates on data quality. While this is done, users will need to be made aware of data unreliability and may require compensatory measures in the form of corresponding batch oriented ETL pipeline that periodically replaces real-time data with complete and accurate data.

6.6.3.3 Publish Facts with Dimensions

Old records of dimensions should be allowed to contain facts that are received early. For example, transaction details for a customer might be available even before a customer or item code is generated. We notice that the facts are available prior to dimensional data and in such scenarios if data for dimensions cannot be determined (due to real-time imperatives) an older version of the dimension must be used or a generic template could be used, otherwise. The dimension data should be posted into the hot (real time) partition as and when revised data is received or through the batch update scheduled for the end of day. Nevertheless, users should be informed that dimensions may contain transient data disparate from the fact data.

6.6.3.4 Preclude Data-Staging

Enterprise information integration (EII) systems stream data directly to the user's interface (screens) from the production sources, thereby precluding its transfer to ETL pipeline's permanent repository. Such circumstances, if extant in your enterprise, should be discussed with the senior management to appropriately assign responsibility for backup, recovery, archival, security, etc. of the uncommitted data.

6.6.3.5 Real-time partitions for Data Visualization

The demands for real-time data have increased with the decreasing time intervals of data transition between DW/Business Intelligence (BI) teams and production transaction processing. A real-time partition design as an extension of traditional data warehouse is a possible solution for resolving the increasing demands for data in real time. This is similar to the architectural construct illustrated in Figure-14 above. The real-time partition needs to conform to the following requisites:

- Contain complete set of activities since the last update of the static data warehouse
- Connect precisely as a true physical fact-table partition to the static DWs grain and content
- Allow continuous trickling of data through simple indexes. The ideal possibility of real-time partition being completely indexed may not be viable in case of relational databases where indexes are logically disparate to the partitioning structure.
- Mapping real-time partition in memory to support high performance queries even without extant indexes.

6.6.3.6 Real-time partition for Transactions

Considering that the granularity of a fact table within the static DW is at transactional level it will contain a single row per transaction. The dimensional structure of real-time partition would be similar to the SDW and may not have any indexes as it requires continuous data loading. The real-time partition would also lack time series as it is supposed to contain only day's data.

For a comparatively large enterprise in the retail business with approximately 10 million daily transactions, the fact table would be sizable in volume. Assuming that the width of each transaction grain is 44 bytes (eight dimensions, three facts and four byte columns) you would have 440 MB of data daily. That would mean, yearly, about 160 GB of raw data in the fact table with heavy indexing and aggregations. However, the daily slice of 440 MB of real-time data should be pinned in memory. The real-time partition may have affinity for fast-loading and demonstrate high speed query performance simultaneously.

6.6.3.7 Real-time partition based on periodic snapshots

If we consider the same example as above but with a monthly granularity then the real-time partition would constitute the current active rolling month. Considering a customer-base of 15 million customers, a 24 month time series would accumulate 360 million rows in its fact table that would again be massively indexed with supporting aggregations. The real-time partition is a snapshot of the currently progressing month that is continuously updated as the month advances. Amendment of fully summative facts and semi-summative balances would be done on availability of respective data. The fact table superset of customer types across the enterprise would be quite narrow with say four facts and five dimensions giving us a real-time partition of 600MB allowing the pinning of the real-time partition in memory. With the rest of the monthly data arriving on the last day of the month, the real-time partition could be merged as the latest month onto the comparatively stable fact table. Then the process could be iterated for the subsequent month by flushing the real-time partition.

Check Your Progress 4

- 1) How is a real-time data warehouse different from a traditional data warehouse?

- 2) Why is CDW a good candidate for real-time data warehousing

6.7 DATA WAREHOUSING AND HADOOP

6.7.1 Introduction

The title of this section may sound confusing at first but as we explore the Hadoop system in detail, we should be able to gain more clarity as to why data warehousing and Hadoop necessitates a combined discussion. Although the data warehousing and Hadoop trend evolved sometime back and has become a mature topic today it is still a continuing trend and therefore, requires an overview to comprehend the concepts involved.

Before we dive into the specifics of the Hadoop system we need to go through a few preliminary concepts and understand them so that we can discuss this topic of data warehousing and Hadoop without any confusion or doubts. We have already studied Data warehouse in the previous chapters above. In this section we will understand the definition and features of Hadoop. We will also compare and contrast Hadoop with a Data warehouse system. We will see how Hadoop can supplement and support a DW. Finally, we will take a look at the advantages and challenges involved in Hadoop supported DW.

6.7.2 What is Hadoop?

Hadoop is an open-source java-oriented framework comprising two fundamental components: i) Storage and ii) Processing and was developed, especially, to deal with massive volumes of data in a distributed computing ecosystem. The storage component of Hadoop is called the Hadoop Distributed File System (HDFS) while the processing component is termed as MapReduce.

Essentially, Hadoop, today, comprises of the following modules:

1. The Hadoop Distributed File System

The HDFS allows data storage with easy accessibility. The data storage resides in an environment guaranteeing *highly reliability* and *highly availability*.

2. MapReduce

MapReduce is principally a combination of two processes. Mapping and Reduction. Data is read from the data sources and converted to a format conducive to data analysts. This is the mapping process. The data is then subjected to mathematical operations like aggregations, grouping, encoding

ranges, etc. also called as reductions and hence the terminology - *reduce*. In the Hadoop framework, it is the MapReduce programming model implementation for processing data on a large scale also termed as *data processing at scale*.

3. Hadoop Common

Hadoop common is a collection of libraries and tools required by other Hadoop modules.

4. YARN

YARN is a platform that orchestrates and manages clustered computing resources and uses them for scheduling user-applications.

5. Hadoop Ozone

Ozone is an object store (a data storage architecture that manages data as objects, through a combination of a globally unique identifier, metadata and the data itself)

6.7.3 Hadoop Architecture

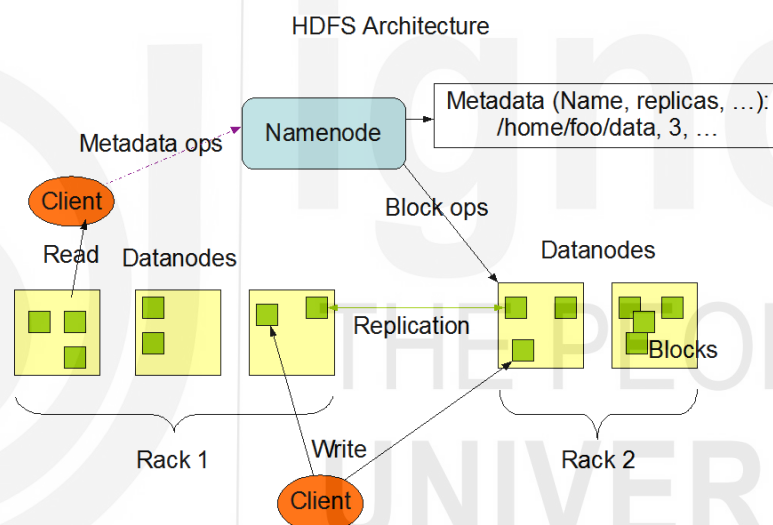


Figure-15. HDFS Architecture for Hadoop 3.3.0
(from Hadoop documentation available at hadoop.apache.org)

The HDFS architecture as shown in Figure-15 implements a master/slave model and has the following components:

NameNode

It is a single node and functions as the master server managing the file-system namespace and regulating client access to files. It handles operations like opening, closing and renaming of files/directories. It also determines the mapping of blocks to DataNodes. NameNode is the arbitrator and repository for all HDFS metadata and its sole existence greatly simplifies the system architecture. The system design is such that user data never flows through the NameNode.

DataNodes

There are multiple DataNodes, usually one per node in the cluster. HDFS allows user data to be stored in files. Files are, internally, split into one or more blocks which are stored in a set of DataNodes and they handle read and write requests from the clients.

Based on instruction from the NameNode, the DataNodes perform block creation, deletion and replication.

NameNodes and DataNodes are software designed to run on commodity machines. Any machine that supports Java can run the NameNode or DataNode as HDFS is built using the Java language. HDFS inherits all the advantages of Java based deployment like portability to a variety of machines.

The File System Namespace

HDFS supports hierarchical file organization and allows creation of directories with files within the directories. It supports file/directory creation, deletion, renaming or moving files/directories within the hierarchy. Additionally, it supports user quotas and access permissions including transparent encryption and snapshots.

Data Replication

HDFS reliably stores very large files across machines in a large cluster. Each file is stored as a sequence of blocks that are replicated for fault tolerance. With the exception of the last block, all blocks are uniform in size. An application can configure the number of replicas of a file. Replication factor is also configurable before and after file creation. Files in HDFS are write-once (except for appends and truncates) and strictly have one writer at any given time. The NameNode makes all decisions pertinent to block replication and periodically receives a Blockreport and a Heartbeat from every DataNode in the cluster.

Hadoop is often used to mean both base modules and sub-modules and also the environment. It may also refer to collection of supplementary software that can be integrated with or function as extensions of Hadoop framework. Some prevalent software are Apache Flume, Apache HBase, Apache Hive, Cloudera Impala, Apache Oozie, Apache Phoenix, Apache Pig, Apache Spark, Apache Sqoop, Apache Storm and Apache ZooKeeper.

6.7.4 Comparison of Hadoop and Data warehouse

Now that we had an overview of Hadoop let us find out its relation with a data warehouse by comparing the two. Data warehouse and Hadoop can be compared on the basis of various characteristics common to both as follows:

6.7.4.1 Data

Data sources can be added or removed in both DWs and Hadoop in addition to allowing heterogeneous data sources. A DW allows consumption of structured data which is transformed to its schema format. Hadoop allows for consumption of both structured and semi-structured data.

Users of Raw data are typically absent in a DW scenario with analysts requiring preprocessed data. Hadoop enables the use of raw data by data engineers and data scientists enabling innovative knowledge discoveries for the enterprise. In case of a DW this advantage is lost during the data transformation process.

Hadoop works on the principle of schema-on-read while a DW works on the principle of schema-on-write. A schema-on-read allows adaptability to the heterogeneous nature of data and therefore, its accommodation for business utilization. Conversely fitting data to the dimensional structure in a DW imposes restrictions on which data can be accommodated.

Hadoop allows distributed data processing at large scale data by virtue of its file system capabilities and architecture supplemented by integrated or add-on toolsets which support the Hadoop ecosystem. A traditional data warehouse architecture would require substantial modifications to its architecture to support a similar set of tools in addition to changing its data repository.

6.7.4.2 Performance

Hadoop tools like Spark, Impala, etc. allow for near-real-time implementations for data propagation across the system allowing data-visualizations on the most current datasets. Ingested data is brought to the analytics stage many times faster in Hadoop when compared to a DW.

6.7.4.3 Governance

Governance in Hadoop is relatively easier due to its highly configurable architecture. Also Hadoop's architectural constructs of data storage, consumption of data in a specific manner (due to its schema-on-read feature) and availability of tools with individual system-wide functionality make governance across the ecosystem easy to manage.

6.7.4.4 Security

Hadoop implements security at block level in addition to offering transparent encryption. Desensitizing data becomes easy due to massive storage and computing capacity of Hadoop. DWs cannot maintain desensitized datasets for longer periods. Also the granularity and methods of security differ vastly when compared to that provided by Hadoop

6.7.4.5 Flexibility

Hadoop is intrinsically flexible in scaling its compute, storage, memory and network resources. This flexibility lends agility to the entire system and data becomes pervasive across the enterprise. Traditional DWs lack this capacity and would involve a lot of cost and efforts to incorporate such flexibility into their architectures.

6.7.5 Conceptual architecture of Hadoop Data Warehouse

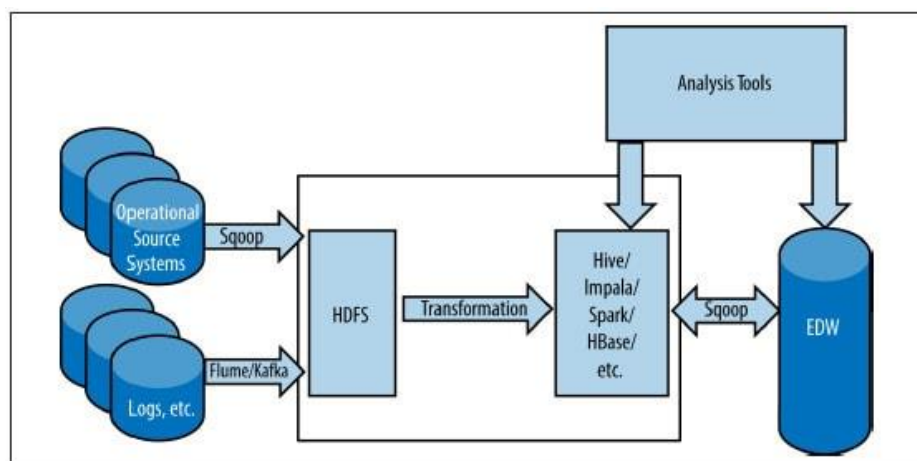


Figure-16. Conceptual Hadoop Data warehouse architecture
(courtesy: Vithal S, article titled “Hadoop Data warehouse Design Considerations” on dwgeek.com)

A Hadoop Data warehouse can be implemented using Apache Hive or Cloudera Impala. Impala offers better performance than Hive due to its massively parallel processing (MPP) capabilities.

The architecture in Figure16 shows a Hadoop supplemented and supported DW. In this scenario, the DW inherits benefits of the Hadoop system augmenting the ETL processes that are at the core of a DW.

On the left side of the figure data sources are shown which could include other heterogeneous sources like sensor data, logs and other structured and semi-structured data. The components of DW in the context of Hadoop are explained as follows:

6.7.5.1 Data Extraction

Tools like Apache Sqoop can be utilized for data extraction from OLTP database sources. Sqoop can also be used to export data to OLTP databases after data processing is completed within the Hadoop system.

6.7.5.2 Data Transformation

Data is ingested into the Hadoop system precluding data staging. This ingested data is transformed and loaded into target data repositories within assigned directories. Tools like Spark, Cloudera Impala, Apache Hive, Apache Pig or MapReduce (the integral processing component of Hadoop) can be used for data transformations.

6.7.5.3 Data Loading

In the context of Hadoop data loading implies data hosting within Hadoop through Apache Hive or Cloudera Impala. Data can be exported to external DWs for ancillary analysis

6.7.5.4 Data Analysis / Visualization

As mentioned above data can be exported for assisting analysis through external DWs or data can be sources directly from the Hadoop system by BI, data analytics and visualization tools. Apache Sqoop can be used for the exporting data assisted through Python or Shell scripts.

6.7.6 Advantages of building a Hadoop Data Warehouse

Hadoop alleviates traditional data warehouse design challenges as follows:

- Heterogeneous data processing at scale
- Improving ETL processing throughput
- Good candidate for near-real-time data warehousing
- Efficient workload management
- Hadoop offers architectural and data level flexibility through scalable and configurable components
- The TCO of the DW system can be reduced through elastic utilization and customizations thereby, increasing the ROI

6.7.7 Challenges of building a Hadoop Data Warehouse

The challenges of a Hadoop based Data warehouse are as follows:

- Permeating the benefits of patterns in raw data to the structured components of a DW
- Data propagation latencies may arise in certain cases where large amounts of data are made available disproportionate to the data being consumed
- Use of Hadoop as an archival repository for historical data may introduce costs for import/export of data if historical data is required.

6.8 DATA WAREHOUSE AUTOMATION

6.8.1 Introduction

Conventional data warehouses or for that matter even some implementations of cloud data warehouses perform repetitive tasks manually with the excuse of business dynamics and constantly changing reporting parameters. This has an adverse effect on the productivity, costs and overall quality expected from the data warehousing platform.

The latest trend is towards data warehouse automation (DWA) which deals with accelerating and automating the data warehouse life-cycle (from conceptualization to implementation). Some enterprises may technically perceive it as automation beginning from analysis of source systems right up to testing and documentation processes.

6.8.2 DWA Maturity

DWA involves the use of sophisticated tools and architectural models to automate planning, design, integration and implementation of DWs through their life-cycle reducing repetitive and time consuming tasks like source analysis, ETL scripts deployment, etc. The automation process requires careful thought for evaluating and selecting the right tools that can drastically improve savings on costs and time at a fraction of TCO of data warehouses.

The functionalities of automation software have matured with explosive growth of data and the race for data-driven insights to leverage businesses. Data explosion has led to the necessity for large-scale data storage and high performance integration platforms. Further, the diversity and velocity of data from various sources (like IoT, social media, etc.) require data processing at an equivalent scale.

6.8.2.1 Maturity Stages of DWA

6.8.2.1.1 DW Architectures and Database management systems (DBMSs)

The earliest data warehouses came into existence with the advent of disk storage that required storing large volumes of data in the form of databases. These requirements stimulated the ideas for relational data and dimension based data marts. Subsequently, SQL enabled Relational DBMSs and proprietary ETL software became available.

6.8.2.1.2 Standardization of DW Architecture

DW Architectures were being formalized and normalized to include automation of source data analysis and components for ingestion and transformation of

heterogeneous data. This would enable the utilization of any data source without worrying about data formats and any specialized treatment for the same.

6.8.2.1.3 Challenges of Enterprise data warehouse and data automation imperative

The pace of changes in the industry posed myriad challenges for enterprises like inefficient metadata management, expensive development and protracted development cycles. Many systems did not integrate efficiently with the underlying components like database management systems, other repositories within the system, etc.

The current DWA software are aligned with business requirements and technological innovations

6.8.2.1.4 Data warehouse automation tools

A DWA tool provides a seamless experience, free from the hassles of coding, for integration and transition of diverse data from its source to a DW and other related components. The tool automates deployment of ETL scripts, batch-processing of data and demonstrates various functions like:

- Processes for high performance ETL and reliable ELT based integration of data
- Modeling of data at source
- Normalized, De-normalized and data structures with multiple dimensions
- Seamless integration with various data sources

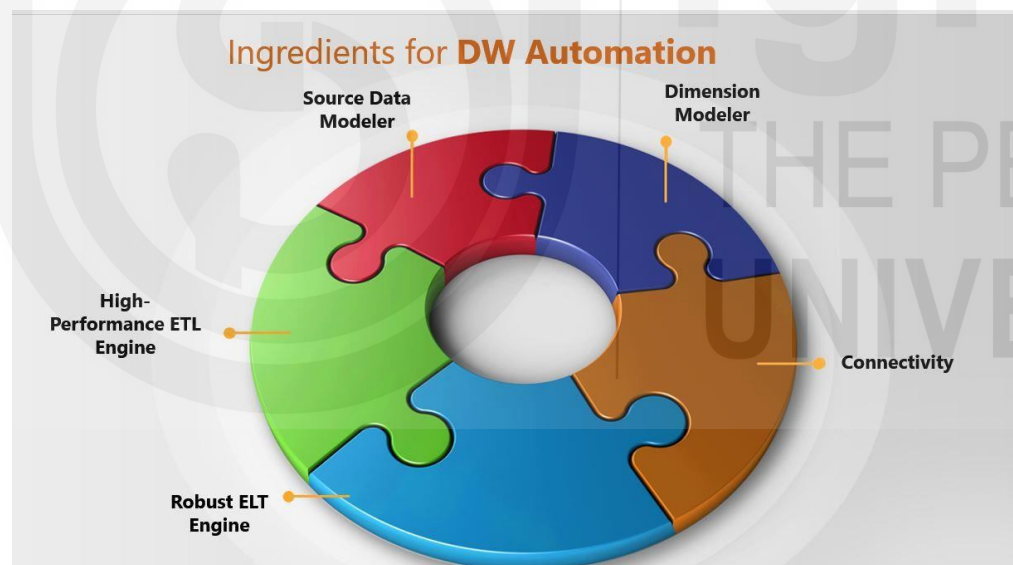


Figure-17. Essential Components of DWA tools

6.8.3 Assessment of DWA preparedness

Enterprises need to evaluate their level of preparedness based on the following needs:

6.8.3.1 DW architecture:

Assess whether your DW architecture is based on a distinctive set of components or whether it uses best-of-breed components and best practices. A precise understanding of the various components that you have implemented and their interactions is of

prime importance to understand the changes required and their impact on your enterprise in terms of cost and efforts.

6.8.3.2 Business requirements:

Assess how business requirements are handled within your organization and the methodologies practiced (waterfall, prototype, agile, etc.) for identifying and understanding them.

6.8.3.3 Policies and Procedures:

Evaluate the structure of organizational processes, compliance requirements, business continuity policies and plans (disaster and recovery management, security, governance etc.), and others. Assess whether the operations are resource and time intensive, vague/detailed, complex or fragile in nature.

6.8.3.4 Data ecosystem maintenance:

Scrutinize whether the maintenance of your data infrastructure is organized, complicated or heavily resource dependent

6.8.3.5 Process and Requirements Persistence:

Evaluate the frequency with which your enterprise changes its processes and requirements or whether it persists for a reasonable amount of time before carefully incorporating changes.

6.8.3.6 Quality and accessibility:

Assess how frequently data is accessed. Also, find out how the delivery of analytical reports is expected whether it is within reasonable time or on an immediate basis. This will let you understand the expectations about data quality, QoS and quality of deliverables.

6.8.4 Advantages of DW automation:

A few significant advantages of DW automation are as follows:

- End-to-end data pipeline acceleration
- Automation of data capture and streaming
- Automation of data management and integration
- Optimization of data propagation paths
- Ensure automatic data flows in their entirety
- Automatic set up of target-oriented data models
- Transformation of data lakes into DWs.

Check Your Progress 5

- 1) How would DWA be advantageous in the case of an enterprise like Amazon?

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6.9 SUMMARY

In this section we followed the trends in data warehousing. We noticed that the trends originated based on industrial implementations of research based solutions offered to challenges of previous generation data warehousing challenges. Therefore, we concluded that those challenges which we face presently and are resolved through research in that area will become trends of the future.

Within the topic of Data Warehousing Trends we had an overview of the following sub-topics:

- Data Lakes

We saw how the concept of data lakes originated with data explosion, termed as Big Data now. We discussed various data lake architectures and learned the best practices for implementing a successful data lake so that it does not turn into a Data Swamp. We also discussed Self-service, a very important component and a design pattern which forms an integral part of any Big Data based implementation.

- Complex Data Marts

In the sub-topic of complex data marts, we studied how complex data marts came into existence. We also discussed its design process compared to a conventional data mart.

- Cloud Data Warehousing

CDW is a comparatively newer trend and with the advent of Big Data the paradigm of DW modeling has changed. We have all the advantages of the cloud to support a cloud hosted data warehouse. We had an overview of the challenges and best practices for implementing a successful CDW

- Real Time Data Warehousing

Real-time data warehousing is a very popular and closely followed trend by major enterprises. In the race for data dominance and biggest market share enterprises want to be able to take decisions based on the most recent data. Instantaneous data accessibility and availability has become the need of the hour with enterprises investing a lot of money and resources to make it a reality. We reviewed a few major proposed architectures that have already been put into practice today. Finally, we discussed the architectural tradeoffs or comprises that are required for deriving the benefits of real-time data availability.

- Data Warehousing and Hadoop

Hadoop was developed to counter the challenges of big data that comes in unstructured or semi-structured forms. The obvious performance and scalability benefits came to the attention of enterprises with large data warehouse implementation. These enterprises thought of utilizing Hadoops toolset to build Hadoop based data warehouses. We studied about the conceptual architectures for the same and discussed the advantages and challenges involved.

- Data Warehouse Automation

Under this sub-topic we had an overview of how DWA has matured over the years and what benefits could be had from automating almost the entire component set of the data warehouse. Best practices were studied including the challenges involved and the benefits possible post DWA.

This unit has touched upon some very important and critical topics that are either already established or would become the mainstay of the future. Students are

requested to study these trends in detail by referring to texts and documentation mentioned below and through their own curious efforts.

6.10 SOLUTIONS/ANSWERS

Check your Progress 1

1. Many challenges have come to the fore-front with increasing volumes and speed of data. Data has increased due to various technological advances resulting in increase in the number of sources and diversity.

Researchers have tried to address myriad challenges subsequent to data explosion and the significant ones being worked upon by maximum researchers are as below:

- Conceptual modeling of DWs and logical data models,
 - Data warehouse loading (data-refreshing),
 - execution efficacy of OLAP queries and data mining algorithms,
 - materialized views,
 - data analysis techniques,
 - metadata management,
 - evolution management of DWs,
 - stream-based, real-time and active data warehouses, and
 - complex data warehousing (for example spatial, XML, object, multimedia)
2. The concept of logical data lakes originated through the implementation of multiple data lakes in a distributed environment. Logical data lakes form an abstraction layer over the implementation of a data lake or data lakes and allow their use without exposing the complexities of location, data sources, target systems, etc. and allow enterprises to perceive data access as an operation on a single entity that spans across multiple locations and brings data from various repositories together. Draw the architectural representation and explain it in your own words. Elaborate on the components and advantages of the concept.
 3. A successful data lake is well cataloged both before and after data ingestion so that data traceability to its source is maintained. Data is well organized and is available through self-service for people across the enterprise. People are aware of data contained within a data lake and utilize it based on their specific needs. Other standards or best practices are adhered to all times like data governance, data quality maintenance, metadata generation, master data management, etc.

Check your progress 2

1. Complex data is data that has no definitive structure or form for example sensor data. Complex data also has high velocity and volumes and does not conform to any standard rules of a schema, technology or function. It is also an attribute of Big data. Complex data forms a major portion of the raw data that is ingested into a data lake and is considered to be of significance for data insights by data engineers and data scientists.

As the race for leveraging businesses based on data-driven insights intensifies, enterprises want to derive the maximum business value hidden within the complex data. In order to do this complex data needs to be processed and converted to usable form. As conventional models are not adequately equipped to

handle complex data that may be unstructured or semi-structured new models are required to efficiently process complex data and make it useful for data analysis.

2. Please see the information related to Data Vault on *wikipedia.org* and frame your answer accordingly. Draw the example illustration and briefly explain the various components of the Data Vault model.

Check your progress 3

1. Enterprises migrate to the cloud due to the following benefits offered by a CDW implementation:

- a. Improved performance

The availability of cloud-based database management software and tools supplemented by sophisticated infrastructure allow performance that is many times faster than on-premises DWs. The ability to easily scale resources based on processing requirements and latencies are drastically reduced compared to alternative implementations of the DW.

- b. Flexibility of cost through flexibility of cloud resources utilization

Cloud implementations allow elastic use of resources on a pay-as-you-use basis which lends flexibility in terms of costs and resource utilization. Conversely, in on-premises based implementations the stakeholders are stuck with the resources and require additional costs to scale their infrastructure resources.

- c. Migration of existing products to the cloud environment

Most enterprises may have previously migrated their ERP systems or other software to the cloud or may be thinking of implementing other projects on the cloud. This makes it all the more imperative to have a CDW to reduce data propagation latencies and integration with the systems already on the cloud.

- d. Derive benefits that are inherent to a cloud platform

A cloud platform offers many benefits based on the myriad services it offers like Software as a service (SaaS), Platform as a service (PaaS), Infrastructure as a service (IaaS), etc. These services can be exploited to derive maximum benefits. For example by exploiting SaaS we can have best of breed software (like ETL tools, database management software, etc.) to support and supplement our Data warehouse implementation.

Check your progress 4

1. Compare the real-time and traditional DWs based on their design, function, performance, data quality and data visualization expectations

For example you could elaborate on the below:

Design components for a traditional DW involve batch-based ETL, data quality, data source scheduling, data completeness is guaranteed through the data processing and data integration components.

For a Real-time DW ETL is not batch-based and uses various other approaches to load new data and update existing data, data quality needs to be compromised as data is consumed immediately as it arrives and waiting for it to be complete in all respects (like foreign key constraints, missing values, etc.) is not permitted. Data processing practices need to be modified to process partial information and later ensure completeness through incremental updates, data integration may require partitioning to hold incomplete information. Data staging needs to be abandoned to eliminate time lost in scrutinizing data.

2. CDW offers various benefits like performance, resource scalability, etc. which make it easy to implement the design changes required for Real-time DW. CDW implementation supports sophisticated tools for data storage, processing and manipulation. CDW also makes data integration easy through its toolset allowing for better data quality and data propagation across the system

Check your progress 5

1. End-to-end data pipeline acceleration would increase the flow of data through its system modules (Inventory, order processing, bill-generation, delivery tracking and order fulfillment and customer feedback

Automation of data capture and streaming would benefit Amazon to collect significant amount of data at various stages from when a customer buys a product to order fulfillment and feedback

Automation of data management and integration would aid Amazon to avoid data duplication, reduce the risks and challenges of data import / export to from other systems, generate data value for business and increase QoS.

Optimization of data propagation paths will ensure data availability with minimum effort and maximum benefit

Amazon can ensure automatic data flows in their entirety i.e. ensure that all the details required for a transaction are completely captured and the data is automatically delivered to the systems requesting it for analysis.

Automatic set up of target-oriented data models would enable Amazon to ensure that data reaches its intended audience through the automatic selection of data source, data processing and integration tools and finally the analytics systems.

As Amazon operates on a very large scale spanning various continents the amount of data generated would be humongous and contained in data lakes. Transformation of data lakes into DWs would enable Amazon to ensure regulatory compliances. It would also allow data localization in the case of countries requiring it for political and other reasons. Through all this the DWs can still be integrated and enjoy the benefits of a data lake ecosystem.

6.11 FURTHER READINGS

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UNIT 7 DATA MINING – AN INTRODUCTION

- 7.0 Introduction
- 7.1 What is Data Mining?
- 7.2 Purpose Of Data Mining
- 7.3 How Does Data Mining Work?
- 7.4 What Types Of Data Can Be Mined?
 - 1.7.1. Database Data
 - 1.7.2. Transactional Data
 - 1.7.3. Other Kinds of Data
- 7.5 Classification of Data Mining Systems
- 7.6 Data Mining Vs Data Warehousing
- 7.7 Data Mining Tools
- 7.8 Applications Of Data Mining
- 7.9 Data Mining Issues

7.0 INTRODUCTION

We are in an age often referred to as the information age in which we have been collecting huge amounts of data. Thanks to sophisticated technologies such as computers, satellites, etc. With the invention of computers and mass digital storage media, we started collecting and storing all sorts of data. Unfortunately, these huge data stored on different storage media became profuse. This chaotic situation of storing the data has led to the creation of database management systems (DBMS). The boosting of database management systems also contributes to massive gathering of data of all types. Today, we have a lot of information than we can handle: from satellite pictures, text reports to business transactions and scientific data and military intelligence. Information retrieval is simply not enough anymore for decision-making. Opposed with huge collections of data, we have generated new requirements for helping us to make better managerial decisions. These requirements may be automatic summarization of data, extraction of the kernel of the information stored, and the discovery of patterns in raw data.

So, Now that we have gone through the introduction part, let's understand what really the concept of data mining is!

7.1 WHAT IS DATA MINING?

Data mining is the process by which the patterns can be found by filtering the data based on a condition of data sorting in order to find something important. At a micro-scale, mining can be defined as an activity that involves gathering data in one place in a structural way. For example, clubbing an Excel Spreadsheet or creating a summarization of the main points of some text.

In other words, the process of data mining can be better explained and defined in the following manner:

- Data mining finds valuable information hidden in large volumes of data.

- Data mining is the analysis of data and the use of software techniques for finding patterns and regularities in sets of data.
- The computer is responsible for finding the patterns by identifying the underlying rules and features in the data.

7.2 PURPOSE OF DATA MINING

The purpose of mining the data can be multi-fold which includes:

Predicting various outcomes;

Modeling target audience;

Collecting the information about the product.

Data mining helps to understand certain aspects of customer behavior. This knowledge allows companies to adapt accordingly and offer the best possible services.

In order to have a thorough understanding of the behavior of the customers, Data mining helps to extract the pattern from the databases. The knowledge thus acquired, allows the companies to offer the best possible services.

7.3 HOW DOES DATA MINING WORK?

Step-wise, the operation of data mining consists of the following elements:

Building target datasets by selecting what kind of data you need;

Preprocessing is the groundwork for the subsequent operations. This process is also known as data exploration.

Preparing the data - a creation of the segmenting rules, cleaning data from noise, handling missing values, performing anomaly checks, and other operations. This stage may also include further data exploration.

7.4 WHAT TYPES OF DATA CAN BE MINED?

As a general technology, data mining can be applied to any kind of data as long as the data are meaningful for a target application. The most basic forms of data for mining applications are database data (Section 1.3.1), data warehouse data (Section 1.3.2), and transactional data (Section 1.3.3). The concepts and techniques presented in this book focus on such data. Data mining can also be applied to other forms of data (e.g., data streams, ordered/sequence data, graph or networked data, spatial data, text data, multimedia data, and the WWW). We present an overview of such data in Section 1.3.4. Techniques for mining of these kinds of data are briefly introduced in Chapter 13. In-depth treatment is considered an advanced topic. Data mining will certainly continue to embrace new data types as they emerge.

1.7.1. Database Data

A database system, also called a database management system (DBMS), consists of a collection of interrelated data, known as a database, and a set of software programs to manage and access the data. The software programs provide mechanisms for defining database structures and data storage; for specifying and managing concurrent, shared, or distributed data access; and for ensuring consistency and security of the information stored despite system crashes or attempts at unauthorized access. A relational database is a collection of tables, each of which is assigned a unique name. Each table consists of a set of attributes (columns or fields) and usually stores a large set of tuples (records or rows). Each tuple in a relational table represents an object identified by a unique key and described by a set of attribute values. A semantic data model, such as an entity-relationship (ER) data model, is often constructed for relational databases. An ER data model represents the database as a set of entities and their relationships.

Example : A relational database for All Electronics. The fictitious All Electronics store is used to illustrate concepts throughout this book. The company is described by the following relation tables: customer, item, employee, and branch. The headers of the tables described here are shown below : (A header is also called the schema of a relation.) The relation customer consists of a set of attributes describing the customer information, including a unique customer identity number (cust ID), customer name, address, age, occupation, annual income, credit information, and category. Similarly, each of the relations item, employee, and branch consists of a set of attributes describing the properties of these entities. Tables can also be used to represent the relationships between or among multiple entities. In our example, these include purchases (customer purchases items, creating a sales transaction handled by an employee), items sold (lists items sold in a given transaction), and works at (employee works at a branch of All Electronics).

Relational data can be accessed by database queries written in a relational query language (e.g., SQL) or with the assistance of graphical user interfaces. A given query is transformed into a set of relational operations, such as join, selection, and projection, and is then optimized for efficient processing. A query allows retrieval of specified subsets of the data. Suppose that your job is to analyze the AllElectronics data. Through the use of relational queries, you can ask things like, “Show me a list of all items that were sold in the last quarter.” Relational languages also use aggregate functions such as sum, avg (average), count, max (maximum), and min (minimum). Using aggregates allows you to ask: “Show me the total sales of the last month, grouped by branch,” or “How many sales transactions occurred in the month of December?” or “Which salesperson had the highest sales?” When mining relational databases, we can go further by searching for trends or data patterns. For example, data mining systems can analyze customer data to predict the credit risk of new customers based on their income, age, and previous credit information. Data mining systems may also detect deviations—that is, items with sales that are far from those expected in comparison with the previous year. Such deviations can then be further investigated. For example, data mining may discover that there has been a change in packaging of an item or a significant increase in price. Relational databases are one of the most commonly available and richest information repositories, and thus they are a major data form in the study of data mining.

1.3.2 Data Warehouses

Suppose that All Electronics is a successful international company with branches around the world. Each branch has its own set of databases. The president of All Electronics has asked

you to provide an analysis of the company's sales per item type per branch for the third quarter. This is a difficult task, particularly since the relevant data are spread out over several databases physically located at numerous sites. If AllElectronics had a data warehouse, this task would be easy. A data warehouse is a repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site. Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing. To facilitate decision making, the data in a data warehouse are organized around major subjects (e.g., customer, item, supplier, and activity). The data are stored to provide information from a historical perspective, such as in the past 6 to 12 months, and are typically summarized. For example, rather than storing the details of each sales transaction, the data warehouse may store a summary of the transactions per item type for each store or, summarized to a higher level, for each sales region. A data warehouse is usually modeled by a multidimensional data structure, called a data cube, in which each dimension corresponds to an attribute or a set of attributes in the schema, and each cell stores the value of some aggregate measure such as count.

Example A data cube for All Electronics. The cube has three dimensions: address (with city values Chicago, New York, Toronto, Vancouver), time (with quarter values Q1, Q2, Q3, Q4), and item (with item type values home entertainment, computer, phone, security). The aggregate value stored in each cell of the cube is sales amount (in thousands). For example, the total sales for the first quarter, Q1, for the items related to security systems in Vancouver is \$400,000, as stored in cell Vancouver, Q1, security. Additional cubes may be used to store aggregate sums over each dimension, corresponding to the aggregate values obtained using different SQL group-bys (e.g., the total sales amount per city and quarter, or per city and item, or per quarter and item, or per each individual dimension). By providing multidimensional data views and the pre computation of summarized data, data warehouse systems can provide inherent support for OLAP. Online analytical processing operations make use of background knowledge regarding the domain of the data being studied to allow the presentation of data at different levels of abstraction. Such operations accommodate different user viewpoints. Examples of OLAP operations include drill-down and roll-up, which allow the user to view the data at differing degrees of summarization. For instance, we can drill down on sales data summarized by quarter to see data summarized by month. Similarly, we can roll up on sales data summarized by city to view data summarized by country. Although data warehouse tools help support data analysis, additional tools for data mining are often needed for in-depth analysis.

1.7.2. Transactional Data

In general, each record in a transactional database captures a transaction, such as a customer's purchase, a flight booking, or a user's clicks on a web page. A transaction typically includes a unique transaction identity number (trans ID) and a list of the items making up the transaction, such as the items purchased in the transaction. A transactional database may have additional tables, which contain other information related to the transactions, such as item description, information about the salesperson or the branch, and so on. Example: A transactional database for All Electronics. Transactions can be stored in a table, with one record per transaction. From the relational database point of view, the sales table in the figure is a nested relation because the attribute list of item IDs contains a set of items. Because most relational database systems do not support nested relational structures, the transactional database

is usually either stored in a flat file in a format similar to the table in Figure 1.8 or unfolded into a standard relation in a format similar to the items sold table in Figure 1.5. As an analyst of All Electronics, you may ask, “Which items sold well together?” This kind of market basket data analysis would enable you to bundle groups of items together as a strategy for boosting sales. For example, given the knowledge that printers are commonly purchased together with computers, you could offer certain printers at a steep discount (or even for free) to customers buying selected computers, in the hopes of selling more computers (which are often more expensive than printers). A traditional database system is not able to perform market basket data analysis. Fortunately, data mining on transactional data can do so by mining frequent item sets, that is, sets of items that are frequently sold together.

1.7.3. Other Kinds of Data

Besides relational database data, data warehouse data, and transaction data, there are many other kinds of data that have versatile forms and structures and rather different semantic meanings. Such kinds of data can be seen in many applications: time-related or sequence data (e.g., historical records, stock exchange data, and time-series and biological sequence data), data streams (e.g., video surveillance and sensor data, which are continuously transmitted), spatial data (e.g., maps), engineering design data (e.g., the design of buildings, system components, or integrated circuits), hypertext and multimedia data (including text, image, video, and audio data), graph and networked data (e.g., social and information networks), and the Web (a huge, widely distributed information repository made available by the Internet). These applications bring about new challenges, like how to handle data carrying special structures (e.g., sequences, trees, graphs, and networks) and specific semantics (such as ordering, image, audio and video contents, and connectivity), and how to mine patterns that carry rich structures and semantics. Various kinds of knowledge can be mined from these kinds of data. Regarding temporal data, we can mine banking data for changing trends, which may aid in the scheduling of bank tellers according to the volume of customer traffic. Stock exchange data can be mined to uncover trends that could help you plan investment strategies (e.g., the best time to purchase All Electronics stock). We could mine computer network data streams to detect intrusions based on the anomaly of message flows, which may be discovered by clustering, dynamic construction of stream models or by comparing the current frequent patterns with those at a previous time.

With spatial data, we may look for patterns that describe changes in metropolitan poverty rates based on city distances from major highways. The relationships among a set of spatial objects can be examined in order to discover which subsets of objects are spatially auto correlated or associated. By mining text data, such as literature on data mining from the past ten years, we can identify the evolution of hot topics in the field. By mining user comments on products (which are often submitted as short text messages), we can assess customer sentiments and understand how well a product is embraced by a market.

From multimedia data, we can mine images to identify objects and classify them by assigning semantic labels or tags. By mining video data of a hockey game, we can detect video sequences corresponding to goals. Web mining can help us learn about the distribution of information on the WWW in general, characterize and classify web pages, and uncover web dynamics and the association and other relationships among

different web pages, users, communities, and web-based activities. It is important to keep in mind that, in many applications, multiple types of data are present. For example, in web mining, there often exist text data and multimedia data (e.g., pictures and videos) on web pages, graph data like web graphs, and map data on some web sites. In bioinformatics, genomic sequences, biological networks, and 3-D spatial structures of genomes may coexist for certain biological objects. Mining multiple data sources of complex data often leads to fruitful findings due to the mutual enhancement and consolidation of such multiple sources. On the other hand, it is also challenging because of the difficulties in data cleaning and data integration, as well as the complex interactions among the multiple sources of such data. While such data require sophisticated facilities for efficient storage, retrieval, and updating, they also provide fertile ground and raise challenging research and implementation issues for data mining.

7.5 CLASSIFICATION OF DATA MINING SYSTEMS

The data mining system can be classified according to the following criteria:

- Database Technology
- Machine Learning
- Other Discipline
- Statistics
- Information Science
- Visualization

Some Other Classification Criteria:

- Classification according to kind of databases mined
- Classification according to kind of knowledge mined
- Classification according to kinds of techniques utilized
- Classification according to applications adapted

Classification according to kind of databases mined

We can classify the data mining system according to kind of databases mined. Database system can be classified according to different criteria such as data models, types of data etc. And the data mining system can be classified accordingly. For example if we classify the database according to data model then we may have a relational, transactional, object- relational, or data warehouse mining system.

Classification according to kind of knowledge mined

We can classify the data mining system according to kind of knowledge mined. It means datamining system are classified on the basis of functionalities such as:

- Characterization
- Discrimination
- Association and Correlation Analysis

- Classification
- Prediction
- Clustering
- Outlier Analysis
- Evolution Analysis

Classification according to kinds of techniques utilized

We can classify the data mining system according to kind of techniques used.
We can describe these techniques according to degree of user interaction involved or the methods of analysis employed.

Classification according to applications adapted

We can classify the data mining system according to application adapted. These applications areas follows:

- Finance
- Telecommunications
- DNA
- Stock Markets
- E-mail.

7.6 DATA MINING VS DATA WAREHOUSING

What is Data warehouse?

A data warehouse is a technique for collecting and managing data from varied sources to provide meaningful business insights. It is a blend of technologies and components which allows the strategic use of data.

Data Warehouse is electronic storage of a large amount of information by a business which is designed for query and analysis instead of transaction processing. It is a process of transforming data into information and making it available to users for analysis.

What Is Data Mining?

Data mining is looking for hidden, valid, and potentially useful patterns in huge data sets. Data Mining is all about discovering unsuspected/ previously unknown relationships amongst the data.

It is a multi-disciplinary skill that uses machine learning, statistics, AI and database technology.

The insights extracted via Data mining can be used for marketing, fraud detection, and scientific discovery, etc.

Data Mining Vs Data Warehouse: Key Differences

Data Mining	Data Warehouse
Data mining is the process of analyzing unknown patterns of data.	A data warehouse is database system which is designed for analytical instead of transactional work.
Data mining is a method of comparing large amounts of data to finding right patterns.	Data warehousing is a method of centralizing data from different sources into one common repository.
Data mining is usually done by business users with the assistance of engineers.	Data warehousing is a process which needs to occur before any data mining can take place.
Data mining is the considered as a process of extracting data from large data sets.	On the other hand, Data warehousing is the process of pooling all relevant data together.
One of the most important benefits of data mining techniques is the detection and identification of errors in the system.	One of the pros of Data Warehouse is its ability to update consistently. That's why it is ideal for the business owner who wants the best and latest features.
Data mining helps to create suggestive patterns of important factors. Like the buying habits of customers, products, sales. So that, companies can make the necessary adjustments in operation and production.	Data Warehouse adds an extra value to operational business systems like CRM systems when the warehouse is integrated.
The Data mining techniques are never 100% accurate and may cause serious consequences in certain conditions.	In the data warehouse, there is great chance that the data which was required for analysis by the organization may not be integrated into the warehouse. It can easily lead to loss of information.
The information gathered based on Data Mining by organizations can be misused against a group of people.	Data warehouses are created for a huge IT project. Therefore, it involves high maintenance system which can impact the revenue of medium to small-scale organizations.
After successful initial queries, users may	Data Warehouse is complicated to

ask more complicated queries which would increase the workload.	implement and maintain.
Organisations can benefit from this analytical tool by equipping pertinent and usable knowledge-based information.	Data warehouse stores a large amount of historical data which helps users to analyze different time periods and trends for making future predictions.
Organisations need to spend lots of their resources for training and Implementation purpose. Moreover, data mining tools work in different manners due to different algorithms employed in their design.	In Data warehouse, data is pooled from multiple sources. The data needs to be cleaned and transformed. This could be a challenge.
The data mining methods are cost-effective and efficient compares to other statistical data applications.	Data warehouse's responsibility is to simplify every type of business data. Most of the work that will be done on user's part is inputting the raw data.
Another critical benefit of data mining techniques is the identification of errors which can lead to losses. Generated data could be used to detect a drop-in sale.	Data warehouse allows users to access critical data from the number of sources in a single place. Therefore, it saves user's time of retrieving data from multiple sources.
Data mining helps to generate actionable strategies built on data insights.	Once you input any information into Data warehouse system, you will unlikely to lose track of this data again. You need to conduct a quick search, helps you to find the right statistic information.

Why use Data Warehouse?

Some most Important reasons for using Data warehouse are:

- Integrates many sources of data and helps to decrease stress on a production system.
- Optimized Data for reading access and consecutive disk scans.
- Data Warehouse helps to protect Data from the source system upgrades.
- Allows users to perform master Data Management.
- Improve data quality in source systems.

Why use Data mining?

Some most important reasons for using Data mining are:

- Establish relevance and relationships amongst data. Use this information to generate profitable insights
- Business can make informed decisions quickly
- Helps to find out unusual shopping patterns in grocery stores.
- Optimize website business by providing customized offers to each visitor.
- Helps to measure customer's response rates in business marketing.
- Creating and maintaining new customer groups for marketing purposes.
- Predict customer defections, like which customers are more likely to switch to another supplier in the nearest future.
- Differentiate between profitable and unprofitable customers.
- Identify all kinds of suspicious behavior, as part of a fraud detection process.

7.7 DATA MINING TOOLS

Data Mining techniques are derived that utilize the domain knowledge from statistical analysis, artificial intelligence, and database systems in order to analyze data in a proper manner in view of different dimensions and perspectives.

Data Mining tools discover patterns or trends from the collection of large sets of data and transforming data into useful information for making decisions.

Ex: Frameworks such as Rstudio or Tableau that allow us to perform different types of data mining analysis.

These tools allow us to implement algorithms like classification and clustering on the dataset and visualize the results.

The Market for Data Mining tool is shining: as per the latest report from Report Linker noted that the market would top **\$1 billion** in sales by **2023**, up from **\$591 million** in **2018**

These are the most popular data mining tools:



1. Orange Data Mining:



Developed at the bioinformatics laboratory at the faculty of computer and information science, Ljubljana University, Slovenia, Orange is a machine learning and data mining software suite. It supports data visualization and is a software-based on components written in Python.

The components of Orange are called "widgets." These widgets range from data visualization and preprocessing to assessing the algorithms and are used for predictive modeling.

Widgets deliver much important functionalities such as:

- Displaying data table and allowing to select features
- Reading of data.
- Training predictors and comparison of learning algorithms
- Data element visualization, etc.

Besides, Orange provides a more interactive and enjoyable atmosphere

Why Orange?

Data coming to orange is formatted quickly to the desired pattern, and the widgets can be easily transferred wherever and whenever needed. Orange is quite interesting to users. Orange allows its users to make smarter decisions in a short time by rapidly comparing and analyzing the data. Data mining can be performed via visual programming or Python scripting. Many analyses are feasible through its visual programming interface(drag and drop connected with widgets)and many visual tools tend to be supported such as bar charts, scatterplots, trees, dendrograms, and heat maps. A substantial amount of widgets(more than 100) tend to be supported.

The instrument has machine learning components, add-ons for bioinformatics and text mining, and it is packed with features for data analytics. This is also used as a python library.

Orange comprises of canvas interface onto which the user places widgets and creates a data analysis workflow. The widget proposes fundamental operations, For example, reading the data, showing a data table, selecting features, training predictors, comparing learning algorithms, visualizing data elements, etc. Orange comes with multiple regression and classification algorithms.

Orange can read documents in native and other data formats. Orange is dedicated to machine learning techniques for classification or supervised data mining. There are two types of objects used in classification: learner and classifiers. Learners consider class-level data and return a classifier. Regression methods are very similar to classification in Orange, and both are designed for supervised data mining and require class-level data. The learning of ensembles combines the predictions of individual models for precision gain. The model can either come from different training data or use different learners on the same sets of data.

2. SAS Data Mining:



SAS stands for Statistical Analysis System. It is a product of the SAS Institute created for analytics and data management. SAS can mine data, change it, manage information from various sources, and analyze statistics. It offers a graphical UI for non-technical users.

SAS data miner allows users to analyze big data and provide accurate insight for timely decision-making purposes. SAS has distributed memory processing architecture that is highly scalable. It is suitable for data mining, optimization, and text mining purposes.

4. Rattle:



Rattle is a data mining tool based on GUI. It uses the R programming language. Rattle combines the power of R with significant data mining features.

5. Rapid Miner:



Rapid Miner is one of the most popular tool for performing predictions. It is written in JAVA programming language. It offers an integrated environment for text mining, deep learning, machine learning, and predictive analysis.

The instrument can be used for a wide range of applications, including company applications, commercial applications, research, education, training, application development, machine learning.

Rapid Miner provides the server on-site as well as in public or private cloud infrastructure. It has a client/server model as its base. A rapid miner comes with template-based frameworks that enable fast delivery with few errors(which are commonly expected in the manual coding writing process)

Oracle BI



Oracle BI is an open source machine learning and data visualization for naïve as well as expert users.

Features:

- Interactive Data Visualization.
- It Offers Interactive data exploration for rapid qualitative analysis with clean visualizations.
- Orange supports hands-on training and visual illustrations of concepts from data science.
- It offers an extensive range of add-ons to data mining from external data sources.

11) KNIME



KNIME is open source software for creating data science applications and services. This Data mining tool helps you to understand data and to design data science workflows.

Features:

- Helps you to build an end to end data science workflows
- Blend data from any source
- Allows you to aggregate, sort, filter, and join data either on your local machine, in-database or in distributed big data environments.
- Build machine learning models for classification, regression, dimension reduction

12) Tanagra

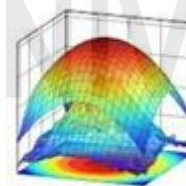


Tanagra is a free to use data mining tool for study and research purposes. It offers various data mining methods from statistical learning, data analysis, and machine learning.

Features:

- Offers easy to use data mining software for researcher and students
- It allows the user to add their data mining methods.

15) Data Melt



DataMelt is a free to use tool for numeric computation, mathematics, data analysis, and data visualization. This program offers you the simplicity of scripting languages, like Python, Ruby, Groovy with the power of hundreds of Java packages.

Features:

- DataMelt offers statistics, analysis of large data volumes, and scientific visualization.
- You can use it with different programming languages on different operating systems.
- It allows you to create high-quality vector-graphics images (EPS, SVG, PDF, etc.), which can be included in LaTeX and another text processor.
- Data Melt offers the usage of scripting languages, which are significantly faster than the standard Python implemented in C.

16) ELKI:



ELKI is an open source data mining tool written in Java. The tool allows us researching algorithms, with an emphasis on unsupervised methods in cluster analysis and outlier detection.

Features:

- ELKI offers an extensive collection of highly parameterizable algorithms
- It allows easy and fair evaluation and benchmarking of algorithms.
- ELKI provides data index structures such as the R*-tree which enhance the process of Data mining

17) SPMF



SPMF is an open-source data mining library written in Java. It is distributed under the GPL license. It allows you to integrate source code with other Java Software.

Features:

- Allows association rule mining
- Supports sequential pattern and sequential rule mining
- Offers High-utility pattern mining,
- Time-series mining.
- Support complex process of Clustering and classification

18) Alteryx



Alteryx is a Business intelligence and analytics solutions for the enterprise. It is a specially designed tool for data analyst and business leaders.

Features:

- Analytics for Midsize Businesses
- It allows for Ad Hoc Analysis.
- Offers fast online Analytical Processing
- Automatic Scheduled Reporting
- Highly customizable Dashboard

19) Enterprise Miner



Enterprise Miner is a SAS software which offers you and cutting-edge algorithms designed to help you solve the most significant challenges and offers the best solutions for your business.

Features:

- Helps you to improve prediction accuracy. Share reliable results
- Easy-to-use GUI and batch processing
- Advanced predictive and descriptive modeling
- Offers Automated scoring
- Automate model deployment and scoring

20) Datawatch



Datawatch Desktop is a Data mining and business intelligence solution. It allows you to focus on real-time data visualization. It offers tools to build and deploy their monitoring and analysis systems without the need to write a single line of code.

Features:

- Drag-and-drop feature allows users to build a customized view of data
- Identify trading anomalies
- Analyze how alternative scenarios will affect performance using historical data

21) Advanced miner



An advanced miner is a useful tool for data processing, analysis, and modeling. Its user-friendly workflow interface allows you to explore various types of data.

Features:

- Extracting and saving data from/to different database systems, files, and data transformations
- Offers various operations on data, like sampling, joining datasets, etc.
- Helps you to build statistical models, variable importance analysis, clustering analysis, etc.
- Easy and effective Models' integration with external IT applications

22) Analytic Solver



Analytic Solver is free to use the point-and-click tool. It allows you to do risk analysis and prescriptive analytics in your browser. It offers full-power Data mining jobs.

Features:

- Helps you to incorporate uncertainty and solve with simulation optimization, stochastic programming, and robust optimization.
- Allows you to define the Monte Carlo simulation model using Excel formulas

23) PolyAnalyst



PolyAnalyst is the Data mining and analytical tool for extracting actionable knowledge hidden and actual structured of the data.

Features:

- Helps you to access data from various sources and merge data from different sources
- You can select from a broad selection of statistical and machine-learning algorithms.

- Offers you to create stuffing report which can be summarized and communicate your insight

24) Civis



Civis empowering you to make informed decisions with data scientist and decision market in mind. It allows your team to collaborate efficiently and find solutions faster.

Features:

- Offers architecture, products, and processes which helps you to protect your data
- You can configure with a library of data ingestion and ETL modules.
- Write code in a script, offers multiple scripts or jobs into a workflow, and define a workflow to run on a schedule.
- Allows you to turn your analysis and models into applications that run on a flexible, production-level infrastructure

25) Viscovery:



Viscovery is a workflow-oriented software suite. It is based on self-organizing maps and multivariate statistics for explorative data mining and predictive modeling. The system excels in intuitive user-guidance, mature implementation.

Features:

- An ideal project environment platform for goal-oriented operation
- Dedicated workflows that which allows you to offer focused navigation
- Clear workflow steps with proven default settings
- Workflow branching allowing generation of model variations
- Functions for integrated documentation and annotation
- Multiple handling tools to facilitate usage

7.8 APPLICATIONS OF DATA MINING

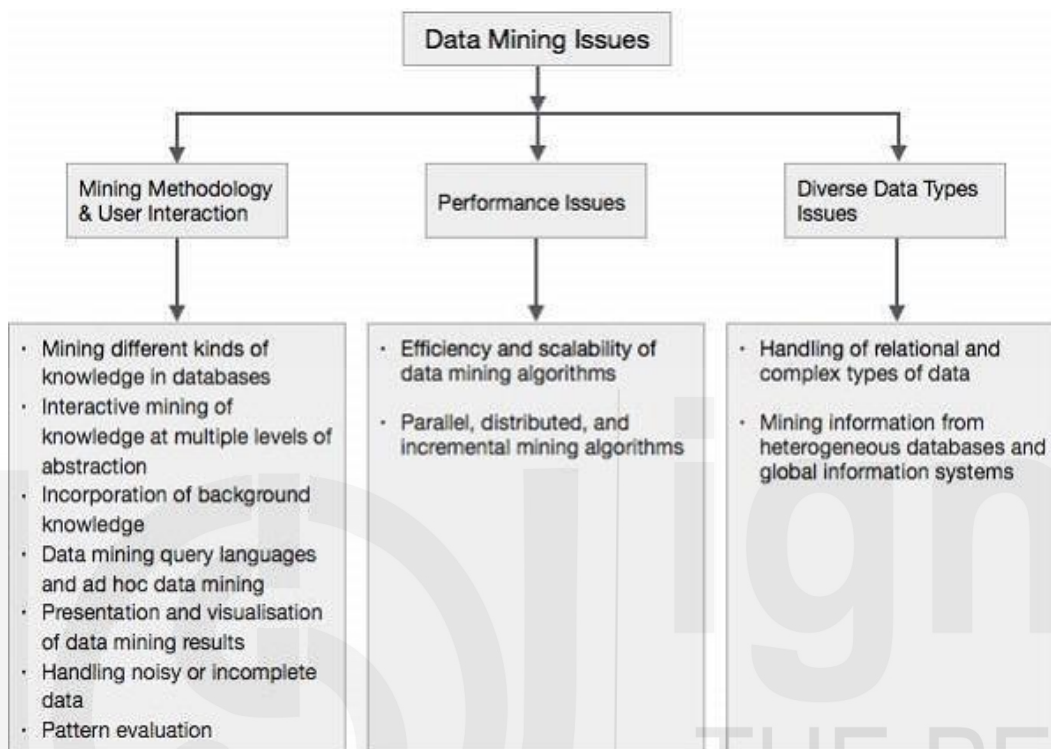
7.9 DATA MINING ISSUES

Data mining is not an easy task, as the algorithms used can get very complex and data is not always available at one place. It needs to be integrated from various

heterogeneous data sources. These factors also create some issues. Here in this tutorial, we will discuss the major issues regarding –

- Mining Methodology and User Interaction
- Performance Issues
- Diverse Data Types Issues

The following diagram describes the major issues.



Mining Methodology and User Interaction Issues

It refers to the following kinds of issues –

- **Mining different kinds of knowledge in databases** – Different users may be interested in different kinds of knowledge. Therefore it is necessary for data mining to cover a broad range of knowledge discovery task.
- **Interactive mining of knowledge at multiple levels of abstraction** – The data mining process needs to be interactive because it allows users to focus the search for patterns, providing and refining data mining requests based on the returned results.
- **Incorporation of background knowledge** – To guide discovery process and to express the discovered patterns, the background knowledge can be used. Background knowledge may be used to express the discovered patterns not only in concise terms but at multiple levels of abstraction.
- **Data mining query languages and ad hoc data mining** – Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.
- **Presentation and visualization of data mining results** – Once the patterns are discovered it needs to be expressed in high level languages, and visual representations. These representations should be easily understandable.

- **Handling noisy or incomplete data** – The data cleaning methods are required to handle the noise and incomplete objects while mining the data regularities. If the data cleaning methods are not there then the accuracy of the discovered patterns will be poor.
- **Pattern evaluation** – The patterns discovered should be interesting because either they represent common knowledge or lack novelty.

Performance Issues

There can be performance-related issues such as –

- **Efficiency and scalability of data mining algorithms** – In order to effectively extract the information from huge amount of data in databases, data mining algorithm must be efficient and scalable.
- **Parallel, distributed, and incremental mining algorithms** – The factors such as huge size of databases, wide distribution of data, and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithms divide the data into partitions which is further processed in a parallel fashion. Then the results from the partitions is merged. The incremental algorithms, update databases without mining the data again from scratch.

Diverse Data Types Issues

- **Handling of relational and complex types of data** – The database may contain complex data objects, multimedia data objects, spatial data, temporal data etc. It is not possible for one system to mine all these kind of data.
- **Mining information from heterogeneous databases and global information systems** – The data is available at different data sources on LAN or WAN. These data source may be structured, semi structured or unstructured. Therefore mining the knowledge from them adds challenges to data mining.

UNIT 8 DATA PREPROCESSING

8.0 Introduction

8.1 Data Cleaning

8.1.1 Missing Values

8.1.2 Noisy Data

8.1.3 Data Cleaning As A Process

8.4 Data Reduction

8.4.1 Data Cube Aggregation

8.4.2 Attribute Subset Selection

8.4.3 Dimensionality Reduction

8.4.4 Numerosity Reduction

8.0 INTRODUCTION

Preprocessing:

Real-world databases are particularly susceptible to noise, incomplete and inconsistent large data sets are popular as one can be a primary, secondary or tertiary source. Low quality data may result in low quality performance, thereby needing additional pre-processing data.

Data Preprocessing Techniques:

- ❖ Data cleaning may be used to prevent noise and to correct incoherence of data.
- ❖ Data aggregation combines data from multiple channels into a cohesive data store, for example a data center.
- ❖ Data transformations may be applied, such as standardization.
- ❖ Data reduction will, for example, reduce data size by the compilation, elimination of duplicate components or clustering; it can operate together.

Preprocessing Necessity:

Typical location properties in vast real-world datasets and databases are incomplete, chaotic and unfulfilling information.

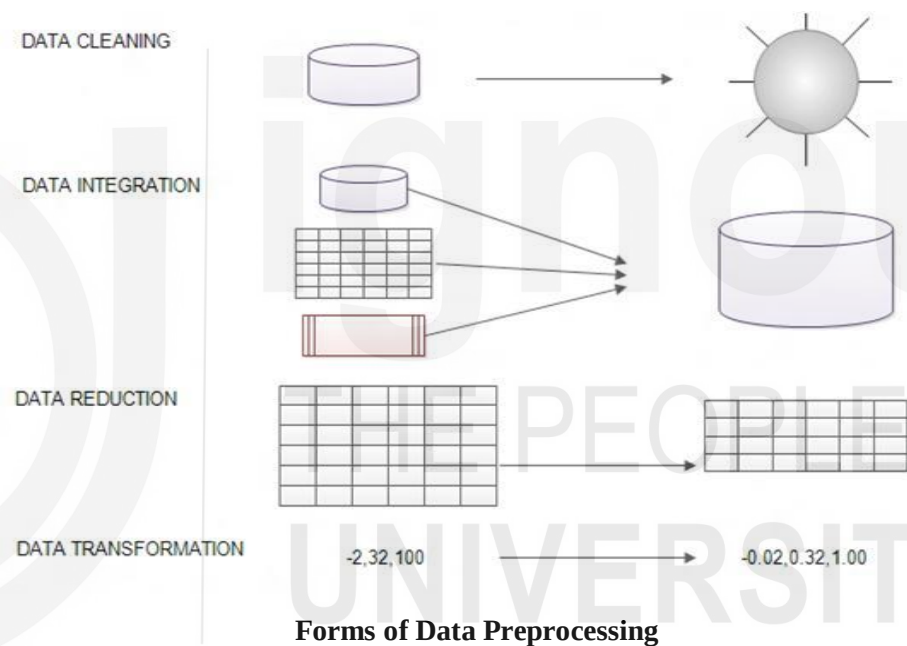
Unfinished data can arise for a number of reasons:

- ❖ Important attributes cannot always be usable.
- ❖ Valid data cannot be recorded because of misunderstanding or equipment failure.
- ❖ Data that disputed other recorded data should have been discarded.
- ❖ Missing data may be inferred, especially for tuples with missing values for certain attributes.
- ❖ Data collection techniques may be unreliable.
- ❖ At the moment of data entry, human or computer errors may have existed.
- ❖ Data processing failures can also occur.
- ❖ There might be technological disadvantages, such as narrow buffer sized for simultaneous data transfer and usage coordination.
- ❖ Data Routines for the cleaning of data are used by filling the missing values.
- ❖ Relieve noise effects, detect or remove outliers and address inconsistencies.
- ❖ Data integration is the hybrid process with many cubes or archives of databases. However, in multiple databases, those characteristics that define a

specific may have separate titles, which lead to inconsistencies and redundancies.

- ❖ Data transformation is a process approach such as standardizations and consolidation that constitutes additional preprocessing processes that contribute to mining process results.
- ❖ Data reduction obtains a simpler data collection image, which is significantly smaller in duration, but provides the same analytical efficiency (or nearly the same). A multitude of methods for data reduction are in use. This includes:
 - a) Data Aggregation (e.g., Creating a data cube).
 - b) Attribute subset selection (By similarity, eliminating unnecessary attributes)
 - c) Dimensionality Reduction (e.g., using encoding systems such as minimal encoding lengths or wavelets).
 - d) Numerosity Reduction (e.g., “Replace” details by alternating, smaller representations such as clusters or parametric structures.
 - e) Generalization (e.g., Data were minimized with the usage of the definition hierarchy)

Data discretization is a method of data reduction that is very useful for the automated generation of idea hierarchies from numerical data.



8.1 DATA CLEANING

The other problem is that findings in the modern world are inconsistent, noisy and inexact. Software – cleaning (or data cleaning) procedures are intended to complete missing values, smooth out sounds as outliers are identified and data errors are corrected.

8.1.1 Missing Values:

Some tuples have no significance mentioned for a range of attributes, such as consumer income. We're going to fill up the vacant values.

The following methods execute missing values over a number of attributes.

- a) **Ignore the tuple:** This is usually done if the class label is missing (assuming the mining activity requires classification). This solution is not quite good

since the tuple contains several attributes with missing values. It is really unfortunate since the percentage of values missed per variable varies greatly.

- b) **Fill in the missing values manually:** This approach is time intensive and might not be feasible due to a wide range of data with many missing values.
- c) **Use a global constant to fill in the missing values:** Replace all missing attributes with the same constant, such as a "unknown" or ∞ .
- d) **Use the attribute mean to fill in the missing value:** Suppose, for instance, that the gross customer income is \$71,000. This importance compensates for the loss of revenue.
- e) **Use the mean attribute for all samples belonging to the same class as the given tuple:** When the borrower is identified by credit danger, replace the missing number with the average income value in the same credit risk category as the tuple.
- f) **Using the most likely meaning to fill in the missing value:** This can be computed using regression, inference-based tools using Bayesian formalism or the introduction of decision tree. For example, in the fixed Decision Tree the use of other market attributes is designed to approximate the missing revenue value.

8.1.2 Noisy Data:

Noisy is a measured random error or variance in the variable. Noise is minimized by techniques of data smoothing.

Binning: Binning uses the "neighborhood" to smooth the storage value of the records. That's the value around it. The sorted values are split into "buckets" and "bins." Since binning methods consult the value community, local smoothing is carried out.

Stored price details (in dollars \$): 2, 4, 6, 8, 10, 12, 14, 16, 18.

Example: Partition in bins (Equal-Frequency):

Bin 1: 2, 4, 6

Bin 2: 8, 10, 12

Bin 3: 14, 16, 18

In the above process, the price data is first sorted and then partitioned into equivalent frequency bins of size 3.

Smoothing by bin mean:

Bin 1: 4, 4, 6

Bin 2: 10, 10, 10

Bin 3: 16, 16, 16

The method assumes that any value in a bin is replaced with the mean value of the bin as it is smoothed by bin. In bin 1, for example the mean of definitions 2, 4 and 6 is 4 $[(2+4+6)/3]$.

Bin borders smoothing:

Bin 1: 2, 2, 4

Bin 2: 8, 12, 12

Bin 3: 14, 14, 18

The maximum and minimum values are defined as the boundary values when the bin boundary is smoothed. Each bin value is then replaced by the closest limit value. The larger the diameter, the more smoothing impact is. Bins will also be of similar width when the range of values in each bin remains unchanged.

Example: Delete the noise with smoothing techniques from the following details:
4, 2, 6, 10, 8, 16, 12, 24, 22, 14, 26.

Stored price details (in dollars \$):

2, 4, 6, 8, 10, 12, 14, 16, 22, 24, 26

Partition in equal – frequency (equi-depth) bins:

Bin 1: 2, 4, 6, 8

Bin 2: 10, 10, 12, 14

Bin 3: 16, 22, 24, 26

Smoothing by bin mean:

Bin 1: 5, 5, 5, 5

Bin 2: 11, 11, 11, 11

Bin 3: 22, 22, 22, 22

Smoothing by bin boundaries:

Bin 1: 2, 2, 2, 8

Bin 2: 10, 10, 14, 14

Bin 3: 16, 16, 16, 26

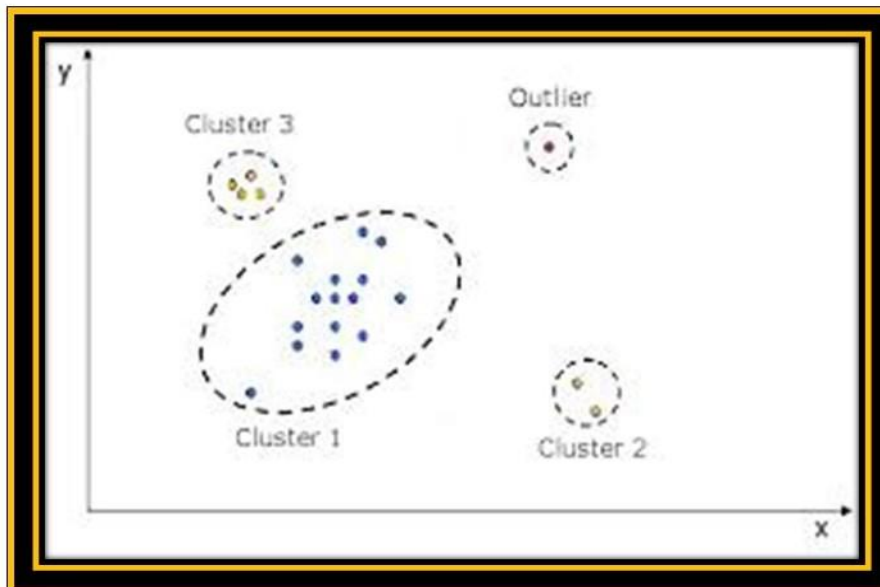
Regression:

By modifying data to perform, for instance for regression, data may be smoothed. Linear regression specifies that two attributes (or variables) be searched for the "right" line so one attribute can be used to predict the other.

Multiple linear regression is a linear regression extension, involving more than two attributes and outcomes that fit onto a multi-dimensional surface.

Clustering:

Clustering Outliers may be detected if the same values are clustered in groups or clusters. Values that go outside the spectrum of clusters can intuitively be considered outliers.



Outliers detected by clustering analysis

Inconsistent Data:

Information stored in the transaction contains inconsistencies. There are inconsistencies where there are conceptual dependence of attributes and incomplete values in data entry. Inconsistencies may be discovered and corrected through manual or information engineering approaches.

8.1.3 Data cleaning as a process:

- ❖ Discrepancy Detection
- ❖ Data Transformation

Discrepancy Detection:

The first step in data cleaning is to identify discrepancies. It considers the meta data awareness and examines the following rules in order to identify the distinction.

Specific guidelines:

Each attribute value must be distinct from each other attribute value.

Consecutive Laws:

If there are no missing values between the variable's lowest and highest values and all values must therefore be unique.

Null Rules:

Define the usage of blanks, question marks, special characters or other strings which can specify null status.

Discrepancy Detection Tools:

Data Scrubbing Tools:

Use specific domain details (e.g. postal address awareness and spelling control) to detect errors and correct details.

Data Auditing Tools:

Analyzes data to classify rules and relations and to evaluate data that violate those parameters.

Data Transformation:

This is the second step of the data cleaning process. After identifying inconsistencies, we have to define (sequence of) transformations and execute to correct them.

Data Transformation Tools:

Data Migration Tools:

Requires fundamental transformations such as replacing the 'gender' string with the 'sex' string to be described.

Extraction Transformation Loading (ETL):

Requires users to make specific user experience transformations (GUI).

Disadvantages of Data Cleaning Process:

- ❖ Due to missing records, analysts can lose track of feasible information. In situations where incomplete findings and outliers have been dropped, this is quite normal.
- ❖ It could contribute to an even bigger issue if automated.
- ❖ It takes time.
- ❖ The procedure is quite expensive.

8.3.1 Data Integration:

It is a methodology that combines data from various platforms into one device. These sources which include many cubes of records, databases or flat files.

The approach to data integration is officially classified as three times <G, S, M> where,

G Stands for the Global Scheme

S Stands for a Heterogeneous Source of a Schema

M Stands for Mapping between source and Global Schema Queries.

There are two main data integration methods: one is a "solid link approach," while the other is a "loose connection approach"

Tight Coupling:

The data warehouse here is regarded as a retrieval of records.

Data from different sources are combined to one physical location via ETL in this pairing (Extraction, Transformation and Loading phase).

Loose Coupling:

In this case the interface will translate the user query into the source database and send the query back to the source databases in order to get the answer.

And the data are only available in the real source repositories.

Data Integration Issues:

During data migration, there are no problems to consider:

- ❖ Schema Integration and Object Matching
- ❖ Redundancy
- ❖ Detection and Resolution of data value conflicts.

These are explained briefly as follows below:

1. Schema Integration and Object Matching:

It can be difficult because in different tables, different individuals can identify the same type. This is linked to the problem of individual identity. Metadata can be used to avoid errors in schema integration. Meta data can also be used for additional data conversation (e.g. where database codes "H" and "S" may be typed in one database, 1 & 2 in a separate database).

2. Redundancy:

That is another serious issue, that if it can be "derived" from another attribute, an attribute (e.g. annual sales) can be redundant. Inconsistencies in the naming of dimensions can also add to the redundancies in the data collection arising from this. A correlation analysis can detect any redundancies, given the two attributes, the test determines how strongly the one attribute indicates the other, based on the data available. The relationship between two attributes, X and Y, can be tested for numerical attributes by calculating the correlation coefficient:

$$r_{X,Y} = (\sum(X - \bar{X})(Y - \bar{Y})) / (n - 1)\sigma_A\sigma_B$$

Where

- ❖ n is the number of tuples
- ❖ $\bar{X}$ means the value of X
- ❖ $\bar{Y}$ means the value of Y
- ❖ σ_A Standard deviation of X
- ❖ σ_B Standard deviation of Y

$$\bar{X} = \frac{\sum X}{n}; \sigma_X = \sqrt{\sum (X - \bar{X})^2 / (n - 1)}$$

$r_{X,Y} > 0$ then X and Y are positively correlated

$r_{X,Y} < 0$ then X and Y are negatively correlated

$r_{X,Y} = 0$ then on correlation between X and Y

Detection and Resolution of Data Value Conflicts:

The third major issue in the integration of data is the detection and resolution of conflicts over data. For instance, the significance of attributes from different sources can vary for the same real person – the world person. This could be connected to changes in image, size or encoding. Room prices in various cities covering not only different currencies, but also different services (such as free breakfast) and taxes for the hotel chain. A lower abstraction level of an attribute can be recorded on one system than the same attribute on another system. Semantic heterogeneity and structure of data face major data integration problems.

A thorough compilation of data from various sources can help to mitigate and avoid the resulting data collection of redundancies and anomalies. This helps to improve the accuracy and speed of the mining process.

8.3.2 Data Transformation:

Data is translated into data mining methods. The data transformation consists of the following steps.

- ❖ Smoothing
- ❖ Aggregation
- ❖ Discretization
- ❖ Attribute Construction
- ❖ Generalization
- ❖ Normalization

Smoothing:

It is a mechanism used to remove data set noise using some algorithms to highlight main data set features. It helps to predict patterns and can be controlled when collecting data to eliminate or reduce any variance or noise.

The concept behind data smoothing is that simple changes that help to predict different trends and patterns can be detected. This helps investors or traders who want to look at a lot of data, which can be difficult to digest for patterns they would not see otherwise.

Aggregation:

The compilation or grouping of data is a way of summer data collection and show. In order to integrate these data sources into the concept of data analysis, the data should be obtained from different data sources. This is a vital step because the accuracy of data analysis depends heavily on the volume and quality of the data used. Reliable high quality and sufficient amounts of data should be collected to achieve the necessary results. Data collection is useful in all aspects including judgments on the financing of commodity, pricing, processes and marketing strategies or business plans. For example, revenue can be aggregated to calculate the monthly and annual total.

Discretization:

It is a way to transform continuous data into a number of short intervals. Many actual data mining activities have ongoing attributes. However, many of the existing data mining systems cannot handle these attributes.

While a data mining task can always handle a continuous attribute, a constant consistency attribute can significantly improve its efficiency by replacements of discrete values.

Attribute Construction:

Where new attributes are created and added to support an attribute set in the mining procedure. This simplifies the original data and improves the efficiency of mining.

Generalization:

Convert low - level data attributes with a hierarchical concept to high-level data attributes. For example, an age in numerical form initially (20, 64) is translated into a categorical sense (Young and Old).

Categorical attributes such as house addresses, for example, may be applied to higher-level definitions such as cities or areas.

Normalization:

All data variables need to be translated into a given set. The standardization approach used is:

- ❖ Min – Max Normalization
- ❖ Z – Score Normalization
- ❖ Decimal Scaling Normalization

Min – Max Normalization:

- ❖ Which translates the initial data linearly.

- ❖ Suppose $\min X$ is the minimum and $\max X$ is the maximum limit of the variable.
- ❖ We've got the formula

$$v' = \frac{v - \min X}{\max X - \min X} (\text{new_max}_X - \text{new_min}_X) + \text{new_min}_X$$

- ❖ where v is the value that you want to plot in the new set.
- ❖ v' is the fresh meaning you get when you normalize the old value.

Example: Assume the attribute revenue minimum and maximum values are 1000 and 16000 respectively. Diagram income in the [0.0, 1.0] range. For salaries a value of 14000 is converted into a minimum – maximum standardization.

$$= \frac{14000 - 1000}{16000 - 1000} (1.0 - 0) + 0 \\ = 0.866$$

Z-Score Normalization:

- ❖ The value of a variable (X) in the Z-Score Normalization or Zero-Median Normalization is uniform according to the mean of the X and its standard deviation.
- ❖ The X value v of the X attribute is computer-standardized to V .

$$v' = \frac{v - \bar{X}}{\sigma X}$$

Where $\bar{X}$ and σ are the mean and standard deviations respectively of the X attribute. This method is helpful if the true minimum and maximum attribute X are not certain or if outliers surpass the minimum – maximum normalization.

Example: Suppose that the average and standard deviation in attribute revenue number are respectively 22,000 and 7,000. In the case of Z-Score standardization, the revenue of 42,000 is transferred to Z-Score.

$$\frac{42000 - 22000}{7000} = 2.85$$

Decimal Scaling Normalization:

- ❖ Standardizes the variable values by changing their decimal position.
- ❖ You can measure the number of points that the decimal point is passed by the absolute maximum value of the X attribute.
- ❖ The value of v of the X attribute is computer-standardized to v' .

$$v' = \frac{v}{10^j}$$

J is such an integer that maximum $(|v'|) < 1$.

Example:

- ❖ Suppose the observed values differ between the lowest - 486 and the highest - 417.
- ❖ The X meaning square is 486. In order to normalize the calculation by decimal scaling, each value must be divided by 1000 so that -486 standardizes -0.486 and 417 to 0.417.

8.4 DATA REDUCTION

Data reduction approaches can be used to simplify the data collection representation, which is much smaller in length and thus maintain accuracy of the original data - the reduced data set mining can be much more efficient, but the analytical output can be the same (or almost the same).

Strategies of Data Reduction:

- ❖ Data cube aggregation, which applies aggregation of data during data cube construction.
- ❖ Set of sub-set attributes, which may be detected and removed by obsolete, weakly significant or redundant measurements.
- ❖ Dimensionality reduction where encoding approaches are used to reduce the data collection size to a minimum.
- ❖ Reduction of numbers when alternative, smaller data representation replaces or predicts information.
- ❖ Discretization and category hierarchy generation where raw data values for the attributes are replaced by ranges or higher logical levels.

8.4.1 Data cube Aggregation:

Data Cube aggregation where a data cube is created using aggregation operations. Data cubes store multidimensional information aggregated. Each cell stores in a multi-dimensional space a data value corresponding to the data point.

For each attribute concept hierarchies may occur, allowing data to be evaluated at multiple abstraction levels. Data cubes provide quick access to pre-computed summary data; this will support both online analytical analysis and data mining.

The cube is available in three forms:

Based cuboid:

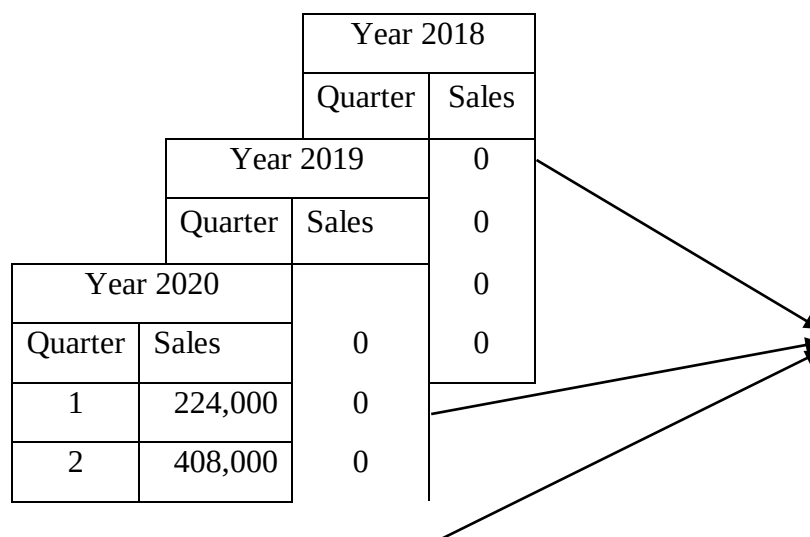
The cube is called the based cuboid, at least in its complexity.

Lattice of cuboid:

Details of the chart of different abstraction layers are often called cubes.

Apex cuboid:

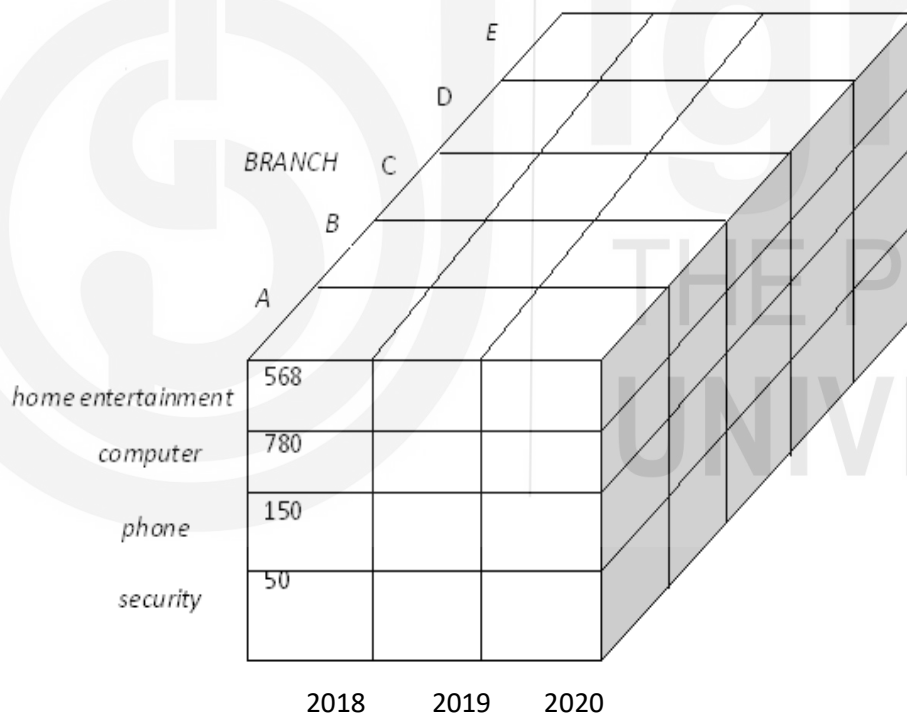
A cube is an apex cuboid in the highest degree of complexity.



3	350,000	0
4	586,000	

Year	Sales
2018	1,568,000
2019	2,356,000
2020	3,594,000

Sales are shown per quarter on the left. The details are aggregated for the annual sales on the right.



A data cube for sales

8.4.2 Attribute Subset Selection:

The selection of a sub-set attribute reduces data by removing redundant attributes or measurements irrelevant. The goal is to find a minimum set of attributes for the selection of a subset attribute. It decreases the number of attributes found in patterns, which promotes comprehension of patterns.

The data set will include a wide range of attributes. Some of them may be outdated or redundant, however. The objective of selecting attributes of sub-sets is to select a minimum number of attributes to lower data collection costs and to reduce loss of

such unwanted attributes. The reduced data collection also enables the discovered pattern to be explained.

Process of Attribute Subset Selection:

The brute force approach can be very expensive, in which data with n attributes can be evaluated by any subset (2^n possible subsets). The simplest way to do this is to use metrics of predictive significance in order to recognise the best or worst attributes. The statistical significance test indicates that the characteristics are separate. This is a greedy approach where the statistically ideal value of the meaning level is determined by 5% and the models are tested time and time again until the p value of all the attributes is less than or equal to the selected meaning level. Attributes with a p -value above the value are discarded. This method is repeated time and time again until all attributes of the data set have a p -value that is less than or equal to their importance. This allows us to obtain reduced data with no irrelevant attributes.

Attribute Subset Selection Methods:

- ❖ Stepwise Forward Selection
- ❖ Stepwise Backward Elimination
- ❖ Combination of Forward Selection and Backward Elimination
- ❖ Decision Tree Introduction

All of the above methods are greedy approaches to the selection of attribute subsets.

Stepwise Forward Selection:

The process starts with a minimum number of empty attributes. The most significant attributes with minimal p -value are selected and added to the minimum list. One attribute in each iteration is applied to the reduced collection.

Stepwise Backward Elimination:

All attributes are considered here in the initial collection of attributes. In each iteration, one attribute is omitted from the set of attributes of p significance.

Combination of Forward Selection and Backward Elimination:

Phase by step, sorting and reverse exclusion are combined so that the necessary attributes are selected more effectively. This is the most common way to collect attributes.

Decision Tree Introduction:

The introduction to the Decision Tree outlines a process flow diagram in which each internal node denotes an attribute test, every branch defines the possible test results and each leaf denotes the class forecast. For each node, the algorithm selects the best attribute to divide the data into different classes. From the data given. A tree is established. The tree contains a collection of attributes from the decreased sub-set attribute. The threshold measurement is used as a stop criterion.

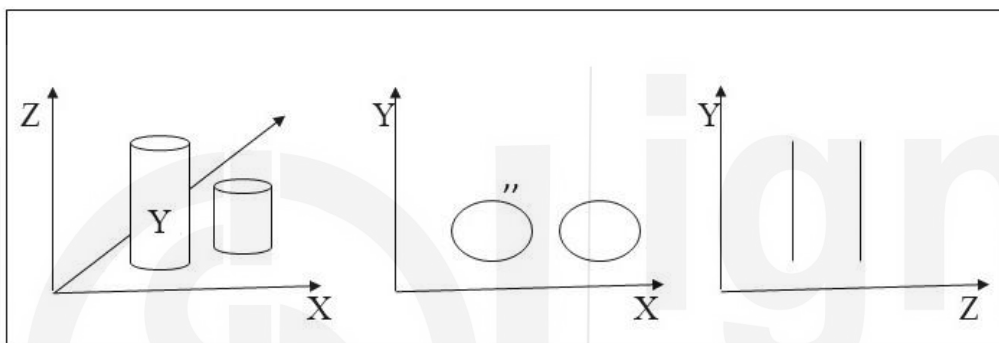
8.4.3 Dimensionality Reduction:

The final definition of machine learning problems is sometimes underlying too many factors. These factors are primarily variables known as features. The higher the number of features, the harder the training set can be imagined and worked on. Most

of these features are frequently connected and therefore redundant. This is where algorithms of dimensional reduction come into play. The reduction in dimension is the method by obtaining a collection of main variables for reducing the number of random variables. It can be divided into feature selection and feature extraction.

Why is reduction in dimensionality significant in Machine Learning:

An simple explanation of reduced dimensionality is given by a basic e-mail classification problem, where we have to classify whether mail is spam or not. There is a wide range of features, including whether the e-mail has a general title, e-mail content, whether an e-mail uses a template, etc. However, both of these functions will overlap. In other cases, a question of moisture and rainfall classification would fall into only one basic characteristic, as the above two correlates strongly. Therefore, in such a problem, we can reduce the number of features. It is difficult to see a problem in 3-D classification if a 2-D can be mapped into a basic 2-D area with the problem of 1-D in a simple diagram. The following figure shows that a 3-D function space is divided into two 1-D function spaces, and the number of characteristics can be further decreased if they are associated.



Dimensionality Reduction

Components of Dimensionality Reduction:

Two-dimensional reduction components are available:

- ❖ Feature Selection
- ❖ Feature Extraction

Feature Selection:

Here we try to fine-tune a subset of the original variables or features array to create a smaller subset that can be used for the query. It normally takes three directions:

- a) **Filter**
- b) **Wrapper**
- c) **Embedded**

Feature Extraction:

This limits data to a lower dimension in a high dimensional space, i.e. space with a limited number of dimensions.

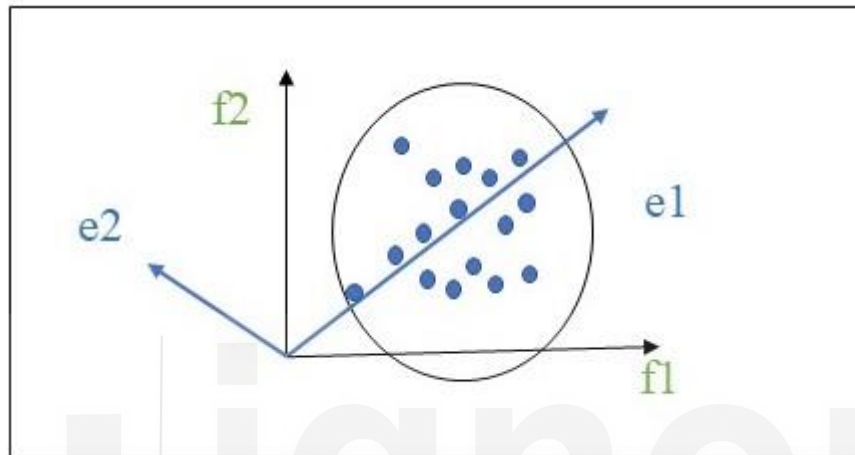
Dimensionality Reduction Methods:

- ❖ Principal Component Analysis (PCA)
- ❖ Linear Discriminant Analysis (LDA)
- ❖ Generalized Discriminant Analysis (GDA)

Dimensionality reduction may be linear or not linear depending on the method employed. Below is the primary linear method, referred to as the principal component analysis or PCA.

Principal Component Analysis (PCA):

Karl Pearson has established this method. Works under the condition that the data variance in the lower dimensional space is mapped to a data in a lower dimensional space.



This includes the following steps:

- ❖ Build a data covariance matrix
- ❖ Calculate the patented vectors in this matrix
- ❖ Self – vector equivalent to the largest own values are used to recreate a significant fraction of the variance of the original results.

We then have a reduced number of proprietary vectors, and the method may have lost some data. The big variances should be maintained and the remaining own vectors preserved.

Advantages and Disadvantages of Dimensionality Reduction:

- ❖ It helps to compact data and also removes disc space.
- ❖ It reduces the time of calculation
- ❖ It also helps, if any, to eliminate redundant features.
- ❖ This can lead to a certain amount of loss of data.
- ❖ PCA continues to recognize ongoing correlations between variables that are often unwanted.
- ❖ If mean and covariance are not sufficient to classify datasets, PCA fails.
- ❖ We cannot see how many of the primary components for maintaining such thumb laws are applied.

8.4.4 Numerosity Reduction:

It is a technique for data reduction that replaces the original information with a smaller kind of data representation. There are two ways to minimize numbers.

- ❖ Parametric Methods
- ❖ Non-Parametric Methods

Parametric Methods:

Data is interpreted by any parametric system model. The model is used to simulate data, which means that only data parameters should be processed rather than actual data. To construct these models, regression and log-linear approaches are used.

Regression:

Regression can be a simple linear regression or a multiple linear regression. If there is only one independent variable, the regression model is referred to as simple linear regression, and if there are more than one independent parameter, such models are called multiple linear regression.

The data is modelled on a straight line in linear regression. For example, the random variable y can be modelled as a linear function of another random variable x , with the equation $y = ax + b$ where the line slopes and y – intercepts are determined by a and b (regression coefficients). Y is modelled as a linear function in multiple linear regression, with two or more (independent) prediction variables.

Log-Linear Model:

For a series of discrete attributes based on a smaller subset of dimensional combinations, a log-linear model can be used to predict the probability of each data point in a multidimensional space. This allows the development of higher dimensional data space from lower dimensional attributes.

The regression as well as the log - linear model on sparse data can be used, but their use can be limited.

Non-Parametric Methods:

These techniques are used to store reduced data representations such as histograms, clusters, samples, and aggregation of data cubes.

Histograms:

Histogram is a frequency overview of the data. It uses binning to estimate the distribution of data that is a popular way of reducing data.

Clustering:

Clustering separates data into groups or clusters. This method divides all data into individual clusters. To keep specifics to a minimum. The representation of cluster data is used to bypass actual data. It also helps to locate outliers of data.

Sampling:

Sampling can be used for data reduction, as a wider range of data can be represented by a much smaller random sample (or subset).

Data cube Aggregation:

Data Cube Aggregation means data transfer to a reduced number of measurements from a complex stage. The resulting data collection is smaller without the information required for the study mission.

Parametric Model:

Regression:

The problem with regression is where the output variable is a real or permanent feature, like "wage" or "weight." There are several different approaches, the simplest

being linear regression. It tries to align the data with the right hyperplane travelling across the dots.

Regression Analysis:

A statistical way to estimate the interaction of dependent variables or criterion variables with one or more separate variables or predictors. The regression analysis explains adjustments for changes in the selected predictors in the parameters. Conditional predictor assumptions - parameters based on which the average value of the dependent variables is given when the individual variables are changed. Three main applications for the regression analysis are predictor strength, effect prediction and trend forecasting.

Types of Regression:

- ❖ Linear regression
- ❖ Logistic regression
- ❖ Polynomial regression
- ❖ Stepwise regression
- ❖ Ridge regression
- ❖ Lasso regression
- ❖ Elastic Net regression

Linear Regression:

This Linear Regression is used for statistical software. Linear regression is a linear approach for modelling an interaction with many predictors and explanatory variables between the criterion or scalar answer. Linear regression depends on the conditional distribution of the probability of the result given the predictor values. There is a chance that linear regression would overfit. The formulation of linear regression is:

$$y' = bX + A$$

Logistic Regression:

It shows that the dependent variable is assessed as a dichotomy. Logistic regression measures the parameters of the logistic model and defines a binomial regression. Logistic regression would be used to discuss data with two possible parameters and the association of predictor criteria. The equation of logistic regression is:

$$l = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Polynomial Regression:

It is used for curvilinear information. The least square method is ideal for polynomial regression. The object of regression analysis is to model the expected value of the dependent variable y in relation to the independent variable X. The equation of polynomial regression is:

$$l = \beta_0 + \beta_1 x_1 + \epsilon$$

Stepwise Regression:

It is used with predictive models to balance regression models. It is done automatically. The variable is added or removed from the explanatory variables collection for each point. The methods of progressive regression are forward collection, retroactive exclusion and bi-directional elimination. The step-by-step regression is:

$$b_{j.std} = b_j (S_x * S_y^{-1})$$

Ridge Regression:

It is a method for studying the effects of multiple regressions. When there is a multi-linearity, the least square figures are unbiased. A certain degree of bias is added to the regression equations, which removes standard errors. The method of ridge regression is:

$$\beta = (X^T X + \lambda * I)^{-1} X^T y$$

Lasso Regression:

An approach to regression analysis that performs both variable selection and regularisation. It uses soft boundaries. It only selects a subset of covariates to use the final construct. The equation of Lasso regression is:

$$N^{-1} \sum_{i=1}^N f(x_i, y_i, \alpha, \beta)$$

Elastic Regression:

It is a regularized approach to regression that integrates lasso and ridge penalties linearly. It supports the creation of vector machines, metrics and portfolio optimization. The penalty's job is as follows:

$$\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$$

Data Discretization and Concept Hierarchy Generation:

Data Discretization methods may be used by dividing the collection of attributes into intervals to minimize the number of values for a given continuous attribute. Interval marks can be used to bypass data values.

Data Discretization features:

- ❖ Leads to a concise
- ❖ Easy – to – use
- ❖ Knowledge – Level representation of mining results

Data Discretization Categories:

- ❖ **Supervised Discretization:** Uses details regarding the class
- ❖ **Unsupervised Discretization or Splitting:** Does not use class knowledge
- ❖ **Top-Down Discretization or Splitting:** Here, the method begins by first finding one or a few points called break points or cut points to separate the whole set of attributes and then performs this recursively over the subsequent intervals.
- ❖ **Bottom – up Discretization or merging:** Here, the method starts by treating all continuous quantities as intervals and then applies this process recursively at the following intervals.

Concept Hierarchy:

A hierarchy of the description defines a discretionary attribute to a certain numerical characteristic. It can be used to reduce data by collecting and replacing low-level concepts such as age-attribute numeric values with higher – low, middle and high levels concepts.

Discretization and concept Hierarchy Generation for Numerical data:

- ❖ Numerical attributes concept hierarchies may be created automatically based on data discretization.
- ❖ Management approaches for qualitative data over definition hierarchy.

- ❖ Binning
- ❖ Histogram Analysis
- ❖ Entropy – Based Discretization
- ❖ χ^2 – Merging
- ❖ Cluster Analysis
- ❖ Discretization by Intuitive Partitioning

Binning:

It's a cutting technique that concentrates on the number of bins listed. Session techniques are also used as arbitrary methods to reduce numbers and generate hierarchical descriptions. These methods can be used recursively to generate Hierarchical definitions on the resulting partitions. It does not use class information and is therefore an arbitrary technique that is not tracked. It is adaptable to the number of containers identified by the customer as outliers.

Histogram Analysis:

Like binning, histogram analysis is an unexpected discretization tool when class information is not used. Histograms partition the value for the disjoint range attribute A named buckets. The histogram analysis algorithm can be applied recursively to each partition to automatically produce a multi-level concept hierarchy that ends after a certain number of concept levels have been achieved.

Entropy – Based Discretization:

It is one of the most commonly used discretization measures. Entropical discretization is a manual top-down separation technique. It addresses the knowledge of class distribution in its assessment and determination of divisions. The system selects the A value in order to discern the numerical function, which has the lowest entropy as a divisive point and corrects the resulting intervals to achieve a hierarchical discernment. This discretization forms a meaning hierarchy for A. Let D consist of a list of attributes and a data tuples defined for the label-class attributes. The simple way to distinguish an A attribute in the entropy set is as follows:

1. Each value of A can be seen as a potential interval or dividing point to separate the A spectrum. In other words, a split point for A divides tuples into two sub-sets in D, corresponding with conditions $A < \text{split point}$ and $A > \text{split point}$ and produces a binary disc recitation.
2. Suppose we divide the tuples into D with A and some split-points. Ideally, the partitioning of this would lead to an exact tuple classification. For instance, if we had two classes, we expect to have all class C1 tuples on one partition and all class C2 tuples on the other.
3. The method for the description of points of division is applied recursively before such stop thresholds are reached, for example when minimum information is fulfilled. The criterion is less than a small one, ϵ or if the number of intervals is greater than a max interval should be required for all candidate split points.

Interval merging by χ^2 Analysis:

- ❖ Chi blends the fastest way to pick the finest.
- ❖ Nearby loops and then mix recurrently in larger intervals. The method is regulated using classes.
- ❖ The Chi merge form is as follows:
- ❖ Initially, each independent value of a numerical attribute A is considered as one interval.
- ❖ For each adjacent pair of intervals, C2 monitoring is performed.

- ❖ Next intervals are combined with the lowest c_2 values, since a couple of low c_2 values show that the distributions are similar.
- ❖ A predefined stop criterion follows the fusion phase recursively.

Cluster Analysis:

Cluster analysis is a traditional technique of discretion. The A values can be split into clusters or groups to determine the numerical A attribute by a clustering algorithm. Clustering can be used to construct a concept hierarchy for A either by a top-down division technique or by a top-down approach in the form of a hierarchy node in each cluster. Each initial cluster or partition can be further broken down into several subclusters that form a lower hierarchical level. Repeated groups of neighbouring clusters form clusters for the bottom-up approach to fusion.

Discretization by intuitive partitioning:

Although the above discretization methods aid in numerical hierarchy, many users want numerical ranges divided into a very uniform, intuitive and natural interpretation of cycles.

The 3-4-5 rule can be used for a fairly uniform, natural appearance of numerical results. The rules generally divide a given data set by level according to the most important digit range into 3, 4 or 5 periods.

The fact is this:

- ❖ When a set of 3,6,7 or 9 is divided into three intervals on the largest digit (3 equal-width intervals for 3,6 and 9; and 3 intervals in the grouping of 2-3-2 for 7)
- ❖ If it covers 2, 4 or 8 separate values at the maximum digit, split the spectrum into 4 periods of equivalent duration.
- ❖ Split the spectrum into five equal-width increments if the largest digit spans 1,5 or 10 separate values.
- ❖ To establish a definition hierarchy for the numerical attribute assigned, the rule can be extended recursively for each interval.
- ❖ Real-world knowledge also includes extremely large positive or negative outer values based on minimum and maximum data values, which can distort any top-down discretionary process.

Concept Hierarchy Generation for categorical data:

Discrete data are categorical data. There is a small (but probably large) amount of different values without an order among the category attribute values. Methods for generating categorical data classification hierarchy.

- ❖ Description of a partial scheme level attribute by customers or experts. Typical hierarchies or measurements are a list of attributes. By specifying a partial or full plan – the user or expert may easily describe a definition hierarchy in a level order of attributes.
- ❖ This classification is basically a manual explanation of the main hierarchy segment by correctly aggregating data for the part of a hierarchy.
- ❖ A complete hierarchy of the definition of a simple value list is difficult to describe in a broad database. Therefore, explicit groupings are practical in deciding on a small number of mid-level outcomes.
- ❖ The user can define a number of attributes that form a description hierarchy but do not specify its partial order. Structure should try to produce the attribute for a meaningful hierarchy of the meanings immediately.

UNIT 9 MINING FREQUENT PATTERNS AND ASSOCIATIONS

9.1 Association Rule Mining

9.1.1. Problem Definition:

9.2 Important Concepts Of Association Rule Mining

9.3 Market Basket Analysis

9.4 Frequent Pattern Mining

9.5 Efficient Frequent Itemset Mining Methods: Finding Frequent Item Sets Using Candidate Generation: The Apriori Algorithm

9.6 Generating Association Rules From Frequent Item-Sets

9.7 Mining Multilevel Association Rules

9.8 Approaches For Mining Multilevel Association Rules

9.9 Mining Multidimensional Association Rules From Relational Databases And Data Warehouses

9.10 Mining Quantitative Association Rules

9.11 From Association Mining To Correlation Analysis

9.1 ASSOCIATION RULE MINING

- Association rule mining is a popular and well researched method for discovering interesting relations between variables in large databases.
- It is intended to identify strong rules discovered in Databases using different measures of interestingness.
- Based on the concept of strong rules, Rakesh Aggarwal et al. introduced association rules.

9.1.1. Problem Definition

The problem of association rule mining is defined as:

Let $I = \{i_1, i_2, \dots, i_n\}$ be a set of n binary attributes called *items*.

Let $D = \{t_1, t_2, \dots, t_m\}$ be a set of transactions called the *database*.

Each transaction in D has a unique transaction ID and contains a subset of the items in I .

A *rule* is defined as an implication of the form $X \Rightarrow Y$

where $X, Y \subseteq I$ and $X \cap Y = \emptyset$.

The sets of items (for short *itemsets*) X and Y are called *antecedent* (left-hand-side or LHS) and *consequent* (right-hand-side or RHS) of the rule respectively.

Example:

To illustrate the concepts, we use a small example from the supermarket domain. The set of items is $I = \{\text{milk, bread, butter, beer}\}$ and a small database containing the items (1 codes presence and 0 absence of an item in a transaction) is shown in the table.

An example rule for the supermarket could be $\{\text{butter, bread}\} \Rightarrow \{\text{milk}\}$ meaning that if butter and bread are bought, customers also buy milk.

Example database with 4 items and 5 transactions

Transaction ID	milk	bread	butter	beer
1	1	1	0	0
2	0	0	1	0
3	0	0	0	1
4	1	1	1	0
5	0	1	0	0

9.2 IMPORTANT CONCEPTS OF ASSOCIATION RULE MINING

- The **support** $\text{supp}(X)$ of an item set X is defined as the proportion of transactions in the data set which contain the item set. In the example database, the item set $\{\text{milk, bread, butter}\}$ has a support of $1/5 = 0.2$ since it occurs in 20% of all transactions (1 out of 5 transactions).

- The **confidence** of a rule is defined

$$\text{conf}(X \Rightarrow Y) = \text{supp}(X \cup Y) / \text{supp}(X).$$

For example, the rule $\{\text{butter, bread}\} \Rightarrow \{\text{milk}\}$ has a confidence of $0.2/0.2 = 1.0$ in the database; which means that for 100% of the transactions containing butter and bread the rule is correct (100% of the times a customer buys butter and bread, milk is bought as well). Confidence can be interpreted as an estimate of the probability $P(Y|X)$, the probability of finding the RHS of the rule in transactions under the condition that these transactions also contain the LHS.

- The **lift** of a rule is defined as

$$\text{lift}(X \Rightarrow Y) = \frac{\text{supp}(X \cup Y)}{\text{supp}(X) \times \text{supp}(Y)}$$

or the ratio of the observed support to that expected if X and Y were independent. The rule $\{\text{milk, bread}\} \Rightarrow \{\text{butter}\}$ has a lift of

$$\frac{0.2}{0.4 \times 0.4} = 1.25.$$

The **conviction** of a rule is defined as

$$\text{conv}(X \Rightarrow Y) = \frac{1 - \text{supp}(Y)}{1 - \text{conf}(X \Rightarrow Y)}$$

The rule $\{\text{milk, bread}\} \Rightarrow \{\text{butter}\}$ has a conviction of $\frac{1 - 0.4}{1 - .5} = 1.2$, and can be interpreted as the ratio of the expected frequency that X occurs without Y (that is to say, the frequency that the rule makes an incorrect prediction) if X and Y were independent divided by the observed frequency of incorrect predictions.

9.3 MARKET BASKET ANALYSIS

This process analyzes customer buying habits by finding associations between the different items that customers place in their shopping baskets. The discovery of such associations can help retailers develop marketing strategies by gaining insight into which items are frequently purchased together by customers. For instance, if customers are buying milk, how likely are they to also buy bread (and what kind of bread) on the same trip to the supermarket. Such information can lead to increased sales by helping retailers do selective marketing and plan their shelf space.

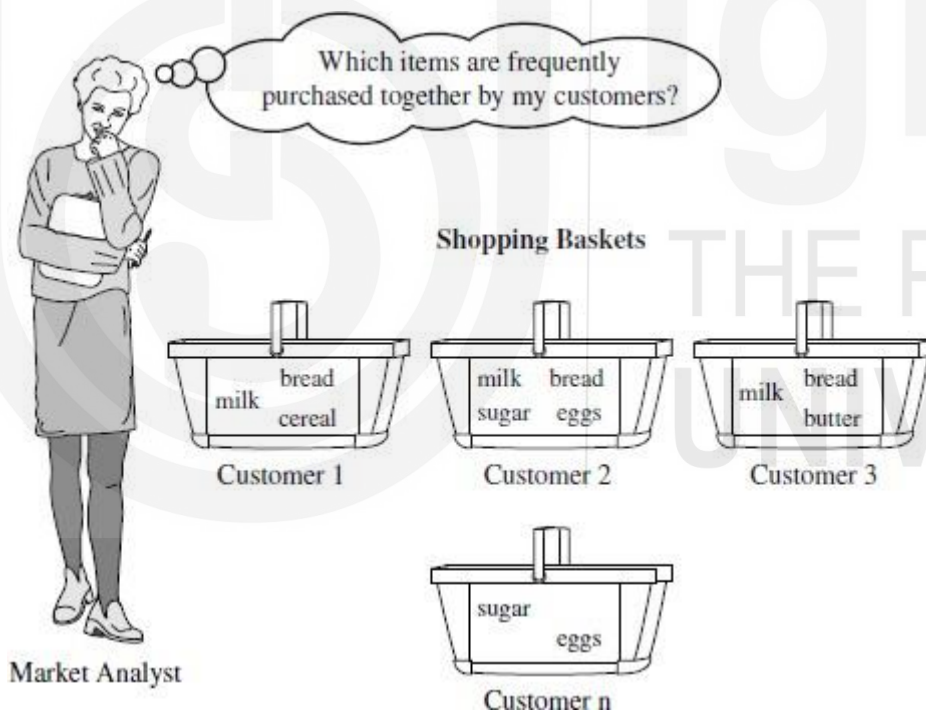


Fig. 9.1. Market Basket Analysis

Example:

If customers who purchase computers also tend to buy anti-virus software at the same time, then placing the hardware display close to the software display may help increase the sales of both items. In an alternative strategy, placing hardware and software at opposite ends of the store may entice customers who purchase such items to pick up other items along the way. For instance, after deciding on an expensive computer, a customer may observe security systems for sale while heading toward the software display to purchase antivirus software and may decide to purchase a home security system as well. Market basket analysis can also help retailers plan

which items to put on sale at reduced prices. If customers tend to purchase computers and printers together, then having a sale on printers may encourage the sale of printers *as well as* computers.

9.4 FREQUENT PATTERN MINING

Frequent pattern mining can be classified in various ways, based on the following criteria:

1. Based on the completeness of patterns to be mined:

- We can mine the complete set of frequent item sets, the closed frequent item sets, and the maximal frequent item sets, given a minimum support threshold.
- We can also mine constrained frequent item sets, approximate frequent item sets, near-match frequent item sets, top-k frequent item sets and so on.

2. Based on the levels of abstraction involved in the rule set:

Some methods for association rule mining can find rules at differing levels of abstraction.

For example, suppose that a set of association rules mined includes the following rules where X is a variable representing a customer:

$$\text{buys}(X, \text{-computer}\parallel) \Rightarrow \text{buys}(X, \text{-HP printer}\parallel) \quad (1)$$

$$\text{buys}(X, \text{-laptop computer}\parallel) \Rightarrow \text{buys}(X, \text{-HP printer}\parallel) \quad (2)$$

In rule (1) and (2), the items bought are referenced at different levels of abstraction (e.g.,

$\text{-computer}\parallel$ is a higher-level abstraction of $\text{-laptop computer}\parallel$).

3. Based on the number of data dimensions involved in the rule:

- If the items or attributes in an association rule reference only one dimension, then it is a single-dimensional association rule.
 $\text{buys}(X, \text{-computer}\parallel) \Rightarrow \text{buys}(X, \text{-antivirus software}\parallel)$
- If a rule references two or more dimensions, such as the dimensions age, income, and buys, then it is a multidimensional association rule. The following rule is an example of a multidimensional rule:
 $\text{age}(X, 30..39\parallel) \wedge \text{income}(X, 42K..48K\parallel) \Rightarrow \text{buys}(X, \text{-high resolution TV}\parallel)$

4. Based on the types of values handled in the rule:

- If a rule involves associations between the presence or absence of items, it is a Boolean association rule.
- If a rule describes associations between quantitative items or attributes, then it is a quantitative association rule.

5. Based on the kinds of rules to be mined:

- Frequent pattern analysis can generate various kinds of rules and other interesting relationships.
- Association rule mining can generate a large number of rules, many of which are redundant or do not indicate a correlation relationship among item sets.
- The discovered associations can be further analyzed to uncover statistical correlations, leading to correlation rules.

6. Based on the kinds of patterns to be mined:

- Many kinds of frequent patterns can be mined from different kinds of data sets.
- Sequential pattern mining searches for frequent subsequences in a sequence data set, where a sequence records an ordering of events.
- For example, with sequential pattern mining, we can study the order in which items are frequently purchased. For instance, customers may tend to first buy a PC, followed by a digital camera, and then a memory card.
- Structured pattern mining searches for frequent substructures in a structured data set.
- Single items are the simplest form of structure.
- Each element of an itemset may contain a subsequence, a sub tree, and so on.
- Therefore, structured pattern mining can be considered as the most general form of frequent pattern mining.

9.5 EFFICIENT FREQUENT ITEMSET MINING METHODS:

FINDING FREQUENT ITEM SETS USING CANDIDATE GENERATION: THE APRIORI ALGORITHM

- Apriori is a seminal algorithm proposed by R. Agrawal and R. Srikant in 1994 for mining frequent itemsets for Boolean association rules.
- The name of the algorithm is based on the fact that the algorithm uses *prior knowledge* of frequent itemset properties.
- Apriori employs an iterative approach known as a *level-wise* search, where k -item sets are used to explore $(k+1)$ -itemsets.
- First, the set of frequent 1-itemsets is found by scanning the database to accumulate the count for each item, and collecting those items that satisfy minimum support. The resulting set is denoted L_1 . Next, L_1 is used to find L_2 , the set of frequent 2-itemsets, which is used to find L_3 , and so on, until no more frequent k -item sets can be found.

- The finding of each L_k requires one full scan of the database.
- A two-step process is followed in Apriori consisting of join and prune action.

Algorithm: Apriori. Find frequent itemsets using an iterative level-wise approach based on candidate generation.

Input:

- D , a database of transactions;
- min_sup , the minimum support count threshold.

Output: L , frequent itemsets in D .

Method:

```

(1)  $L_1 = \text{find\_frequent\_1-itemsets}(D)$ ;
(2) for  $(k = 2; L_{k-1} \neq \phi; k++)$  {
(3)    $C_k = \text{apriori\_gen}(L_{k-1})$ ;
(4)   for each transaction  $t \in D$  { // scan  $D$  for counts
(5)      $C_t = \text{subset}(C_k, t)$ ; // get the subsets of  $t$  that are candidates
(6)     for each candidate  $c \in C_t$ 
(7)        $c.\text{count}++$ ;
(8)   }
(9)    $L_k = \{c \in C_k | c.\text{count} \geq min\_sup\}$ 
(10) }
(11) return  $L = \cup_k L_k$ ;

procedure apriori_gen( $L_{k-1}$ :frequent  $(k-1)$ -itemsets)
(1) for each itemset  $l_1 \in L_{k-1}$ 
(2)   for each itemset  $l_2 \in L_{k-1}$ 
(3)     if  $(l_1[1] = l_2[1]) \wedge (l_1[2] = l_2[2]) \wedge \dots \wedge (l_1[k-2] = l_2[k-2]) \wedge (l_1[k-1] < l_2[k-1])$  then {
(4)        $c = l_1 \bowtie l_2$ ; // join step: generate candidates
(5)       if has_infrequent_subset( $c, L_{k-1}$ ) then
(6)         delete  $c$ ; // prune step: remove unfruitful candidate
(7)       else add  $c$  to  $C_k$ ;
(8)     }
(9) return  $C_k$ ;

procedure has_infrequent_subset( $c$ : candidate  $k$ -itemset;
                                 $L_{k-1}$ : frequent  $(k-1)$ -itemsets); // use prior knowledge
(1) for each  $(k-1)$ -subset  $s$  of  $c$ 
(2)   if  $s \notin L_{k-1}$  then
(3)     return TRUE;
(4) return FALSE;

```

Fig. 9.2. Pseudo-code for Apriori Algorithm

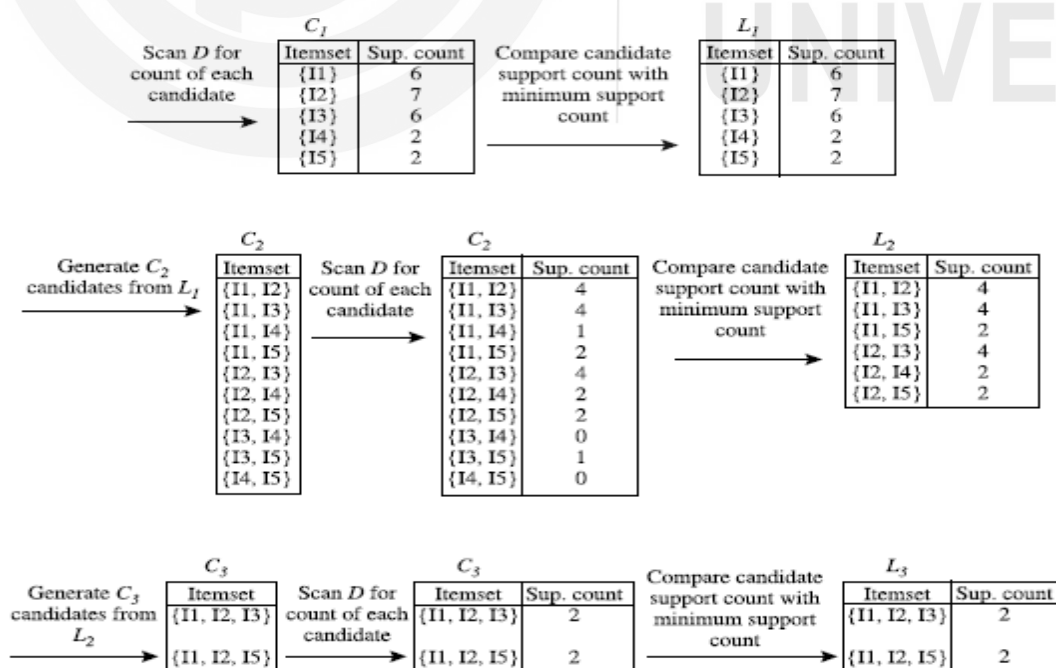
Example:

TID	List of item IDs
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

There are nine transactions in this database, that is, $|D| = 9$.

Steps:

1. In the first iteration of the algorithm, each item is a member of the set of candidate1- itemsets, C1. The algorithm simply scans all of the transactions in order to count the number of occurrences of each item.
2. Suppose that the minimum support count required is 2, that is, $\min \text{sup} = 2$. The set of frequent 1-itemsets, L1, can then be determined. It consists of the candidate 1-itemsets satisfying minimum support. In our example, all of the candidates in C1 satisfy minimum support.
3. To discover the set of frequent 2-itemsets, L2, the algorithm uses the join L1 on L1 to generate a candidate set of 2-itemsets, C2. No candidates are removed from C2 during the prune step because each subset of the candidates is also frequent.
4. Next, the transactions in D are scanned and the support count of each candidate itemset in C2 is accumulated.
5. The set of frequent 2-itemsets, L2, is then determined, consisting of those candidate2- item sets in C2 having minimum support.
6. The generation of the set of candidate 3-itemsets, C3, From the join step, we first get $C3 = L2 \times L2 = (\{I1, I2, I3\}, \{I1, I2, I5\}, \{I1, I3, I5\}, \{I2, I3, I4\}, \{I2, I3, I5\}, \{I2, I4, I5\})$. Based on the Apriori property that all subsets of a frequent item-set must also be frequent, we can determine that the four latter candidates cannot possibly be frequent.
7. The transactions in D are scanned in order to determine L3, consisting of those candidate3-itemsets in C3 having minimum support.
8. The algorithm uses $L3 \times L3$ to generate a candidate set of 4-itemsets, C4.



Generation of candidate itemsets and frequent itemsets, where the minimum support count is 2.

- (a) Join: $C_3 = L_2 \bowtie L_2 = \{\{I1, I2\}, \{I1, I3\}, \{I1, I5\}, \{I2, I3\}, \{I2, I4\}, \{I2, I5\}\} \bowtie \{\{I1, I2\}, \{I1, I3\}, \{I1, I5\}, \{I2, I3\}, \{I2, I4\}, \{I2, I5\}\}$
 $= \{\{I1, I2, I3\}, \{I1, I2, I5\}, \{I1, I3, I5\}, \{I2, I3, I4\}, \{I2, I3, I5\}, \{I2, I4, I5\}\}.$
- (b) Prune using the Apriori property: All nonempty subsets of a frequent itemset must also be frequent. Do any of the candidates have a subset that is not frequent?
- The 2-item subsets of $\{I1, I2, I3\}$ are $\{I1, I2\}$, $\{I1, I3\}$, and $\{I2, I3\}$. All 2-item subsets of $\{I1, I2, I3\}$ are members of L_2 . Therefore, keep $\{I1, I2, I3\}$ in C_3 .
 - The 2-item subsets of $\{I1, I2, I5\}$ are $\{I1, I2\}$, $\{I1, I5\}$, and $\{I2, I5\}$. All 2-item subsets of $\{I1, I2, I5\}$ are members of L_2 . Therefore, keep $\{I1, I2, I5\}$ in C_3 .
 - The 2-item subsets of $\{I1, I3, I5\}$ are $\{I1, I3\}$, $\{I1, I5\}$, and $\{I3, I5\}$. $\{I3, I5\}$ is not a member of L_2 , and so it is not frequent. Therefore, remove $\{I1, I3, I5\}$ from C_3 .
 - The 2-item subsets of $\{I2, I3, I4\}$ are $\{I2, I3\}$, $\{I2, I4\}$, and $\{I3, I4\}$. $\{I3, I4\}$ is not a member of L_2 , and so it is not frequent. Therefore, remove $\{I2, I3, I4\}$ from C_3 .
 - The 2-item subsets of $\{I2, I3, I5\}$ are $\{I2, I3\}$, $\{I2, I5\}$, and $\{I3, I5\}$. $\{I3, I5\}$ is not a member of L_2 , and so it is not frequent. Therefore, remove $\{I2, I3, I5\}$ from C_3 .
 - The 2-item subsets of $\{I2, I4, I5\}$ are $\{I2, I4\}$, $\{I2, I5\}$, and $\{I4, I5\}$. $\{I4, I5\}$ is not a member of L_2 , and so it is not frequent. Therefore, remove $\{I2, I4, I5\}$ from C_3 .
- (c) Therefore, $C_3 = \{\{I1, I2, I3\}, \{I1, I2, I5\}\}$ after pruning.

Generation and pruning of candidate 3-itemsets, C_3 , from L_2 using the Apriori property.

Fig. 9.3. Working of Apriori Algorithm

9.6 GENERATING ASSOCIATION RULES FROM FREQUENT ITEM-SETS

Once the frequent item-sets from transactions in a database D have been found, it is straightforward to generate strong association rules from them.

$$\text{confidence}(A \Rightarrow B) = P(B|A) = \frac{\text{support_count}(A \cup B)}{\text{support_count}(A)}.$$

The conditional probability is expressed in terms of itemset support count, where $\text{support_count}(A \cup B)$ is the number of transactions containing the itemsets $A \cup B$, and $\text{support_count}(A)$ is the number of transactions containing the itemset A . Based on this equation, association rules can be generated as follows:

- For each frequent itemset l , generate all nonempty subsets of l .
- For every nonempty subset s of l , output the rule " $s \Rightarrow (l - s)$ " if $\frac{\text{support_count}(l)}{\text{support_count}(s)} \geq \text{min_conf}$, where min_conf is the minimum confidence threshold.

Example:

Generating association rules. Let's try an example based on the transactional data for *AllElectronics* shown in Table 5.1. Suppose the data contain the frequent itemset $l = \{I1, I2, I5\}$. What are the association rules that can be generated from l ? The nonempty subsets of l are $\{I1, I2\}$, $\{I1, I5\}$, $\{I2, I5\}$, $\{I1\}$, $\{I2\}$, and $\{I5\}$. The resulting association rules are as shown below, each listed with its confidence:

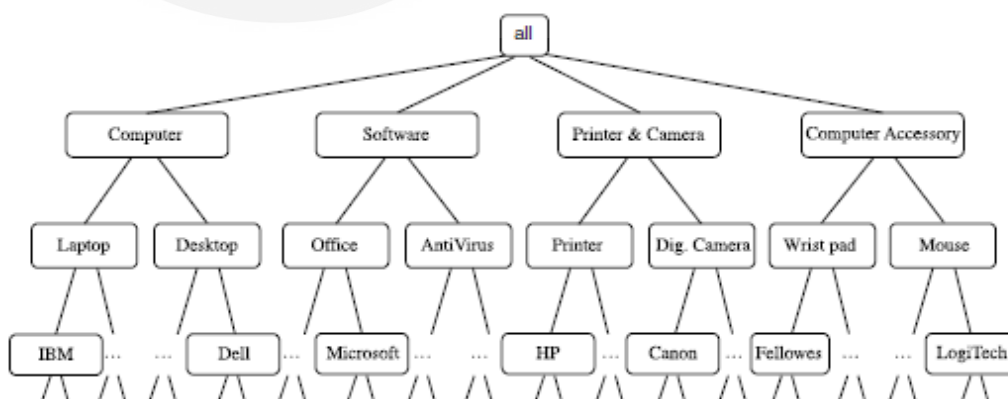
$I1 \wedge I2 \Rightarrow I5,$	$\text{confidence} = 2/4 = 50\%$
$I1 \wedge I5 \Rightarrow I2,$	$\text{confidence} = 2/2 = 100\%$
$I2 \wedge I5 \Rightarrow I1,$	$\text{confidence} = 2/2 = 100\%$
$I1 \Rightarrow I2 \wedge I5,$	$\text{confidence} = 2/6 = 33\%$
$I2 \Rightarrow I1 \wedge I5,$	$\text{confidence} = 2/7 = 29\%$
$I5 \Rightarrow I1 \wedge I2,$	$\text{confidence} = 2/2 = 100\%$

9.7 MINING MULTILEVEL ASSOCIATION RULES

- For many applications, it is difficult to find strong associations among data items at lower primitive levels of abstraction due to the sparsity of data at those levels.
- Strong associations discovered at high levels of abstraction may represent commonsenseknowledge.
- Therefore, data mining systems should provide capabilities for mining association rulesat multiple levels of abstraction, with sufficient flexibility for easy traversal among different abstraction spaces.
- Association rules generated from mining data at multiple levels of abstraction are called multiple-level or multilevel association rules.
- Multilevel association rules can be mined efficiently using concept hierarchies under a support-confidence framework.
- In general, a top-down strategy is employed, where counts are accumulated for the calculation of frequent itemsets at each concept level, starting at the concept level 1 and working downward in the hierarchy toward the more specific concept levels, until no more frequent itemsets can be found.

A concept hierarchy defines a sequence of mappings from a set of low-level concepts to higher level, more general concepts. Data can be generalized by replacing low-level concepts within the data by their higher-level concepts, or ancestors, from a concept hierarchy.

TID	Items Purchased
T100	IBM-ThinkPad-T40/2373, HP-Photosmart-7660
T200	Microsoft-Office-Professional-2003, Microsoft-Plus!-Digital-Media
T300	Logitech-MX700-Cordless-Mouse, Fellowes-Wrist-Rest
T400	Dell-Dimension-XPS, Canon-PowerShot-S400
T500	IBM-ThinkPad-R40/P4M, Symantec-Norton-Antivirus-2003
...	...



A concept hierarchy for AllElectronics computer items.

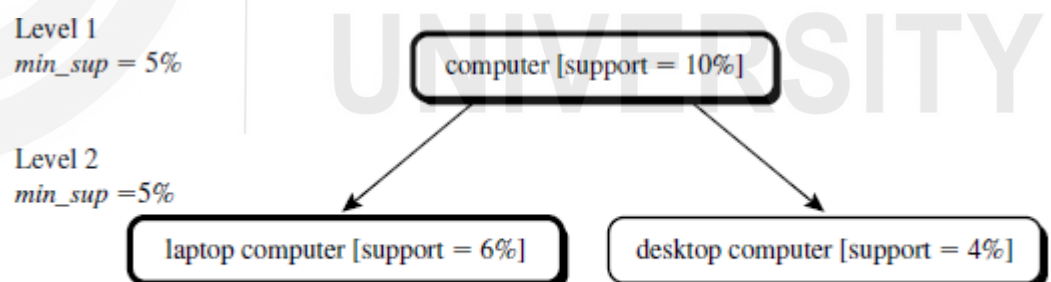
The concept hierarchy has five levels, respectively referred to as levels 0 to 4, starting with level 0 at the root node for all.

- Here, Level 1 includes computer, software, printer & camera, and computer accessory.
- Level 2 includes laptop computer, desktop computer, office software, antivirus software
- Level 3 includes IBM desktop computer. . . Microsoft office software and so on.
- Level 4 is the most specific abstraction level of this hierarchy.

9.8 APPROACHES FOR MINING MULTILEVEL ASSOCIATION RULES

1. Uniform Minimum Support:

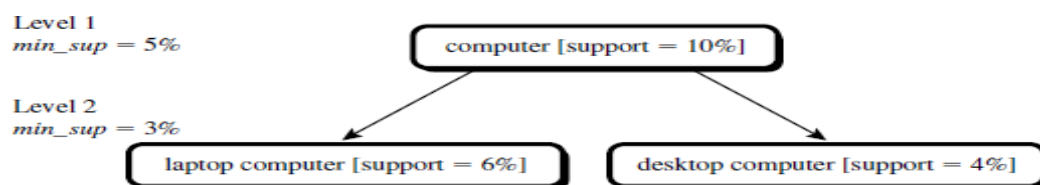
- The same minimum support threshold is used when mining at each level of abstraction.
- When a uniform minimum support threshold is used, the search procedure is simplified.
- The method is also simple in that users are required to specify only one minimum support threshold.
- The uniform support approach, however, has some difficulties. It is unlikely that items at lower levels of abstraction will occur as frequently as those at higher levels of abstraction.
- If the minimum support threshold is set too high, it could miss some meaningful associations occurring at low abstraction levels. If the threshold is set too low, it may generate many uninteresting associations occurring at high abstraction levels.



Multilevel mining with uniform support.

2. Reduced Minimum Support

- Each level of abstraction has its own minimum support threshold.
- The deeper the level of abstraction, the smaller the corresponding threshold is.
- For example, the minimum support thresholds for levels 1 and 2 are 5% and 3%, respectively. In this way, computer, laptop computer, and desktop computer are all considered frequent.



3. Group-Based Minimum Support:

- Because users or experts often have insight as to which groups are more important than others, it is sometimes more desirable to set up user-specific, item, or group based minimal support thresholds when mining multilevel rules.
- For example, a user could set up the minimum support thresholds based on product price, or on items of interest, such as by setting particularly low support thresholds for laptop computers and flash drives in order to pay particular attention to the association patterns containing items in these categories.

9.9 MINING MULTIDIMENSIONAL ASSOCIATION RULES FROM RELATIONAL DATABASES AND DATA WAREHOUSES

- Single dimensional or intra dimensional association rule contains a single distinct predicate (e.g., buys) with multiple occurrences i.e., the predicate occurs more than once within the rule.
 $\text{buys}(X, \text{-digital camera}\parallel) \Rightarrow \text{buys}(X, \text{-HP printer}\parallel)$
- Association rules that involve two or more dimensions or predicates can be referred to as multidimensional association rules.
 $\text{age}(X, \text{"20...29"}) \wedge \text{occupation}(X, \text{"student"}) \Rightarrow \text{buys}(X, \text{"laptop"})$
- Above Rule contains three predicates (age, occupation, and buys), each of which occurs only once in the rule. Hence, we say that it has no repeated predicates.
- Multidimensional association rules with no repeated predicates are called inter dimensional association rules.
- We can also mine multidimensional association rules with repeated predicates, which contain multiple occurrences of some predicates. These rules are called hybrid-dimensional association rules. An example of such a rule is the following, where the predicate buys is repeated:
 $\text{age}(X, \text{-20...29}\parallel) \wedge \text{buys}(X, \text{-laptop}\parallel) \Rightarrow \text{buys}(X, \text{-HP printer}\parallel)$

9.10 MINING QUANTITATIVE ASSOCIATION RULES

- Quantitative association rules are multidimensional association rules in which the numeric attributes are *dynamically* discretized during the mining process so as to satisfy some mining criteria, such as maximizing the confidence or compactness of the rules mined.

- In this section, we focus specifically on how to mine quantitative association rules having two quantitative attributes on the left-hand side of the rule and one categorical attribute on the right-hand side of the rule. That is

$A_{quan1} \wedge A_{quan2} \Rightarrow A_{cat}$

Where A_{quan1} and A_{quan2} are tests on quantitative attribute interval

A_{cat} tests a categorical attribute from the task-relevant data.

- Such rules have been referred to as two-dimensional quantitative association rules, because they contain two quantitative dimensions.
- For instance, suppose you are curious about the association relationship between pairs of quantitative attributes, like customer age and income, and the type of television (such as *high-definition TV*, i.e., *HDTV*) that customers like to buy.

An example of such a 2-D quantitative association rule is

$age(X, -30 \dots 39) \wedge income(X, -42K \dots 48K) \Rightarrow buys(X, -HDTV)$

9.11 FROM ASSOCIATION MINING TO CORRELATION ANALYSIS

- A correlation measure can be used to augment the support-confidence framework for association rules. This leads to *correlation rules* of the form
 $A \Rightarrow B [support, confidence, correlation]$
- That is, a correlation rule is measured not only by its support and confidence but also by the correlation between itemsets A and B . There are many different correlation measures from which to choose. In this section, we study various correlation measures to determine which would be good for mining large data sets.
- Lift is a simple correlation measure that is given as follows. The occurrence of itemset A is independent of the occurrence of itemset B if $P(A \cup B) = P(A)P(B)$; otherwise, itemsets A and B are dependent and correlated as events. This definition can easily be extended to more than two itemsets.

The lift between the occurrence of A and B can be measured by computing

$$lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)}.$$

- If the $lift(A, B)$ is less than 1, then the occurrence of A is negatively correlated with the occurrence of B .
- If the resulting value is greater than 1, then A and B are positively correlated, meaning that the occurrence of one implies the occurrence of the other.
- If the resulting value is equal to 1, then A and B are independent and there is no correlation between them.

UNIT 10 CLASSIFICATION

- 10.0 Introduction
 - 10.1 Definition Of Classification
 - 10.2 General Approaches To Solving A Classification Problem
 - 10.3 Evaluation Of Classifiers
 - 10.4 Classification Techniques
 - 10.5 Decision Trees – Decision Tree Construction
 - 10.6 Methods For Expressing Attribute Test Conditions
-

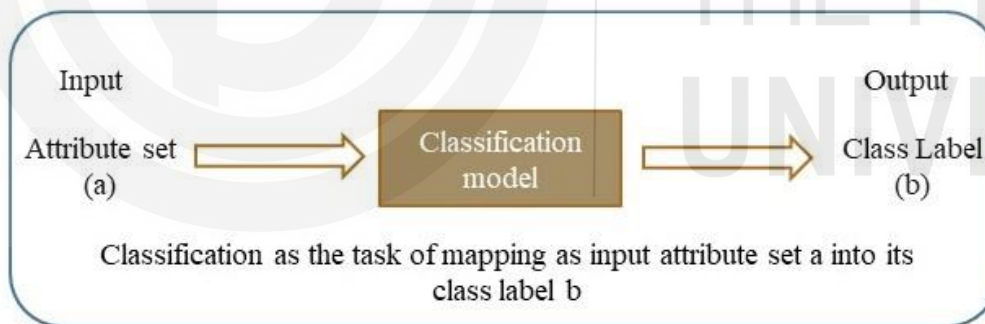
10.0 INTRODUCTION

Data mining:

Classification is a single step in the data mining process. It is used for organizing objects based on some key features. Several approaches such as the nearest neighbor classification, Decision Tree Learning and Support Vector Machines are employed for data mining classification.

10.1 DEFINITION OF CLASSIFICATION

A classification problem requires that examples be classified into one of two or more classes. A classification can include reliable or discrete input variables. A two-class dilemma is also regarded as a challenge between two classes or binary classification. A multi-class grouping problem is often called a two-group problem. Classification is the activity of learning a target function "c," which maps each attribute to one of the class labels "b".



The attribute set 'a' can contain a number of attributes and can contain binary, categorical and continuous attributes. The class 'b' mark must have a distinct, i.e. binary or categorical attribute (nominal or ordinal).

Classification Models:

Descriptive Modeling:

It is a model of classification used to summarize data.

Predictive Modeling:

It is a classification model used to predict the class mark of unknown data.

Applications of Classification Models:

- ❖ Spam detection e-mails based on the header and content of the document.
- ❖ Classification of galaxies according to their types.
- ❖ Classification of students according to their qualifications.
- ❖ Patients are classified according to their medical history.
- ❖ Classification can be used for the approval of credit.

10.2 GENERAL APPROACHES TO SOLVING A CLASSIFICATION PROBLEM

- ❖ A classification technique is a systematic way to construct classification models based on data collection.
- ❖ Examples include decision tree classification machines, rule-based classification machines, neural networks, vector support machines and Bayes naive classifiers.
- ❖ Each technique uses the learning algorithm to find a template that best matches the relationship between the attribute set and the class mark of the data input.
- ❖ A training array consists of records that identify class marks. The training test is used for the creation of a classification model for the test set. The test collection consists of records with an unspecified class mark
- ❖ The estimation of the results of the classification model would be based on the number of test records correctly and incorrectly estimated by the model.
- ❖ These numbers are presented in a table called an uncertainty matrix.

10.3 EVALUATION OF CLASSIFIERS

It's probably wrong if it's easy.

You could have attacked a few data problems if you are new to a data science course or try to gather the bases yourself. You also played with various classifiers, feature sets, and perhaps different sets of parameters, etc. Finally, you can think about the evaluation: how well does your classification work?

I presume that you already know the basic evaluation methodology: you do not know how to assess a training set, maybe cross validation, and maybe even trust bars or even t-tests. It's Data Science 101, which you already know right now. Ok, that's not what I'm talking about. I don't talk about that. I don't talk about that. What I'm talking about is something more simple but harder: how do you calculate your classifier's performance?

The obvious response is to use accuracy: accurate classification of the number of instances. You have a classifier which takes test examples and takes classes for each class. His guess in any test example is either correct or wrong. You basically calculate how many right choices your classification makes, the difference between the number and accuracy of your classification is the total number of test examples. It's just so easy. Most findings of study are accurate and many practical ventures accurate. That's the default metric. What's the fuck with that? Precision is a simple measure that misleads several issues in the real world. Indeed, "but it succeeded in the laboratory, why does it not work in the field?" experience is best achieved by using accuracy as an evaluation metric.

What's wrong with accuracy?

Let us begin with a single two-class domain to illustrate the issues. Each classifier sees examples for this domain from the two classes and offers one of two possible rulings: Y or N. Every decision can be made with a test set and a particular classifier:

1. a positive example that has been graded as positive. This is a genuinely positive development.
2. A good example that is misclassified as negative. That's a wrong negative.
3. a negative case that was labelled as negative. That's a real negative.
4. A bad example is misclassified as good. That's a false positive thing.

A two-to-two matrix can be generated with the columns as true classes and the rows as hypothesized classes. It looks as follows:

"confusion matrix"

or

"contingency table"

		Actual	
		+	-
Predicted	Y	True positives	False positives
	N	False negatives	True negatives

Entries are counts of correct classifications and counts of errors

Each case contributes exactly to the count in a single cell. This is your approach to your classification.

Precision is just the diagonal number divided by the total:

$$\text{Accuracy} = \frac{\text{True positives} + \text{True negatives}}{\text{True positives} + \text{False positives} + \text{True negatives} + \text{False negatives}}$$

What is wrong with it?

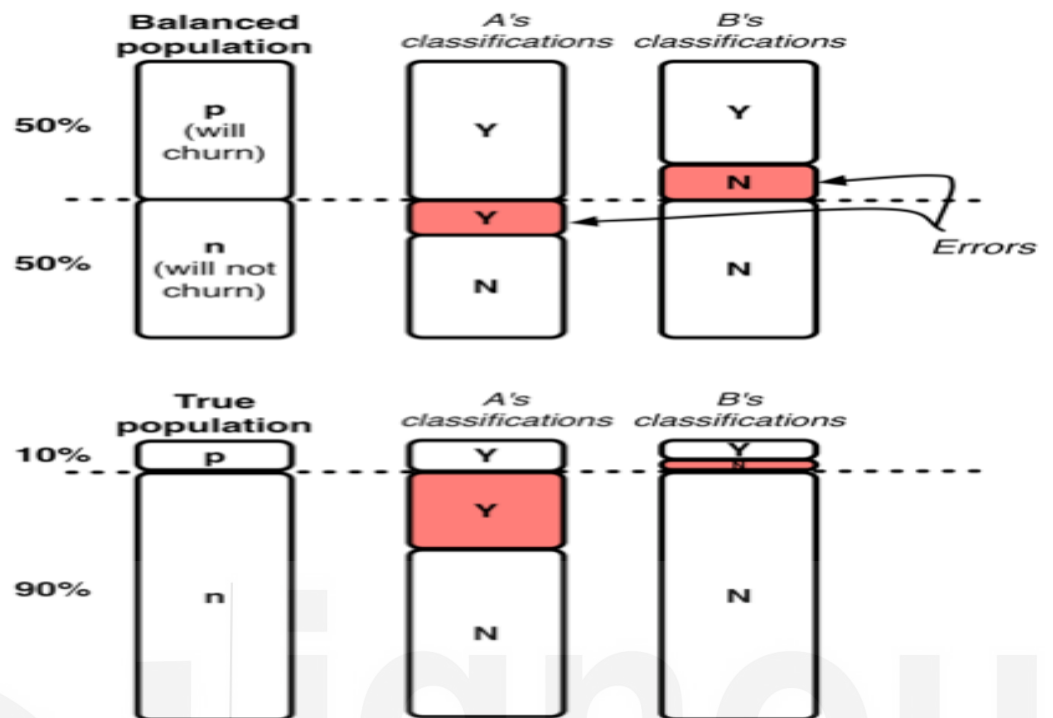
The issue of class imbalance:

Accuracy only includes all examples and records a percentage of correct answers. Accuracy can be great when working with balanced (or roughly balanced) datasets. The higher you get from 50/50, the greater the precision. Consider a 99:1 divided negative to positive dataset. Only guessing that the majority class yields a 99% correct description!

Unbalanced realms are the law in the real world, not the exception! I have never handled a balanced dataset outside academia. The explanation is very simple. Machine learning is also used in the processing of populations consisting of a large number of negative (uninteresting) examples and a small number of positive examples (interesting, alarming).

For eg, fraud detection (1% is an immense problem; 0.1% is closer), screening for HIV (the incidence in the USA is around ~0.4%), prediction of disc drive failure (~1%

a year), click-through response rates (0.09%), development error rates (0.1%) - the list goes a long way. Unbalance is anywhere. Everywhere.



Why is this difficult assessment? An example is shown in the left diagram. We want to see which customers will probably churn (cancel their contracts) soon.

The top half reveals the trainers. The population has been balanced, so half is positive and half is negative. Two classifiers, A and B, have been qualified. They each make the same error percentage, so their accuracy in this particular data set is the same as seen in the top half by the red areas.

In the lower half the same two classifiers appear in a test setting, where the population is now 10% (will churn) to 90% (will not churn). That's far more practical. Notice that A is even worse now because its mistakes are false positives.

Two lessons are drawn from this example. One is that you need to know more than what accuracy tells you about your classifier. The other is that, regardless of the population in which you train, you still have to test the population you care for.

The question of error costs:

A second issue remains. Take the realm of click-through advertisement replies. The estimated click-through rate (previous class) is around 0.09 percent. That means that a classifier saying, "No, the consumer won't press," has a 99.91% accuracy. What is wrong with this? What is wrong with it? Nothing, strictly speaking! This is excellent performance—if you cared about precision.

The problem is that certain classes are far more important than others. In this scenario, you think much more for those who press 0.09% than about the 99.91% who don't. More specifically, it is not a big mistake to show an ad to certain people who don't click on it (a False Positive in the matrix above), but it isn't bad for people who want to click on an ad to not show it (a False Negative) is much worse. After all, aiming at the right people for the ad is the whole reason that you create a classifier first.

In several domains, different error costs are normal. In medical diagnoses, where there is no cancer, the expense is different from the lack of a true event. The cost of finding an insignificant document in document classification is different from the cost of losing an important document, and so on.

What about collections of public data?

You may have found that you've been working with databases from somewhere like the UCI Dataset Repository that do not seem to have these problems. The classes are always reasonably balanced, and it seems accurate and unproblematic when error costs are given. While these data sets may have originated with real issues, they have been artificially cleaned, balanced and provided with accurate error costs in many cases. Contest managers must be able to judge and announce a definite winner, so the judging criteria are squarely defined beforehand.

Other measurements

At this point, you are certain that the precision of plain classification is a poor measurement for real-world domains. Instead, what do you use? Many other assessment criteria were created. It is important to note that and way of synthesizing the confusion matrix is simply different.

		Actual	
		+	-
Predicted	Y	True positives	False positives
	N	False negatives	True negatives

Here are some metrics that you can probably find:

- **True positive rate** = $TP/(TP+FN) = 1 - \text{false negative rate}$
- **False positive rate** = $FP/(FP+TN) = 1 - \text{true negative rate}$
- **sensitivity** = true positive rate
- **specificity** = true negative rate
- **positive predictive value** = $TP/(TP+FP)$
- **recall** = $TP / (TP+FN) = \text{true positive rate}$
- **precision** = $TP / (TP+FP)$

F-score is the harmonic mean of precision and recall:

$$F = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

G-score is the geometric mean of precision and recall:

$$G = \sqrt{\text{precision} \cdot \text{recall}}$$

Two other steps are worth noting, but their precise meanings are too complex to clarify. Cohen's Kappa is an agreement metric that takes into account the amount of agreement that chance can expect. It does this by taking into consideration class imbalances and the propensity of the classifier to vote yes or no. The White Man U test is a general measure of the classifier's separation ability. It tests how well the classifier can classify positive instances over negative cases.

Expected cost:

Which is the right number to calculate, given all these numbers? We will avoid the usual statistical response ("It depends"), and reply: none of the above-mentioned.

The uncertainty matrix involves frequencies of the four results. The most accurate solution to the problem is to measure the estimated costs by using these four numbers (or equivalently, expected benefit). You should note that the normal form of an anticipated value is to take and calculate the likelihood of each circumstance by its value. We translate the numbers in the above matrix to cost and gain for each of the four. The number is the final calculation:

Planned costs = $p(p)$ for cost(p) + $p(n)$ for cost(s) (n)

Here $p(p)$ and $p(n)$ are each class's probabilities, also known as the class priors. Cost(p) is shorthand for the cost of a positive example. What is this? What is this? We need to divide it into two instances: if we get it right (a real good one), and if we go wrong (a false negative).

$\text{cost}(p) = p(\text{true positive}) \text{value}(\text{true positive}) + p(\text{false negative}) \text{value added cost}(\text{false negative})$

Likewise, for the negative class: it is a true negative, if we have it correctly, otherwise it is a false positive. When we extend the entire cost equation, we get the messy (but hopefully comprehensible) equation:

Expected cost = $p(p) \times [p(\text{true positive}) \times \text{benefit}(\text{true positive})$

$+ p(\text{false negative}) \times \text{cost}(\text{false negative})]$

$+ p(n) \times [p(\text{true negative}) \times \text{benefit}(\text{true negative})$

$+ p(\text{false positive}) \times \text{cost}(\text{false positive})]$

This is much more accurate than precision, since it divides each of the four options for any choice and multiplies it by the costs or benefits associated with that decision. You can estimate the priors $p(p)$ and $p(n)$ directly from your results. The four $p(\text{true positive})$ expressions you obtain from your classifier; specifically, from the confusion matrix which comes from a test sample classifier. The value cost () and advantage () are extrinsic values which cannot be extracted from data. They quantify the actual effects of the decisions of the classifier. These must be calculated on the basis of expert domain information.

Ideally, you can test a classifier if you want its output to be reduced to a single number. If you have a set of classificatory and want to select the best, this method allows you to precisely measure the performance of each classifier.

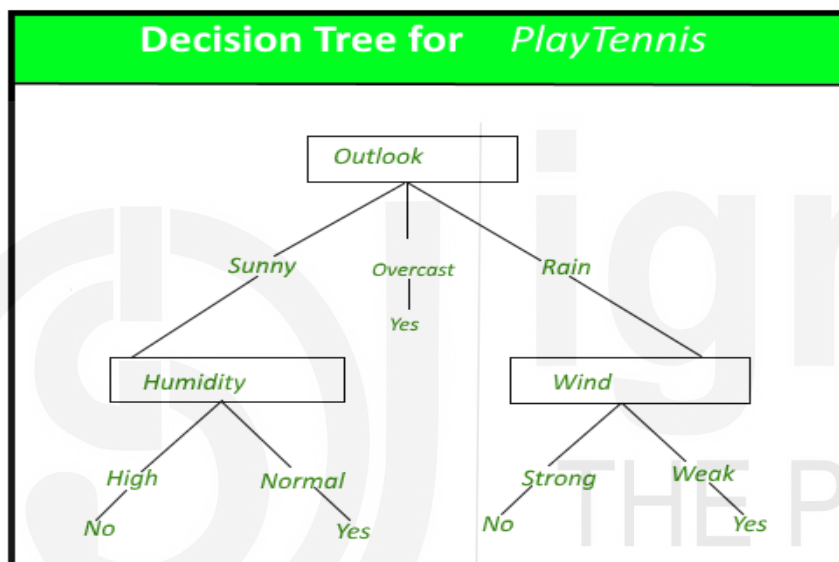
10.4 CLASSIFICATION TECHNIQUES

Classification is a technique used to predict group membership for data cases through data mining (machine learning). There are many methods of classification that can be used for classification purposes. We present the basic classification techniques in this chapter. Later on, we spoke about some key classification methods, including Bayesian networks, decision tree inference, neighbour classification, and SVMs, which have strengths, limitations, possible implementations and problems with their solution. The purpose of this study is to provide an exhaustive analysis of various classification techniques in machine learning. This work would allow both academia and new academics to further consolidate the foundation of the classification methods. Classification is used to detect which category is connected to each data instance within a given dataset. It is used to classify data according to certain constraints into various groups. For classification a variety of significant classification algorithms are used, such as C4.5, ID3, k-nearest neighbour classifier, Naive Bayes, SVM and ANN. A classification methodology generally follows three classification approaches: statistical, machine learning, and neural network. This paper offers an inclusive

10.5 DISCISION TREES – DECISION TREE CONSTRUCTION

Decision Tree:

Decision tree is the most effective and common prediction and classification method. A decision tree is a flux chart-like tree structure, where each internal node indicates an attribute test, each branch is the product of the test, and each leaf node has a class name.



Construction of Decision Tree:

A tree can be "learned" by dividing the source into subsets based on a value test attribute. This process is replicated in a recursive way, called recursive partitioning on each derived subset. The recursion is done when the subset at a node has all the same value as the target variable or when splitting doesn't add any more value to the forecasts. The construction of a decision tree classification involves no domain knowledge or parameter setting, so it is ideal for exploration of explorative knowledge. High dimensional data can be handled by decision trees. The decision tree classifier has excellent accuracy in general. Induction of decision tree is a standard inductive way of learning classification information.

Decision Tree Representation:

Decision trees identify instances by sorting them from the root to a leaf node that provides the instance classification. An instance is classified by starting from the tree root node, evaluating the specified attribute by this node, then moving down the tree branch corresponding to the attribute's value, as shown in the figure above. This process is repeated for the subtree that is rooted in the new node.

The decision tree in the figure above classifies the tennis court and the leaf classification as suitable for a certain morning. (In this situation, yes or no). For example, the instance

(Outlook = Rain, Temperature = Hot, Humidity = High, Wind = Strong)

The left branch of this decision tree will be sorted down and therefore classified as a negative case.

In other words, we can assume that the decision tree represents a breakdown of constraints on instance attribute values.

(Outlook = Sunny $\wedge$ Humidity = Normal) $\vee$ (Outlook = Overcast) $\vee$ (Outlook = Rain $\wedge$ Wind = Weak)

Strengths and Weakness of Decision Tree approach:

The strengths of decision tree approaches are: • Decision trees can produce comprehensible rules.

- ❖ Classification of decision trees without much computation.
- ❖ Decision trees can accommodate both categorical and continuous variables.
- ❖ Decision trees clearly show which fields for prediction or classification are most important.
- ❖ The drawbacks of decision tree methods: • Decision trees are less suited to estimate tasks where the goal is to predict a constant attribute value.
- ❖ The decision trees are vulnerable to mistakes in many class problems and relatively limited numbers of training instances.
- ❖ Decision tree can be costly to train computationally. The decision-making method is computationally costly. Each splitting field must be sorted at each node before the best split can be identified. Combinations of fields are used in some algorithms and search must be made for optimal combined weights. Pruning algorithms can also be costly as many sub-trees of candidates have to be created and compared.

Measures of for selecting the Best Spilt:

There are several tests to decide the best way to split data.

Let $P(i|t)$ indicate the fraction of Class I records at a t node. Measures to pick the best split also rely on the child's nodes' degree of impurity. The less impurity, the more the class distribution distorted. For eg, the class node (0.1) has no impurity, while the class node (0.5,0.5) has the highest impurity.

$$Entropy(t) = - \sum_{i=0}^{c-1} p(i|t) \log_2 p(i|t)$$

$$Gini(t) = 1 - \sum_{i=0}^{c-1} [p(i|t)]^2$$

$$Classification\ error(t) = 1 - \max[p(i|t)]$$

Splitting of Nominal Attributes:

For a nominal attribute, one-to-one mapping, you can have either binary or multi splits.

For equal distributions, the calculation of the Gini index is the same. It is preferable to have a smaller overall GINI index, but you cannot exclude single sections of the index. In our case, it's the multi-way split.

Splitting of Continuous Attributes:

To obtain a continuous property from a continuous attribute, we make a selection.

The values in our example have an increasing order of distinctness.

Between two positions of values that are adjacent to each other, is equal to the mean of the values on either side of that one.

The GINI index must be calculated for any attribute split split. The smaller GIN index can be used as a range if continuous is chosen.

Model Overfitting:

A classification model is made up of two groups of errors: misspellings and misclassifications.

- ❖ Training errors
- ❖ Generalization errors

Training Errors:

The percentage of training records that have errors due to mal-classification. or "To give you an example, a record in the training data has already existed in the test data, but the class prediction was incorrect." Errors of this kind are referred to as training errors.

Generalization Errors:

The model's forecasted variance. Let's assume for example the class label for the test data is under the last data point is identified, but it is the opposite of what is observed. It is difficult to estimate exactly how many errors of this kind are known as Extrapolation errors.

A good model should have a low number of faults while being easy to learn and validate.

It takes a great deal of trial and error when the tree is small. a suboptimal relationship between the model and the data.

The error rate increases as the tree grows, but the training error decreases. Using all available training data will lead to the phenomenon known as overfitting.

In this example, the tree with less nodes, there are more errors in training, and less in testing.

Methods for selecting the best split:

Unless a random state function is defined, train test split divides arrays into random subsets. The optimal split for training and research is said to be 80:20. You will have to change it according to the data set size and the complexity of the parameter.

Algorithm for Decision Tree Induction:

Bayesian Classification:

Baye's Theorem:

The theorem of Bayes named after British mathematician Thomas Bayes from the 18th century is a mathematical method to determine the probability of a condition. Conditional probability is the probability of an outcome, based on a prior result. The theorem of Bayes offers a way to revisit current predictions or hypotheses (update probabilities) with new or additional proof. The Bayes theorem can be used in finance to assess the danger of lending capital to potential lenders.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Naive-Bayesian Classification:

In statistics, Bayes naive classifiers are a family of simple "probabilistic classifiers" based on the theorem of Bayes, which has clear independence assumptions between characteristics. They are among the simplest Bayesian network models, but they achieve higher accuracy levels along with kernel density estimates. Classification systems from Naïve Bayes are very scalable and require a number of parameters in linear numbers in a learning problem.

Bayesian Belief Networks:

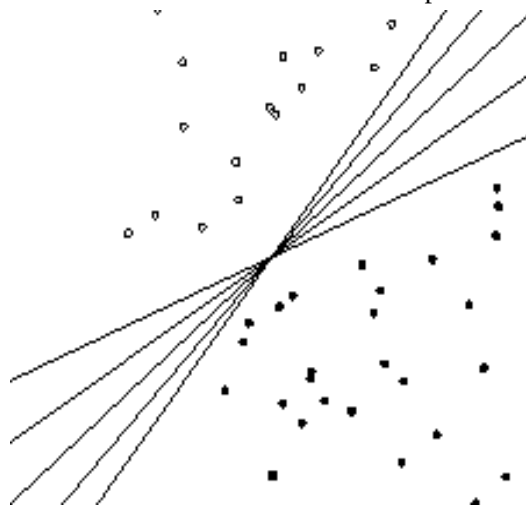
A Bayesian network (also known as a Bayes network, a belief network, or a decision-making network) is a probable graphical model representing a number of variables and their dependence through a directed acyclic graph (DAG). Bayesian networks are suitable for an incident that has happened and are likely to contribute to one of the potential established triggers.

Support Vector Machines:

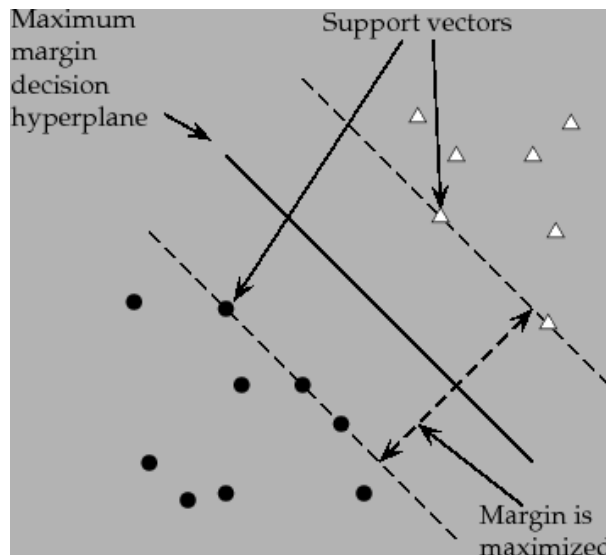
Support Vector Machine or SVM is a simple but effective algorithm for supervised machine learning that can be used to create both regression and classification models. SVM algorithm for both linear and non-linear separable datasets will work well.

Support vector machines: The linearly separable case

In this part, we show that Naive Bayes and Roccio are two study methods for linear classifiers, perhaps the largest text classifiers group. To simplify the discussion, this section only covers two classifiers and a linear classifier is defined when the linear combination of characteristics with one threshold is compared to a two classifier.



There are an infinite number of hyperplanes that separate two linearly separable classes.



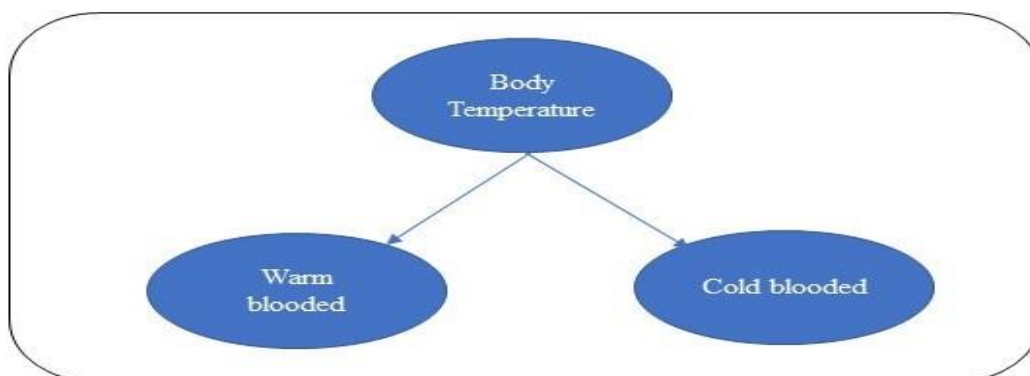
The support vectors are the 5 points right up against the margin of the classifier. There are several possible linear separators in two-class, separable training data sets, like the one in Figure. An intuitively clear boundary between the data items of the two classes appears better than one that approaches examples of one or both classes very closely. Although some learning methods such as perceptron algorithms find just any linear separator, others, such as Naive Bayes, look for some criteria for the best linear separator. In particular, the SVM determines the requirement for a decision-making surface that is as far removed as possible from any data point. The distance between the decision surface and the nearest data point decides the classifier's margin. This construction method necessarily implies that the SVM decision function is defined entirely by a (usually small) data sub-set defining the separator location. These points are known as support vectors (in a vector space, a point can be thought of as a vector between the origin and that point). The figure indicates the sample problem margin and support vectors. Other data points do not play a role in deciding the chosen decision surface.

10.6 METHODS FOR EXPRESSING ATTRIBUTE TEST CONDITIONS

The methods used to express test conditions for attributes are as follows. You are You are:

Binary Attribute:

The binary attribute test condition generates two results as shown below.



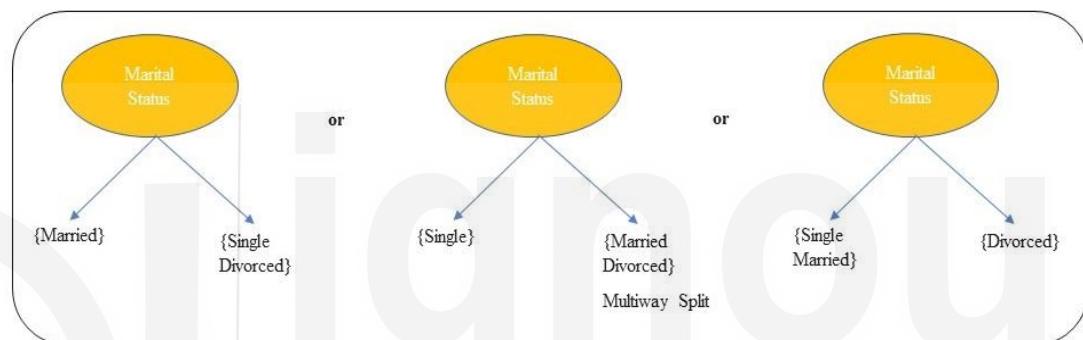
Nominal Attributes:

Classification

As many values may exist for a nominal attribute, its test conditions can be expressed in two respects as shown below.



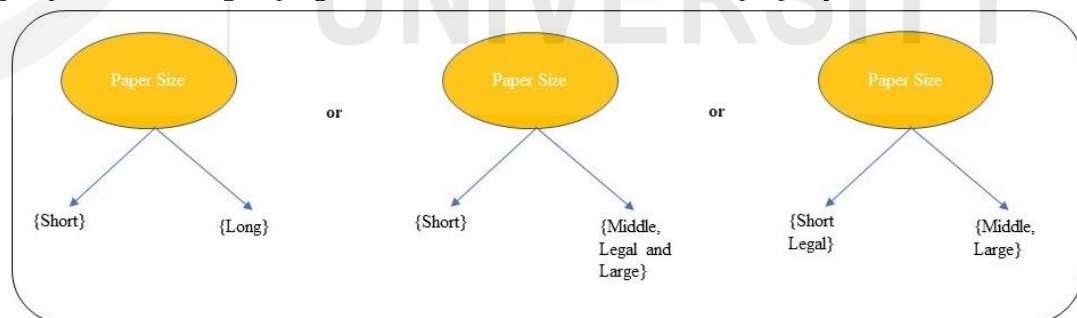
The number of outcomes for a multi-way division depends on the number of different values for the attribute.



Some algorithms, like CART, only support binary divisions. In this case, the k -attribute values can be split into $2k-1-1$. For instance, the marital status is a value of three attributes, we can divide it in $2 \times 3 - 1 - 1$; that means 3 different ways.

Ordinal Attribute:

It can also create splits in binary or multiple ways. Ordinal attribute values may be grouped unless the grouping violates the attribute value's order property.



Different ways of grouping ordinal attribute values

In the example above, condition A and condition B fulfil order, but condition C violates the property of the order.

Continuous Attributes:

The test condition may be expressed as comparison ($A < v$) or ($A \geq v$) test with binary results, or a query in the range of results with the results $v_i \leq A < v_{i+1}$ for $i=1,2,\dots,k$.

UNIT 11 CLUSTERING

Clustering: Problem Definition:

A group of comparable data objects is classed as a cluster in clustering. A collection of data is referred to as a group. The cluster analysis divides data into groupings based on how closely they resemble one another. A label is assigned to a group once the data has been classified into several groups. By doing the classification, it aids in adaptation to shifts.

Clustering is an unsupervised Machine Learning-based Algorithm that divides a set of data points into clusters so that the objects are all part of the same group. Using clustering, data can be subdivided into smaller, more manageable groups. Data from each of these subgroups are grouped together into a single cluster. Because our client data has been segmented into groups, we're in a better position to determine who could benefit most from this offering.

- ❖ To put it simply, a cluster is a group of related things.
- ❖ It's a grouping of objects where the distance between any two members is less than the distance between any two people.
- ❖ A multidimensional space segment with a high density of objects that is related to the other segments.

Data mining Cluster Analysis: What Is It?

Data mining's Cluster Analysis refers to the process of identifying groupings of items that are similar in some respects but differ from one another in other respects.

A cluster is a grouping of comparable items that belongs to the same category.

A set of items in which the distance between any two things in the cluster is less than the distance between any two objects in the cluster and any object not placed inside the cluster.

High-density region related to other regions in multidimensional space.

Clustering is a technique used to turn a collection of unrelated abstract objects into a set of related classes.

It is a technique for dividing large collections of data or objects into smaller groups called clusters.

As a stand-alone tool, it provides insight into data distribution and can be used as a pre-processing step for other algorithms or as a pre-processing step in its own right.

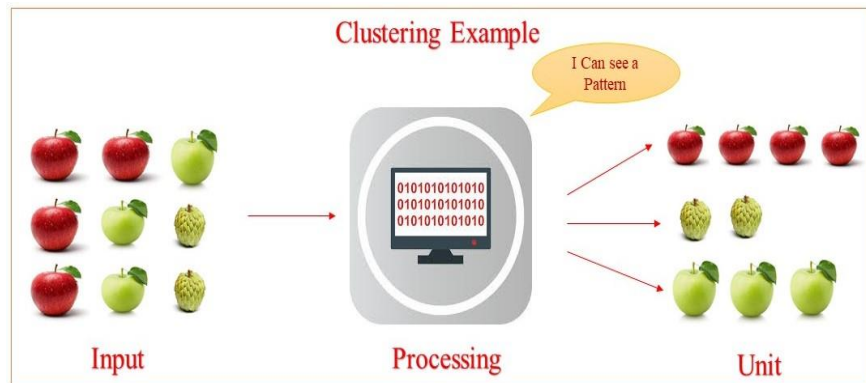


Fig: Example of Clustering

Uses for cluster analysis in data mining:

Data clustering analysis has a wide range of applications, including image processing, data analysis, pattern identification, and market research. With the use of Data clustering, firms can find new client groupings in their database. Buying patterns can also be used to classify data.

Data mining clustering aids in the classification of animals and plants by utilizing comparable functions or genes in biology. Finding out more about how species are structured becomes easier with this information. Data mining clustering identifies geographic areas. There are regions in the earth observation database that are similar to each other.

In the city, a particular type of dwelling is included in a specific neighborhoods. Information can be discovered more easily by classifying online files using clustering in data mining. Additionally, it's employed in detection software. If a credit card is being used fraudulently, the pattern of deceit can be identified using clustering in data mining.

To better understand each cluster, this information is useful. Using this tool, one can gain insight into the distribution of data and use it to do data mining tasks.

A cluster's data objects can be thought of as one big group.

While performing cluster analysis, we first divide the data set into groups. It allocates levels to groups based on data similarities.

An advantage of over-classification is that it is flexible and can assist identify crucial qualities that set differentiating groups apart.

Applications of cluster analysis in Data Mining:

- ❖ Clustering analysis is widely utilized in a variety of fields, including data analysis, market research, pattern identification, and image processing.
- ❖ It aids marketers in identifying distinct customer segments based on their customers' purchasing habits. They are able to classify their clients.
- ❖ It aids data discovery by assigning documents on the internet.
- ❖ For example, credit card fraud detection relies on clustering.
- ❖ Cluster analysis is a data mining function that provides insight into data distribution so that the properties of each cluster may be analysed.
- ❖ It can be used in biology to figure out plant and animal taxonomies, classify genes with similar functions, and have a better understanding of population structure.
- ❖ Earth observation databases use this data to identify similar land regions and to group houses in a city based on house type, value, and geographic position.

Clustering Requirements for Data Mining:

Its multiplicity of uses has made clustering analysis an emerging data mining topic. Due to the recent development of a variety of data clustering toolsets and their extensive use in a variety of fields, such as image processing and computational biology to mobile communication and medicine these methods are likely to become more prominent in the future. It's impossible to standardise data clustering algorithms, which is the fundamental problem. Advanced algorithms may work well with some data sets, but they fail or perform poorly when used with other sets. Despite numerous attempts to standardize algorithms that work effectively in all scenarios, little progress has been accomplished. So far, a slew of clustering utilities have been proposed. However, each method has pros and cons and isn't applicable in all real-world scenarios.

Interpretability:

Clustering should produce useable, intelligible, and interpretable results.

Clustering is a powerful tool.

Facilitates handling of stale or corrupted data:

Most of the time, the data is jumbled and unorganized. Because it can't be analyzed fast, data mining relies heavily on clustering. By organizing the data into groups of related data objects, grouping can provide some structure to the data. It becomes easier for the data specialist to process the data and also to uncover new things in the process of processing data.

High Dimensional:

Using data clustering, you'll be able to work with large and little data of any dimension.

Discovery of attribute clusters based on shape:

The technique for clustering detects groups of arbitrary shapes. There's also a little cluster here with a spherical form.

Usability of algorithms while dealing with various types of data:

Using clustering methods, a wide range of data types can be analyzed. Binary data, categorical data, and interval-based data are examples of possible data types.

Clustering Scalability:

Dealing with a large database is commonplace. The method must be scalable in order to deal with large databases.

Categorization of Major Clustering Methods:

The following categories describe several clustering techniques

- ❖ Partitioning Method
- ❖ Hierarchical Method
- ❖ Density-based Method
- ❖ Grid-Based Method
- ❖ Model-Based Method
- ❖ Constraint-based Method

Partitioning Method:

Let's say "a" partitioning is done on the database's "b" objects in this method. Each partition and $a < b$ will be used to represent a cluster. After the classification of items, C is the number of groups. There are a few requirements that must be met with this Partitioning Clustering Method, and they are as follows:

- ❖ Only one group should have a single objective at any given time.
- ❖ There should be no organization that does not have a single goal.

When using a Partitioning Clustering Method, keep in mind the following considerations:

- ❖ If we provide the number of partitions up front, there will be an initial partitioning (say m).
- ❖ Iterative relocation is a strategy that involves moving an object from one group to another in order to enhance partitioning.

K-Means Algorithm:

Clustering by K-Means is a machine learning or data science problem-solving technique that utilizes an unsupervised learning method. This article will teach you about the K-means clustering technique, how it works, and how to use it in Python.

What is K-Means Algorithm:

Using K-Means Clustering, an approach known as Unsupervised Learning, the unlabeled dataset can be divided into several clusters for analysis. For example, if $K=2$, then there will be two clusters, and if $K=3$, then there will be three clusters, and so on. K indicates the number of pre-defined clusters that need to be produced in the process.

With this approach, the unlabeled dataset is divided into as many clusters as possible, so that each dataset only belongs to one group.

There is no requirement for training because it automatically discovers the groups in the unlabeled dataset and clusters the data into them.

Each cluster has a corresponding centroid in this centroid-based approach. This algorithm's primary goal is to reduce the total distance between each data point and the clusters to which it belongs.

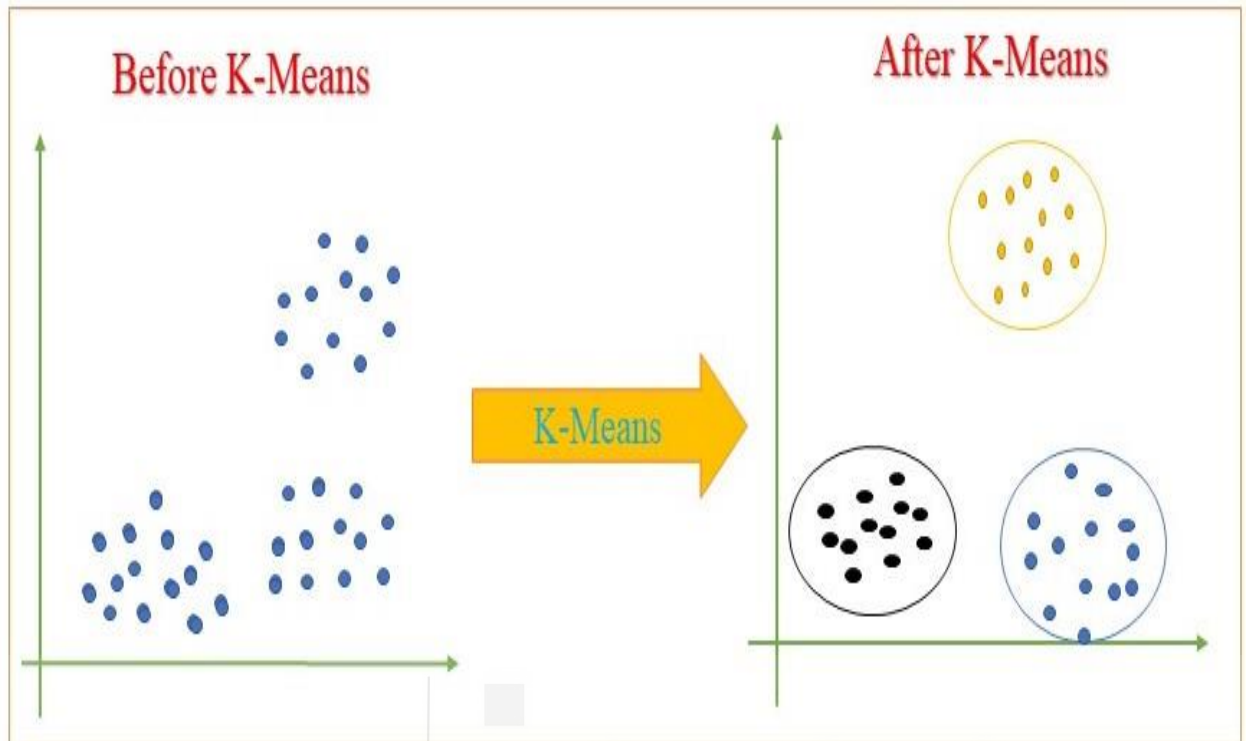
The algorithm starts with an unlabeled dataset, separates it into k clusters, and then continues the procedure until no better clusters can be found. To make this algorithm work, the value of k must be known beforehand.

k-means clustering has two primary functions.

- ❖ Iteratively arrive at the optimal value for K centre points or centroids.
- ❖ Each data point is assigned to the k -center that is the closest to it. A cluster is formed by data points that are close to a given k -center.

Consequently, data points with certain commonality are found in each cluster, and these clusters are separated from one another.

The K-means Clustering Algorithm's operation is illustrated in the diagram below:

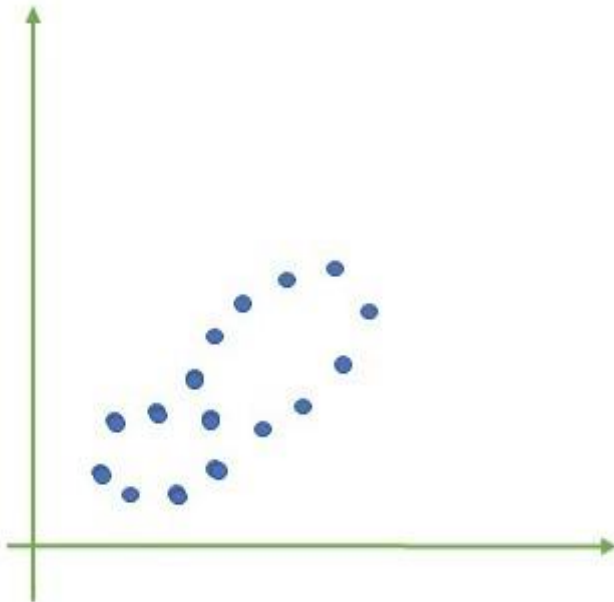


What is the K-Means Algorithm and How Does It Work?

- ❖ The number of clusters will be determined by selecting K.
- ❖ Precisely where you want to place K points or centroids is up to you. A dataset other than the one used as input can be used.
- ❖ Create the predefined K clusters by assigning each data point to its nearest centroid.
- ❖ Determine the standard deviation and re-locate the cluster centroid if necessary.
- ❖ Once again, reassign each data point to the cluster's new nearest centroid.
- ❖ FINISH is the next step if there are any reassignments. Otherwise, move to step 4.
- ❖ The prototype has been completed.

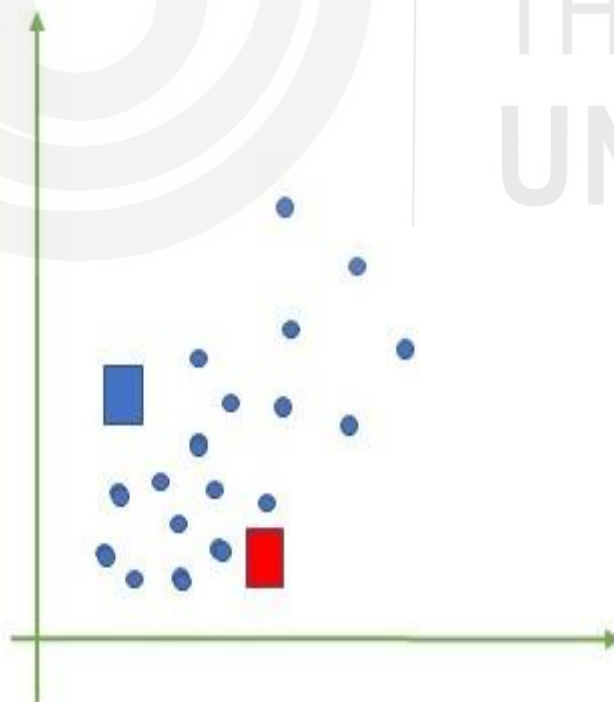
Let's take a closer look at the preceding phases by examining the accompanying diagrams:

Assume M1 and M2 are two variables. Below you'll find the x-y axis scatter plot for these two variables.



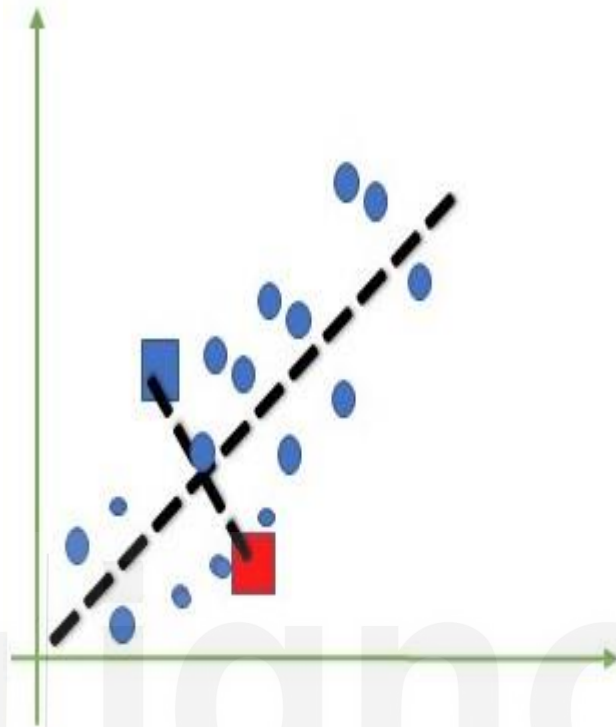
In order to identify the dataset and sort it into various groups, we'll use cluster number k , or $K=2$. This means that we'll make an effort to divide the datasets into two distinct groups.

To create the cluster, we'll need to pick k randomly distributed points (or centroid points). These points might be taken from the dataset or be taken from anywhere else. Here we are selecting k points from a dataset that does not include the two points below. Take a look at the image below:

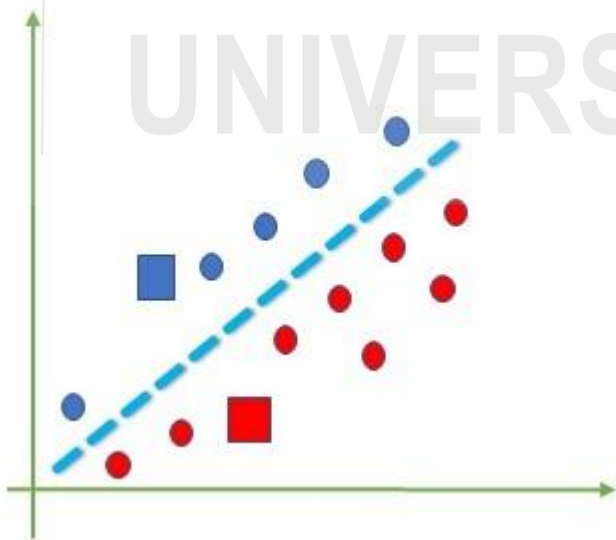


We will now give a K -point or centroid to each scatter plot data point. We'll figure it out by using some of the maths we've learned about measuring

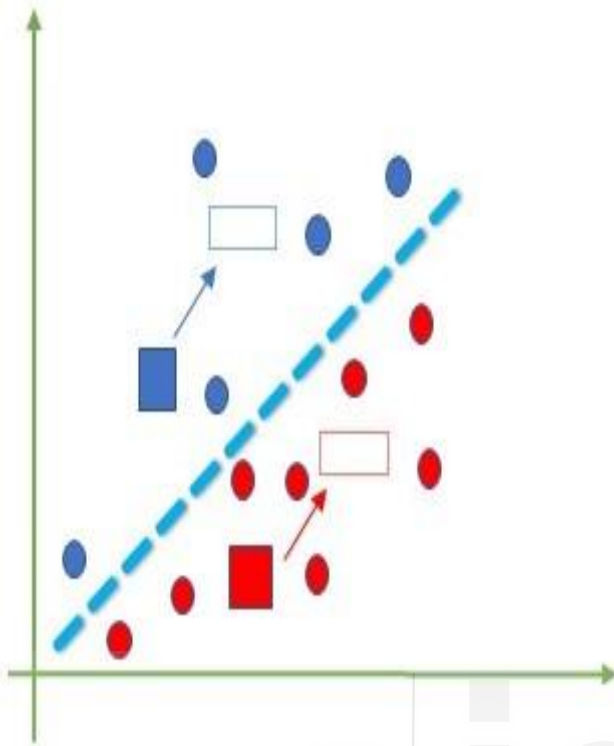
distances between points. As a result, we'll use the median of the two centroids as our point of departure. Take a look at the image below:



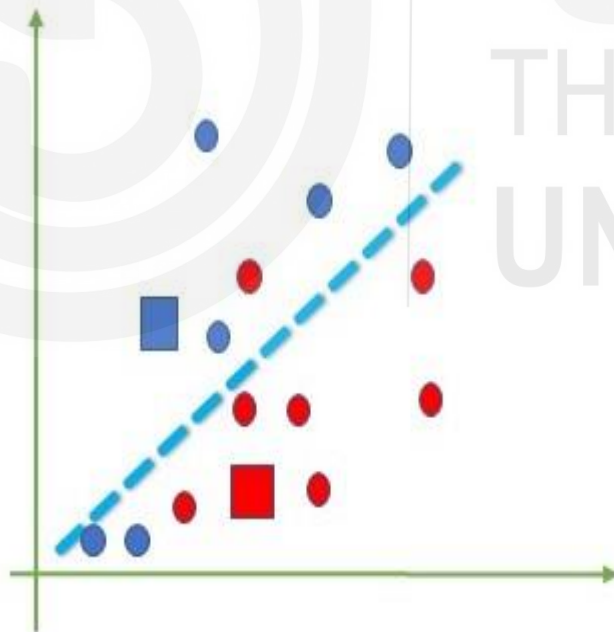
The image above shows that the points on the left side of the line are near the K1 centroid, which is blue, and the points on the right of the line are near the yellow centroid, which is yellow. To make things easier to understand, let's use blue and yellow to colour them.



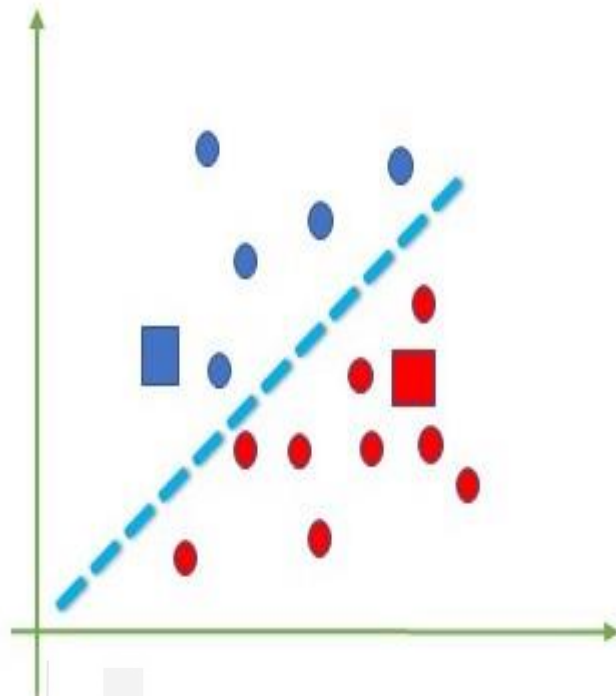
To discover the closest cluster, we'll choose a new centroid and go through the process again. New centroids will be found by computing the gravity centres of the existing centroids, and then selecting new centroids as follows:



Each data point will be assigned to the new centroid at this point. To do this, we'll use the same method we used to locate the median line. The median will look like the image below:

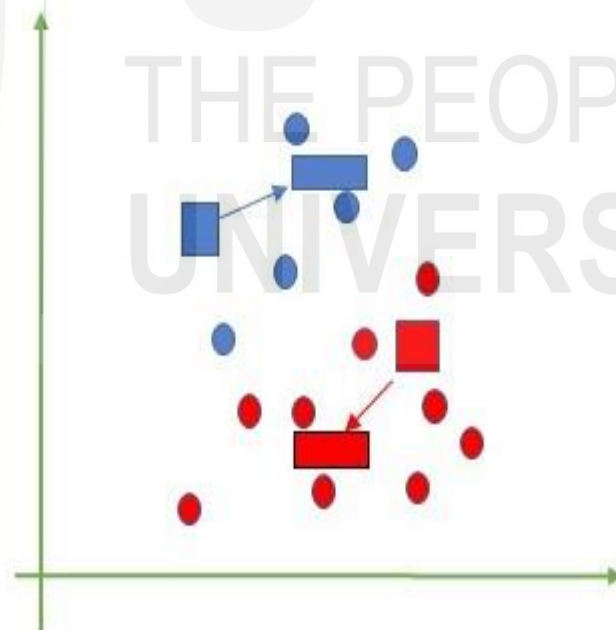


Two blue points and one yellow point may be seen above. The blue points are directly across from each other. This means that new centroids will be created using these three locations.

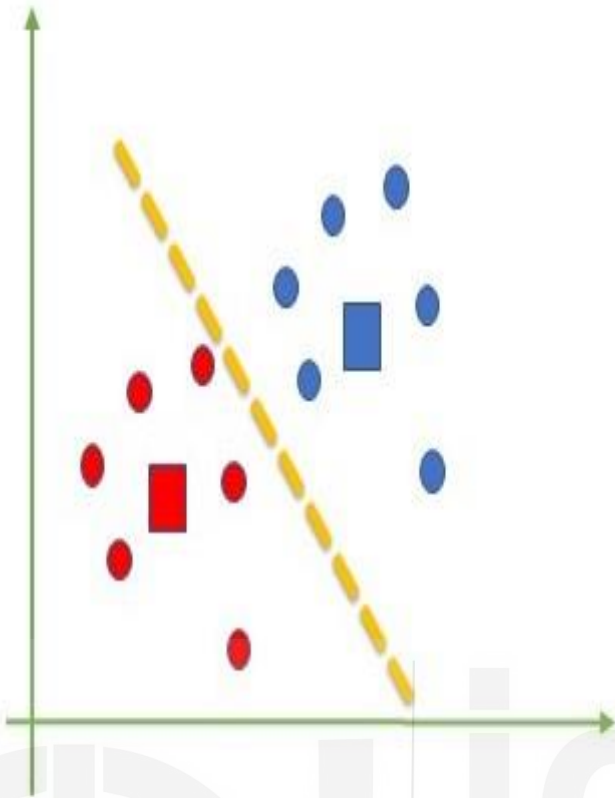


Because of the reassignment, we will return to step 4, which is the search for new centroids or K-points.

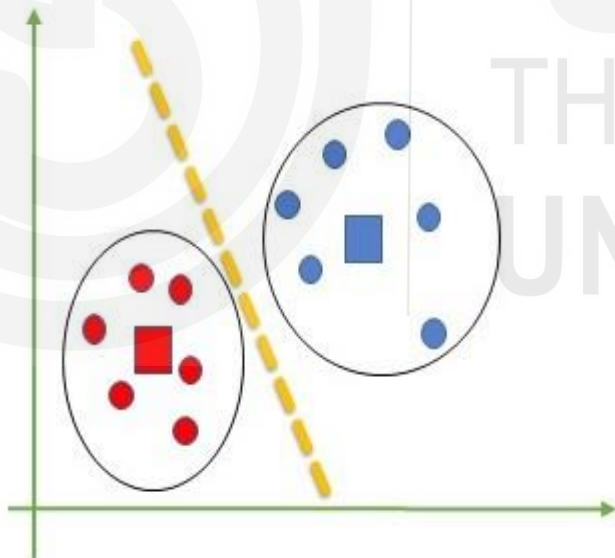
As indicated in the next image, we'll go through the process of identifying the centre of gravity for each of our centroids a second time.



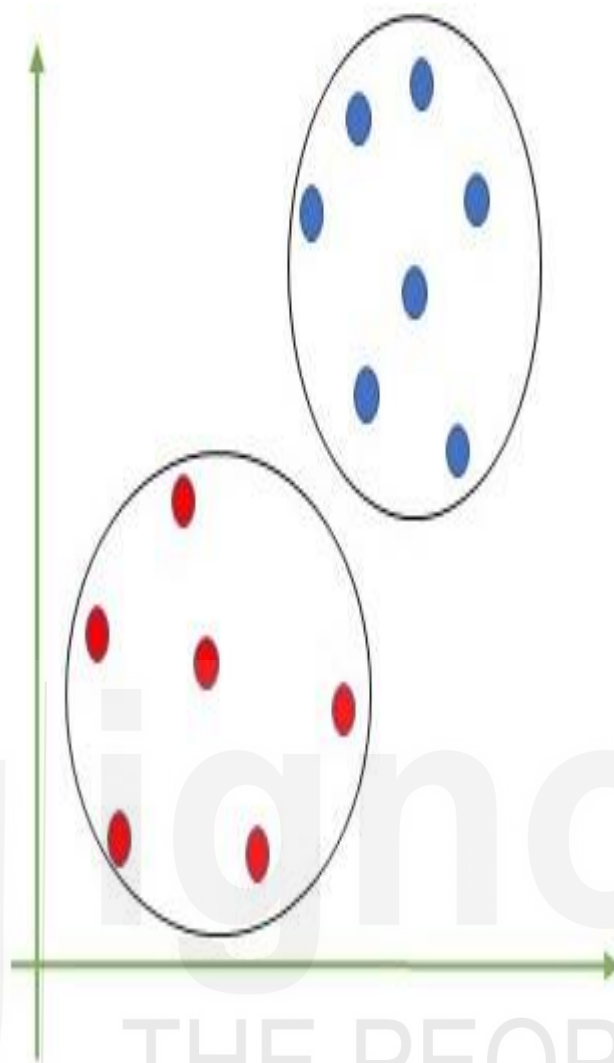
With the new centroids, we will redraw the median line and redistribute the data points. As a result, here is what the image will look like:



Since no dissimilar data points exist on either side of the line, our model has been built, as seen in the graphic above: Take a look at the image below:



Now that our model is complete, we can eliminate the centroids we previously thought existed, leaving us with the two final clusters displayed in the graphic below:

**K-Medoids:**

It's similar to the k-means clustering problem in that it uses k-medoids. The PAM algorithm was developed by Leonard Kaufman and Peter J. Rousseeuw, hence the name. K-means and K-medoids, two partitional algorithms, reduce the distance between cluster points and the central point of a cluster. Using this method, actual data points are selected as the cluster centers (medoids or examples), as opposed to using the k-means algorithm, which does not require the cluster centre to be an input data point. As a result, the cluster centers are easier to interpret (it is the average between the points in the cluster). As a result, unlike with k-means, the dissimilarity metrics used with k-medoids aren't restricted to Euclidean distance. Instead than minimizing the sum of squared Euclidean distances, k-medoids minimize the pairwise dissimilarities as opposed to the total.

The k-medoids approach is a classic clustering technique that separates an object data set into n clusters, with a fixed number of clusters k . (which implies that the programmer must specify k before the execution of a k-medoids algorithm). Some methods, such as the silhouette approach, can be used to assess a k value's "goodness."

Cluster medoid objects are those that have an average dissimilarity to all other items in the cluster of one or less.

Hierarchical Method:

This method decomposes a set of data items into a hierarchy. Depending on how the hierarchical breakdown is generated, we can put hierarchical approaches into different categories. In this case, there are two methods to choose from.

- ❖ Agglomerative Approach
- ❖ Divisive Approach

Agglomerative Approach:

The bottom-up approach is another term for this strategy. At the outset, all of the teams are divided up. When all the groups have been merged, or the criterion for termination has been satisfied, the process continues merging them.

There are two methods for improving the quality of hierarchical clustering in data mining:

- ❖ At each level of hierarchical clustering, the object's links should be thoroughly examined.
- ❖ For the integration of hierarchical agglomeration, one can utilize a hierarchical agglomerative algorithm. As part of this method, micro-clusters are formed first. When data objects are micro clusters, they are subjected to macro clustering.

Divisive Approach:

With the Divisive technique, you work your way down, rather than starting at the top. All of the data items are initially maintained in a single cluster in this method. The continual iteration splits the group into smaller clusters. Iteration will continue until the condition of termination is met using the constant iteration method. After a group is divided or merged, it's impossible to go back and undo the changes.

Key Issues in Hierarchical Clustering:

Lack of a Global Objective Function:

The K-Means algorithm has a global objective function, but agglomerative hierarchical clustering approaches cluster on a local level. As a result, this technique has a distinct advantage over global functions in terms of cost and complexity.

Ability to Handle Different cluster Sizes:

To deal with merged clusters of varied sizes, we need a solution.

Merging Decisions Are Final:

One disadvantage of this method is that after two clusters have been merged, they can no longer be divided up for a better union in the future.

Density-based Method:

Density is the primary consideration in this Data Mining clustering approach. This clustering method is based on the concept of mass. This clustering technique causes the cluster to grow indefinitely. In the group's radius, each data point should have at least one number of points.

Grid-Based Method:

This Grid-Based Clustering Method makes use of a grid by grouping together the clustered objects. You can create a Grid Structure by breaking down the object space into discrete cells.

Advantage of Grid-based clustering method:

- ❖ The processing time is substantially faster with this method than with another, and so it can save time.
- ❖ The number of cells in the quantized space in each dimension determines how this method works.

Model-Based Method:

This approach hypothesizes a model for each cluster in order to locate the data that fits the hypothesis the best. This technique uses a clustering of the density function to find the clusters. It depicts the data's spatial distribution.

In addition, using this method, you may compute the number of clusters based on standard statistics, while also accounting for outliers and noise. As a result, the clustering methods it produces are solid.

Constraint-based Method:

This technique uses user- or application-oriented constraints to cluster data. When we talk about constraints, we're referring to things like user expectations or desirable clustering qualities. Constraints allow us to communicate with the

clustering process in an interactive manner. Both the user and the application have the ability to specify constraints.

Strengths and weakness of Hierarchical Clustering:

Heterogeneous clustering has the advantage of being both simple to understand and simple to implement. Among its flaws are the fact that it rarely offers the best solution, the large number of arbitrary decisions it requires, the inability to work with missing data, the inability to work well with mixed data types, the inability to handle very large data sets, and the common misinterpretation of the dendrogram are all problems. It's best to use latent class analysis as an alternative to other methods.

Outlier Analysis:

In statistics, an outlier is a data point that deviates significantly from the norm. Measuring and execution errors are two possible causes. Outlier analysis, also known as outlier mining, is the process of examining data that stands out for whatever reason. It's impossible to do data analysis without encountering outliers. An outlier is a data point that deviates significantly from the norm. As a result, an outlier is critical since it indicates an experimentation flaw. There are numerous applications for outliers, such as identifying fraud and creating new market trends.

Outliers are frequently mistaken for random fluctuations in the data. Outliers, on the other hand, are distinct from random noise in the following way:

- ❖ An outlier is a point of observation that differs from the rest because of its location.
- ❖ To better detect outliers, noise should be eliminated.

Data mining outliers can have a number of different causes.

Outliers in Data Mining can have a variety of causes. Following are a few of these contributing factors:

- ❖ Identifying financial fraud such as credit card hacking or other similar scams makes use of this technology.
- ❖ It's utilized to keep track of a customer's changing purchase habits.
- ❖ It's used to find and report human-made mistakes in typing.
- ❖ It's utilized for troubleshooting and identifying problems with machines and systems.

In Data Mining, why should outliers be handled?

Data Mining must deal with outliers for a variety of reasons. These are a few of the explanations:

The outcomes of databases are impacted by outliers.

- ❖ Outliers frequently produce good or useful discoveries and conclusions, allowing researchers to identify various patterns or trends.
- ❖ Even in the world of study, outliers can be useful. They can be a lifesaver when doing research.
- ❖ Data mining's most important subfield is outlier analysis.

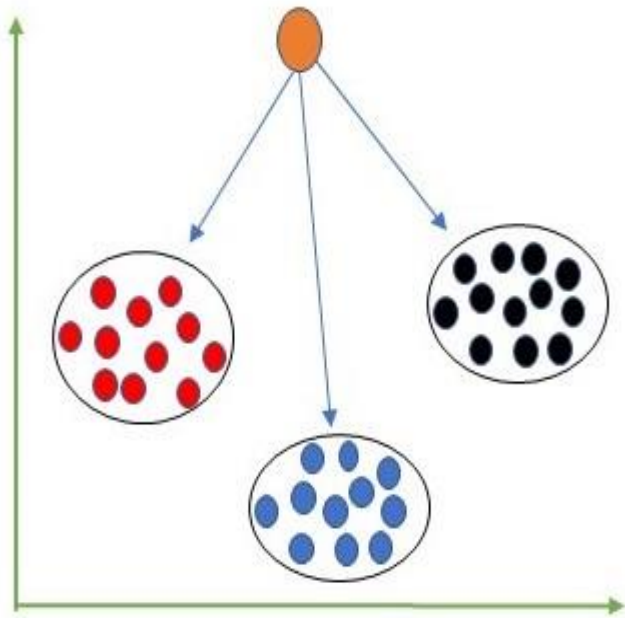
Uses for Detecting Outliers in Data Mining:

In Data Mining, it is common to utilize Outlier Detection to find anomalies. In data mining, it's utilized to find patterns or trends. The following are some examples of where outlier detection is put to use in data mining:

- ❖ Fraud Detection
- ❖ Telecom Fraud Detection
- ❖ Intrusion Detection in Cyber Security
- ❖ Medical Analysis
- ❖ Environment Monitoring such as Cyclone, Tsunami, Floods, Drought and so on
- ❖ Noticing unforeseen entries in Databases

Why Outlier Analysis:

For the most part, data mining algorithms ignore outliers like noise and exceptions. However, in certain applications like fraud detection, uncommon occurrences can be just as fascinating as the more common ones, therefore performing an outlier analysis becomes critical.



Detecting Outlier:

Each cluster has a mean value in the K-Means clustering method. Things are grouped together based on how near their arithmetic mean values are to one another. First, we must establish the threshold value so that any data point further away from its nearest cluster is considered an outlier for the purposes of this analysis. After that, we'll need to figure out how far the test data are from the mean of each cluster. The data will be labelled as an outlier if the distance between it and its nearest cluster exceeds a certain limit.

Detecting outliers using various methods:

Detecting outliers uses three main strategies. These are the strategies I've come up with:

The Statistical Approach

The Distance Based Approach

The Deviation Based Approach

Techniques for Detecting Outliers:

Detecting outliers involves using one of four methods:

- ❖ Numeric
- ❖ Z-Score
- ❖ DBSCAN
- ❖ Isolation forest

Numeric:

In a one-dimensional feature space, this is the simplest approach. An IQR (Inter Quartile Range) methodology is used to find outliers in this method. A

nonparametric property of this model is that it does not account for repeated measurements.

Z-Score:

Parametric outlier detection using Z-score is possible in a single or small feature space. To use Z-score, the data must have a normal distribution with standard deviation of one standard deviation. The following formula is used to determine Z_i :

$$Z_i = (x_i - \mu) / \sigma$$

DBSCAN:

The DBSCAN clustering technology is used in this Outlier Detection methodology. DBSCAN is a nonparametric outlier detection approach based on density in a one- or multi-dimensional feature space. All of the data points are categorised as either Core Points, Border Points, or Noise Points.

Isolation forest:

When dealing with large datasets in a one- or multi-dimensional feature space, Isolation Forest is a nonparametric method that is especially useful. Isolation numbers, which can be defined as the number of splits required to isolate a data point, are critical concepts in this strategy.

Unit 12: Text and Web Mining

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12.0 TEXT MINING: INTRODUCTION

It is important to understand the language and its evolution. The text mining and Natural Language Processing (NLP) are connected now. NLP here stands for natural language processing and moving forward, the various application of NLP in the industry and its different components can be explored. The success of human race is because of the ability to communicate and share information now that is where the concept of language comes in however many such standards came up resulting in many such language with each language having its own set of basic shapes called alphabets and the combination of alphabets resulted in verse and the combination of these words arranged meaningfully resulted in the formation of a sentence now each language has a set of rules that is used while developing these sentences and these set of rules are also known as grammar not coming to today's world that is the 21st century according to the industry estimates only 21 percent of the available data is present in the structured format data is being generated.

12.1 OBJECTIVES

After going through this unit, you should be able to:

- understand the purpose of Text Mining.
- describe the general goals of Text Mining.
- discuss the evolution of Text Mining.
- describe various data analysis techniques performed by the Text Mining on the web.
- list, discuss an and
- describe various structures of operating system.

12.2 TEXT DATA ANALYSIS AND INFORMATION RETREIVAL

The tweets or messages on WhatsApp Facebook Instagram or through text messages and the majority of this data exists in the textual form which is highly unstructured in nature now in order to produce significant and actionable insights from the text data it is important to get acquainted with the techniques of text analysis so let's understand what is text analysis or text mining now it is the process of deriving meaningful information from natural language text and text mining usually involves the process of structuring the input text deriving patterns within the structured data and finally evaluating the interpreted output compared with the kind of data stored in database text is unstructured amorphous and difficult to deal with algorithmically nevertheless in the modern culture text is the most common vehicle for the formal exchange of information now as text mining refers to the process of arriving high-quality information from text the overall goal here is to turn the text into data for analysis. Text mining has various areas to explore as shown in fig1

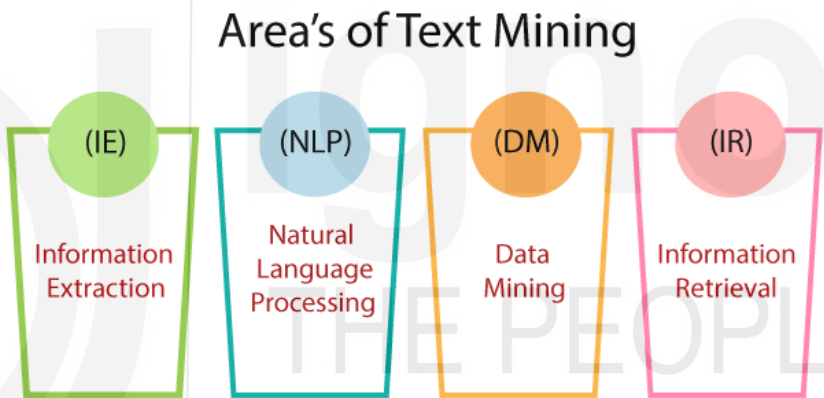


Figure1: Areas of Text Mining

The tweets or messages on WhatsApp Facebook Instagram or through text messages and most data exists in the textual form which is highly unstructured in nature now in order to produce significant and actionable insights from the text data it is important to get acquainted with the techniques of text analysis so let's understand what is text analysis or text mining now it is the process of deriving meaningful information from natural language text and text mining usually involves the process of structuring the input text deriving patterns within the structured data and finally evaluating the interpreted output compared with the kind of data stored in database text. Text mining refers to the process of arriving high-quality information from text the overall goal here is to turn the text into data for analysis.

Information Extraction is the techniques of taking out the information from the unstructured text data or semi-structured data contains in the electronic documents. The process identifies the entities, then classify them and store in the databases from the unstructured text documents.

Natural Language Processing (NLP): The human language which can be found in WhatsApp chats, blogs, social media reviews or any reviews which are written in any

offline documents. This is done by the application of NLP or natural language processing so let's understand what is natural language processing so NLP refers to the artificial intelligence method of communicating with an intelligent system using natural language by utilizing NLP and its components one can organize the massive chunks of textual data perform numerous or automated tasks and solve a wide range of problems such as automatic summarization machine translation named entity recognition speech recognition and topic segmentation. So, let's understand the basic structure of an NLP application considering the chat pod here as an example we can see first we have the NLP layer which is connected with the knowledge base and the data storage now the knowledge base containing logs, a large history of all the chats which are used to train the particular algorithm and again we have the data storage where we have the interaction history and the analytics of that interaction which in turn helps the NLP layer to generate the meaningful output. So, now if we have a look at the various applications of NLP first, we have sentimental analysis now this is a field where NLP is used heavily, we have speech recognition now here we are also talking about the voice assistance like google assistant cortana and the siri.

Data Mining is the process of extracting significant data from standard language text. It uses various techniques of grouping the similar kind of data into clusters. It helps to predict the values or data after training the model from the previously recorded or historical data. Data mining technique is used to classify using variety of classification techniques or predictive modelling. It can predict the marketing trends and human marketing behavior, customer experience hence helping the business tycoons to build strategy for business and markets.

Text Data Analytics

First, let's define the term text mining and the term text for analytics. if you look at the two phrases literally mining emphasizes more on the process, so it gives us an algorithm a very good view of the problem analytics on the other hand emphasizes more on the result. Text mining and analytics implies to turn text data into high quality information or actionable knowledge so in both cases we have the problem of dealing with a lot of text data. The consumer is making decision to buy a product by learning from others' experience from social media reviews. The text mining supplies knowledge for optimal decision making.

We can utilize text mining in any industry, which means the data can be in any format, such as Word, PDF, XML, JSON, or anything else. Now we'll look at how text mining works. Text mining is a semi-automated procedure that necessitates the collection of data from a variety of sources. The textual data is in an unstructured format and is a free-flowing data, we cannot immediately apply data mining techniques to it, so we must first organize that thing. Once we develop a framework for the textual data, the next extremely crucial step is to structure it. After converting unstructured data into structured data, the next step is to cleanse the data. The next step is to clean the data and prepare it for analysis. Cleaning is the act of removing specific terms, such as stop words, and then removing any irrelevant words that may be included in the data. Once that is done, the data is available for analysis. Create a document matrix for a data analysis.

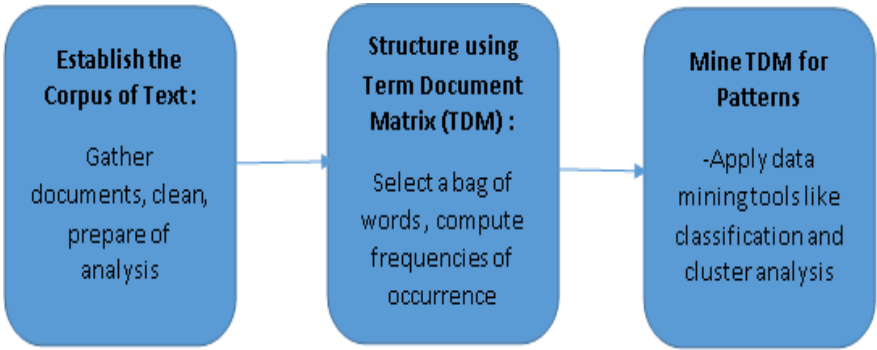


Figure 1: Text Data Analysis

Once the data matrix is developed from the input documents. The words found in those documents depending on the type of data various clustering techniques of machine learning can be applied.

Check Your Progress 1

- 1) Mention few applications of Text Mining?
.....
.....
.....
- 2) What are the techniques used to analyze the web usage patterns?
.....
.....
.....

12.3 DIMENSIONALITY REDUCTION OF TEXT

The conventional way of dealing with text is to collect information from various sources like text, pdf files, database files etc. After the preprocessing and cleaning of data, the analysis of data is done. The extraction of the summary of the whole content automatically is called Text Summarization. Another technique, Text Categorization existed to group the text predefined by users. The process of segmentation of text into relevant groups is called text clusters. Since, the text is quite huge and enormous to analyze so to reduce the size of the text, the dimensions of the text can be reduced. The text dimensions can be reduced by taking out the words order and their meaning. So, it can be analyzed into two ways.

- a) Syntax Processing: The syntactical processing of the text can be achieved by tokenization and removal of stop words.
- b) Semantic Processing: It means retaining the meaning of a text. The different techniques are applied for the same. A sentence has a main logical concept called predicate. The arguments of the predicate can be extracted from other parts of the sentence. The identification of the predicate and the arguments for that predicate is known as semantic role labeling.

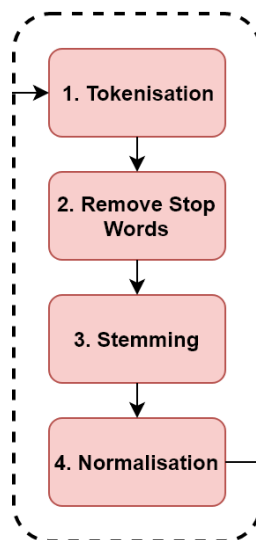


Figure 2: Text Preprocessing

Text Pre-processing: The Text mining is the process of cleaning up the data. It involves the steps of clean up. It means of removing of unnecessary text or data. The removal of ads from the web pages. It involves the conversion of cleaned text into binary formats.

- Tokenization: It converts the text into small words called tokens separated by white spaces. Transferring strings into a single textual token. Segmentation of running text into sentences and words.
- Cutting a text into pieces called *tokens*.
- Removing certain characters, such as punctuation.

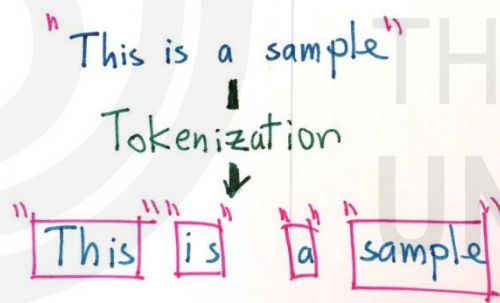


Figure 3: Tokenization

After the process of tokenization, the data are grouped into grams.

Bigrams: Bi means two so, grouping of two consecutive data together is called bigrams.

Trigrams: Tri means three, grouping of three consecutive data together is called trigrams. Similarly, it can be done,

N-grams: The groups of N consecutive data together form N-grams.

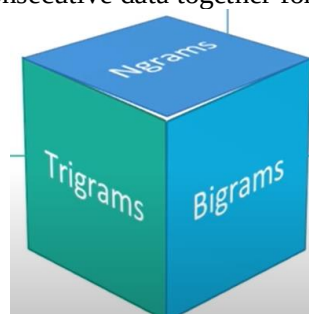


Figure 4: N-grams

POS Tagging (Part of Speech Tagging): It is the step of tagging a label or word to every token generated in the previous step. It is the identification of entity.

- According to the role of a word in a sentence, it can be tagged as a noun, verb, adjective, adverb, preposition, etc.
- Correct tags such as nouns, verbs, adjectives, etc. should be assigned.

Vector from text: Creating vectors from the text ensuring that it should not result in sparse matrix as it involves high computational costs. The important point to note here is that whatever technique we apply for dimension reduction from the text the meaning or the information of the text should not be lost. Once the text preprocessing is done then the next step is to transform the data in some structured form of a data base. The cleaned text is then transformed into generation of attributes. It can be achieved by two approaches as follows, first is

1) Bag of words:

2) Vector Space

For example, there are sample of reviews of a comedy movie so, the reviews of the viewer can be:

Review 1: This movie has a good sense of humor

Review 2: This movie is boring and long

Review 3: It is a one-time watch movie

You can easily observe three different opinions of three different viewers. You can see thousands of reviews about a movie on the internet. All these users generated text can help us out to takeout some interpretation in gauging that how a movie has performed. The above three reviews mentioned above cannot be given to the machine learning engine to analyze positive or negative reviews. So, we apply some text filtering techniques like Bag of words.

Bag of words (BOW): It is the kind of a model in which the text is written in the form of numbers. It can be represented as represent a sentence as a bag of words vector (a string of numbers).

First, the vocabulary of all the unique words in above three reviews. The vocabulary ‘This’, ‘movie’, ‘has’, ‘a’, ‘good’, ‘sense’, ‘of’, ’humor’, ‘the’, ‘is’, ‘long’, ‘boring’, ‘and’ ‘it’, ‘one-time’, ‘watch’.

	1 Thi s	2 Movi e	3 Ha s	4 a	5 goo d	6 sens e	7 o f	8 humo r	9 i s	10	11 Borin g	12 an d	13 it	14 One - time	15 watc h	Lengt h of the review s (in words)
R 1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	8
R 2	1	1	0	0	0	0	0	0	1	1	1	1	0	0	0	6
R 3	0	0	0	0	0	0	0	0	1	0	0	0	1	1	1	4

Vector of R1: [1 1 1 1 1 1 1 1 0 0 0 0 0 0 0]

Vector of R2: [0 1 0 0 0 0 0 1 1 1 1 0 0 0]

Vector of R3 : [0 0 0 0 0 0 0 1 0 0 0 1 1 1]

The basic idea of BoW model is to identify proper nouns, persons, locations, organizations, named identity.

Term Frequency – Inverse Document Frequency (TF-IDF): It is a measure to represent the count or the number of occurrences in the document for any term. Let's say the term is represented as t , and document as d .

TF-IDF can be represented as the number of terms in the document d divided by the total number of terms in the document. So, every document will have its own TF score.

$$tf_{t,d} = \frac{n_{t,d}}{\text{Number of terms in the document}}$$

For example, look at the review of the movie again taken in the previous statement.

Let's calculate the TF for the review R1

Review 1: This movie has a good sense of humor

In this you will observe that the frequency of each word mentioned is as follows.

Vocabulary: 'This', 'movie', 'has', 'a', 'good', 'sense', 'of', 'humor'.

Number of words in Review 1 = 8.

TF for the word 'this' = (number of times 'this' appears in review 1) / (number of terms in review 1) = 1/8

Similarly, $TF('movie') = 1/8$ and so on.

Inverse Document Frequency (IDF): It is a measure to find out the occurrence of the words in number of documents. To compute the significance of the word we need to find out the TF and IDF both the measures.

$$idf_t = \log \frac{\text{number of documents}}{\text{number of documents with term 't'}}$$

For example, IDF for the Review 1 is:

$$IDF('this') = \log (\text{number of documents/number of documents containing the word 'this'}) = \log (3/2) = 0.18$$

Similarly,

$$IDF('movie') = \log (3/3) = 0$$

$$IDF('is') = \log (3/2) = 0.18$$

Hence, we can observe that the words with less score like 0 contains less importance than the high frequency words in occurring in many documents. We can now compute the TF-IDF score for each word in the corpus. Words with a higher score are more important, and those with a lower score are less important:

$$(tf_idf)_{t,d} = tf_{t,d} * idf_t$$

The TF-IDF score for every word in Review 1:

$$TF-IDF ('this', \text{Review 1}) = TF ('this', \text{Review 1}) * IDF('this') = 1/8 * 0.18 = 0.0225$$

Similarly,

$$TF-IDF ('movie', \text{Review 1}) = 1/8 * 0 = 0$$

It has been observed that TF-IDF also gives larger values for less frequent words and is high when both IDF and TF values are high i.e., the word is rare in all the documents combined but frequent in a single document.

- C. Feature Selection: It is also called attribute selection. It is the process of selecting a significant attribute. It is an important step in the creation of a model. The features selected helps to remove redundant features from the model.
- D. Data Mining: The data mining techniques clustering and classification helps in prediction and generating the patterns.
- E. Evaluate: The evaluation techniques help to evaluate the model. The model with better accuracy yields optimum results.

Check Your Progress 1

- 1) Calculate the TF-IDF(‘good’ , Customer_Review 2) Review is : This product has good quality and good fit.
.....
.....
.....
- 2) What is the difference between Stemming and Lemmatization in Text Mining?
.....
.....
.....

12.4 TEXT MINING APPROACHES

The approaches of Text Mining are like a container of text documents and extraction of concepts is like a black box technique, from every document so much useful information can be taken out. The need to extract deep meaning from documents is important in terms of utilizing information to build up knowledge. The technology advancement in the field of text mining enables the task easier. There are various text mining techniques.

- 1) Numericizing Text
The challenges in performing text data analysis:
 - a) *Large number of documents:* There are large text documents containing huge amount of information. The objective is to find out the concept of the document, which can only be extracted from the randomly selected few documents. So, we cannot say how much would be the accuracy of the analysis as there would be large number of attributes.
 - b) *Removal of few characters, rare words, numbers etc.* The processing or indexing of the document needs preprocessing or cleaning of data. The words which appear in the document for very less time or infrequent can be removed.
 - c) *Removal of Stop words:* In text mining data analysis the number of words frequency is counted, then the number of times the word is appearing is counted. The documents

are classified based on the frequencies. In English language the stop words here referred to “the”, “a”, “of”, “an”, “since”. These are words which are used very often to frame the sentence but provide less or unique meaning to the analysts.

- Adopting pre-defined stop words.

Sample text with Stop Words	Without Stop Words
TextMining – A technique of data mining for analysis of web data	TextMining, technique, datamining, analysis, web, data
The movie was awesome	Movie, awesome
The product quality is bad	Product, quality, bad

d) *Synonyms Identification*: Similarly, for the text mining analysis the synonyms of the words are identified. These words are then combined for indexing.

e) *Stemming Algorithms*: The stemming algorithms is another preprocessing step before indexing of documents in text mining enables to reduction of irrelevant words from the main word in the text.

- Slicing the end or the beginning of words.
- Intent of removing affixes.



Lemmaization

- Reduction of a word to its base form.
- Grouping together different forms of the same word.



f) *Support for different languages* is quite significant in text mining process.

Check Your Progress 2

- 1) What is the difference between Stemming and Lemmatization in Text Mining?
-
-
-

12.4 WEB MINING

Web mining as the name suggests that it involves the mining of web data. The extraction of information from websites uses data mining techniques. It is an application based on data mining techniques. The parameters generally to be mined in web pages are hyperlinks, text or content of web pages, linked user activity between web pages of the same website or among different websites. All user activities are stored in a web server log file.

Web Mining can be referred as discovering interesting and useful information from Web content and usage

- 12.4.1 Web Mining performs various tasks such as:
- 1) Generating patterns existing in some websites, like customer buying behavior or navigation of web sites.
- 2) The web mining helps to retrieve faster results of the queries or the search text posted on the search engines like Google, Yahoo etc.
- 3) The ability to classify web documents according to the search performed on the ecommerce websites helps to increase businesses and transactions.

- 12.4.2 Web Mining Features
- Web search, e.g. Google, Yahoo, MSN, Ask, Froogle (comparison shopping), job ads (Flipdog)
 - The web mining is not like relation, it has text content and linkage structure.
 - On the www the user generated data is increasing rapidly. So, Googles’ usage logs are very huge in size. Data generated per day on google can be compared with the largest data warehouse unit.
 - Web mining can react in real-time with dynamic patterns generated on the web. In this no direct human interaction is involved.
 - Web Server: It maintains the entry of web log pages in the log file. This web log entries helps to identify the loyal or potential customers from ecommerce website or companies.
 - Web page is considered as a graph like structure, where pages are considered as nodes, hyperlinks as edges.
Pages = nodes, hyperlinks = edges
 - Ignore content
 - Directed graph
 - High linkage
 - 8-10 links/page on average
 - Power-law degree distribution

Applications of Web Mining in eCommerce can be observed as

- Recommendations: e.g. Netflix, Amazon
- improving conversion rate: next best product to offer
- Advertising, e.g. Google AdSense
- Fraud detection: click fraud detection,
- Improving Web site design and performance

There are three types of web mining as shown in the following figure 5.



Figure 5: Three types of Web Mining (source: internet)

12.5 WEB CONTENT MINING

Web content mining is used for taking out information, data from web pages. It extracts the information and knowledge from the content of web pages. In web content mining each page is considered as an individual document. The Hypertext markup language (HTML) provides the logical layout and structure of the webpage from where the lot of useful information can be extracted. This extracted data provides easy level aggregation of web pages.

- The usefulness of web content mining is that it helps in distinguish topics or heading content of the web pages. You might have observed whenever you write a search text on the google search engine, you get a list of suggestions. It is internally performed by web content mining.
- Web content mining scans or reads the images or text and group of web pages. The input queries posted by the users and display the result of search engines. For example, if a user wants to search some information about some hotel in a particular region, the list will be displayed.

12.5 WEB STRUCTURE MINING

Web structure mining helps the web pages to extract knowledge. This task is not easy as the challenge in accessing the unstructured web data link is huge requires lot of computational costs. Since, the web data is unstructured automatic generation of knowledge or pattern is difficult.

It is used to find out the link structural parameters of hyperlink. It helps to find out the network link or if the hyperlink is directly associated with the data. The social network analysis in web structure mining treats web as a directed graph, in which the webpages are as vertices associated with the hyperlinks.

12.6 MINING MULTIMEDIA DATA ON THE WEB

The websites are flooded with the multimedia data like, video, audio, images, and graphs. This multimedia data has different characteristics. The videos, images, audio, and pictures have different methods of archiving and retrieving the information. The multimedia data on the web has different properties this is the reason the typical multimedia data mining techniques cannot be applied. This web-based multimedia has texts and links. The text and links are the important features of the multimedia data to organize web pages. The better organization of web pages help in effective search operation. The web page layout mining can be applied to segregate the web pages into the set of multimedia semantic blocks from non-multimedia web pages. There are few web-based mining terminologies and algorithms to understand.

PageRank: This measure is used to count the number of pages the webpage is connected to other websites. It gives the importance of the webpage. The Google search engine uses the algorithm PageRank and rank the web page very significant if is frequently connected with the other webpages on the social network. It works on the concept of probability distribution representing the likelihood that a person on random click would reach to any page. It is assumed the equal distribution in the beginning of the computational process. This measure works on iterations. Iterating or repetition of page ranking process would help rank the web page closely reflecting to its true value.

HITS: This measure is used to rate the webpage. It is developed by Jon Kleinberg. It uses hubs and authorities to be determined from a web page. Hubs and Authorities defines a recursive relationship between web pages.

- This algorithm helps in web link structure and speeds up the search operation of a web page. Given a query to a Search Engine, the set of highly relevant web pages are called Roots. They are potential Authorities.
- Pages that are not very relevant but point to pages in the Root are called Hubs. Thus, an Authority is a page that many hubs link to whereas a Hub is a page that links to many authorities.

Page Layout analysis: It extracts and maintains the page-to-block, block-to-page relationships from link structure of web pages.

Vision page segmentation (VIPS) algorithm: It first extracts all the suitable blocks from the HTML Document Object Model (DOM) tree, and then it finds the separators between these blocks. Here separators denote the horizontal or vertical lines in a Web page that visually cross with no blocks. Based on these separators, the semantic tree of the Web page is constructed. A Web page can be represented as a set of blocks (leaf nodes of the semantic tree). Compared with DOM-based methods, the segments obtained by VIPS are more semantically aggregated. Noisy information, such as navigation, advertisement, and decoration can be easily removed because these elements are often placed in certain positions on a page. Contents with different topics are distinguished as separate blocks.

You can understand simply by considering following points,

- The web page contains links and links contained in different semantic blocks point to pages of different topics.
 - Calculate the significance of web page using algorithms PageRank or HITS.
- Split pages into semantic blocks

Apply link analysis on semantic block level. For example, in the below figure 1, it is clearly shown. From Figure 1, we can see the links in different blocks point to the pages with different topics. In this example, one link points to a page about entertainment and another link points to a page about sports.



Figure 6: example of a sample web page (new.yahoo.com), showing web page with different semantic blocks (red, green, and brown rectangular boxes). Every block has different importance in the web page. The links in different blocks points to the pages with different topics.

To analyze the web page containing multimedia data there is a technique known as Link analysis. It uses two most significant algorithms PageRank and HITS to analyze the significance of web pages. This technique uses each page as a single node in the web graph. But since, web page with multimedia has lot of data and links. So, cannot be considered as a single node in the graph. So, in this case the web page is partitioned into blocks using vision page segmentation also called VIPS algorithm. So, now after

extracting all the required information the semantic graph can be developed over world wide web in which each node represents a semantic topic or semantic structure of the web page.

VIPS algorithm helps in determining the text for web pages. This is the closely related text that provides content or text description of web pages and used to build image index. The web image search can then be performed using any traditional search technique. Google, Yahoo still uses this approach to search web image page.

Block-level Link analysis: The block-to-block model is quite useful for web image retrieval and web page categorization. It uses kinds of relationships, i.e., block-to-page and page-to-block. Let's see some definitions. Let P denote the set of all the web pages,

$$P = \{p_1, p_2, \dots, p_k\}, \text{ where } k \text{ is the number of web pages.}$$

Let B denote the set of all the blocks,

$$B = \{b_1, b_2, \dots, b_n\}, \text{ where } n \text{ is the number of blocks.}$$

It is important to note that, for each block there is only one page that contains that block. $b_i \in p_j$ means the block i is contained in the page j.

Block-Based Link Structure Analysis: This can be explained using matrix notations. Consider Z is the block-to-page matrix with dimension $n \times k$. Z can be formally defined as follows:

$$Z_{ij} = \begin{cases} 1/s_i & \text{if there is a link from block } i \text{ to page } j \\ 0 & \text{otherwise} \end{cases}$$

where s_i is the number of pages that block i links to. Z_{ij} can also be viewed as a probability of jumping from block i to page j.

The block-to-page relationship gives a more accurate and robust representation of the link structures of the web unlikely, HITS as at times it deviates from the web text information. It is used to organize the web image pages. The image graph deduced can be used to achieve high-quality web image clustering results. The web page graph for web image can be constructed by considering measuring which tells the relationship between blocks and images, block-to-image, and image-to-block. Likewise, page-to-block and block-to-pages.

12.7 AUTOMATIC CLASSIFICATION OF WEB DOCUMENTS

The categorization of web pages into the respective subjects or domains is called classification of web documents. For example, in the following figure, it has shown various categories like, books, electronics etc. let's say you are doing online shopping on the amazon website and there are so many webpages so when you search for electronics the respective web page containing the information of electronics is displayed. This is the classification of products which is done on the textual and image contents.



Figure 7: Types of web documents containing different types of data.

The problem with the classification of web documents is that every time the model is to be constructed by applying some algorithms to classify the document is mammoth task. The large number of unorganized web pages may have redundant documents.

The automated document classification of web pages is based on the textual content. The model requires initial training phase of document classifiers for each category based on training examples.

In the figure below it is shown that the documents can be collected from different sources. After the collection of documents data cleansing is performed using extraction transformation and loading techniques. The documents can be grouped according to the similarity measure (grouping of the documents according to the similarity between the documents) and TF-IDF. The machine learning model is created and executed, and different clusters are generated.

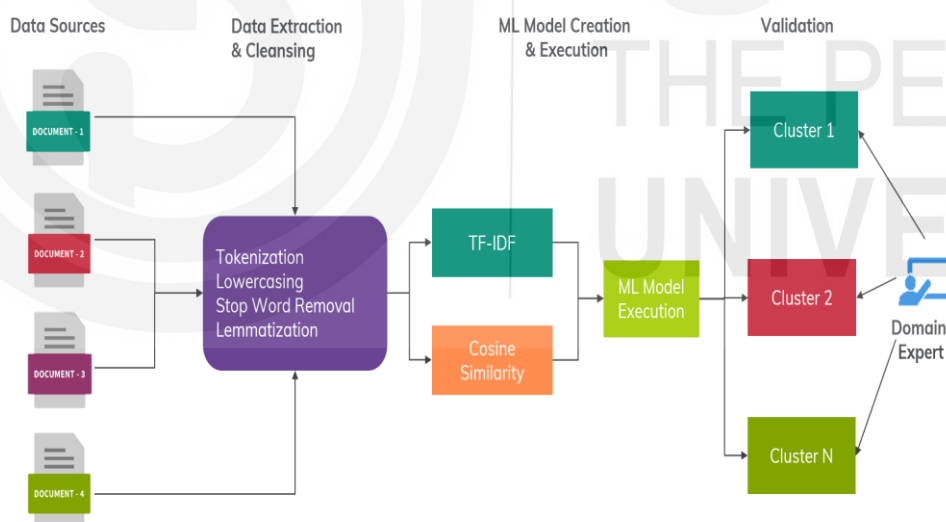


Figure 8: Automatic Classification of Web documents

Automated document classification identifies the documents and groups the relevant documents without any external efforts. There are various tools available in the market like RapidMiner, Azure, Machine Learning Studio, Amazon Sage maker, KNIME and Python. The trained model automatically reads the data from documents (PDF, DOC, PPT) and classifies the data according to the category of the document. This trained model is already trained with the Machine Learning and Natural Language Processing techniques. There are domain experts who perform this task efficiently.

Benefits of Automatic Document Classification System

- 1) It is more efficient system of classification as produces improved accuracy of results and speed up the process of classification.
- 2) The system incurs in less operational costs
- 3) Easy data store and retrieval.
- 4) It organizes the files and documents in a better streamlined way.

12.8 WEB USAGE MINING

Web Usage Mining: This term web usage mining is used to mine the data of web log records. It helps to find out the navigation and access patterns of web pages. This mining helps to study the user access pattern of web pages. It uses webserver log files to mine or track the behavior of the user. It helps to know the interests or the perception of the user, while the web structure mining and content mining focusses more on the intention of the user. After analyzing and exploring the regularities in web blogs, can identify potential customers for ecommerce. It enhances the quality and delivery of internet connection and the improves web server system performance.

The web log file contains all the information and data pertaining to web sites and user activities. The URL has a lot of information related with IP address of the request generated and at what time the request initiated. Like Amazon which is an ecommerce website where millions of transactions take place. So, huge number of web log entries are recorded gigabytes every day. So, it is quite important to have efficient web mining techniques to generate relevant information in less time.

The web mining technique involves taking out knowledge from the raw data stored in web log files. So, these raw data needs preprocessing cleaning, transformation and loading of data to retrieve and analyze important information. The web log data is ready after preprocessing of data.

The second source of information is URL (Uniform Resource Locator), it contains IP address, time, and web page content. Using the information of web log, a multidimensional cube is constructed. so that the knowledge can be extracted from various dimensions. The analysis of OLAP cube helped to identify top number of users and top users accessing the web page. Now the data mining techniques can be applied to find out the necessary information about the user accessing behavior of web pages and using association rule mining a pattern can be generated. The web log data can be used majorly for tracking and analyzing the user behavior on the social network.

The weblog files helped used to analyze system performance, web caching, identifying web page, understanding the nature of web traffic, understanding the nature of web traffic, purpose of that network behavior. For example, some studies have proposed adaptive sites: websites that improve themselves by learning from user access patterns. Weblog analysis may also help build customized Web services for individual users.

Check Your Progress 3

1) What web log file contains?

.....

2) Show the diagram for Image mining process with brief explanation?

.....

1.12 SOLUTIONS/ANSWERS

Check Your Progress 1

Answer 1: Challenges in Web Mining:

- The web page link structure is quite complex to analyze as the web page is linked with many more other web pages. There exists lot of documents in the digital library of the web. The data in this library is not organized.
- The web data is uploaded on the web pages dynamically on regular basis.
- Diversity of client networks having different interests, backgrounds, and usage purposes. The network is growing rapidly.
- Another challenge is to extract the relevant data for subject or domain or user.

Answer 2: Techniques used to analyze the web usage patterns are as follows:

- 1) Session and web page visitor analysis: The web log file contains the record of users visiting web pages, frequency of visit, days, and the duration for how long the user stays on the web page.
- 2) OLAP (Online Analytical Processing): OLAP can be performed on different parts of log related data in a certain interval of time.
- 3) Web Structure Mining: It produces the structural summary of the web pages. It identifies the web page and indirect or direct link of that page with others. It helps the companies to identify the commercial link of business websites.

Answer 3: Application of Web Mining:

- Digital Marketing
- Data analysis on website and application accomplishment.
- User behavior analysis
- Advertising and campaign accomplishment analysis.

Check Your Progress 2

Answer 1: The main difference between BLHITS (Block HITS) and HITS

BLHITS	HITS
Links are from blocks to pages	Links from pages to pages
Root is top ranked blocks	Root is top ranked pages
Analyses only top ranked block links	Analyses all the links of all the pages
Content analysis at block level	Content analysis at page level

Check Your Progress 3

Answer 1: Weblog data can be used with Web content and Web linkage structure mining to aid Web page ranking, Web document classification, and the creation of a multilayered Web information base, because it provides information about what kind of users will visit what types of Web pages. The mining of a user's interactions with the web is a particularly intriguing use of Web usage mining.

Answer 2:

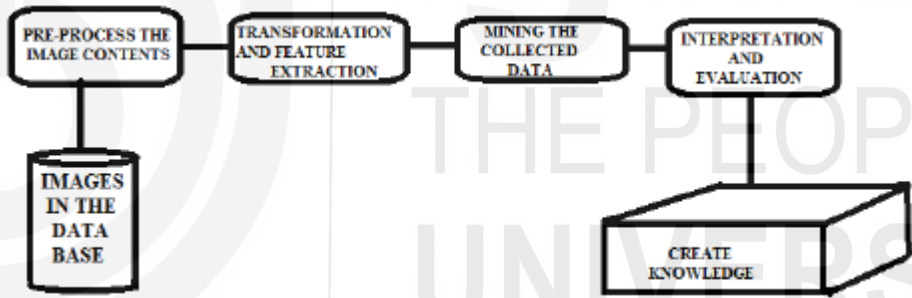


Figure : Image Web Mining